

College Physics 2e

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CHAPTER 33

Particle Physics

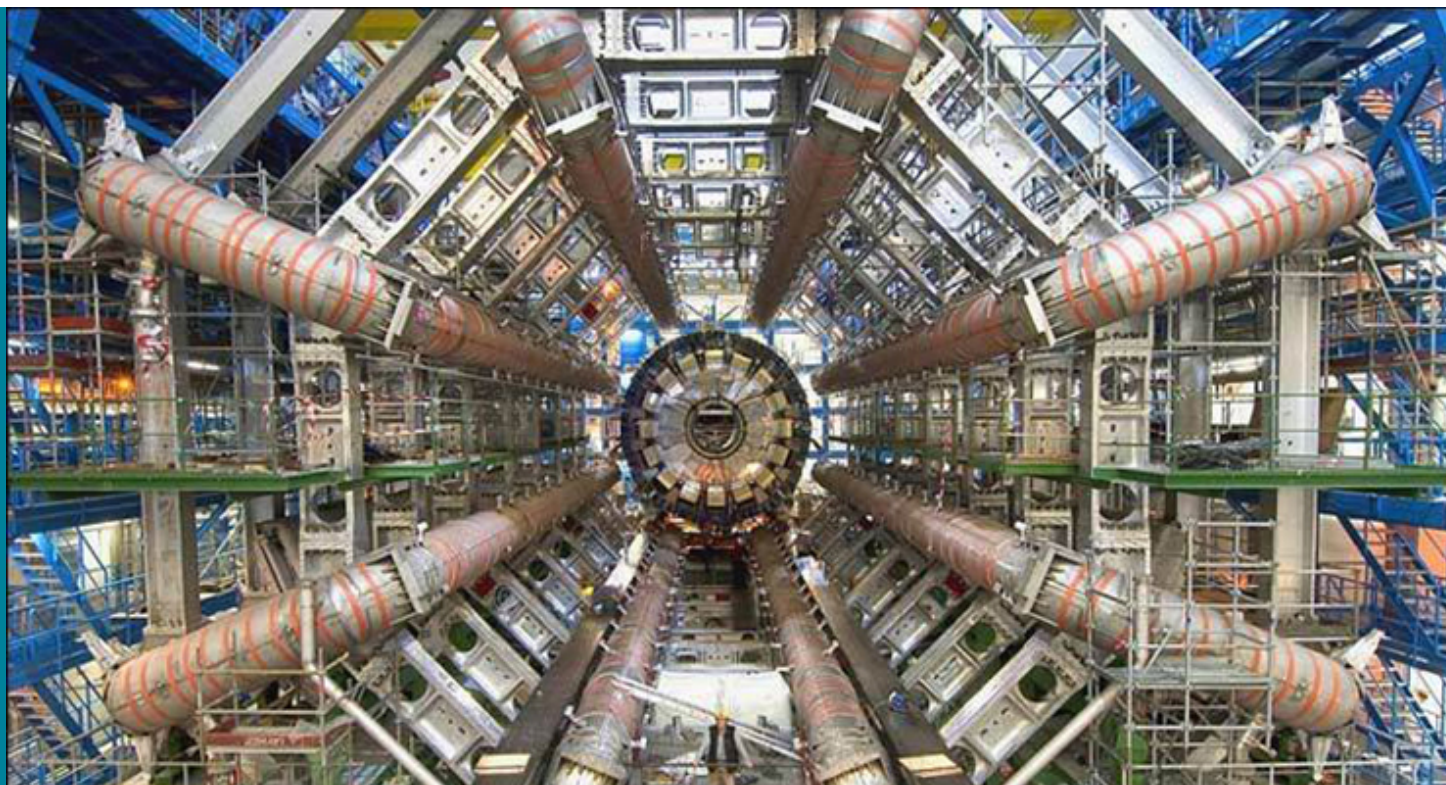


FIGURE 33.1 Part of the Large Hadron Collider at CERN, on the border of Switzerland and France. The LHC is a particle accelerator, designed to study fundamental particles. (credit: Image Editor, Flickr)

CHAPTER OUTLINE

33.1 The Yukawa Particle and the Heisenberg Uncertainty Principle Revisited

33.2 The Four Basic Forces

33.3 Accelerators Create Matter from Energy

33.4 Particles, Patterns, and Conservation Laws

33.5 Quarks: Is That All There Is?

33.6 GUTs: The Unification of Forces

INTRODUCTION TO PARTICLE PHYSICS [Learning Objectives](#)

By the end of this section, you will be able to:

Following ideas remarkably similar to those of the ancient Greeks, we continue to look for smaller and smaller structures in nature, hoping ultimately to find and understand the most fundamental building blocks that exist. Atomic physics deals with the smallest units of elements and compounds. In its study, we have found a relatively small number of atoms with systematic properties that explained a tremendous range of phenomena. Nuclear physics is concerned with the nuclei of atoms and their substructures. Here, a smaller number of components—the proton and neutron—make up all nuclei. Exploring the systematic behavior of their interactions has revealed even more about matter, forces, and energy. **Particle physics** deals with the substructures of atoms and nuclei and is particularly aimed at finding those truly fundamental particles that have no further substructure. Just as in atomic and nuclear physics, we have found a complex array of particles and properties with systematic characteristics analogous to the periodic table and the chart of nuclides. An underlying structure is apparent, and there is some reason to think that we are finding particles that have no substructure. Of course, we have been in similar situations

before. For example, atoms were once thought to be the ultimate substructure. Perhaps we will find deeper and deeper structures and never come to an ultimate substructure. We may never really know, as indicated in [Figure 33.2](#).

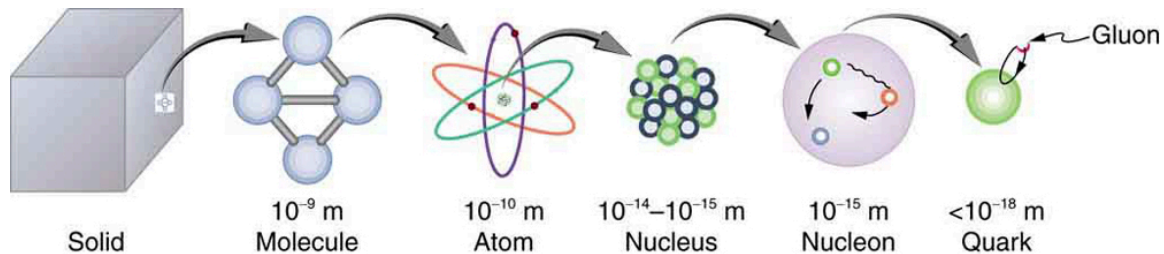


FIGURE 33.2 The properties of matter are based on substructures called molecules and atoms. Atoms have the substructure of a nucleus with orbiting electrons, the interactions of which explain atomic properties. Protons and neutrons, the interactions of which explain the stability and abundance of elements, form the substructure of nuclei. Protons and neutrons are not fundamental—they are composed of quarks. Like electrons and a few other particles, quarks may be the fundamental building blocks of all there is, lacking any further substructure. But the story is not complete, because quarks and electrons may have substructure smaller than details that are presently observable.

This chapter covers the basics of particle physics as we know it today. An amazing convergence of topics is evolving in particle physics. We find that some particles are intimately related to forces, and that nature on the smallest scale may have its greatest influence on the large-scale character of the universe. It is an adventure exceeding the best science fiction because it is not only fantastic, it is real.

33.1 The Yukawa Particle and the Heisenberg Uncertainty Principle Revisited

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Define Yukawa particle.
- State the Heisenberg uncertainty principle.
- Describe pion.
- Estimate the mass of a pion.
- Explain meson.

Particle physics as we know it today began with the ideas of Hideki Yukawa in 1935. Physicists had long been concerned with how forces are transmitted, finding the concept of fields, such as electric and magnetic fields to be very useful. A field surrounds an object and carries the force exerted by the object through space. Yukawa was interested in the strong nuclear force in particular and found an ingenious way to explain its short range. His idea is a blend of particles, forces, relativity, and quantum mechanics that is applicable to all forces. Yukawa proposed that force is transmitted by the exchange of particles (called carrier particles). The field consists of these carrier particles.

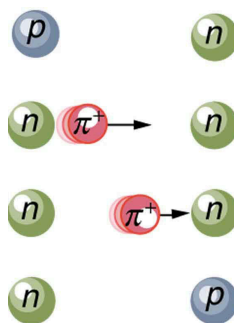


FIGURE 33.3 The strong nuclear force is transmitted between a proton and neutron by the creation and exchange of a pion. The pion is created through a temporary violation of conservation of mass-energy and travels from the proton to the neutron and is recaptured. It is not directly observable and is called a virtual particle. Note that the proton and neutron change identity in the process. The range of the force is limited by the fact that the pion can only exist for the short time allowed by the Heisenberg uncertainty principle. Yukawa used the finite range of the strong nuclear force to estimate the mass of the pion; the shorter the range, the larger the mass of the carrier particle.

Specifically for the strong nuclear force, Yukawa proposed that a previously unknown particle, now called a **pion**, is exchanged between nucleons, transmitting the force between them. [Figure 33.3](#) illustrates how a pion would carry a

force between a proton and a neutron. The pion has mass and can only be created by violating the conservation of mass-energy. This is allowed by the Heisenberg uncertainty principle if it occurs for a sufficiently short period of time. As discussed in [Probability: The Heisenberg Uncertainty Principle](#) the Heisenberg uncertainty principle relates the uncertainties ΔE in energy and Δt in time by

$$\Delta E \Delta t \geq \frac{h}{4\pi}, \quad 33.1$$

where h is Planck's constant. Therefore, conservation of mass-energy can be violated by an amount ΔE for a time $\Delta t \approx \frac{h}{4\pi\Delta E}$ in which time no process can detect the violation. This allows the temporary creation of a particle of mass m , where $\Delta E = mc^2$. The larger the mass and the greater the ΔE , the shorter is the time it can exist. This means the range of the force is limited, because the particle can only travel a limited distance in a finite amount of time. In fact, the maximum distance is $d \approx c\Delta t$, where c is the speed of light. The pion must then be captured and, thus, cannot be directly observed because that would amount to a permanent violation of mass-energy conservation. Such particles (like the pion above) are called **virtual particles**, because they cannot be directly observed but their *effects* can be directly observed. Realizing all this, Yukawa used the information on the range of the strong nuclear force to estimate the mass of the pion, the particle that carries it. The steps of his reasoning are approximately retraced in the following worked example:



EXAMPLE 33.1

Calculating the Mass of a Pion

Taking the range of the strong nuclear force to be about 1 fermi (10^{-15} m), calculate the approximate mass of the pion carrying the force, assuming it moves at nearly the speed of light.

Strategy

The calculation is approximate because of the assumptions made about the range of the force and the speed of the pion, but also because a more accurate calculation would require the sophisticated mathematics of quantum mechanics. Here, we use the Heisenberg uncertainty principle in the simple form stated above, as developed in [Probability: The Heisenberg Uncertainty Principle](#). First, we must calculate the time Δt that the pion exists, given that the distance it travels at nearly the speed of light is about 1 fermi. Then, the Heisenberg uncertainty principle can be solved for the energy ΔE , and from that the mass of the pion can be determined. We will use the units of MeV/c^2 for mass, which are convenient since we are often considering converting mass to energy and vice versa.

Solution

The distance the pion travels is $d \approx c\Delta t$, and so the time during which it exists is approximately

$$\begin{aligned} \Delta t &\approx \frac{d}{c} = \frac{10^{-15} \text{ m}}{3.0 \times 10^8 \text{ m/s}} \\ &\approx 3.3 \times 10^{-24} \text{ s}. \end{aligned} \quad 33.2$$

Now, solving the Heisenberg uncertainty principle for ΔE gives

$$\Delta E \approx \frac{h}{4\pi\Delta t} \approx \frac{6.63 \times 10^{-34} \text{ J} \cdot \text{s}}{4\pi(3.3 \times 10^{-24} \text{ s})}. \quad 33.3$$

Solving this and converting the energy to MeV gives

$$\Delta E \approx (1.6 \times 10^{-11} \text{ J}) \frac{1 \text{ MeV}}{1.6 \times 10^{-13} \text{ J}} = 100 \text{ MeV}. \quad 33.4$$

Mass is related to energy by $\Delta E = mc^2$, so that the mass of the pion is $m = \Delta E/c^2$, or

$$m \approx 100 \text{ MeV}/c^2. \quad 33.5$$

Discussion

This is about 200 times the mass of an electron and about one-tenth the mass of a nucleon. No such particles were known at the time Yukawa made his bold proposal.

Yukawa's proposal of particle exchange as the method of force transfer is intriguing. But how can we verify his proposal if we cannot observe the virtual pion directly? If sufficient energy is in a nucleus, it would be possible to free the pion—that is, to create its mass from external energy input. This can be accomplished by collisions of energetic particles with nuclei, but energies greater than 100 MeV are required to conserve both energy and momentum. In 1947, pions were observed in cosmic-ray experiments, which were designed to supply a small flux of high-energy protons that may collide with nuclei. Soon afterward, accelerators of sufficient energy were creating pions in the laboratory under controlled conditions. Three pions were discovered, two with charge and one neutral, and given the symbols π^+ , π^- , and π^0 , respectively. The masses of π^+ and π^- are identical at 139.6 MeV/ c^2 , whereas π^0 has a mass of 135.0 MeV/ c^2 . These masses are close to the predicted value of 100 MeV/ c^2 and, since they are intermediate between electron and nucleon masses, the particles are given the name **meson** (now an entire class of particles, as we shall see in [Particles, Patterns, and Conservation Laws](#)).

The pions, or π -mesons as they are also called, have masses close to those predicted and feel the strong nuclear force. Another previously unknown particle, now called the muon, was discovered during cosmic-ray experiments in 1936 (one of its discoverers, Seth Neddermeyer, also originated the idea of implosion for plutonium bombs). Since the mass of a muon is around 106 MeV/ c^2 , at first it was thought to be the particle predicted by Yukawa. But it was soon realized that muons do not feel the strong nuclear force and could not be Yukawa's particle. Their role was unknown, causing the respected physicist I. I. Rabi to comment, "Who ordered that?" This remains a valid question today. We have discovered hundreds of subatomic particles; the roles of some are only partially understood. But there are various patterns and relations to forces that have led to profound insights into nature's secrets.

33.2 The Four Basic Forces

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- State the four basic forces.
- Explain the Feynman diagram for the exchange of a virtual photon between two positive charges.
- Define QED.
- Describe the Feynman diagram for the exchange of a between a proton and a neutron.

As first discussed in [Problem-Solving Strategies](#) and mentioned at various points in the text since then, there are only four distinct basic forces in all of nature. This is a remarkably small number considering the myriad phenomena they explain. Particle physics is intimately tied to these four forces. Certain fundamental particles, called carrier particles, carry these forces, and all particles can be classified according to which of the four forces they feel. The table given below summarizes important characteristics of the four basic forces.

Force	Approximate relative strength	Range	+/- ¹	Carrier particle
Gravity	10^{-38}	∞	+ only	Graviton (conjectured)
Electromagnetic	10^{-2}	∞	+/-	Photon (observed)
Weak force	10^{-13}	$< 10^{-18}$ m	+/-	W^+ , W^- , Z^0 (observed ²)
Strong force	1	$< 10^{-15}$ m	+/-	Gluons (conjectured ³)

TABLE 33.1 Properties of the Four Basic Forces

1 + attractive; - repulsive; + /- both.

2 Predicted by theory and first observed in 1983.

3 Eight proposed—indirect evidence of existence. Underlie meson exchange.

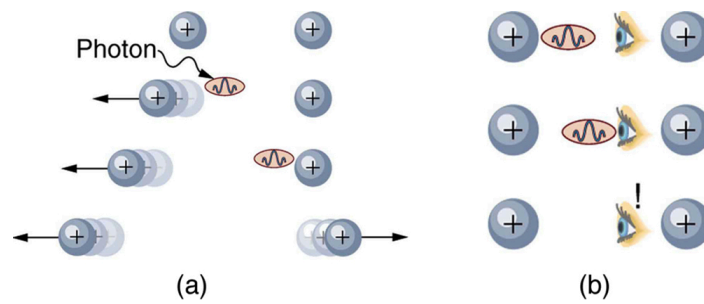


FIGURE 33.4 The first image shows the exchange of a virtual photon transmitting the electromagnetic force between charges, just as virtual pion exchange carries the strong nuclear force between nucleons. The second image shows that the photon cannot be directly observed in its passage, because this would disrupt it and alter the force. In this case it does not get to the other charge.

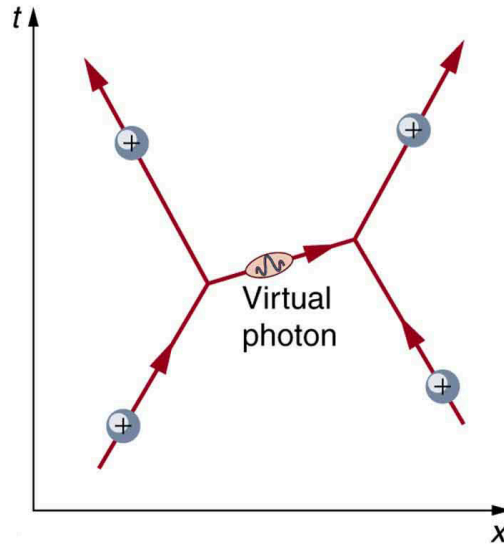


FIGURE 33.5 The Feynman diagram for the exchange of a virtual photon between two positive charges illustrates how the electromagnetic force is transmitted on a quantum mechanical scale. Time is graphed vertically while the distance is graphed horizontally. The two positive charges are seen to be repelled by the photon exchange.

Although these four forces are distinct and differ greatly from one another under all but the most extreme circumstances, we can see similarities among them. (In [GUTs: the Unification of Forces](#), we will discuss how the four forces may be different manifestations of a single unified force.) Perhaps the most important characteristic among the forces is that they are all transmitted by the exchange of a carrier particle, exactly like what Yukawa had in mind for the strong nuclear force. Each carrier particle is a virtual particle—it cannot be directly observed while transmitting the force. [Figure 33.4](#) shows the exchange of a virtual photon between two positive charges. The photon cannot be directly observed in its passage, because this would disrupt it and alter the force.

[Figure 33.5](#) shows a way of graphing the exchange of a virtual photon between two positive charges. This graph of time versus position is called a **Feynman diagram**, after the brilliant American physicist Richard Feynman (1918–1988) who developed it.

[Figure 33.6](#) is a Feynman diagram for the exchange of a virtual pion between a proton and a neutron representing the same interaction as in [Figure 33.3](#). Feynman diagrams are not only a useful tool for visualizing interactions at the quantum mechanical level, they are also used to calculate details of interactions, such as their strengths and probability of occurring. Feynman was one of the theorists who developed the field of **quantum electrodynamics** (QED), which is the quantum mechanics of electromagnetism. QED has been spectacularly successful in describing electromagnetic interactions on the submicroscopic scale. Feynman was an inspiring teacher, had a colorful personality, and made a profound impact on generations of physicists. He shared the 1965 Nobel Prize with Julian Schwinger and S. I. Tomonaga for work in QED with its deep implications for particle physics.

Why is it that particles called gluons are listed as the carrier particles for the strong nuclear force when, in [The Yukawa Particle and the Heisenberg Uncertainty Principle Revisited](#), we saw that pions apparently carry that force? The answer is that pions are exchanged but they have a substructure and, as we explore it, we find that the strong

force is actually related to the indirectly observed but more fundamental **gluons**. In fact, all the carrier particles are thought to be fundamental in the sense that they have no substructure. Another similarity among carrier particles is that they are all bosons (first mentioned in [Patterns in Spectra Reveal More Quantization](#)), having integral intrinsic spins.

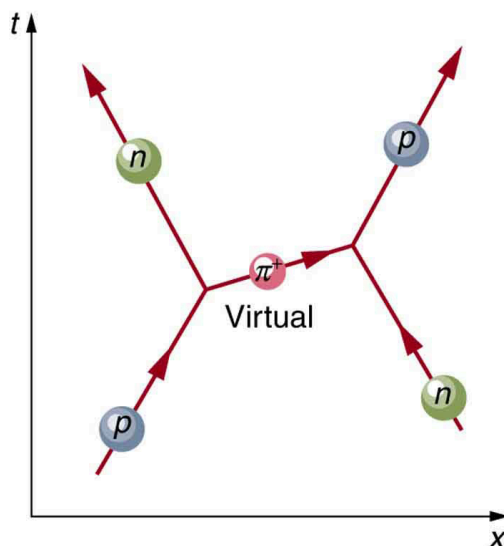


FIGURE 33.6 The image shows a Feynman diagram for the exchange of a π^+ between a proton and a neutron, carrying the strong nuclear force between them. This diagram represents the situation shown more pictorially in [Figure 33.4](#).

There is a relationship between the mass of the carrier particle and the range of the force. The photon is massless and has energy. So, the existence of (virtual) photons is possible only by virtue of the Heisenberg uncertainty principle and can travel an unlimited distance. Thus, the range of the electromagnetic force is infinite. This is also true for gravity. It is infinite in range because its carrier particle, the graviton, has zero rest mass. (Gravity is the most difficult of the four forces to understand on a quantum scale because it affects the space and time in which the others act. But gravity is so weak that its effects are extremely difficult to observe quantum mechanically. We shall explore it further in [General Relativity and Quantum Gravity](#)). The W^+ , W^- , and Z^0 particles that carry the weak nuclear force have mass, accounting for the very short range of this force. In fact, the W^+ , W^- , and Z^0 are about 1000 times more massive than pions, consistent with the fact that the range of the weak nuclear force is about 1/1000 that of the strong nuclear force. Gluons are actually massless, but since they act inside massive carrier particles like pions, the strong nuclear force is also short ranged.

The relative strengths of the forces given in the [Table 33.1](#) are those for the most common situations. When particles are brought very close together, the relative strengths change, and they may become identical at extremely close range. As we shall see in [GUTs: the Unification of Forces](#), carrier particles may be altered by the energy required to bring particles very close together—in such a manner that they become identical.

33.3 Accelerators Create Matter from Energy

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- State the principle of a cyclotron.
- Explain the principle of a synchrotron.
- Describe the voltage needed by an accelerator between accelerating tubes.
- State Fermilab's accelerator principle.

Before looking at all the particles we now know about, let us examine some of the machines that created them. The fundamental process in creating previously unknown particles is to accelerate known particles, such as protons or electrons, and direct a beam of them toward a target. Collisions with target nuclei provide a wealth of information, such as information obtained by Rutherford using energetic helium nuclei from natural α radiation. But if the energy of the incoming particles is large enough, new matter is sometimes created in the collision. The more energy input or ΔE , the more matter m can be created, since $m = \Delta E/c^2$. Limitations are placed on what can occur by known

conservation laws, such as conservation of mass-energy, momentum, and charge. Even more interesting are the unknown limitations provided by nature. Some expected reactions do occur, while others do not, and still other unexpected reactions may appear. New laws are revealed, and the vast majority of what we know about particle physics has come from accelerator laboratories. It is the particle physicist's favorite indoor sport, which is partly inspired by theory.

Early Accelerators

An early accelerator is a relatively simple, large-scale version of the electron gun. The **Van de Graaff** (named after the Dutch physicist), which you have likely seen in physics demonstrations, is a small version of the ones used for nuclear research since their invention for that purpose in 1932. For more, see [Figure 33.7](#). These machines are electrostatic, creating potentials as great as 50 MV, and are used to accelerate a variety of nuclei for a range of experiments. Energies produced by Van de Graaffs are insufficient to produce new particles, but they have been instrumental in exploring several aspects of the nucleus. Another, equally famous, early accelerator is the **cyclotron**, invented in 1930 by the American physicist, E. O. Lawrence (1901–1958). For a visual representation with more detail, see [Figure 33.8](#). Cyclotrons use fixed-frequency alternating electric fields to accelerate particles. The particles spiral outward in a magnetic field, making increasingly larger radius orbits during acceleration. This clever arrangement allows the successive addition of electric potential energy and so greater particle energies are possible than in a Van de Graaff. Lawrence was involved in many early discoveries and in the promotion of physics programs in American universities. He was awarded the 1939 Nobel Prize in Physics for the cyclotron and nuclear activations, and he has an element and two major laboratories named for him.

A **synchrotron** is a version of a cyclotron in which the frequency of the alternating voltage and the magnetic field strength are increased as the beam particles are accelerated. Particles are made to travel the same distance in a shorter time with each cycle in fixed-radius orbits. A ring of magnets and accelerating tubes, as shown in [Figure 33.9](#), are the major components of synchrotrons. Accelerating voltages are synchronized (i.e., occur at the same time) with the particles to accelerate them, hence the name. Magnetic field strength is increased to keep the orbital radius constant as energy increases. High-energy particles require strong magnetic fields to steer them, so superconducting magnets are commonly employed. Still limited by achievable magnetic field strengths, synchrotrons need to be very large at very high energies, since the radius of a high-energy particle's orbit is very large. Radiation caused by a magnetic field accelerating a charged particle perpendicular to its velocity is called **synchrotron radiation** in honor of its importance in these machines. Synchrotron radiation has a characteristic spectrum and polarization, and can be recognized in cosmic rays, implying large-scale magnetic fields acting on energetic and charged particles in deep space. Synchrotron radiation produced by accelerators is sometimes used as a source of intense energetic electromagnetic radiation for research purposes.



FIGURE 33.7 An artist's rendition of a Van de Graaff generator.

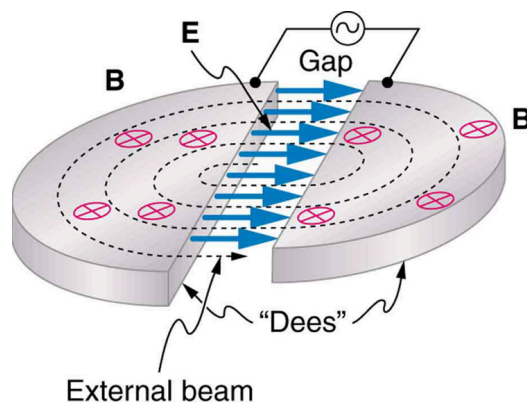


FIGURE 33.8 Cyclotrons use a magnetic field to cause particles to move in circular orbits. As the particles pass between the plates of the Dees, the voltage across the gap is oscillated to accelerate them twice in each orbit.

Modern Behemoths and Colliding Beams

Physicists have built ever-larger machines, first to reduce the wavelength of the probe and obtain greater detail, then to put greater energy into collisions to create new particles. Each major energy increase brought new information, sometimes producing spectacular progress, motivating the next step. One major innovation was driven by the desire to create more massive particles. Since momentum needs to be conserved in a collision, the particles created by a beam hitting a stationary target should recoil. This means that part of the energy input goes into recoil kinetic energy, significantly limiting the fraction of the beam energy that can be converted into new particles. One solution to this problem is to have head-on collisions between particles moving in opposite directions. **Colliding beams** are made to meet head-on at points where massive detectors are located. Since the total incoming momentum is zero, it is possible to create particles with momenta and kinetic energies near zero. Particles with masses equivalent to twice the beam energy can thus be created. Another innovation is to create the antimatter counterpart of the beam particle, which thus has the opposite charge and circulates in the opposite direction in the same beam pipe. For a schematic representation, see [Figure 33.10](#).

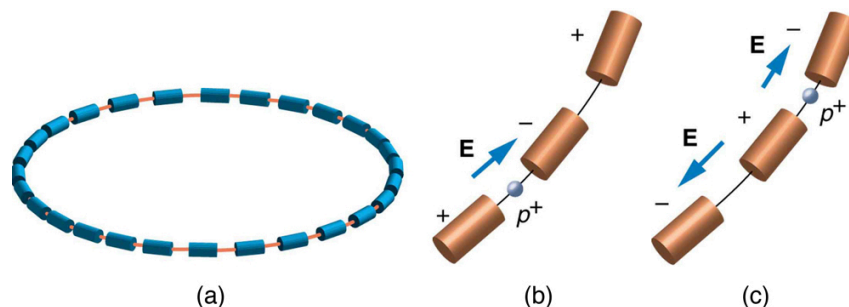


FIGURE 33.9 (a) A synchrotron has a ring of magnets and accelerating tubes. The frequency of the accelerating voltages is increased to cause the beam particles to travel the same distance in shorter time. The magnetic field should also be increased to keep each beam burst traveling in a fixed-radius path. Limits on magnetic field strength require these machines to be very large in order to accelerate particles to very high energies. (b) A positive particle is shown in the gap between accelerating tubes. (c) While the particle passes through the tube, the potentials are reversed so that there is another acceleration at the next gap. The frequency of the reversals needs to be varied as the particle is accelerated to achieve successive accelerations in each gap.

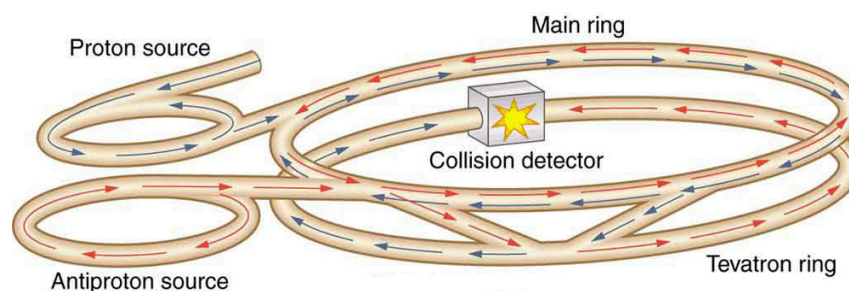


FIGURE 33.10 This schematic shows the two rings of Fermilab's accelerator and the scheme for colliding protons and antiprotons (not to scale).

Detectors capable of finding the new particles in the spray of material that emerges from colliding beams are as

impressive as the accelerators. While the Fermilab Tevatron had proton and antiproton beam energies of about 1 TeV, so that it can create particles up to $2 \text{ TeV}/c^2$, the Large Hadron Collider (LHC) at the European Center for Nuclear Research (CERN) has achieved beam energies of 3.5 TeV, so that it has a 7-TeV collision energy; CERN hopes to double the beam energy in 2014. The now-canceled Superconducting Super Collider was being constructed in Texas with a design energy of 20 TeV to give a 40-TeV collision energy. It was to be an oval 30 km in diameter. Its cost as well as the politics of international research funding led to its demise.

In addition to the large synchrotrons that produce colliding beams of protons and antiprotons, there are other large electron-positron accelerators. The oldest of these was a straight-line or **linear accelerator**, called the Stanford Linear Accelerator (SLAC), which accelerated particles up to 50 GeV as seen in [Figure 33.11](#). Positrons created by the accelerator were brought to the same energy and collided with electrons in specially designed detectors. Linear accelerators use accelerating tubes similar to those in synchrotrons, but aligned in a straight line. This helps eliminate synchrotron radiation losses, which are particularly severe for electrons made to follow curved paths. CERN had an electron-positron collider appropriately called the Large Electron-Positron Collider (LEP), which accelerated particles to 100 GeV and created a collision energy of 200 GeV. It was 8.5 km in diameter, while the SLAC machine was 3.2 km long.

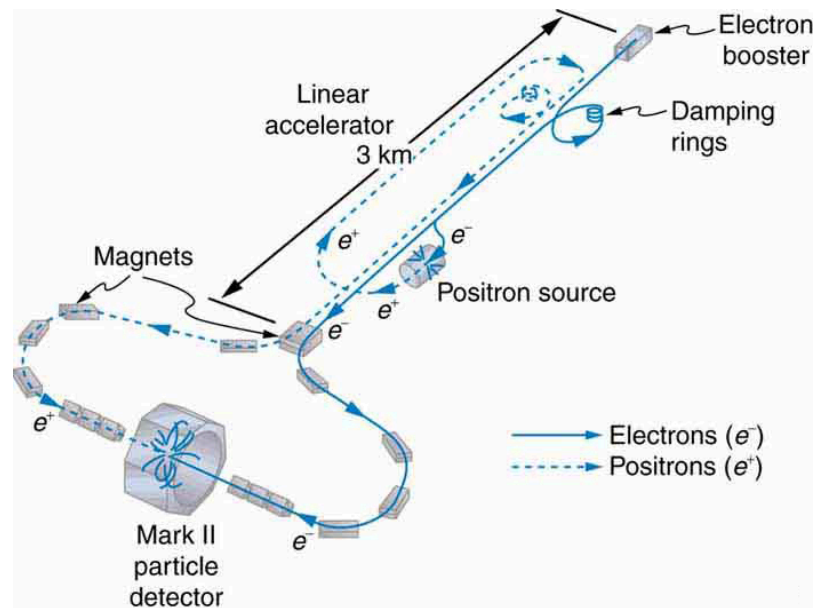


FIGURE 33.11 The Stanford Linear Accelerator was 3.2 km long and had the capability of colliding electron and positron beams. SLAC was also used to probe nucleons by scattering extremely short wavelength electrons from them. This produced the first convincing evidence of a quark structure inside nucleons in an experiment analogous to those performed by Rutherford long ago.



EXAMPLE 33.2

Calculating the Voltage Needed by the Accelerator Between Accelerating Tubes

A linear accelerator designed to produce a beam of 800-MeV protons has 2000 accelerating tubes. What average voltage must be applied between tubes (such as in the gaps in [Figure 33.9](#)) to achieve the desired energy?

Strategy

The energy given to the proton in each gap between tubes is $PE_{\text{elec}} = qV$ where q is the proton's charge and V is the potential difference (voltage) across the gap. Since $q = q_e = 1.6 \times 10^{-19} \text{ C}$ and $1 \text{ eV} = (1 \text{ V})(1.6 \times 10^{-19} \text{ C})$, the proton gains 1 eV in energy for each volt across the gap that it passes through. The AC voltage applied to the tubes is timed so that it adds to the energy in each gap. The effective voltage is the sum of the gap voltages and equals 800 MV to give each proton an energy of 800 MeV.

Solution

There are 2000 gaps and the sum of the voltages across them is 800 MV; thus,

$$V_{\text{gap}} = \frac{800 \text{ MV}}{2000} = 400 \text{ kV}.$$

33.6

Discussion

A voltage of this magnitude is not difficult to achieve in a vacuum. Much larger gap voltages would be required for higher energy, such as those at the 50-GeV SLAC facility. Synchrotrons are aided by the circular path of the accelerated particles, which can orbit many times, effectively multiplying the number of accelerations by the number of orbits. This makes it possible to reach energies greater than 1 TeV.

33.4 Particles, Patterns, and Conservation Laws

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Define matter and antimatter.
- Outline the differences between hadrons and leptons.
- State the differences between mesons and baryons.

In the early 1930s only a small number of subatomic particles were known to exist—the proton, neutron, electron, photon and, indirectly, the neutrino. Nature seemed relatively simple in some ways, but mysterious in others. Why, for example, should the particle that carries positive charge be almost 2000 times as massive as the one carrying negative charge? Why does a neutral particle like the neutron have a magnetic moment? Does this imply an internal structure with a distribution of moving charges? Why is it that the electron seems to have no size other than its wavelength, while the proton and neutron are about 1 fermi in size? So, while the number of known particles was small and they explained a great deal of atomic and nuclear phenomena, there were many unexplained phenomena and hints of further substructures.

Things soon became more complicated, both in theory and in the prediction and discovery of new particles. In 1928, the British physicist P.A.M. Dirac (see [Figure 33.12](#)) developed a highly successful relativistic quantum theory that laid the foundations of quantum electrodynamics (QED). His theory, for example, explained electron spin and magnetic moment in a natural way. But Dirac's theory also predicted negative energy states for free electrons. By 1931, Dirac, along with Oppenheimer, realized this was a prediction of positively charged electrons (or positrons). In 1932, American physicist Carl Anderson discovered the positron in cosmic ray studies. The positron, or e^+ , is the same particle as emitted in β^+ decay and was the first antimatter that was discovered. In 1935, Yukawa predicted pions as the carriers of the strong nuclear force, and they were eventually discovered. Muons were discovered in cosmic ray experiments in 1937, and they seemed to be heavy, unstable versions of electrons and positrons. After World War II, accelerators energetic enough to create these particles were built. Not only were predicted and known particles created, but many unexpected particles were observed. Initially called elementary particles, their numbers proliferated to dozens and then hundreds, and the term “particle zoo” became the physicist's lament at the lack of simplicity. But patterns were observed in the particle zoo that led to simplifying ideas such as quarks, as we shall soon see.



FIGURE 33.12 P.A.M. Dirac's theory of relativistic quantum mechanics not only explained a great deal of what was known, it also predicted antimatter. (credit: Cambridge University, Cavendish Laboratory)

Matter and Antimatter

The positron was only the first example of antimatter. Every particle in nature has an antimatter counterpart, although some particles, like the photon, are their own antiparticles. Antimatter has charge opposite to that of matter (for example, the positron is positive while the electron is negative) but is nearly identical otherwise, having the same mass, intrinsic spin, half-life, and so on. When a particle and its antimatter counterpart interact, they annihilate one another, usually totally converting their masses to pure energy in the form of photons as seen in [Figure 33.13](#). Neutral particles, such as neutrons, have neutral antimatter counterparts, which also annihilate when they interact. Certain neutral particles are their own antiparticle and live correspondingly short lives. For example, the neutral pion π^0 is its own antiparticle and has a half-life about 10^{-8} shorter than π^+ and π^- , which are each other's antiparticles. Without exception, nature is symmetric—all particles have antimatter counterparts. For example, antiprotons and antineutrons were first created in accelerator experiments in 1956 and the antiproton is negative. Antihydrogen atoms, consisting of an antiproton and antielectron, were observed in 1995 at CERN, too. It is possible to contain large-scale antimatter particles such as antiprotons by using electromagnetic traps that confine the particles within a magnetic field so that they don't annihilate with other particles. However, particles of the same charge repel each other, so the more particles that are contained in a trap, the more energy is needed to power the magnetic field that contains them. It is not currently possible to store a significant quantity of antiprotons. At any rate, we now see that negative charge is associated with both low-mass (electrons) and high-mass particles (antiprotons) and the apparent asymmetry is not there. But this knowledge does raise another question—why is there such a predominance of matter and so little antimatter? Possible explanations emerge later in this and the next chapter.

Hadrons and Leptons

Particles can also be revealingly grouped according to what forces they feel between them. All particles (even those that are massless) are affected by gravity, since gravity affects the space and time in which particles exist. All charged particles are affected by the electromagnetic force, as are neutral particles that have an internal distribution of charge (such as the neutron with its magnetic moment). Special names are given to particles that feel the strong and weak nuclear forces. **Hadrons** are particles that feel the strong nuclear force, whereas **leptons** are particles that do not. The proton, neutron, and the pions are examples of hadrons. The electron, positron, muons, and neutrinos are examples of leptons, the name meaning low mass. Leptons feel the weak nuclear force. In fact, all particles feel the weak nuclear force. This means that hadrons are distinguished by being able to feel both the strong and weak nuclear forces.

[Table 33.2](#) lists the characteristics of some of the most important subatomic particles, including the directly observed carrier particles for the electromagnetic and weak nuclear forces, all leptons, and some hadrons. Several hints related to an underlying substructure emerge from an examination of these particle characteristics. Note that the carrier particles are called **gauge bosons**. First mentioned in [Patterns in Spectra Reveal More Quantization](#), a

boson is a particle with zero or an integer value of intrinsic spin (such as $s = 0, 1, 2, \dots$), whereas a **fermion** is a particle with a half-integer value of intrinsic spin ($s = 1/2, 3/2, \dots$). Fermions obey the Pauli exclusion principle whereas bosons do not. All the known and conjectured carrier particles are bosons.

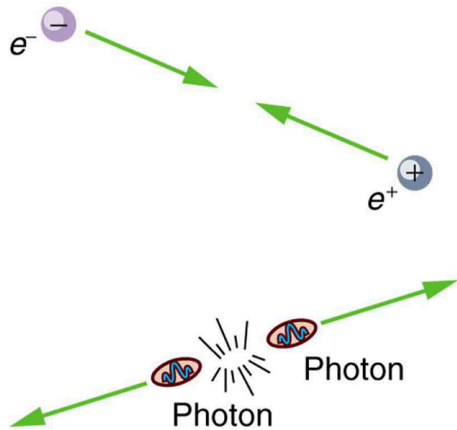


FIGURE 33.13 When a particle encounters its antiparticle, they annihilate, often producing pure energy in the form of photons. In this case, an electron and a positron convert all their mass into two identical energy rays, which move away in opposite directions to keep total momentum zero as it was before. Similar annihilations occur for other combinations of a particle with its antiparticle, sometimes producing more particles while obeying all conservation laws.

Category	Particle name	Symbol	Antiparticle	Rest mass (MeV/c ²)	<i>B</i>	<i>L_e</i>	<i>L_μ</i>	<i>L_τ</i>	<i>S</i>	Lifetime ⁴ (s)
Gauge	Photon	γ	Self	0	0	0	0	0	0	Stable
Bosons	<i>W</i>	W^+	W^-	80.39×10^3	0	0	0	0	0	1.6×10^{-25}
	<i>Z</i>	Z^0	Self	91.19×10^3	0	0	0	0	0	1.32×10^{-25}
Leptons	Electron	e^-	e^+	0.511	0	± 1	0	0	0	Stable
	Neutrino (<i>e</i>)	ν_e	$\bar{\nu}_e$	0(7.0eV) ⁵	0	± 1	0	0	0	Stable
	Muon	μ^-	μ^+	105.7	0	0	± 1	0	0	2.20×10^{-6}
	Neutrino (μ)	ν_μ	$\bar{\nu}_\mu$	0(< 0.27)	0	0	± 1	0	0	Stable
	Tau	τ^-	τ^+	1777	0	0	0	± 1	0	2.91×10^{-13}
	Neutrino (τ)	ν_τ	$\bar{\nu}_\tau$	0(< 31)	0	0	0	± 1	0	Stable
Hadrons (selected)										
Mesons	Pion	π^+	π^-	139.6	0	0	0	0	0	2.60×10^{-8}
		π^0	Self	135.0	0	0	0	0	0	8.4×10^{-17}

TABLE 33.2 Selected Particle Characteristics⁷

⁴ Lifetimes are traditionally given as $t_{1/2}/0.693$ (which is $1/\lambda$, the inverse of the decay constant).
⁵ Neutrino masses may be zero. Experimental upper limits are given in parentheses.
⁷ The lower of the $\mp$ or $\pm$ symbols are the values for antiparticles.

Category	Particle name	Symbol	Antiparticle	Rest mass (MeV/ c^2)	B	L_e	L_μ	L_τ	S	Lifetime ⁴ (s)
	Kaon	K^+	K^-	493.7	0	0	0	0	± 1	1.24×10^{-8}
		K^0	$\bar{K}^0$	497.6	0	0	0	0	± 1	0.90×10^{-10}
	Eta	η^0	Self	547.9	0	0	0	0	0	2.53×10^{-19}

(many other mesons known)

Baryons	Proton	p	$\bar{p}$	938.3	± 1	0	0	0	0	Stable ⁶
	Neutron	n	$\bar{n}$	939.6	± 1	0	0	0	0	882
	Lambda	Λ^0	$\bar{\Lambda}^0$	1115.7	± 1	0	0	0	∓ 1	2.63×10^{-10}
	Sigma	Σ^+	$\bar{\Sigma}^-$	1189.4	± 1	0	0	0	∓ 1	0.80×10^{-10}
		Σ^0	$\bar{\Sigma}^0$	1192.6	± 1	0	0	0	∓ 1	7.4×10^{-20}
		Σ^-	$\bar{\Sigma}^+$	1197.4	± 1	0	0	0	∓ 1	1.48×10^{-10}
	Xi	Ξ^0	$\bar{\Xi}^0$	1314.9	± 1	0	0	0	∓ 2	2.90×10^{-10}
		Ξ^-	$\bar{\Xi}^+$	1321.7	± 1	0	0	0	∓ 2	1.64×10^{-10}
	Omega	Ω^-	Ω^+	1672.5	± 1	0	0	0	∓ 3	0.82×10^{-10}

(many other baryons known)

TABLE 33.2 Selected Particle Characteristics⁷

All known leptons are listed in the table given above. There are only six leptons (and their antiparticles), and they seem to be fundamental in that they have no apparent underlying structure. Leptons have no discernible size other than their wavelength, so that we know they are pointlike down to about 10^{-18} m. The leptons fall into three families, implying three conservation laws for three quantum numbers. One of these was known from β decay, where the existence of the electron's neutrino implied that a new quantum number, called the **electron family number** L_e is conserved. Thus, in β decay, an antielectron's neutrino $\bar{\nu}_e$ must be created with $L_e = -1$ when an electron with $L_e = +1$ is created, so that the total remains 0 as it was before decay.

Once the muon was discovered in cosmic rays, its decay mode was found to be

$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu,$$

33.7

6 Experimental lower limit is $>5 \times 10^{32}$ for proposed mode of decay.

which implied another “family” and associated conservation principle. The particle ν_μ is a muon’s neutrino, and it is created to conserve **muon family number** L_μ . So muons are leptons with a family of their own, and **conservation of total** L_μ also seems to be obeyed in many experiments.

More recently, a third lepton family was discovered when τ particles were created and observed to decay in a manner similar to muons. One principal decay mode is

$$\tau^- \rightarrow \mu^- + \bar{\nu}_\mu + \nu_\tau. \quad 33.8$$

Conservation of total L_τ seems to be another law obeyed in many experiments. In fact, particle experiments have found that lepton family number is not universally conserved, due to neutrino “oscillations,” or transformations of neutrinos from one family type to another.

Mesons and Baryons

Now, note that the hadrons in the table given above are divided into two subgroups, called mesons (originally for medium mass) and baryons (the name originally meaning large mass). The division between mesons and baryons is actually based on their observed decay modes and is not strictly associated with their masses. **Mesons** are hadrons that can decay to leptons and leave no hadrons, which implies that mesons are not conserved in number. **Baryons** are hadrons that always decay to another baryon. A new physical quantity called **baryon number** B seems to always be conserved in nature and is listed for the various particles in the table given above. Mesons and leptons have $B = 0$ so that they can decay to other particles with $B = 0$. But baryons have $B = +1$ if they are matter, and $B = -1$ if they are antimatter. The **conservation of total baryon number** is a more general rule than first noted in nuclear physics, where it was observed that the total number of nucleons was always conserved in nuclear reactions and decays. That rule in nuclear physics is just one consequence of the conservation of the total baryon number.

Forces, Reactions, and Reaction Rates

The forces that act between particles regulate how they interact with other particles. For example, pions feel the strong force and do not penetrate as far in matter as do muons, which do not feel the strong force. (This was the way those who discovered the muon knew it could not be the particle that carries the strong force—its penetration or range was too great for it to be feeling the strong force.) Similarly, reactions that create other particles, like cosmic rays interacting with nuclei in the atmosphere, have greater probability if they are caused by the strong force than if they are caused by the weak force. Such knowledge has been useful to physicists while analyzing the particles produced by various accelerators.

The forces experienced by particles also govern how particles interact with themselves if they are unstable and decay. For example, the stronger the force, the faster they decay and the shorter is their lifetime. An example of a nuclear decay via the strong force is ${}^8\text{Be} \rightarrow \alpha + \alpha$ with a lifetime of about 10^{-16} s. The neutron is a good example of decay via the weak force. The process $n \rightarrow p + e^- + \bar{\nu}_e$ has a longer lifetime of 882 s. The weak force causes this decay, as it does all β decay. An important clue that the weak force is responsible for β decay is the creation of leptons, such as e^- and $\bar{\nu}_e$. None would be created if the strong force was responsible, just as no leptons are created in the decay of ${}^8\text{Be}$. The systematics of particle lifetimes is a little simpler than nuclear lifetimes when hundreds of particles are examined (not just the ones in the table given above). Particles that decay via the weak force have lifetimes mostly in the range of 10^{-16} to 10^{-12} s, whereas those that decay via the strong force have lifetimes mostly in the range of 10^{-16} to 10^{-23} s. Turning this around, if we measure the lifetime of a particle, we can tell if it decays via the weak or strong force.

Yet another quantum number emerges from decay lifetimes and patterns. Note that the particles Λ , Σ , Ξ , and Ω decay with lifetimes on the order of 10^{-10} s (the exception is Σ^0 , whose short lifetime is explained by its particular quark substructure.), implying that their decay is caused by the weak force alone, although they are hadrons and feel the strong force. The decay modes of these particles also show patterns—in particular, certain decays that should be possible within all the known conservation laws do not occur. Whenever something is possible in physics, it will happen. If something does not happen, it is forbidden by a rule. All this seemed strange to those studying these particles when they were first discovered, so they named a new quantum number **strangeness**, given the symbol S in the table given above. The values of strangeness assigned to various particles are based on the decay systematics. It is found that **strangeness is conserved by the strong force**, which governs the production of most of

these particles in accelerator experiments. However, **strangeness is not conserved by the weak force**. This conclusion is reached from the fact that particles that have long lifetimes decay via the weak force and do not conserve strangeness. All of this also has implications for the carrier particles, since they transmit forces and are thus involved in these decays.



EXAMPLE 33.3

Calculating Quantum Numbers in Two Decays

(a) The most common decay mode of the Ξ^- particle is $\Xi^- \rightarrow \Lambda^0 + \pi^-$. Using the quantum numbers in the table given above, show that strangeness changes by 1, baryon number and charge are conserved, and lepton family numbers are unaffected.

(b) Is the decay $K^+ \rightarrow \mu^+ + \nu_\mu$ allowed, given the quantum numbers in the table given above?

Strategy

In part (a), the conservation laws can be examined by adding the quantum numbers of the decay products and comparing them with the parent particle. In part (b), the same procedure can reveal if a conservation law is broken or not.

Solution for (a)

Before the decay, the Ξ^- has strangeness $S = -2$. After the decay, the total strangeness is -1 for the Λ^0 , plus 0 for the π^- . Thus, total strangeness has gone from -2 to -1 or a change of $+1$. Baryon number for the Ξ^- is $B = +1$ before the decay, and after the decay the Λ^0 has $B = +1$ and the π^- has $B = 0$ so that the total baryon number remains $+1$. Charge is -1 before the decay, and the total charge after is also $0 - 1 = -1$. Lepton numbers for all the particles are zero, and so lepton numbers are conserved.

Discussion for (a)

The Ξ^- decay is caused by the weak interaction, since strangeness changes, and it is consistent with the relatively long 1.64×10^{-10} -s lifetime of the Ξ^- .

Solution for (b)

The decay $K^+ \rightarrow \mu^+ + \nu_\mu$ is allowed if charge, baryon number, mass-energy, and lepton numbers are conserved. Strangeness can change due to the weak interaction. Charge is conserved as $s \rightarrow d$. Baryon number is conserved, since all particles have $B = 0$. Mass-energy is conserved in the sense that the K^+ has a greater mass than the products, so that the decay can be spontaneous. Lepton family numbers are conserved at 0 for the electron and tau family for all particles. The muon family number is $L_\mu = 0$ before and $L_\mu = -1 + 1 = 0$ after. Strangeness changes from $+1$ before to $0 + 0$ after, for an allowed change of 1 . The decay is allowed by all these measures.

Discussion for (b)

This decay is not only allowed by our reckoning, it is, in fact, the primary decay mode of the K^+ meson and is caused by the weak force, consistent with the long 1.24×10^{-8} -s lifetime.

There are hundreds of particles, all hadrons, not listed in [Table 33.2](#), most of which have shorter lifetimes. The systematics of those particle lifetimes, their production probabilities, and decay products are completely consistent with the conservation laws noted for lepton families, baryon number, and strangeness, but they also imply other quantum numbers and conservation laws. There are a finite, and in fact relatively small, number of these conserved quantities, however, implying a finite set of substructures. Additionally, some of these short-lived particles resemble the excited states of other particles, implying an internal structure. All of this jigsaw puzzle can be tied together and explained relatively simply by the existence of fundamental substructures. Leptons seem to be fundamental structures. Hadrons seem to have a substructure called quarks. [Quarks: Is That All There Is?](#) explores the basics of the underlying quark building blocks.



FIGURE 33.14 Murray Gell-Mann (b. 1929) proposed quarks as a substructure of hadrons in 1963 and was already known for his work on the concept of strangeness. Although quarks have never been directly observed, several predictions of the quark model were quickly confirmed, and their properties explain all known hadron characteristics. Gell-Mann was awarded the Nobel Prize in 1969. (credit: Luboš Motl)

33.5 Quarks: Is That All There Is?

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Define fundamental particle.
- Describe quark and antiquark.
- List the flavors of quark.
- Outline the quark composition of hadrons.
- Determine quantum numbers from quark composition.

Quarks have been mentioned at various points in this text as fundamental building blocks and members of the exclusive club of truly elementary particles. Note that an elementary or **fundamental particle** has no substructure (it is not made of other particles) and has no finite size other than its wavelength. This does not mean that fundamental particles are stable—some decay, while others do not. Keep in mind that *all* leptons seem to be fundamental, whereas *no* hadrons are fundamental. There is strong evidence that **quarks** are the fundamental building blocks of hadrons as seen in [Figure 33.15](#). Quarks are the second group of fundamental particles (leptons are the first). The third and perhaps final group of fundamental particles is the carrier particles for the four basic forces. Leptons, quarks, and carrier particles may be all there is. In this module we will discuss the quark substructure of hadrons and its relationship to forces as well as indicate some remaining questions and problems.

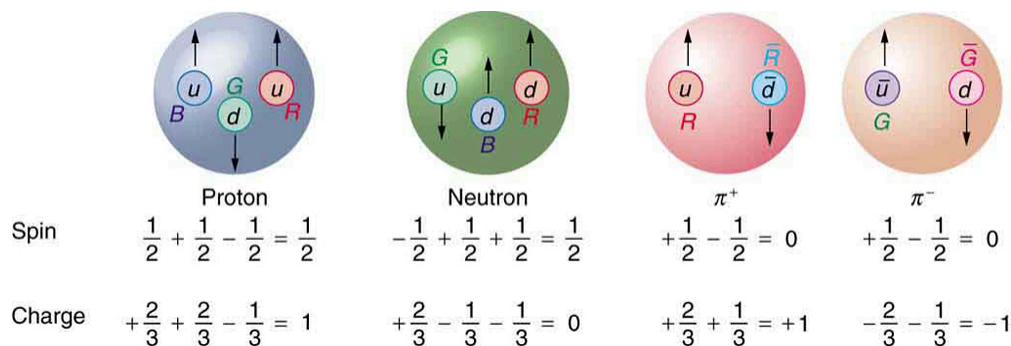


FIGURE 33.15 All baryons, such as the proton and neutron shown here, are composed of three quarks. All mesons, such as the pions

shown here, are composed of a quark-antiquark pair. Arrows represent the spins of the quarks, which, as we shall see, are also colored. The colors are such that they need to add to white for any possible combination of quarks.

Conception of Quarks

Quarks were first proposed independently by American physicists Murray Gell-Mann and George Zweig in 1963. Their quaint name was taken by Gell-Mann from a James Joyce novel—Gell-Mann was also largely responsible for the concept and name of strangeness. (Whimsical names are common in particle physics, reflecting the personalities of modern physicists.) Originally, three quark types—or **flavors**—were proposed to account for the then-known mesons and baryons. These quark flavors are named **up** (u), **down** (d), and **strange** (s). All quarks have half-integral spin and are thus fermions. All mesons have integral spin while all baryons have half-integral spin. Therefore, mesons should be made up of an even number of quarks while baryons need to be made up of an odd number of quarks. [Figure 33.15](#) shows the quark substructure of the proton, neutron, and two pions. The most radical proposal by Gell-Mann and Zweig is the fractional charges of quarks, which are $\pm(\frac{2}{3})q_e$ and $(\frac{1}{3})q_e$, whereas all directly observed particles have charges that are integral multiples of q_e . Note that the fractional value of the quark does not violate the fact that the e is the smallest unit of charge that is observed, because a free quark cannot exist. [Table 33.3](#) lists characteristics of the six quark flavors that are now thought to exist. Discoveries made since 1963 have required extra quark flavors, which are divided into three families quite analogous to leptons.

How Does it Work?

To understand how these quark substructures work, let us specifically examine the proton, neutron, and the two pions pictured in [Figure 33.15](#) before moving on to more general considerations. First, the proton p is composed of the three quarks uud , so that its total charge is $+(\frac{2}{3})q_e + (\frac{2}{3})q_e - (\frac{1}{3})q_e = q_e$, as expected. With the spins aligned as in the figure, the proton's intrinsic spin is $+(\frac{1}{2}) + (\frac{1}{2}) - (\frac{1}{2}) = (\frac{1}{2})$, also as expected. Note that the spins of the up quarks are aligned, so that they would be in the same state except that they have different colors (another quantum number to be elaborated upon a little later). Quarks obey the Pauli exclusion principle. Similar comments apply to the neutron n , which is composed of the three quarks udd . Note also that the neutron is made of charges that add to zero but move internally, producing its well-known magnetic moment. When the neutron β^- decays, it does so by changing the flavor of one of its quarks. Writing neutron β^- decay in terms of quarks,

$$n \rightarrow p + \beta^- + \bar{\nu}_e \text{ becomes } udd \rightarrow uud + \beta^- + \bar{\nu}_e. \quad 33.9$$

We see that this is equivalent to a down quark changing flavor to become an up quark:

$$d \rightarrow u + \beta^- + \bar{\nu}_e \quad 33.10$$

Name	Symbol	Antiparticle	Spin	Charge	B ⁸	S	c	b	t	Mass (GeV/ c^2) ⁹
Up	u	$\bar{u}$	1/2	$\pm\frac{2}{3}q_e$	$\pm\frac{1}{3}$	0	0	0	0	0.005
Down	d	$\bar{d}$	1/2	$\mp\frac{1}{3}q_e$	$\pm\frac{1}{3}$	0	0	0	0	0.008
Strange	s	$\bar{s}$	1/2	$\mp\frac{1}{3}q_e$	$\pm\frac{1}{3}$	∓ 1	0	0	0	0.50
Charmed	c	$\bar{c}$	1/2	$\pm\frac{2}{3}q_e$	$\pm\frac{1}{3}$	0	± 1	0	0	1.6

TABLE 33.3 Quarks and Antiquarks¹⁰

⁸ B is baryon number, S is strangeness, c is charm, b is bottomness, t is topness.

⁹ Values are approximate, are not directly observable, and vary with model.

¹⁰ The lower of the $\pm$ symbols are the values for antiquarks.

Name	Symbol	Antiparticle	Spin	Charge	B ⁸	S	c	b	t	Mass (GeV/ c^2) ⁹
Bottom	b	$\bar{b}$	1/2	$\mp \frac{1}{3}q_e$	$\pm \frac{1}{3}$	0	0	∓ 1	0	5
Top	t	$\bar{t}$	1/2	$\pm \frac{2}{3}q_e$	$\pm \frac{1}{3}$	0	0	0	± 1	173

TABLE 33.3 Quarks and Antiquarks¹⁰

Particle	Quark Composition
Mesons	
π^+	$u\bar{d}$
π^-	$\bar{u}d$
π^0	$u\bar{u}, d\bar{d}$ mixture ¹¹
η^0	$u\bar{u}, d\bar{d}$ mixture ¹²
K^0	$d\bar{s}$
$\bar{K}^0$	$\bar{d}s$
K^+	$u\bar{s}$
K^-	$\bar{u}s$
J/ψ	$c\bar{c}$
Υ	$b\bar{b}$
Baryons ^{13, 14}	
p	uud
n	udd
Δ^0	udd
Δ^+	uud
Δ^-	ddd
Δ^{++}	uuu
Λ^0	uds

TABLE 33.4 Quark Composition of

11 These two mesons are different mixtures, but each is its own antiparticle, as indicated by its quark composition.

12 These two mesons are different mixtures, but each is its own antiparticle, as indicated by its quark composition.

13 Antibaryons have the antiquarks of their counterparts. The antiproton $\bar{p}$ is $\bar{u}\bar{u}\bar{d}$, for example.14 Baryons composed of the same quarks are different states of the same particle. For example, the Δ^+ is an excited state of the proton.

Particle	Quark Composition
Σ^0	uds
Σ^+	uus
Σ^-	dds
Ξ^0	uss
Ξ^-	dss
Ω^-	sss

TABLE 33.4 Quark Composition of Selected Hadrons¹⁵

This is an example of the general fact that **the weak nuclear force can change the flavor of a quark**. By general, we mean that any quark can be converted to any other (change flavor) by the weak nuclear force. Not only can we get $d \rightarrow u$, we can also get $u \rightarrow d$. Furthermore, the strange quark can be changed by the weak force, too, making $s \rightarrow u$ and $s \rightarrow d$ possible. This explains the violation of the conservation of strangeness by the weak force noted in the preceding section. Another general fact is that **the strong nuclear force cannot change the flavor of a quark**.

Again, from [Figure 33.15](#), we see that the π^+ meson (one of the three pions) is composed of an up quark plus an antidown quark, or $u\bar{d}$. Its total charge is thus $+\left(\frac{2}{3}\right)q_e + \left(-\frac{1}{3}\right)q_e = q_e$, as expected. Its baryon number is 0, since it has a quark and an antiquark with baryon numbers $+\left(\frac{1}{3}\right) - \left(\frac{1}{3}\right) = 0$. The π^+ half-life is relatively long since, although it is composed of matter and antimatter, the quarks are different flavors and the weak force should cause the decay by changing the flavor of one into that of the other. The spins of the u and $\bar{d}$ quarks are antiparallel, enabling the pion to have spin zero, as observed experimentally. Finally, the π^- meson shown in [Figure 33.15](#) is the antiparticle of the π^+ meson, and it is composed of the corresponding quark antiparticles. That is, the π^+ meson is $u\bar{d}$, while the π^- meson is $\bar{u}d$. These two pions annihilate each other quickly, because their constituent quarks are each other's antiparticles.

Two general rules for combining quarks to form hadrons are:

1. Baryons are composed of three quarks, and antibaryons are composed of three antiquarks.
2. Mesons are combinations of a quark and an antiquark.

One of the clever things about this scheme is that only integral charges result, even though the quarks have fractional charge.

All Combinations are Possible

All quark combinations are possible. [Table 33.4](#) lists some of these combinations. When Gell-Mann and Zweig proposed the original three quark flavors, particles corresponding to all combinations of those three had not been observed. The pattern was there, but it was incomplete—much as had been the case in the periodic table of the elements and the chart of nuclides. The Ω^- particle, in particular, had not been discovered but was predicted by quark theory. Its combination of three strange quarks, sss , gives it a strangeness of -3 (see [Table 33.2](#)) and other predictable characteristics, such as spin, charge, approximate mass, and lifetime. If the quark picture is complete, the Ω^- should exist. It was first observed in 1964 at Brookhaven National Laboratory and had the predicted characteristics as seen in [Figure 33.16](#). The discovery of the Ω^- was convincing indirect evidence for the existence of the three original quark flavors and boosted theoretical and experimental efforts to further explore particle physics in terms of quarks.

PATTERNS AND PUZZLES: ATOMS, NUCLEI, AND QUARKS

Patterns in the properties of atoms allowed the periodic table to be developed. From it, previously unknown elements were predicted and observed. Similarly, patterns were observed in the properties of nuclei, leading to the chart of nuclides and successful predictions of previously unknown nuclides. Now with particle physics, patterns imply a quark substructure that, if taken literally, predicts previously unknown particles. These have now been observed in another triumph of underlying unity.

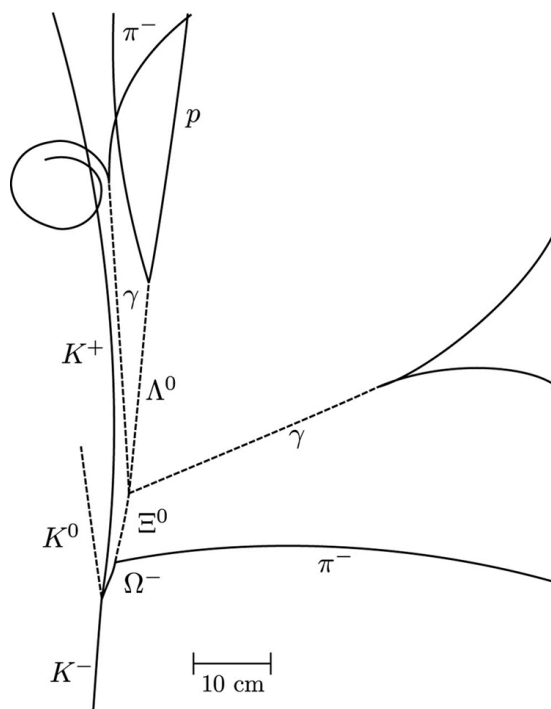


FIGURE 33.16 The image relates to the discovery of the Ω^- . It is a secondary reaction in which an accelerator-produced K^- collides with a proton via the strong force and conserves strangeness to produce the Ω^- with characteristics predicted by the quark model. As with other predictions of previously unobserved particles, this gave a tremendous boost to quark theory. (credit: Brookhaven National Laboratory)



EXAMPLE 33.4

Quantum Numbers From Quark Composition

Verify the quantum numbers given for the Ξ^0 particle in [Table 33.2](#) by adding the quantum numbers for its quark composition as given in [Table 33.4](#).

Strategy

The composition of the Ξ^0 is given as uss in [Table 33.4](#). The quantum numbers for the constituent quarks are given in [Table 33.3](#). We will not consider spin, because that is not given for the Ξ^0 . But we can check on charge and the other quantum numbers given for the quarks.

Solution

The total charge of uss is $+\left(\frac{2}{3}\right)q_e - \left(\frac{1}{3}\right)q_e - \left(\frac{1}{3}\right)q_e = 0$, which is correct for the Ξ^0 . The baryon number is $+\left(\frac{1}{3}\right) + \left(\frac{1}{3}\right) + \left(\frac{1}{3}\right) = 1$, also correct since the Ξ^0 is a matter baryon and has $B = 1$, as listed in [Table 33.2](#). Its strangeness is $S = 0 - 1 - 1 = -2$, also as expected from [Table 33.2](#). Its charm, bottomness, and topness are 0, as are its lepton family numbers (it is not a lepton).

Discussion

This procedure is similar to what the inventors of the quark hypothesis did when checking to see if their solution to the puzzle of particle patterns was correct. They also checked to see if all combinations were known, thereby predicting the previously unobserved Ω^- as the completion of a pattern.

Now, Let Us Talk About Direct Evidence

At first, physicists expected that, with sufficient energy, we should be able to free quarks and observe them directly. This has not proved possible. There is still no direct observation of a fractional charge or any isolated quark. When large energies are put into collisions, other particles are created—but no quarks emerge. There is nearly direct evidence for quarks that is quite compelling. By 1967, experiments at SLAC scattering 20-GeV electrons from protons had produced results like Rutherford had obtained for the nucleus nearly 60 years earlier. The SLAC scattering experiments showed unambiguously that there were three pointlike (meaning they had sizes considerably smaller than the probe's wavelength) charges inside the proton as seen in [Figure 33.17](#). This evidence made all but the most skeptical admit that there was validity to the quark substructure of hadrons.

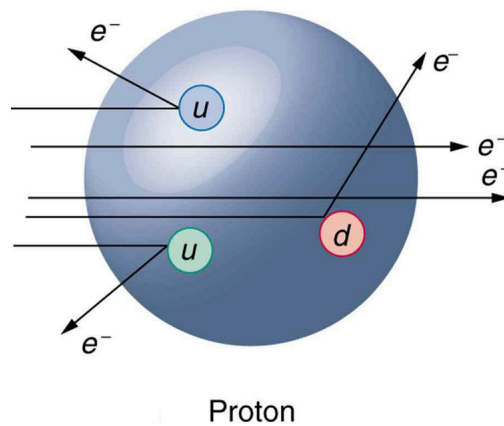


FIGURE 33.17 Scattering of high-energy electrons from protons at facilities like SLAC produces evidence of three point-like charges consistent with proposed quark properties. This experiment is analogous to Rutherford's discovery of the small size of the nucleus by scattering α particles. High-energy electrons are used so that the probe wavelength is small enough to see details smaller than the proton.

More recent and higher-energy experiments have produced jets of particles in collisions, highly suggestive of three quarks in a nucleon. Since the quarks are very tightly bound, energy put into separating them pulls them only so far apart before it starts being converted into other particles. More energy produces more particles, not a separation of quarks. Conservation of momentum requires that the particles come out in jets along the three paths in which the quarks were being pulled. Note that there are only three jets, and that other characteristics of the particles are consistent with the three-quark substructure.

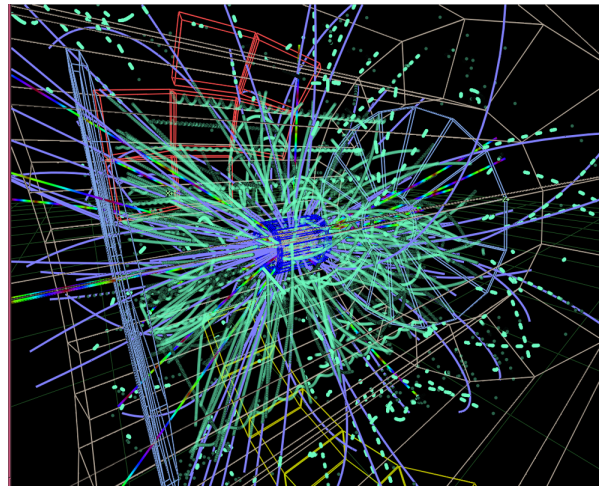


FIGURE 33.18 Simulation of a proton-proton collision at 14-TeV center-of-mass energy in the ALICE detector at CERN LHC. The lines follow

particle trajectories and the cyan dots represent the energy depositions in the sensitive detector elements. (credit: Matevž Tadel)

Quarks Have Their Ups and Downs

The quark model actually lost some of its early popularity because the original model with three quarks had to be modified. The up and down quarks seemed to compose normal matter as seen in [Table 33.4](#), while the single strange quark explained strangeness. Why didn't it have a counterpart? A fourth quark flavor called **charm** (c) was proposed as the counterpart of the strange quark to make things symmetric—there would be two normal quarks (u and d) and two exotic quarks (s and c). Furthermore, at that time only four leptons were known, two normal and two exotic. It was attractive that there would be four quarks and four leptons. The problem was that no known particles contained a charmed quark. Suddenly, in November of 1974, two groups (one headed by C. C. Ting at Brookhaven National Laboratory and the other by Burton Richter at SLAC) independently and nearly simultaneously discovered a new meson with characteristics that made it clear that its substructure is $c\bar{c}$. It was called J by one group and psi (ψ) by the other and now is known as the J/ψ meson. Since then, numerous particles have been discovered containing the charmed quark, consistent in every way with the quark model. The discovery of the J/ψ meson had such a rejuvenating effect on quark theory that it is now called the November Revolution. Ting and Richter shared the 1976 Nobel Prize.

History quickly repeated itself. In 1975, the tau (τ) was discovered, and a third family of leptons emerged as seen in [Table 33.2](#). Theorists quickly proposed two more quark flavors called **top** (t) or truth and **bottom** (b) or beauty to keep the number of quarks the same as the number of leptons. And in 1976, the upsilon (Υ) meson was discovered and shown to be composed of a bottom and an antibottom quark or $b\bar{b}$, quite analogous to the J/ψ being $c\bar{c}$ as seen in [Table 33.4](#). Being a single flavor, these mesons are sometimes called bare charm and bare bottom and reveal the characteristics of their quarks most clearly. Other mesons containing bottom quarks have since been observed. In 1995, two groups at Fermilab confirmed the top quark's existence, completing the picture of six quarks listed in [Table 33.3](#). Each successive quark discovery—first c , then b , and finally t —has required higher energy because each has higher mass. Quark masses in [Table 33.3](#) are only approximately known, because they are not directly observed. They must be inferred from the masses of the particles they combine to form.

What's Color got to do with it?—A Whiter Shade of Pale

As mentioned and shown in [Figure 33.15](#), quarks carry another quantum number, which we call **color**. Of course, it is not the color we sense with visible light, but its properties are analogous to those of three primary and three secondary colors. Specifically, a quark can have one of three color values we call **red** (R), **green** (G), and **blue** (B) in analogy to those primary visible colors. Antiquarks have three values we call **antired or cyan** ($\bar{R}$), **antigreen or magenta** ($\bar{G}$), and **antiblu or yellow** ($\bar{B}$) in analogy to those secondary visible colors. The reason for these names is that when certain visual colors are combined, the eye sees white. The analogy of the colors combining to white is used to explain why baryons are made of three quarks, why mesons are a quark and an antiquark, and why we cannot isolate a single quark. The force between the quarks is such that their combined colors produce white. This is illustrated in [Figure 33.19](#). A baryon must have one of each primary color or RGB, which produces white. A meson must have a primary color and its anticolor, also producing white.

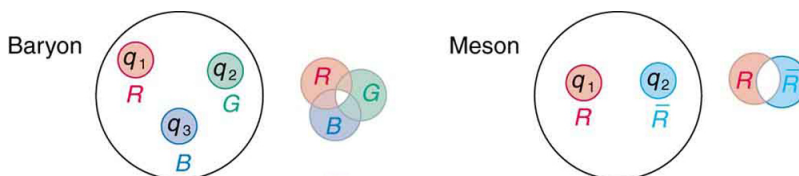


FIGURE 33.19 The three quarks composing a baryon must be RGB, which add to white. The quark and antiquark composing a meson must be a color and anticolor, here $R\bar{R}$ also adding to white. The force between systems that have color is so great that they can neither be separated nor exist as colored.

Why must hadrons be white? The color scheme is intentionally devised to explain why baryons have three quarks and mesons have a quark and an antiquark. Quark color is thought to be similar to charge, but with more values. An ion, by analogy, exerts much stronger forces than a neutral molecule. When the color of a combination of quarks is white, it is like a neutral atom. The forces a white particle exerts are like the polarization forces in molecules, but in hadrons these leftovers are the strong nuclear force. When a combination of quarks has color other than white, it exerts *extremely* large forces—even larger than the strong force—and perhaps cannot be stable or permanently

separated. This is part of the **theory of quark confinement**, which explains how quarks can exist and yet never be isolated or directly observed. Finally, an extra quantum number with three values (like those we assign to color) is necessary for quarks to obey the Pauli exclusion principle. Particles such as the Ω^- , which is composed of three strange quarks, sss , and the Δ^{++} , which is three up quarks, uuu , can exist because the quarks have different colors and do not have the same quantum numbers. Color is consistent with all observations and is now widely accepted. Quark theory including color is called **quantum chromodynamics** (QCD), also named by Gell-Mann.

The Three Families

Fundamental particles are thought to be one of three types—leptons, quarks, or carrier particles. Each of those three types is further divided into three analogous families as illustrated in [Figure 33.20](#). We have examined leptons and quarks in some detail. Each has six members (and their six antiparticles) divided into three analogous families. The first family is normal matter, of which most things are composed. The second is exotic, and the third more exotic and more massive than the second. The only stable particles are in the first family, which also has unstable members.

Always searching for symmetry and similarity, physicists have also divided the carrier particles into three families, omitting the graviton. Gravity is special among the four forces in that it affects the space and time in which the other forces exist and is proving most difficult to include in a Theory of Everything or TOE (to stub the pretension of such a theory). Gravity is thus often set apart. It is not certain that there is meaning in the groupings shown in [Figure 33.20](#), but the analogies are tempting. In the past, we have been able to make significant advances by looking for analogies and patterns, and this is an example of one under current scrutiny. There are connections between the families of leptons, in that the τ decays into the μ and the μ into the e . Similarly for quarks, the higher families eventually decay into the lowest, leaving only u and d quarks. We have long sought connections between the forces in nature. Since these are carried by particles, we will explore connections between gluons, $W^\pm$ and Z^0 , and photons as part of the search for unification of forces discussed in [GUTs: The Unification of Forces](#).

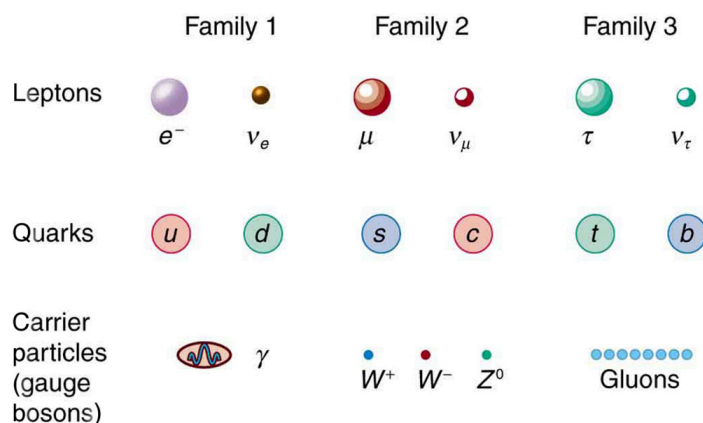


FIGURE 33.20 The three types of particles are leptons, quarks, and carrier particles. Each of those types is divided into three analogous families, with the graviton left out.

33.6 GUTs: The Unification of Forces

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- State the grand unified theory.
- Explain the electroweak theory.
- Define gluons.
- Describe the principle of quantum chromodynamics.
- Define the standard model.

Present quests to show that the four basic forces are different manifestations of a single unified force follow a long tradition. In the 19th century, the distinct electric and magnetic forces were shown to be intimately connected and are now collectively called the electromagnetic force. More recently, the weak nuclear force has been shown to be connected to the electromagnetic force in a manner suggesting that a theory may be constructed in which all four forces are unified. Certainly, there are similarities in how forces are transmitted by the exchange of carrier particles, and the carrier particles themselves (the gauge bosons in [Table 33.2](#)) are also similar in important ways. The

analogy to the unification of electric and magnetic forces is quite good—the four forces are distinct under normal circumstances, but there are hints of connections even on the atomic scale, and there may be conditions under which the forces are intimately related and even indistinguishable. The search for a correct theory linking the forces, called the **Grand Unified Theory (GUT)**, is explored in this section in the realm of particle physics. [Frontiers of Physics](#) expands the story in making a connection with cosmology, on the opposite end of the distance scale.

[Figure 33.21](#) is a Feynman diagram showing how the weak nuclear force is transmitted by the carrier particle Z^0 , similar to the diagrams in [Figure 33.5](#) and [Figure 33.6](#) for the electromagnetic and strong nuclear forces. In the 1960s, a gauge theory, called **electroweak theory**, was developed by Steven Weinberg, Sheldon Glashow, and Abdus Salam and proposed that the electromagnetic and weak forces are identical at sufficiently high energies. One of its predictions, in addition to describing both electromagnetic and weak force phenomena, was the existence of the W^+ , W^- , and Z^0 carrier particles. Not only were three particles having spin 1 predicted, the mass of the W^+ and W^- was predicted to be $81 \text{ GeV}/c^2$, and that of the Z^0 was predicted to be $90 \text{ GeV}/c^2$. (Their masses had to be about 1000 times that of the pion, or about $100 \text{ GeV}/c^2$, since the range of the weak force is about 1000 times less than the strong force carried by virtual pions.) In 1983, these carrier particles were observed at CERN with the predicted characteristics, including masses having the predicted values as seen in [Table 33.2](#). This was another triumph of particle theory and experimental effort, resulting in the 1984 Nobel Prize to the experiment's group leaders Carlo Rubbia and Simon van der Meer. Theorists Weinberg, Glashow, and Salam had already been honored with the 1979 Nobel Prize for other aspects of electroweak theory.

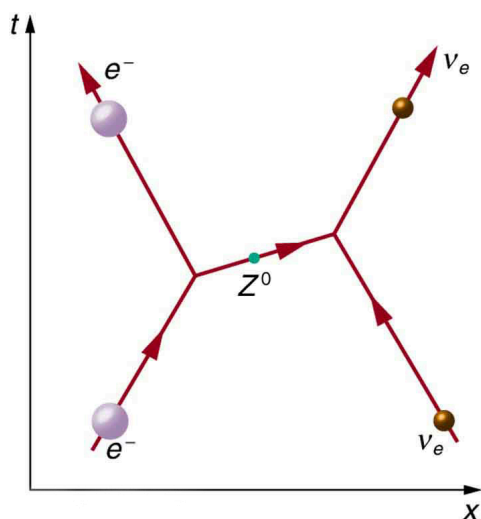


FIGURE 33.21 The exchange of a virtual Z^0 carries the weak nuclear force between an electron and a neutrino in this Feynman diagram. The Z^0 is one of the carrier particles for the weak nuclear force that has now been created in the laboratory with characteristics predicted by electroweak theory.

Although the weak nuclear force is very short ranged ($< 10^{-18} \text{ m}$, as indicated in [Table 33.1](#)), its effects on atomic levels can be measured given the extreme precision of modern techniques. Since electrons spend some time in the nucleus, their energies are affected, and spectra can even indicate new aspects of the weak force, such as the possibility of other carrier particles. So systems many orders of magnitude larger than the range of the weak force supply evidence of electroweak unification in addition to evidence found at the particle scale.

Gluons (g) are the proposed carrier particles for the strong nuclear force, although they are not directly observed. Like quarks, gluons may be confined to systems having a total color of white. Less is known about gluons than the fact that they are the carriers of the weak and certainly of the electromagnetic force. QCD theory calls for eight gluons, all massless and all spin 1. Six of the gluons carry a color and an anticolor, while two do not carry color, as illustrated in [Figure 33.22\(a\)](#). There is indirect evidence of the existence of gluons in nucleons. When high-energy electrons are scattered from nucleons and evidence of quarks is seen, the momenta of the quarks are smaller than they would be if there were no gluons. That means that the gluons carrying force between quarks also carry some momentum, inferred by the already indirect quark momentum measurements. At any rate, the gluons carry color charge and can change the colors of quarks when exchanged, as seen in [Figure 33.22\(b\)](#). In the figure, a red down quark interacts with a green strange quark by sending it a gluon. That gluon carries red away from the down quark

and leaves it green, because it is an $R\bar{G}$ (red-antigreen) gluon. (Taking antigreen away leaves you green.) Its antigreenness kills the green in the strange quark, and its redness turns the quark red.

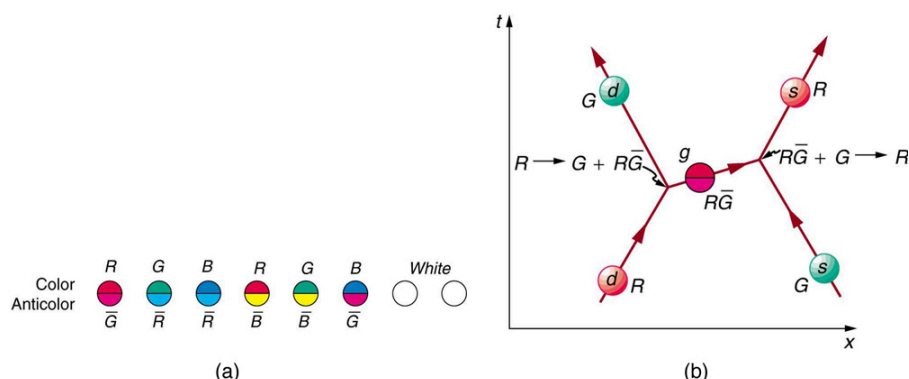


FIGURE 33.22 In figure (a), the eight types of gluons that carry the strong nuclear force are divided into a group of six that carry color and a group of two that do not. Figure (b) shows that the exchange of gluons between quarks carries the strong force and may change the color of a quark.

The strong force is complicated, since observable particles that feel the strong force (hadrons) contain multiple quarks. [Figure 33.23](#) shows the quark and gluon details of pion exchange between a proton and a neutron as illustrated earlier in [Figure 33.3](#) and [Figure 33.6](#). The quarks within the proton and neutron move along together exchanging gluons, until the proton and neutron get close together. As the u quark leaves the proton, a gluon creates a pair of virtual particles, a d quark and a $\bar{d}$ antiquark. The d quark stays behind and the proton turns into a neutron, while the u and $\bar{d}$ move together as a π^+ ([Table 33.4](#) confirms the $u\bar{d}$ composition for the π^+ .) The $\bar{d}$ annihilates a d quark in the neutron, the u joins the neutron, and the neutron becomes a proton. A pion is exchanged and a force is transmitted.

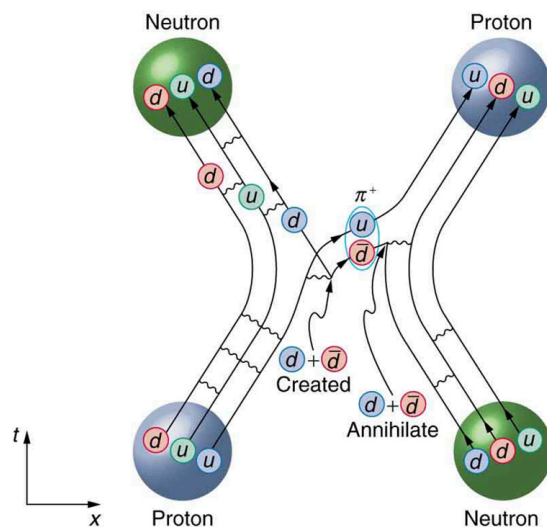


FIGURE 33.23 This Feynman diagram is the same interaction as shown in [Figure 33.6](#), but it shows the quark and gluon details of the strong force interaction.

It is beyond the scope of this text to go into more detail on the types of quark and gluon interactions that underlie the observable particles, but the theory (**quantum chromodynamics** or QCD) is very self-consistent. So successful have QCD and the electroweak theory been that, taken together, they are called the **Standard Model**. Advances in knowledge are expected to modify, but not overthrow, the Standard Model of particle physics and forces.

MAKING CONNECTIONS: UNIFICATION OF FORCES

Grand Unified Theory (GUT) is successful in describing the four forces as distinct under normal circumstances, but connected in fundamental ways. Experiments have verified that the weak and electromagnetic force become identical at very small distances and provide the GUT description of the carrier particles for the forces. GUT

predicts that the other forces become identical under conditions so extreme that they cannot be tested in the laboratory, although there may be lingering evidence of them in the evolution of the universe. GUT is also successful in describing a system of carrier particles for all four forces, but there is much to be done, particularly in the realm of gravity.

How can forces be unified? They are definitely distinct under most circumstances, for example, being carried by different particles and having greatly different strengths. But experiments show that at extremely small distances, the strengths of the forces begin to become more similar. In fact, electroweak theory's prediction of the W^+ , W^- , and Z^0 carrier particles was based on the strengths of the two forces being identical at extremely small distances as seen in [Figure 33.24](#). As discussed in case of the creation of virtual particles for extremely short times, the small distances or short ranges correspond to the large masses of the carrier particles and the correspondingly large energies needed to create them. Thus, the energy scale on the horizontal axis of [Figure 33.24](#) corresponds to smaller and smaller distances, with 100 GeV corresponding to approximately, 10^{-18} m for example. At that distance, the strengths of the EM and weak forces are the same. To test physics at that distance, energies of about 100 GeV must be put into the system, and that is sufficient to create and release the W^+ , W^- , and Z^0 carrier particles. At those and higher energies, the masses of the carrier particles becomes less and less relevant, and the Z^0 in particular resembles the massless, chargeless, spin 1 photon. In fact, there is enough energy when things are pushed to even smaller distances to transform the, and Z^0 into massless carrier particles more similar to photons and gluons. These have not been observed experimentally, but there is a prediction of an associated particle called the **Higgs boson**. The mass of this particle is not predicted with nearly the certainty with which the mass of the W^+ , W^- , and Z^0 particles were predicted, but it was hoped that the Higgs boson could be observed at the now-canceled Superconducting Super Collider (SSC). Ongoing experiments at the Large Hadron Collider at CERN have presented some evidence for a Higgs boson with a mass of 125 GeV, and there is a possibility of a direct discovery during 2012. The existence of this more massive particle would give validity to the theory that the carrier particles are identical under certain circumstances.

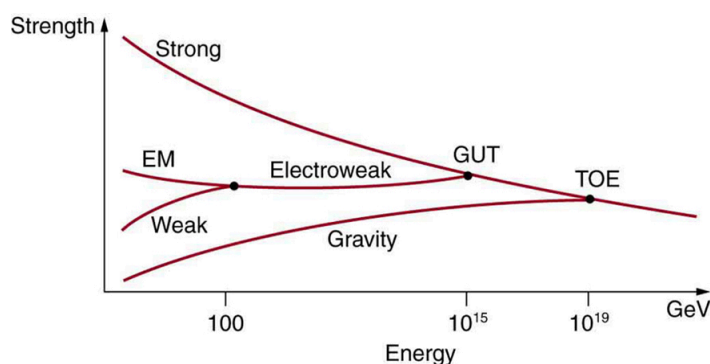


FIGURE 33.24 The relative strengths of the four basic forces vary with distance and, hence, energy is needed to probe small distances. At ordinary energies (a few eV or less), the forces differ greatly as indicated in [Table 33.1](#). However, at energies available at accelerators, the weak and EM forces become identical, or unified. Unfortunately, the energies at which the strong and electroweak forces become the same are unreachable even in principle at any conceivable accelerator. The universe may provide a laboratory, and nature may show effects at ordinary energies that give us clues about the validity of this graph.

The small distances and high energies at which the electroweak force becomes identical with the strong nuclear force are not reachable with any conceivable human-built accelerator. At energies of about 10^{14} GeV (16,000 J per particle), distances of about 10^{-30} m can be probed. Such energies are needed to test theory directly, but these are about 10^{10} higher than the proposed giant SSC would have had, and the distances are about 10^{-12} smaller than any structure we have direct knowledge of. This would be the realm of various GUTs, of which there are many since there is no constraining evidence at these energies and distances. Past experience has shown that any time you probe so many orders of magnitude further (here, about 10^{12}), you find the unexpected. Even more extreme are the energies and distances at which gravity is thought to unify with the other forces in a TOE. Most speculative and least constrained by experiment are TOEs, one of which is called **Superstring theory**. Superstrings are entities that are 10^{-35} m in scale and act like one-dimensional oscillating strings and are also proposed to underlie all particles, forces, and space itself.

At the energy of GUTs, the carrier particles of the weak force would become massless and identical to gluons. If that happens, then both lepton and baryon conservation would be violated. We do not see such violations, because we do not encounter such energies. However, there is a tiny probability that, at ordinary energies, the virtual particles that violate the conservation of baryon number may exist for extremely small amounts of time (corresponding to very small ranges). All GUTs thus predict that the proton should be unstable, but would decay with an extremely long lifetime of about 10^{31} y. The predicted decay mode is

$$p \rightarrow \pi^0 + e^+, \text{ (proposed proton decay)} \quad 33.11$$

which violates both conservation of baryon number and electron family number. Although 10^{31} y is an extremely long time (about 10^{21} times the age of the universe), there are a lot of protons, and detectors have been constructed to look for the proposed decay mode as seen in [Figure 33.25](#). It is somewhat comforting that proton decay has not been detected, and its experimental lifetime is now greater than 5×10^{32} y. This does not prove GUTs wrong, but it does place greater constraints on the theories, benefiting theorists in many ways.

From looking increasingly inward at smaller details for direct evidence of electroweak theory and GUTs, we turn around and look to the universe for evidence of the unification of forces. In the 1920s, the expansion of the universe was discovered. Thinking backward in time, the universe must once have been very small, dense, and extremely hot. At a tiny fraction of a second after the fabled Big Bang, forces would have been unified and may have left their fingerprint on the existing universe. This, one of the most exciting forefronts of physics, is the subject of [Frontiers of Physics](#).

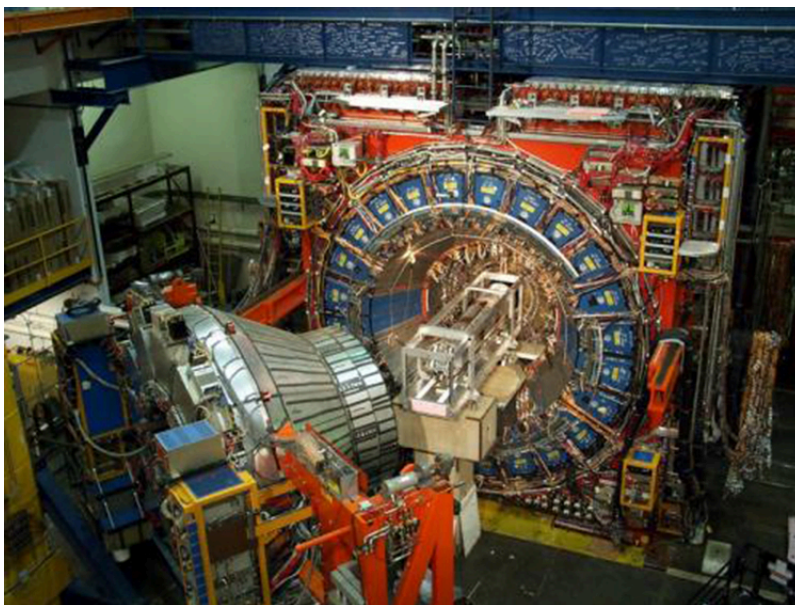


FIGURE 33.25 In the Tevatron accelerator at Fermilab, protons and antiprotons collide at high energies, and some of those collisions could result in the production of a Higgs boson in association with a W boson. When the W boson decays to a high-energy lepton and a neutrino, the detector triggers on the lepton, whether it is an electron or a muon. (credit: D. J. Miller)

Glossary

baryon number a conserved physical quantity that is zero for mesons and leptons and ± 1 for baryons and antibaryons, respectively

baryons hadrons that always decay to another baryon

boson particle with zero or an integer value of intrinsic spin

bottom a quark flavor

charm a quark flavor, which is the counterpart of the strange quark

colliding beams head-on collisions between particles moving in opposite directions

color a quark flavor

conservation of total baryon number a general rule based on the observation that the total number of nucleons was always conserved in nuclear reactions and decays

conservation of total electron family number a general rule stating that the total electron family number stays the same through an interaction

conservation of total muon family number a general rule stating that the total muon family number stays the same through an interaction

cyclotron accelerator that uses fixed-frequency alternating electric fields and fixed magnets to accelerate particles in a circular spiral path

down the second-lightest of all quarks

electron family number the number ± 1 that is assigned to all members of the electron family, or the number 0 that is assigned to all particles not in the electron family

electroweak theory theory showing connections between EM and weak forces

fermion particle with a half-integer value of intrinsic spin

Feynman diagram a graph of time versus position that describes the exchange of virtual particles between subatomic particles

flavors quark type

fundamental particle particle with no substructure

gauge boson particle that carries one of the four forces

gluons eight proposed particles which carry the strong force

gluons exchange particles, analogous to the exchange of photons that gives rise to the electromagnetic force between two charged particles

grand unified theory theory that shows unification of the strong and electroweak forces

hadrons particles that feel the strong nuclear force

Higgs boson a massive particle that, if observed,

would give validity to the theory that carrier particles are identical under certain circumstances

leptons particles that do not feel the strong nuclear force

linear accelerator accelerator that accelerates particles in a straight line

meson hadrons that can decay to leptons and leave no hadrons

meson particle whose mass is intermediate between the electron and nucleon masses

muon family number the number ± 1 that is assigned to all members of the muon family, or the number 0 that is assigned to all particles not in the muon family

particle physics the study of and the quest for those truly fundamental particles having no substructure

pion particle exchanged between nucleons, transmitting the force between them

quantum chromodynamics quark theory including color

quantum chromodynamics the governing theory of connecting quantum number color to gluons

quantum electrodynamics the theory of electromagnetism on the particle scale

quark an elementary particle and a fundamental constituent of matter

standard model combination of quantum chromodynamics and electroweak theory

strange the third lightest of all quarks

strangeness a physical quantity assigned to various particles based on decay systematics

superstring theory a theory of everything based on vibrating strings some 10^{-35} m in length

synchrotron a version of a cyclotron in which the frequency of the alternating voltage and the magnetic field strength are increased as the beam particles are accelerated

synchrotron radiation radiation caused by a magnetic field accelerating a charged particle perpendicular to its velocity

tau family number the number ± 1 that is assigned to all members of the tau family, or the number 0 that is assigned to all particles not in the tau family

theory of quark confinement explains how quarks can exist and yet never be isolated or directly observed

top a quark flavor

up the lightest of all quarks

Van de Graaff early accelerator: simple, large-scale version of the electron gun

virtual particles particles which cannot be directly observed but their effects can be directly observed

Section Summary

33.1 The Yukawa Particle and the Heisenberg Uncertainty Principle Revisited

- Yukawa's idea of virtual particle exchange as the carrier of forces is crucial, with virtual particles being formed in temporary violation of the conservation of mass-energy as allowed by the Heisenberg uncertainty principle.

33.2 The Four Basic Forces

- The four basic forces and their carrier particles are summarized in the [Table 33.1](#).
- Feynman diagrams are graphs of time versus position and are highly useful pictorial representations of particle processes.
- The theory of electromagnetism on the particle scale is called quantum electrodynamics (QED).

33.3 Accelerators Create Matter from Energy

- A variety of particle accelerators have been used to explore the nature of subatomic particles and to test predictions of particle theories.
- Modern accelerators used in particle physics are either large synchrotrons or linear accelerators.
- The use of colliding beams makes much greater energy available for the creation of particles, and collisions between matter and antimatter allow a greater range of final products.

33.4 Particles, Patterns, and Conservation Laws

- All particles of matter have an antimatter counterpart that has the opposite charge and certain other quantum numbers as seen in [Table 33.2](#). These matter-antimatter pairs are otherwise very similar but will annihilate when brought together. Known particles can be divided into three major groups—leptons, hadrons, and carrier particles (gauge bosons).
- Leptons do not feel the strong nuclear force and are further divided into three groups—electron family designated by electron family number L_e ; muon family designated by muon family number L_μ ; and tau family designated by tau family number L_τ . The family numbers are not universally conserved due to neutrino oscillations.
- Hadrons are particles that feel the strong nuclear

force and are divided into baryons, with the baryon family number B being conserved, and mesons.

33.5 Quarks: Is That All There Is?

- Hadrons are thought to be composed of quarks, with baryons having three quarks and mesons having a quark and an antiquark.
- The characteristics of the six quarks and their antiquark counterparts are given in [Table 33.3](#), and the quark compositions of certain hadrons are given in [Table 33.4](#).
- Indirect evidence for quarks is very strong, explaining all known hadrons and their quantum numbers, such as strangeness, charm, topness, and bottomness.
- Quarks come in six flavors and three colors and occur only in combinations that produce white.
- Fundamental particles have no further substructure, not even a size beyond their de Broglie wavelength.
- There are three types of fundamental particles—leptons, quarks, and carrier particles. Each type is divided into three analogous families as indicated in [Figure 33.20](#).

33.6 GUTs: The Unification of Forces

- Attempts to show unification of the four forces are called Grand Unified Theories (GUTs) and have been partially successful, with connections proven between EM and weak forces in electroweak theory.
- The strong force is carried by eight proposed particles called gluons, which are intimately connected to a quantum number called color—their governing theory is thus called quantum chromodynamics (QCD). Taken together, QCD and the electroweak theory are widely accepted as the Standard Model of particle physics.
- Unification of the strong force is expected at such high energies that it cannot be directly tested, but it may have observable consequences in the as-yet unobserved decay of the proton and topics to be discussed in the next chapter. Although unification of forces is generally anticipated, much remains to be done to prove its validity.

Conceptual Questions

33.3 Accelerators Create Matter from Energy

1. The total energy in the beam of an accelerator is far greater than the energy of the individual beam particles. Why isn't this total energy available to create a single extremely massive particle?
2. Synchrotron radiation takes energy from an accelerator beam and is related to acceleration. Why would you expect the problem to be more severe for electron accelerators than proton accelerators?
3. What two major limitations prevent us from building high-energy accelerators that are physically small?
4. What are the advantages of colliding-beam accelerators? What are the disadvantages?

33.4 Particles, Patterns, and Conservation Laws

5. Large quantities of antimatter isolated from normal matter should behave exactly like normal matter. An antiatom, for example, composed of positrons, antiprotons, and antineutrons should have the same atomic spectrum as its matter counterpart. Would you be able to tell it is antimatter by its emission of antiphotons? Explain briefly.
6. Massless particles are not only neutral, they are chargeless (unlike the neutron). Why is this so?
7. Massless particles must travel at the speed of light, while others cannot reach this speed. Why are all massless particles stable? If evidence is found that neutrinos spontaneously decay into other particles, would this imply they have mass?
8. When a star erupts in a supernova explosion, huge numbers of electron neutrinos are formed in nuclear reactions. Such neutrinos from the 1987A supernova in the relatively nearby Magellanic Cloud were observed within hours of the initial brightening, indicating they traveled to earth at approximately the speed of light. Explain how this data can be used to set an upper limit on the mass of the neutrino, noting that if the mass is small the neutrinos could travel very close to the speed of light and have a reasonable energy (on the order of MeV).
9. Theorists have had spectacular success in predicting previously unknown particles. Considering past theoretical triumphs, why should we bother to perform experiments?
10. What lifetime do you expect for an antineutron isolated from normal matter?
11. Why does the η^0 meson have such a short lifetime compared to most other mesons?
12. (a) Is a hadron always a baryon?
(b) Is a baryon always a hadron?
(c) Can an unstable baryon decay into a meson, leaving no other baryon?
13. Explain how conservation of baryon number is responsible for conservation of total atomic mass (total number of nucleons) in nuclear decay and reactions.

33.5 Quarks: Is That All There Is?

14. The quark flavor change $d \rightarrow u$ takes place in β^- decay. Does this mean that the reverse quark flavor change $u \rightarrow d$ takes place in β^+ decay? Justify your response by writing the decay in terms of the quark constituents, noting that it looks as if a proton is converted into a neutron in β^+ decay.
15. Explain how the weak force can change strangeness by changing quark flavor.
16. Beta decay is caused by the weak force, as are all reactions in which strangeness changes. Does this imply that the weak force can change quark flavor? Explain.
17. Why is it easier to see the properties of the c , b , and t quarks in mesons having composition W^- or $t\bar{t}$ rather than in baryons having a mixture of quarks, such as udb ?
18. How can quarks, which are fermions, combine to form bosons? Why must an even number combine to form a boson? Give one example by stating the quark substructure of a boson.
19. What evidence is cited to support the contention that the gluon force between quarks is greater than the strong nuclear force between hadrons? How is this related to color? Is it also related to quark confinement?
20. Discuss how we know that π -mesons (π^+ , π , π^0) are not fundamental particles and are not the

basic carriers of the strong force.

21. An antibaryon has three antiquarks with colors $\bar{R}\bar{G}\bar{B}$. What is its color?
22. Suppose leptons are created in a reaction. Does this imply the weak force is acting? (for example, consider β decay.)
23. How can the lifetime of a particle indicate that its decay is caused by the strong nuclear force? How can a change in strangeness imply which force is responsible for a reaction? What does a change in quark flavor imply about the force that is responsible?
24. (a) Do all particles having strangeness also have at least one strange quark in them?
(b) Do all hadrons with a strange quark also have nonzero strangeness?
25. The sigma-zero particle decays mostly via the reaction $\Sigma^0 \rightarrow \Lambda^0 + \gamma$. Explain how this decay and the respective quark compositions imply that the Σ^0 is an excited state of the Λ^0 .
26. What do the quark compositions and other quantum numbers imply about the relationships

between the Δ^+ and the proton? The Δ^0 and the neutron?

27. Discuss the similarities and differences between the photon and the Z^0 in terms of particle properties, including forces felt.
28. Identify evidence for electroweak unification.
29. The quarks in a particle are confined, meaning individual quarks cannot be directly observed. Are gluons confined as well? Explain

33.6 GUTs: The Unification of Forces

30. If a GUT is proven, and the four forces are unified, it will still be correct to say that the orbit of the moon is determined by the gravitational force. Explain why.
31. If the Higgs boson is discovered and found to have mass, will it be considered the ultimate carrier of the weak force? Explain your response.
32. Gluons and the photon are massless. Does this imply that the W^+ , W^- , and Z^0 are the ultimate carriers of the weak force?

Problems & Exercises

33.1 The Yukawa Particle and the Heisenberg Uncertainty Principle Revisited

1. A virtual particle having an approximate mass of $10^{14} \text{ GeV}/c^2$ may be associated with the unification of the strong and electroweak forces. For what length of time could this virtual particle exist (in temporary violation of the conservation of mass-energy as allowed by the Heisenberg uncertainty principle)?
2. Calculate the mass in GeV/c^2 of a virtual carrier particle that has a range limited to 10^{-30} m by the Heisenberg uncertainty principle. Such a particle might be involved in the unification of the strong and electroweak forces.
3. Another component of the strong nuclear force is transmitted by the exchange of virtual K -mesons. Taking K -mesons to have an average mass of $495 \text{ MeV}/c^2$, what is the approximate range of this component of the strong force?

33.2 The Four Basic Forces

4. (a) Find the ratio of the strengths of the weak and electromagnetic forces under ordinary circumstances.

(b) What does that ratio become under circumstances in which the forces are unified?

5. The ratio of the strong to the weak force and the ratio of the strong force to the electromagnetic force become 1 under circumstances where they are unified. What are the ratios of the strong force to those two forces under normal circumstances?

33.3 Accelerators Create Matter from Energy

6. At full energy, protons in the 2.00-km-diameter Fermilab synchrotron travel at nearly the speed of light, since their energy is about 1000 times their rest mass energy.
(a) How long does it take for a proton to complete one trip around?
(b) How many times per second will it pass through the target area?
7. Suppose a W^- created in a bubble chamber lives for $5.00 \times 10^{-25} \text{ s}$. What distance does it move in this time if it is traveling at $0.900 c$? Since this distance is too short to make a track, the presence of the W^- must be inferred from its decay products. Note that the time is longer

than the given W^- lifetime, which can be due to the statistical nature of decay or time dilation.

8. What length track does a π^+ traveling at $0.100c$ leave in a bubble chamber if it is created there and lives for 2.60×10^{-8} s? (Those moving faster or living longer may escape the detector before decaying.)
9. The 3.20-km-long SLAC produces a beam of 50.0-GeV electrons. If there are 15,000 accelerating tubes, what average voltage must be across the gaps between them to achieve this energy?
10. Because of energy loss due to synchrotron radiation in the LHC at CERN, only 5.00 MeV is added to the energy of each proton during each revolution around the main ring. How many revolutions are needed to produce 7.00-TeV (7000 GeV) protons, if they are injected with an initial energy of 8.00 GeV?
11. A proton and an antiproton collide head-on, with each having a kinetic energy of 7.00 TeV (such as in the LHC at CERN). How much collision energy is available, taking into account the annihilation of the two masses? (Note that this is not significantly greater than the extremely relativistic kinetic energy.)
12. When an electron and positron collide at the SLAC facility, they each have 50.0 GeV kinetic energies. What is the total collision energy available, taking into account the annihilation energy? Note that the annihilation energy is insignificant, because the electrons are highly relativistic.

33.4 Particles, Patterns, and Conservation Laws

13. The π^0 is its own antiparticle and decays in the following manner: $\pi^0 \rightarrow \gamma + \gamma$. What is the energy of each γ ray if the π^0 is at rest when it decays?
14. The primary decay mode for the negative pion is $\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$. What is the energy release in MeV in this decay?
15. The mass of a theoretical particle that may be associated with the unification of the electroweak and strong forces is $10^{14} \text{ GeV}/c^2$.
 - (a) How many proton masses is this?
 - (b) How many electron masses is this? (This indicates how extremely relativistic the

accelerator would have to be in order to make the particle, and how large the relativistic quantity γ would have to be.)

16. The decay mode of the negative muon is $\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu$.
 - (a) Find the energy released in MeV.
 - (b) Verify that charge and lepton family numbers are conserved.
17. The decay mode of the positive tau is $\tau^+ \rightarrow \mu^+ + \nu_\mu + \bar{\nu}_\tau$.
 - (a) What energy is released?
 - (b) Verify that charge and lepton family numbers are conserved.
 - (c) The τ^+ is the antiparticle of the τ^- . Verify that all the decay products of the τ^+ are the antiparticles of those in the decay of the τ^- given in the text.
18. The principal decay mode of the sigma zero is $\Sigma^0 \rightarrow \Lambda^0 + \gamma$.
 - (a) What energy is released?
 - (b) Considering the quark structure of the two baryons, does it appear that the Σ^0 is an excited state of the Λ^0 ?
 - (c) Verify that strangeness, charge, and baryon number are conserved in the decay.
 - (d) Considering the preceding and the short lifetime, can the weak force be responsible? State why or why not.
19. (a) What is the uncertainty in the energy released in the decay of a π^0 due to its short lifetime?
 (b) What fraction of the decay energy is this, noting that the decay mode is $\pi^0 \rightarrow \gamma + \gamma$ (so that all the π^0 mass is destroyed)?
20. (a) What is the uncertainty in the energy released in the decay of a τ^- due to its short lifetime?
 (b) Is the uncertainty in this energy greater than or less than the uncertainty in the mass of the tau neutrino? Discuss the source of the uncertainty.

33.5 Quarks: Is That All There Is?

21. (a) Verify from its quark composition that the Δ^+ particle could be an excited state of the proton.
 (b) There is a spread of about 100 MeV in the decay energy of the Δ^+ , interpreted as uncertainty due to its short lifetime. What is its approximate lifetime?
 (c) Does its decay proceed via the strong or

weak force?

- 22.** Accelerators such as the Triangle Universities Meson Facility (TRIUMF) in British Columbia produce secondary beams of pions by having an intense primary proton beam strike a target. Such “meson factories” have been used for many years to study the interaction of pions with nuclei and, hence, the strong nuclear force. One reaction that occurs is $\pi^+ + p \rightarrow \Delta^{++} \rightarrow \pi^+ + p$, where the Δ^{++} is a very short-lived particle. The graph in [Figure 33.26](#) shows the probability of this reaction as a function of energy. The width of the bump is the uncertainty in energy due to the short lifetime of the Δ^{++} .
- (a) Find this lifetime.
- (b) Verify from the quark composition of the particles that this reaction annihilates and then recreates a d quark and a $\bar{d}$ antiquark by writing the reaction and decay in terms of quarks.
- (c) Draw a Feynman diagram of the production and decay of the Δ^{++} showing the individual quarks involved.

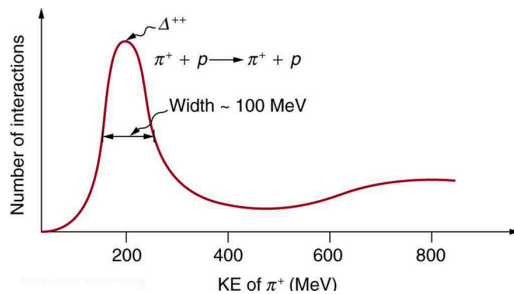


FIGURE 33.26 This graph shows the probability of an interaction between a π^+ and a proton as a function of energy. The bump is interpreted as a very short lived particle called a Δ^{++} . The approximately 100-MeV width of the bump is due to the short lifetime of the Δ^{++} .

- 23.** The reaction $\pi^+ + p \rightarrow \Delta^{++}$ (described in the preceding problem) takes place via the strong force. (a) What is the baryon number of the Δ^{++} particle? (b) Draw a Feynman diagram of the reaction showing the individual quarks involved.
- 24.** One of the decay modes of the omega minus is $\Omega^- \rightarrow \Xi^0 + \pi^-$.
- (a) What is the change in strangeness?
- (b) Verify that baryon number and charge are conserved, while lepton numbers are unaffected.
- (c) Write the equation in terms of the constituent quarks, indicating that the weak force is responsible.
- 25.** Repeat the previous problem for the decay mode $\Omega^- \rightarrow \Lambda^0 + K^-$.
- 26.** One decay mode for the eta-zero meson is $\eta^0 \rightarrow \gamma + \gamma$.
- (a) Find the energy released.
- (b) What is the uncertainty in the energy due to the short lifetime?
- (c) Write the decay in terms of the constituent quarks.
- (d) Verify that baryon number, lepton numbers, and charge are conserved.
- 27.** One decay mode for the eta-zero meson is $\eta^0 \rightarrow \pi^0 + \pi^0$.
- (a) Write the decay in terms of the quark constituents.
- (b) How much energy is released?
- (c) What is the ultimate release of energy, given the decay mode for the pi zero is $\pi^0 \rightarrow \gamma + \gamma$?
- 28.** Is the decay $n \rightarrow e^+ + e^-$ possible considering the appropriate conservation laws? State why or why not.
- 29.** Is the decay $\mu^- \rightarrow e^- + \nu_e + \nu_\mu$ possible considering the appropriate conservation laws? State why or why not.
- 30.** (a) Is the decay $\Lambda^0 \rightarrow n + \pi^0$ possible considering the appropriate conservation laws? State why or why not. (b) Write the decay in terms of the quark constituents of the particles.
- 31.** (a) Is the decay $\Sigma^- \rightarrow n + \pi^-$ possible considering the appropriate conservation laws? State why or why not. (b) Write the decay in terms of the quark constituents of the particles.
- 32.** The only combination of quark colors that produces a white baryon is RGB . Identify all the color combinations that can produce a white meson.
- 33.** (a) Three quarks form a baryon. How many combinations of the six known quarks are there if all combinations are possible? (b) This number is less than the number of known baryons. Explain why.
- 34.** (a) Show that the conjectured decay of the proton, $p \rightarrow \pi^0 + e^+$, violates conservation of baryon number and conservation of lepton number. (b) What is the analogous decay process for the antiproton?
- 35.** Verify the quantum numbers given for the Ω^+ in [Table 33.2](#) by adding the quantum numbers for

its quark constituents as inferred from [Table 33.4](#).

36. Verify the quantum numbers given for the proton and neutron in [Table 33.2](#) by adding the quantum numbers for their quark constituents as given in [Table 33.4](#).
37. (a) How much energy would be released if the proton did decay via the conjectured reaction $p \rightarrow \pi^0 + e^+$?
(b) Given that the π^0 decays to two γ s and that the e^+ will find an electron to annihilate, what total energy is ultimately produced in proton decay?
(c) Why is this energy greater than the proton's total mass (converted to energy)?
38. (a) Find the charge, baryon number, strangeness, charm, and bottomness of the J/ψ particle from its quark composition.
(b) Do the same for the Υ particle.
39. There are particles called D -mesons. One of them is the D^+ meson, which has a single positive charge and a baryon number of zero, also the value of its strangeness, topness, and bottomness. It has a charm of +1. What is its quark configuration?
40. There are particles called bottom mesons or B -mesons. One of them is the B^- meson, which has a single negative charge; its baryon number is zero, as are its strangeness, charm, and topness. It has a bottomness of -1 . What is its quark configuration?
41. (a) What particle has the quark composition $\bar{u}\bar{u}\bar{d}$?
(b) What should its decay mode be?
42. (a) Show that all combinations of three quarks produce integral charges. Thus baryons must have integral charge.
(b) Show that all combinations of a quark and an antiquark produce only integral charges. Thus mesons must have integral charge.

33.6 GUTs: The Unification of Forces

43. Integrated Concepts

The intensity of cosmic ray radiation decreases rapidly with increasing energy, but there are occasionally extremely energetic cosmic rays that create a shower of radiation from all the particles they create by striking a nucleus in the atmosphere as seen in the figure given below. Suppose a cosmic ray particle having an energy

of 10^{10} GeV converts its energy into particles with masses averaging $200 \text{ MeV}/c^2$. (a) How many particles are created? (b) If the particles rain down on a 1.00-km^2 area, how many particles are there per square meter?

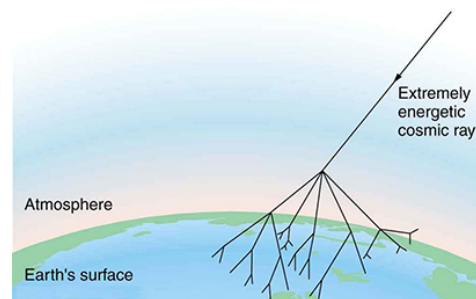


FIGURE 33.27 An extremely energetic cosmic ray creates a shower of particles on earth. The energy of these rare cosmic rays can approach a joule (about 10^{10} GeV) and, after multiple collisions, huge numbers of particles are created from this energy. Cosmic ray showers have been observed to extend over many square kilometers.

44. Integrated Concepts

Assuming conservation of momentum, what is the energy of each γ ray produced in the decay of a neutral at rest pion, in the reaction $\pi^0 \rightarrow \gamma + \gamma$?

45. Integrated Concepts

What is the wavelength of a 50-GeV electron, which is produced at SLAC? This provides an idea of the limit to the detail it can probe.

46. Integrated Concepts

(a) Calculate the relativistic quantity $\gamma = \frac{1}{\sqrt{1-v^2/c^2}}$ for 1.00-TeV protons produced at Fermilab. (b) If such a proton created a π^+ having the same speed, how long would its life be in the laboratory? (c) How far could it travel in this time?

47. Integrated Concepts

The primary decay mode for the negative pion is $\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$. (a) What is the energy release in MeV in this decay? (b) Using conservation of momentum, how much energy does each of the decay products receive, given the π^- is at rest when it decays? You may assume the muon antineutrino is massless and has momentum $p = E/c$, just like a photon.

48. Integrated Concepts

Plans for an accelerator that produces a secondary beam of K -mesons to scatter from nuclei, for the purpose of studying the strong force, call for them to have a kinetic energy of

500 MeV. (a) What would the relativistic quantity $\gamma = \frac{1}{\sqrt{1-v^2/c^2}}$ be for these particles? (b) How long would their average lifetime be in the laboratory? (c) How far could they travel in this time?

49. Integrated Concepts

Suppose you are designing a proton decay experiment and you can detect 50 percent of the proton decays in a tank of water. (a) How many kilograms of water would you need to see one decay per month, assuming a lifetime of 10^{31} y? (b) How many cubic meters of water is this? (c) If the actual lifetime is 10^{33} y, how long would you have to wait on an average to see a single proton decay?

50. Integrated Concepts

In supernovas, neutrinos are produced in huge amounts. They were detected from the 1987A supernova in the Magellanic Cloud, which is about 120,000 light years away from the Earth (relatively close to our Milky Way galaxy). If neutrinos have a mass, they cannot travel at the speed of light, but if their mass is small, they can get close. (a) Suppose a neutrino with a $7\text{-eV}/c^2$ mass has a kinetic energy of 700 keV. Find the relativistic quantity $\gamma = \frac{1}{\sqrt{1-v^2/c^2}}$ for it. (b) If the neutrino leaves the 1987A supernova at the same time as a photon and both travel to Earth, how much sooner does the photon arrive? This is not a large time difference, given that it is impossible to know which neutrino left with which photon and the poor efficiency of the neutrino detectors. Thus, the fact that neutrinos were observed within hours of the brightening of the supernova only places an upper limit on the

neutrino's mass. (Hint: You may need to use a series expansion to find v for the neutrino, since its γ is so large.)

51. Construct Your Own Problem

Consider an ultrahigh-energy cosmic ray entering the Earth's atmosphere (some have energies approaching a joule). Construct a problem in which you calculate the energy of the particle based on the number of particles in an observed cosmic ray shower. Among the things to consider are the average mass of the shower particles, the average number per square meter, and the extent (number of square meters covered) of the shower. Express the energy in eV and joules.

52. Construct Your Own Problem

Consider a detector needed to observe the proposed, but extremely rare, decay of an electron. Construct a problem in which you calculate the amount of matter needed in the detector to be able to observe the decay, assuming that it has a signature that is clearly identifiable. Among the things to consider are the estimated half life (long for rare events), and the number of decays per unit time that you wish to observe, as well as the number of electrons in the detector substance.

53. Critical Thinking (a) How much mass is converted to energy when a proton and antiproton annihilate each other? (b) How much energy is produced in the conversion above? (c) If energy were to produce a proton-antiproton pair, how much energy would this take? (d) What characteristic other than one being antimatter is different for a proton and an antiproton?

CHAPTER 34

Frontiers of Physics



FIGURE 34.1 This galaxy is ejecting huge jets of matter, powered by an immensely massive black hole at its center. (credit: X-ray: NASA/CXC/CfA/R. Kraft et al.)

CHAPTER OUTLINE

- 34.1 Cosmology and Particle Physics**
- 34.2 General Relativity and Quantum Gravity**
- 34.3 Superstrings**
- 34.4 Dark Matter and Closure**
- 34.5 Complexity and Chaos**
- 34.6 High-temperature Superconductors**
- 34.7 Some Questions We Know to Ask**

INTRODUCTION TO FRONTIERS OF PHYSICS Frontiers are exciting. There is mystery, surprise, adventure, and discovery. The satisfaction of finding the answer to a question is made keener by the fact that the answer always leads to a new question. The picture of nature becomes more complete, yet nature retains its sense of mystery and never loses its ability to awe us. The view of physics is beautiful looking both backward and forward in time. What marvelous patterns we have discovered. How clever nature seems in its rules and connections. How awesome. And we continue looking ever deeper and ever further, probing the basic structure of matter, energy, space, and time and wondering about the scope of the universe, its beginnings and future.

You are now in a wonderful position to explore the forefronts of physics, both the new discoveries and the unanswered questions. With the concepts, qualitative and quantitative, the problem-solving skills, the feeling for connections among topics, and all the rest you have mastered, you can more deeply appreciate and enjoy the brief treatments that follow. Years from now you will still enjoy the quest with an insight all the greater for your efforts.

34.1 Cosmology and Particle Physics

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Discuss the expansion of the universe.
- Explain the Big Bang.

Look at the sky on some clear night when you are away from city lights. There you will see thousands of individual stars and a faint glowing background of millions more. The Milky Way, as it has been called since ancient times, is an arm of our galaxy of stars—the word *galaxy* coming from the Greek word *galaxias*, meaning milky. We know a great deal about our Milky Way galaxy and of the billions of other galaxies beyond its fringes. But they still provoke wonder and awe (see [Figure 34.2](#)). And there are still many questions to be answered. Most remarkable when we view the universe on the large scale is that once again explanations of its character and evolution are tied to the very small scale. Particle physics and the questions being asked about the very small scales may also have their answers in the very large scales.



FIGURE 34.2 Take a moment to contemplate these clusters of galaxies, photographed by the Hubble Space Telescope. Trillions of stars linked by gravity in fantastic forms, glowing with light and showing evidence of undiscovered matter. What are they like, these myriad stars? How did they evolve? What can they tell us of matter, energy, space, and time? (credit: NASA, ESA, K. Sharon (Tel Aviv University) and E. Ofek (Caltech))

As has been noted in numerous Things Great and Small vignettes, this is not the first time the large has been explained by the small and vice versa. Newton realized that the nature of gravity on Earth that pulls an apple to the ground could explain the motion of the moon and planets so much farther away. Minute atoms and molecules explain the chemistry of substances on a much larger scale. Decays of tiny nuclei explain the hot interior of the Earth. Fusion of nuclei likewise explains the energy of stars. Today, the patterns in particle physics seem to be explaining the evolution and character of the universe. And the nature of the universe has implications for unexplored regions of particle physics.

Cosmology is the study of the character and evolution of the universe. What are the major characteristics of the universe as we know them today? First, there are approximately 10^{11} galaxies in the observable part of the universe. An average galaxy contains more than 10^{11} stars, with our Milky Way galaxy being larger than average, both in its number of stars and its dimensions. Ours is a spiral-shaped galaxy with a diameter of about 100,000 light years and a thickness of about 2000 light years in the arms with a central bulge about 10,000 light years across. The Sun lies about 30,000 light years from the center near the galactic plane. There are significant clouds of gas, and there is a halo of less-dense regions of stars surrounding the main body. (See [Figure 34.3](#).) Evidence strongly suggests the existence of a large amount of additional matter in galaxies that does not produce light—the mysterious dark matter we shall later discuss.

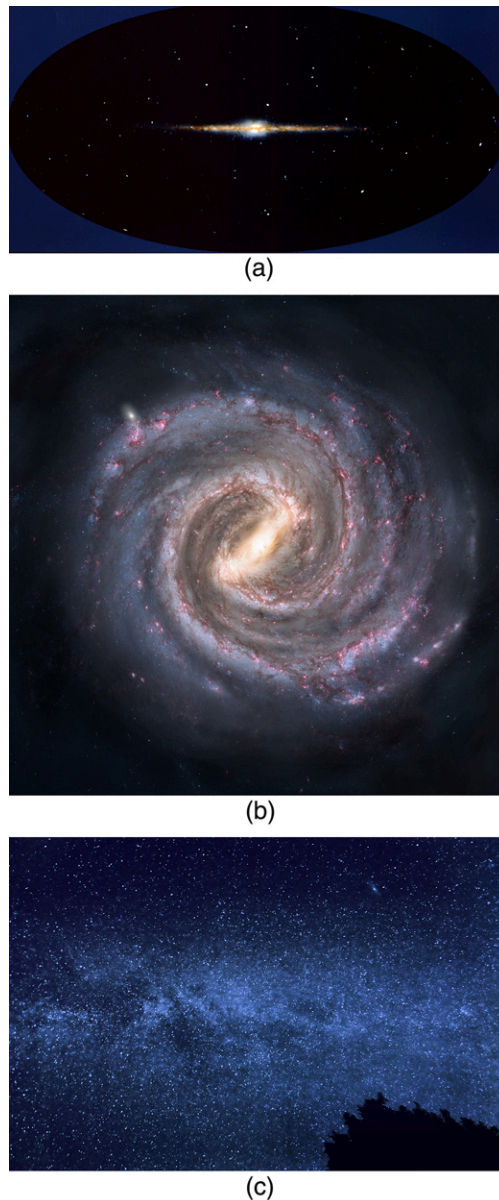


FIGURE 34.3 The Milky Way galaxy is typical of large spiral galaxies in its size, its shape, and the presence of gas and dust. We are fortunate to be in a location where we can see out of the galaxy and observe the vastly larger and fascinating universe around us. (a) Side view. (b) View from above. (c) The Milky Way as seen from Earth. (credits: (a) NASA, (b) Nick Risinger, (c) Andy)

Distances are great even within our galaxy and are measured in light years (the distance traveled by light in one year). The average distance between galaxies is on the order of a million light years, but it varies greatly with galaxies forming clusters such as shown in [Figure 34.2](#). The Magellanic Clouds, for example, are small galaxies close to our own, some 160,000 light years from Earth. The Andromeda galaxy is a large spiral galaxy like ours and lies 2 million light years away. It is just visible to the naked eye as an extended glow in the Andromeda constellation. Andromeda is the closest large galaxy in our local group, and we can see some individual stars in it with our larger telescopes. The most distant known galaxy is 14 billion light years from Earth—a truly incredible distance. (See [Figure 34.4](#).)

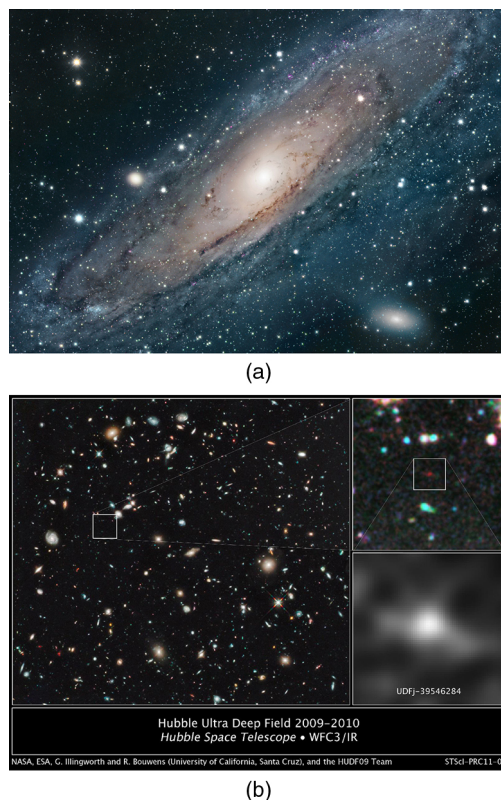


FIGURE 34.4 (a) Andromeda is the closest large galaxy, at 2 million light years distance, and is very similar to our Milky Way. The blue regions harbor young and emerging stars, while dark streaks are vast clouds of gas and dust. A smaller satellite galaxy is clearly visible. (b) The box indicates what may be the most distant known galaxy, estimated to be 13 billion light years from us. It exists in a much older part of the universe. (credit: NASA, ESA, G. Illingworth (University of California, Santa Cruz), R. Bouwens (University of California, Santa Cruz and Leiden University), and the HUDF09 Team)

Consider the fact that the light we receive from these vast distances has been on its way to us for a long time. In fact, the time in years is the same as the distance in light years. For example, the Andromeda galaxy is 2 million light years away, so that the light now reaching us left it 2 million years ago. If we could be there now, Andromeda would be different. Similarly, light from the most distant galaxy left it 14 billion years ago. We have an incredible view of the past when looking great distances. We can try to see if the universe was different then—if distant galaxies are more tightly packed or have younger-looking stars, for example, than closer galaxies, in which case there has been an evolution in time. But the problem is that the uncertainties in our data are great. Cosmology is almost typified by these large uncertainties, so that we must be especially cautious in drawing conclusions. One consequence is that there are more questions than answers, and so there are many competing theories. Another consequence is that any hard data produce a major result. Discoveries of some importance are being made on a regular basis, the hallmark of a field in its golden age.

Perhaps the most important characteristic of the universe is that all galaxies except those in our local cluster seem to be moving away from us at speeds proportional to their distance from our galaxy. It looks as if a gigantic explosion, universally called the **Big Bang**, threw matter out some billions of years ago. This amazing conclusion is based on the pioneering work of Edwin Hubble (1889–1953), the American astronomer. In the 1920s, Hubble first demonstrated conclusively that other galaxies, many previously called nebulae or clouds of stars, were outside our own. He then found that all but the closest galaxies have a red shift in their hydrogen spectra that is proportional to their distance. The explanation is that there is a **cosmological red shift** due to the expansion of space itself. The photon wavelength is stretched in transit from the source to the observer. Double the distance, and the red shift is doubled. While this cosmological red shift is often called a Doppler shift, it is not—space itself is expanding. There is no center of expansion in the universe. All observers see themselves as stationary; the other objects in space appear to be moving away from them. Hubble was directly responsible for discovering that the universe was much larger than had previously been imagined and that it had this amazing characteristic of rapid expansion.

Universal expansion on the scale of galactic clusters (that is, galaxies at smaller distances are not uniformly

receding from one another) is an integral part of modern cosmology. For galaxies farther away than about 50 Mly (50 million light years), the expansion is uniform with variations due to local motions of galaxies within clusters. A representative recession velocity v can be obtained from the simple formula

$$v = H_0 d, \quad 34.1$$

where d is the distance to the galaxy and H_0 is the **Hubble constant**. The Hubble constant is a central concept in cosmology. Its value is determined by taking the slope of a graph of velocity versus distance, obtained from red shift measurements, such as shown in [Figure 34.5](#). We shall use an approximate value of $H_0 = 20$ (km/s)/Mly. Thus, $v = H_0 d$ is an average behavior for all but the closest galaxies. For example, a galaxy 100 Mly away (as determined by its size and brightness) typically moves away from us at a speed of $v = (20 \text{ (km/s)/Mly})(100 \text{ Mly}) = 2000 \text{ km/s}$. There can be variations in this speed due to so-called local motions or interactions with neighboring galaxies. Conversely, if a galaxy is found to be moving away from us at speed of 100,000 km/s based on its red shift, it is at a distance

$d = v/H_0 = (10,000 \text{ km/s}) / (20 \text{ (km/s)/Mly}) = 5000 \text{ Mly} = 5 \text{ Gly}$ or $5 \times 10^9 \text{ ly}$. This last calculation is approximate, because it assumes the expansion rate was the same 5 billion years ago as now. A similar calculation in Hubble's measurement changed the notion that the universe is in a steady state.

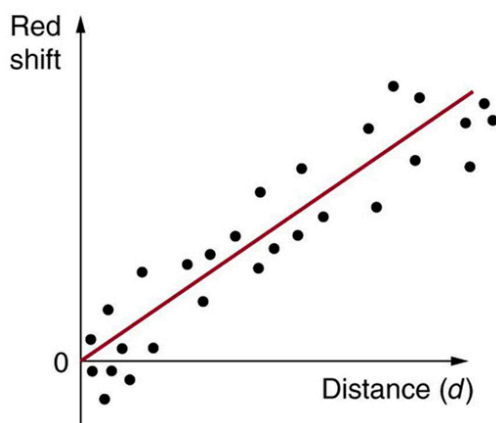


FIGURE 34.5 This graph of red shift versus distance for galaxies shows a linear relationship, with larger red shifts at greater distances, implying an expanding universe. The slope gives an approximate value for the expansion rate. (credit: John Cub).

One of the most intriguing developments recently has been the discovery that the expansion of the universe may be *faster now* than in the past, rather than slowing due to gravity as expected. Various groups have been looking, in particular, at supernovas in moderately distant galaxies (less than 1 Gly) to get improved distance measurements. Those distances are larger than expected for the observed galactic red shifts, implying the expansion was slower when that light was emitted. This has cosmological consequences that are discussed in [Dark Matter and Closure](#). The first results, published in 1999, are only the beginning of emerging data, with astronomy now entering a data-rich era.

[Figure 34.6](#) shows how the recession of galaxies looks like the remnants of a gigantic explosion, the famous Big Bang. Extrapolating backward in time, the Big Bang would have occurred between 13 and 15 billion years ago when all matter would have been at a point. Questions instantly arise. What caused the explosion? What happened before the Big Bang? Was there a before, or did time start then? Will the universe expand forever, or will gravity reverse it into a Big Crunch? And is there other evidence of the Big Bang besides the well-documented red shifts?

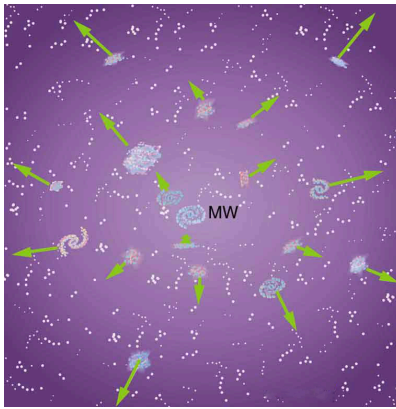


FIGURE 34.6 Galaxies are flying apart from one another, with the more distant moving faster as if a primordial explosion expelled the matter from which they formed. The most distant known galaxies move nearly at the speed of light relative to us.

The Russian-born American physicist George Gamow (1904–1968) was among the first to note that, if there was a Big Bang, the remnants of the primordial fireball should still be evident and should be blackbody radiation. Since the radiation from this fireball has been traveling to us since shortly after the Big Bang, its wavelengths should be greatly stretched. It will look as if the fireball has cooled in the billions of years since the Big Bang. Gamow and collaborators predicted in the late 1940s that there should be blackbody radiation from the explosion filling space with a characteristic temperature of about 7 K. Such blackbody radiation would have its peak intensity in the microwave part of the spectrum. (See [Figure 34.7](#).) In 1964, Arno Penzias and Robert Wilson, two American scientists working with Bell Telephone Laboratories on a low-noise radio antenna, detected the radiation and eventually recognized it for what it is.

[Figure 34.7\(b\)](#) shows the spectrum of this microwave radiation that permeates space and is of cosmic origin. It is the most perfect blackbody spectrum known, and the temperature of the fireball remnant is determined from it to be 2.725 ± 0.002 K. The detection of what is now called the **cosmic microwave background** (CMBR) was so important (generally considered as important as Hubble's detection that the galactic red shift is proportional to distance) that virtually every scientist has accepted the expansion of the universe as fact. Penzias and Wilson shared the 1978 Nobel Prize in Physics for their discovery.

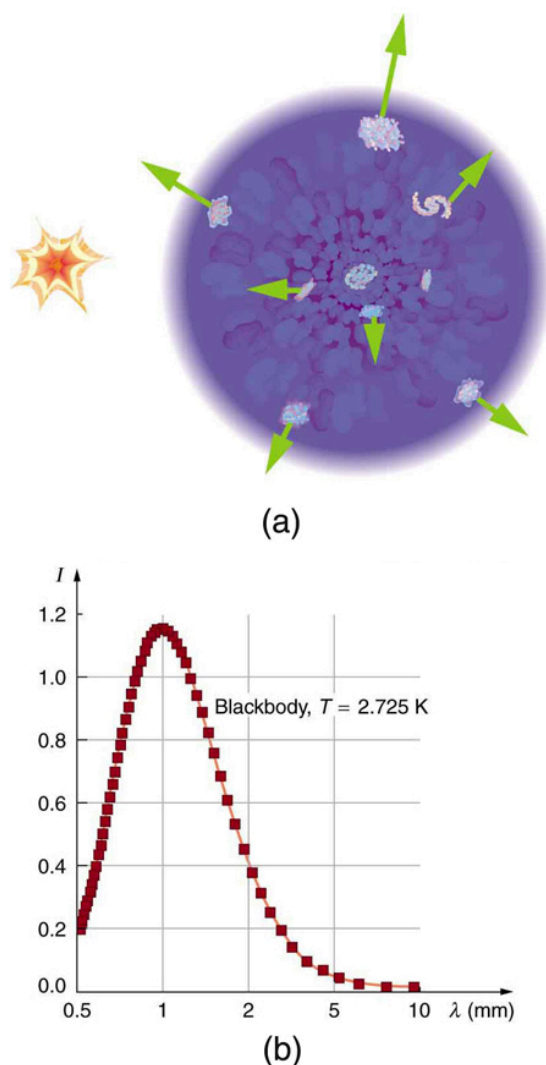


FIGURE 34.7 (a) The Big Bang is used to explain the present observed expansion of the universe. It was an incredibly energetic explosion some 10 to 20 billion years ago. After expanding and cooling, galaxies form inside the now-cold remnants of the primordial fireball. (b) The spectrum of cosmic microwave radiation is the most perfect blackbody spectrum ever detected. It is characteristic of a temperature of 2.725 K, the expansion-cooled temperature of the Big Bang's remnant. This radiation can be measured coming from any direction in space not obscured by some other source. It is compelling evidence of the creation of the universe in a gigantic explosion, already indicated by galactic red shifts.

MAKING CONNECTIONS: COSMOLOGY AND PARTICLE PHYSICS

There are many connections of cosmology—by definition involving physics on the largest scale—with particle physics—by definition physics on the smallest scale. Among these are the dominance of matter over antimatter, the nearly perfect uniformity of the cosmic microwave background, and the mere existence of galaxies.

Matter versus antimatter We know from direct observation that antimatter is rare. The Earth and the solar system are nearly pure matter. Space probes and cosmic rays give direct evidence—the landing of the Viking probes on Mars would have been spectacular explosions of mutual annihilation energy if Mars were antimatter. We also know that most of the universe is dominated by matter. This is proven by the lack of annihilation radiation coming to us from space, particularly the relative absence of 0.511-MeV γ rays created by the mutual annihilation of electrons and positrons. It seemed possible that there could be entire solar systems or galaxies made of antimatter in perfect symmetry with our matter-dominated systems. But the interactions between stars and galaxies would sometimes bring matter and antimatter together in large amounts. The annihilation radiation they would produce is simply not observed. Antimatter in nature is created in particle collisions and in β^+ decays, but only in small amounts that quickly annihilate, leaving almost pure matter surviving.

Particle physics seems symmetric in matter and antimatter. Why isn't the cosmos? The answer is that particle physics is not quite perfectly symmetric in this regard. The decay of one of the neutral K -mesons, for example, preferentially creates more matter than antimatter. This is caused by a fundamental small asymmetry in the basic forces. This small asymmetry produced slightly more matter than antimatter in the early universe. If there was only one part in 10^9 more matter (a small asymmetry), the rest would annihilate pair for pair, leaving nearly pure matter to form the stars and galaxies we see today. So the vast number of stars we observe may be only a tiny remnant of the original matter created in the Big Bang. Here at last we see a very real and important asymmetry in nature. Rather than be disturbed by an asymmetry, most physicists are impressed by how small it is. Furthermore, if the universe were completely symmetric, the mutual annihilation would be more complete, leaving far less matter to form us and the universe we know.

How can something so old have so few wrinkles? A troubling aspect of cosmic microwave background radiation (CMBR) was soon recognized. True, the CMBR verified the Big Bang, had the correct temperature, and had a blackbody spectrum as expected. But the CMBR was *too* smooth—it looked identical in every direction. Galaxies and other similar entities could not be formed without the existence of fluctuations in the primordial stages of the universe and so there should be hot and cool spots in the CMBR, nicknamed wrinkles, corresponding to dense and sparse regions of gas caused by turbulence or early fluctuations. Over time, dense regions would contract under gravity and form stars and galaxies. Why aren't the fluctuations there? (This is a good example of an answer producing more questions.) Furthermore, galaxies are observed very far from us, so that they formed very long ago. The problem was to explain how galaxies could form so early and so quickly after the Big Bang if its remnant fingerprint is perfectly smooth. The answer is that if you look very closely, the CMBR is not perfectly smooth, only extremely smooth.

A satellite called the Cosmic Background Explorer (COBE) carried an instrument that made very sensitive and accurate measurements of the CMBR. In April of 1992, there was extraordinary publicity of COBE's first results—there were small fluctuations in the CMBR. Further measurements were carried out by experiments including NASA's Wilkinson Microwave Anisotropy Probe (WMAP), which launched in 2001. Data from WMAP provided a much more detailed picture of the CMBR fluctuations. (See [Figure 34.7](#).) These amount to temperature fluctuations of only $200\ \mu\text{K}$ out of $2.7\ \text{K}$, better than one part in 1000. The WMAP experiment will be followed up by the European Space Agency's Planck Surveyor, which launched in 2009.

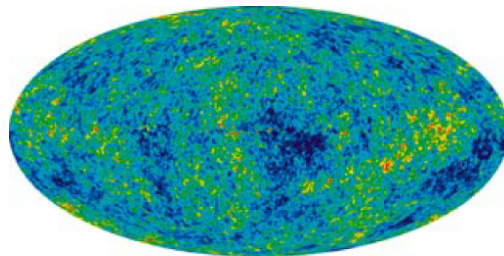


FIGURE 34.8 This map of the sky uses color to show fluctuations, or wrinkles, in the cosmic microwave background observed with the WMAP spacecraft. The Milky Way has been removed for clarity. Red represents higher temperature and higher density, while blue is lower temperature and density. The fluctuations are small, less than one part in 1000, but these are still thought to be the cause of the eventual formation of galaxies. (credit: NASA/WMAP Science Team)

Let us now examine the various stages of the overall evolution of the universe from the Big Bang to the present, illustrated in [Figure 34.9](#). Note that scientific notation is used to encompass the many orders of magnitude in time, energy, temperature, and size of the universe. Going back in time, the two lines approach but do not cross (there is no zero on an exponential scale). Rather, they extend indefinitely in ever-smaller time intervals to some infinitesimal point.

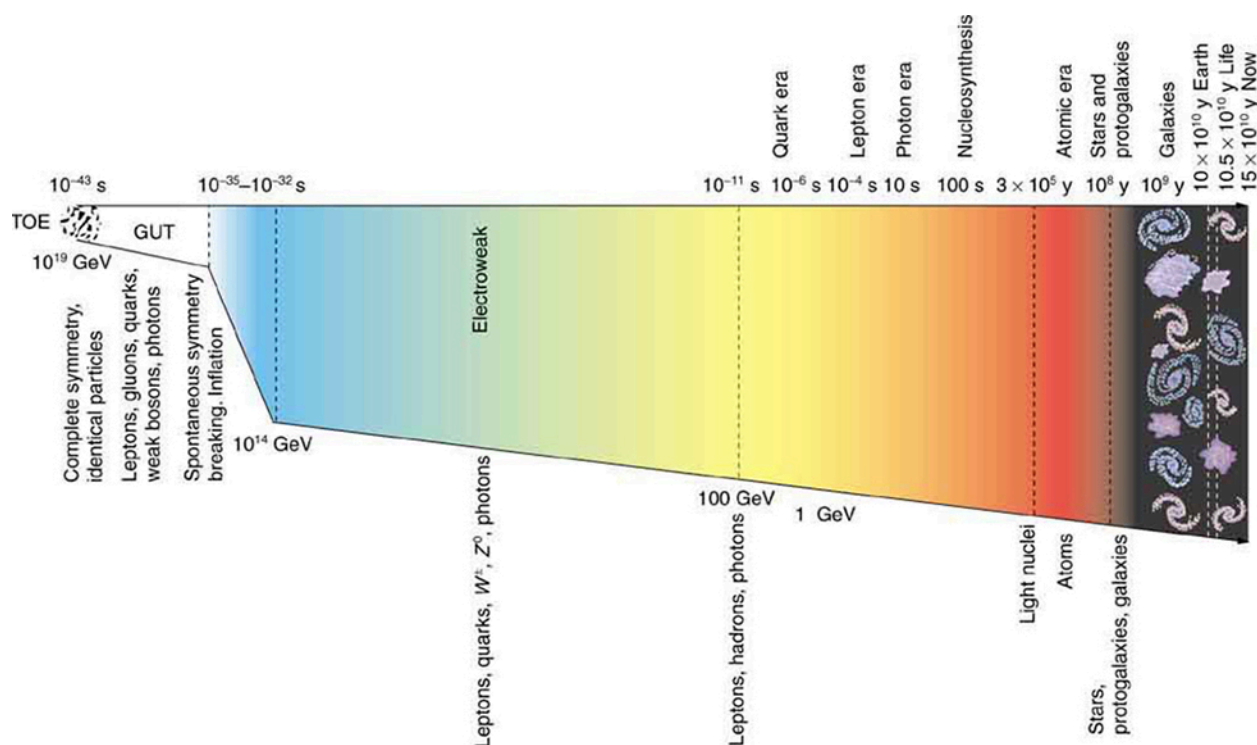


FIGURE 34.9 The evolution of the universe from the Big Bang onward is intimately tied to the laws of physics, especially those of particle physics at the earliest stages. The universe is relativistic throughout its history. Theories of the unification of forces at high energies may be verified by their shaping of the universe and its evolution.

Going back in time is equivalent to what would happen if expansion stopped and gravity pulled all the galaxies together, compressing and heating all matter. At a time long ago, the temperature and density were too high for stars and galaxies to exist. Before then, there was a time when the temperature was too great for atoms to exist. And farther back yet, there was a time when the temperature and density were so great that nuclei could not exist. Even farther back in time, the temperature was so high that average kinetic energy was great enough to create short-lived particles, and the density was high enough to make this likely. When we extrapolate back to the point of $W^\pm$ and Z^0 production (thermal energies reaching 1 TeV, or a temperature of about 10^{15} K), we reach the limits of what we know directly about particle physics. This is at a time about 10^{-12} s after the Big Bang. While 10^{-12} s may seem to be negligibly close to the instant of creation, it is not. There are important stages before this time that are tied to the unification of forces. At those stages, the universe was at extremely high energies and average particle separations were smaller than we can achieve with accelerators. What happened in the early stages before 10^{-12} s is crucial to all later stages and is possibly discerned by observing present conditions in the universe. One of these is the smoothness of the CMBR.

Names are given to early stages representing key conditions. The stage before 10^{-11} s back to 10^{-34} s is called the **electroweak epoch**, because the electromagnetic and weak forces become identical for energies above about 100 GeV. As discussed earlier, theorists expect that the strong force becomes identical to and thus unified with the electroweak force at energies of about 10^{14} GeV. The average particle energy would be this great at 10^{-34} s after the Big Bang, if there are no surprises in the unknown physics at energies above about 1 TeV. At the immense energy of 10^{14} GeV (corresponding to a temperature of about 10^{26} K), the $W^\pm$ and Z^0 carrier particles would be transformed into massless gauge bosons to accomplish the unification. Before 10^{-34} s back to about 10^{-43} s, we have Grand Unification in the **GUT epoch**, in which all forces except gravity are identical. At 10^{-43} s, the average energy reaches the immense 10^{19} GeV needed to unify gravity with the other forces in TOE, the Theory of Everything. Before that time is the **TOE epoch**, but we have almost no idea as to the nature of the universe then, since we have no workable theory of quantum gravity. We call the hypothetical unified force **superforce**.

Now let us imagine starting at TOE and moving forward in time to see what type of universe is created from various events along the way. As temperatures and average energies decrease with expansion, the universe reaches the stage where average particle separations are large enough to see differences between the strong and electroweak

forces (at about 10^{-35} s). After this time, the forces become distinct in almost all interactions—they are no longer unified or symmetric. This transition from GUT to electroweak is an example of **spontaneous symmetry breaking**, in which conditions spontaneously evolved to a point where the forces were no longer unified, breaking that symmetry. This is analogous to a phase transition in the universe, and a clever proposal by American physicist Alan Guth in the early 1980s ties it to the smoothness of the CMBR. Guth proposed that spontaneous symmetry breaking (like a phase transition during cooling of normal matter) released an immense amount of energy that caused the universe to expand extremely rapidly for the brief time from 10^{-35} s to about 10^{-32} s. This expansion may have been by an incredible factor of 10^{50} or more in the size of the universe and is thus called the **inflationary scenario**. One result of this inflation is that it would stretch the wrinkles in the universe nearly flat, leaving an extremely smooth CMBR. While speculative, there is as yet no other plausible explanation for the smoothness of the CMBR. Unless the CMBR is not really cosmic but local in origin, the distances between regions of similar temperatures are too great for any coordination to have caused them, since any coordination mechanism must travel at the speed of light. Again, particle physics and cosmology are intimately entwined. There is little hope that we may be able to test the inflationary scenario directly, since it occurs at energies near 10^{14} GeV, vastly greater than the limits of modern accelerators. But the idea is so attractive that it is incorporated into most cosmological theories.

Characteristics of the present universe may help us determine the validity of this intriguing idea. Additionally, the recent indications that the universe's expansion rate may be *increasing* (see [Dark Matter and Closure](#)) could even imply that we are *in* another inflationary epoch.

It is important to note that, if conditions such as those found in the early universe could be created in the laboratory, we would see the unification of forces directly today. The forces have not changed in time, but the average energy and separation of particles in the universe have. As discussed in [The Four Basic Forces](#), the four basic forces in nature are distinct under most circumstances found today. The early universe and its remnants provide evidence from times when they were unified under most circumstances.

34.2 General Relativity and Quantum Gravity

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the effect of gravity on light.
- Discuss black hole.
- Explain quantum gravity.

When we talk of black holes or the unification of forces, we are actually discussing aspects of general relativity and quantum gravity. We know from [Special Relativity](#) that relativity is the study of how different observers measure the same event, particularly if they move relative to one another. Einstein's theory of **general relativity** describes all types of relative motion including accelerated motion and the effects of gravity. General relativity encompasses special relativity and classical relativity in situations where acceleration is zero and relative velocity is small compared with the speed of light. Many aspects of general relativity have been verified experimentally, some of which are better than science fiction in that they are bizarre but true. **Quantum gravity** is the theory that deals with particle exchange of gravitons as the mechanism for the force, and with extreme conditions where quantum mechanics and general relativity must both be used. A good theory of quantum gravity does not yet exist, but one will be needed to understand how all four forces may be unified. If we are successful, the theory of quantum gravity will encompass all others, from classical physics to relativity to quantum mechanics—truly a Theory of Everything (TOE).

General Relativity

Einstein first considered the case of no observer acceleration when he developed the revolutionary special theory of relativity, publishing his first work on it in 1905. By 1916, he had laid the foundation of general relativity, again almost on his own. Much of what Einstein did to develop his ideas was to mentally analyze certain carefully and clearly defined situations—doing this is to perform a **thought experiment**. [Figure 34.10](#) illustrates a thought experiment like the ones that convinced Einstein that light must fall in a gravitational field. Think about what a person feels in an elevator that is accelerated upward. It is identical to being in a stationary elevator in a gravitational field. The feet of a person are pressed against the floor, and objects released from hand fall with

identical accelerations. In fact, it is not possible, without looking outside, to know what is happening—acceleration upward or gravity. This led Einstein to correctly postulate that acceleration and gravity will produce identical effects in all situations. So, if acceleration affects light, then gravity will, too. [Figure 34.10](#) shows the effect of acceleration on a beam of light shone horizontally at one wall. Since the accelerated elevator moves up during the time light travels across the elevator, the beam of light strikes low, seeming to the person to bend down. (Normally a tiny effect, since the speed of light is so great.) The same effect must occur due to gravity, Einstein reasoned, since there is no way to tell the effects of gravity acting downward from acceleration of the elevator upward. Thus gravity affects the path of light, even though we think of gravity as acting between masses and photons are massless.

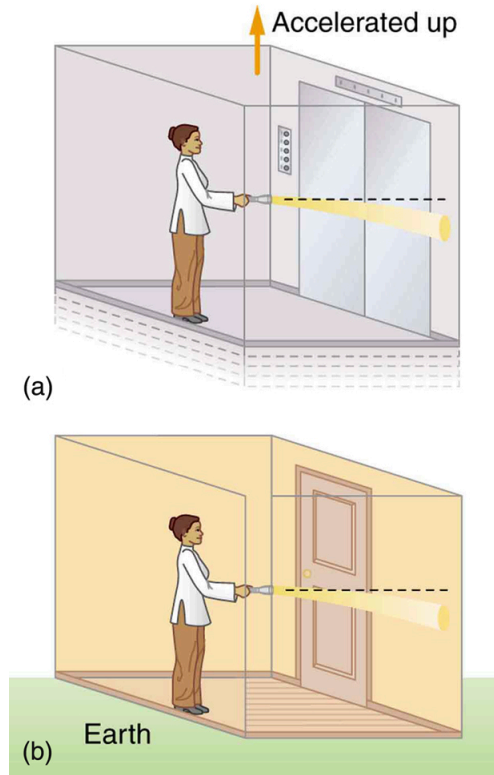


FIGURE 34.10 (a) A beam of light emerges from a flashlight in an upward-accelerating elevator. Since the elevator moves up during the time the light takes to reach the wall, the beam strikes lower than it would if the elevator were not accelerated. (b) Gravity has the same effect on light, since it is not possible to tell whether the elevator is accelerating upward or acted upon by gravity.

Einstein's theory of general relativity got its first verification in 1919 when starlight passing near the Sun was observed during a solar eclipse. (See [Figure 34.11](#).) During an eclipse, the sky is darkened and we can briefly see stars. Those in a line of sight nearest the Sun should have a shift in their apparent positions. Not only was this shift observed, but it agreed with Einstein's predictions well within experimental uncertainties. This discovery created a scientific and public sensation. Einstein was now a folk hero as well as a very great scientist. The bending of light by matter is equivalent to a bending of space itself, with light following the curve. This is another radical change in our concept of space and time. It is also another connection that any particle with mass or energy (massless photons) is affected by gravity.

There are several current forefront efforts related to general relativity. One is the observation and analysis of gravitational lensing of light. Another is analysis of the definitive proof of the existence of black holes. Direct observation of gravitational waves or moving wrinkles in space is being searched for. Theoretical efforts are also being aimed at the possibility of time travel and wormholes into other parts of space due to black holes.

Gravitational lensing As you can see in [Figure 34.11](#), light is bent toward a mass, producing an effect much like a converging lens (large masses are needed to produce observable effects). On a galactic scale, the light from a distant galaxy could be “lensed” into several images when passing close by another galaxy on its way to Earth. Einstein predicted this effect, but he considered it unlikely that we would ever observe it. A number of cases of this effect have now been observed; one is shown in [Figure 34.12](#). This effect is a much larger scale verification of general relativity. But such gravitational lensing is also useful in verifying that the red shift is proportional to

distance. The red shift of the intervening galaxy is always less than that of the one being lensed, and each image of the lensed galaxy has the same red shift. This verification supplies more evidence that red shift is proportional to distance. Confidence that the multiple images are not different objects is bolstered by the observations that if one image varies in brightness over time, the others also vary in the same manner.

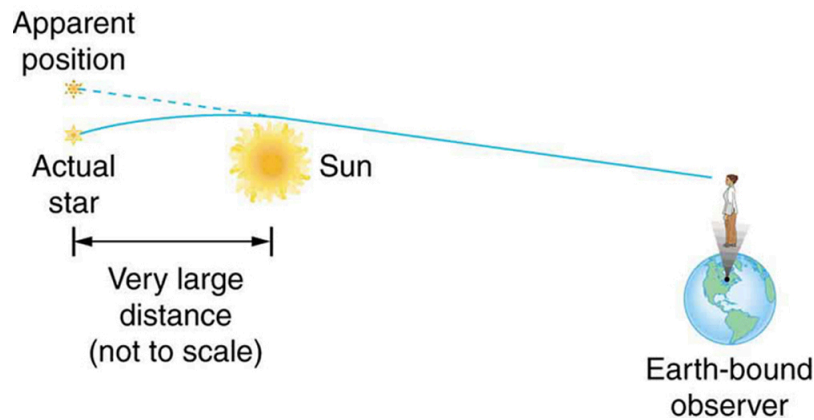


FIGURE 34.11 This schematic shows how light passing near a massive body like the Sun is curved toward it. The light that reaches the Earth then seems to be coming from different locations than the known positions of the originating stars. Not only was this effect observed, the amount of bending was precisely what Einstein predicted in his general theory of relativity.

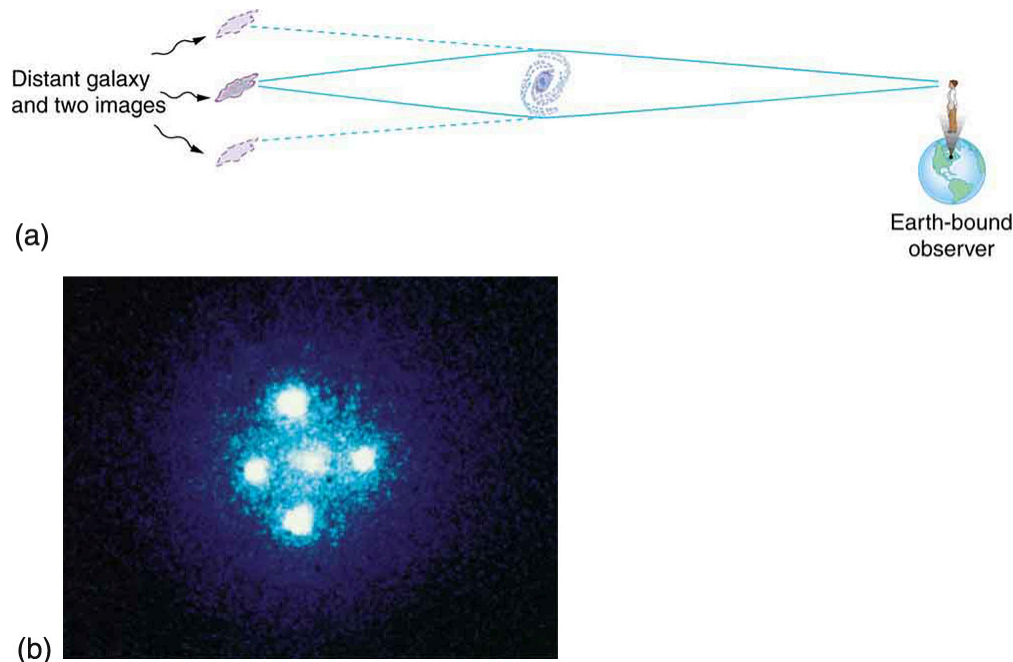


FIGURE 34.12 (a) Light from a distant galaxy can travel different paths to the Earth because it is bent around an intermediary galaxy by gravity. This produces several images of the more distant galaxy. (b) The images around the central galaxy are produced by gravitational lensing. Each image has the same spectrum and a larger red shift than the intermediary. (credit: NASA, ESA, and STScI)

Black holes Black holes are objects having such large gravitational fields that things can fall in, but nothing, not even light, can escape. Bodies, like the Earth or the Sun, have what is called an **escape velocity**. If an object moves straight up from the body, starting at the escape velocity, it will just be able to escape the gravity of the body. The greater the acceleration of gravity on the body, the greater is the escape velocity. As long ago as the late 1700s, it was proposed that if the escape velocity is greater than the speed of light, then light cannot escape. Simon Laplace (1749–1827), the French astronomer and mathematician, even incorporated this idea of a dark star into his writings. But the idea was dropped after Young’s double slit experiment showed light to be a wave. For some time, light was thought not to have particle characteristics and, thus, could not be acted upon by gravity. The idea of a black hole was very quickly reincarnated in 1916 after Einstein’s theory of general relativity was published. It is now thought that black holes can form in the supernova collapse of a massive star, forming an object perhaps 10 km across and having a mass greater than that of our Sun. It is interesting that several prominent physicists who

worked on the concept, including Einstein, firmly believed that nature would find a way to prohibit such objects.

Black holes are difficult to observe directly, because they are small and no light comes directly from them. In fact, no light comes from inside the **event horizon**, which is defined to be at a distance from the object at which the escape velocity is exactly the speed of light. The radius of the event horizon is known as the **Schwarzschild radius** R_S and is given by

$$R_S = \frac{2GM}{c^2}, \quad 34.2$$

where G is the universal gravitational constant, M is the mass of the body, and c is the speed of light. The event horizon is the edge of the black hole and R_S is its radius (that is, the size of a black hole is twice R_S). Since G is small and c^2 is large, you can see that black holes are extremely small, only a few kilometers for masses a little greater than the Sun's. The object itself is inside the event horizon.

Physics near a black hole is fascinating. Gravity increases so rapidly that, as you approach a black hole, the tidal effects tear matter apart, with matter closer to the hole being pulled in with much more force than that only slightly farther away. This can pull a companion star apart and heat inflowing gases to the point of producing X rays. (See [Figure 34.13](#).) We have observed X rays from certain binary star systems that are consistent with such a picture. This is not quite proof of black holes, because the X rays could also be caused by matter falling onto a neutron star. These objects were first discovered in 1967 by the British astrophysicists, Jocelyn Bell and Anthony Hewish.

Neutron stars are literally a star composed of neutrons. They are formed by the collapse of a star's core in a supernova, during which electrons and protons are forced together to form neutrons (the reverse of neutron β decay). Neutron stars are slightly larger than a black hole of the same mass and will not collapse further because of resistance by the strong force. However, neutron stars cannot have a mass greater than about eight solar masses or they must collapse to a black hole. With recent improvements in our ability to resolve small details, such as with the orbiting Chandra X-ray Observatory, it has become possible to measure the masses of X-ray-emitting objects by observing the motion of companion stars and other matter in their vicinity. What has emerged is a plethora of X-ray-emitting objects too massive to be neutron stars. This evidence is considered conclusive and the existence of black holes is widely accepted. These black holes are concentrated near galactic centers.

We also have evidence that supermassive black holes may exist at the cores of many galaxies, including the Milky Way. Such a black hole might have a mass millions or even billions of times that of the Sun, and it would probably have formed when matter first coalesced into a galaxy billions of years ago. Supporting this is the fact that very distant galaxies are more likely to have abnormally energetic cores. Some of the moderately distant galaxies, and hence among the younger, are known as **quasars** and emit as much or more energy than a normal galaxy but from a region less than a light year across. Quasar energy outputs may vary in times less than a year, so that the energy-emitting region must be less than a light year across. The best explanation of quasars is that they are young galaxies with a supermassive black hole forming at their core, and that they become less energetic over billions of years. In closer superactive galaxies, we observe tremendous amounts of energy being emitted from very small regions of space, consistent with stars falling into a black hole at the rate of one or more a month. The Hubble Space Telescope (1994) observed an accretion disk in the galaxy M87 rotating rapidly around a region of extreme energy emission. (See [Figure 34.13](#).) A jet of material being ejected perpendicular to the plane of rotation gives further evidence of a supermassive black hole as the engine.

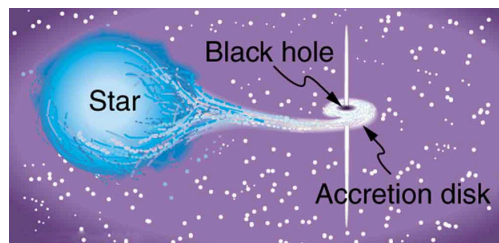


FIGURE 34.13 A black hole is shown pulling matter away from a companion star, forming a superheated accretion disk where X rays are emitted before the matter disappears forever into the hole. The in-fall energy also ejects some material, forming the two vertical spikes. (See also the photograph in the Introduction to Frontiers of Physics.) There are several X-ray-emitting objects in space that are consistent with this picture and are likely to be black holes.

Gravitational waves If a massive object distorts the space around it, like the foot of a water bug on the surface of a

pond, then movement of the massive object should create waves in space like those on a pond. **Gravitational waves** are mass-created distortions in space that propagate at the speed of light and are predicted by general relativity. Since gravity is by far the weakest force, extreme conditions are needed to generate significant gravitational waves. Gravity near binary neutron star systems is so great that significant gravitational wave energy is radiated as the two neutron stars orbit one another. American astronomers, Joseph Taylor and Russell Hulse, measured changes in the orbit of such a binary neutron star system. They found its orbit to change precisely as predicted by general relativity, a strong indication of gravitational waves, and were awarded the 1993 Nobel Prize. American physicist Joseph Weber pioneered direct detection of gravitational waves in the 1960s, but did not succeed in observing conclusive events. Gravitational wave signals were first observed directly on September 14, 2015 by the LIGO (Laser Interferometer Gravitational Wave Observatory) system's twin laser interferometer detectors, one in the state of Washington, and another in Louisiana (See [Figure 34.15](#)). On February 11, 2016, LIGO and the VIRGO (Variability of Irradiance and Gravitational Oscillations) facility in Italy, with a single detector, announced the first confirmed observation of gravitational waves from colliding black holes. This confirms a key prediction of Einstein's theory of general relativity and provides the first direct evidence that black holes merge. Since its first historic detection, LIGO has discovered many more gravitational wave events generated by merging black holes, colliding neutron stars, and neutron star/black hole mergers.

Quantum Gravity

Black holes radiate Quantum gravity is important in those situations where gravity is so extremely strong that it has effects on the quantum scale, where the other forces are ordinarily much stronger. The early universe was such a place, but black holes are another. The first significant connection between gravity and quantum effects was made by the Russian physicist Yakov Zel'dovich in 1971, and other significant advances followed from the British physicist Stephen Hawking. (See [Figure 34.16](#).) These two showed that black holes could radiate away energy by quantum effects just outside the event horizon (nothing can escape from inside the event horizon). Black holes are, thus, expected to radiate energy and shrink to nothing, although extremely slowly for most black holes. The mechanism is the creation of a particle-antiparticle pair from energy in the extremely strong gravitational field near the event horizon. One member of the pair falls into the hole and the other escapes, conserving momentum. (See [Figure 34.17](#).) When a black hole loses energy and, hence, rest mass, its event horizon shrinks, creating an even greater gravitational field. This increases the rate of pair production so that the process grows exponentially until the black hole is nuclear in size. A final burst of particles and γ rays ensues. This is an extremely slow process for black holes about the mass of the Sun (produced by supernovas) or larger ones (like those thought to be at galactic centers), taking on the order of 10^{67} years or longer! Smaller black holes would evaporate faster, but they are only speculated to exist as remnants of the Big Bang. Searches for characteristic γ -ray bursts have produced events attributable to more mundane objects like neutron stars accreting matter.

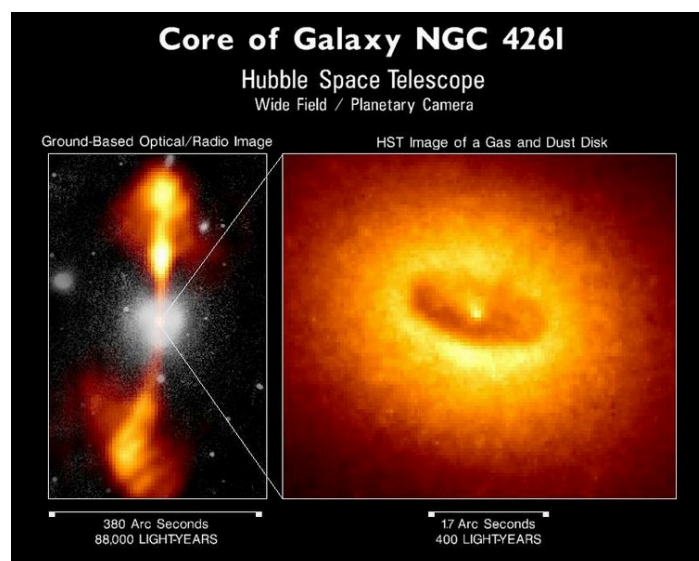


FIGURE 34.14 This Hubble Space Telescope photograph shows the extremely energetic core of the NGC 4261 galaxy. With the superior resolution of the orbiting telescope, it has been possible to observe the rotation of an accretion disk around the energy-producing object as well as to map jets of material being ejected from the object. A supermassive black hole is consistent with these observations, but other

possibilities are not quite eliminated. (credit: NASA and ESA)



FIGURE 34.15 The control room of the LIGO gravitational wave detector. Gravitational waves will cause extremely small vibrations in a mass in this detector, which will be detected by laser interferometer techniques. Such detection in coincidence with other detectors and with astronomical events, such as supernovas, would provide direct evidence of gravitational waves. (credit: Tobin Fricke)



FIGURE 34.16 Stephen Hawking (b. 1942) has made many contributions to the theory of quantum gravity. Hawking is a long-time survivor of ALS and has produced popular books on general relativity, cosmology, and quantum gravity. (credit: Lwp Kommunikáció)

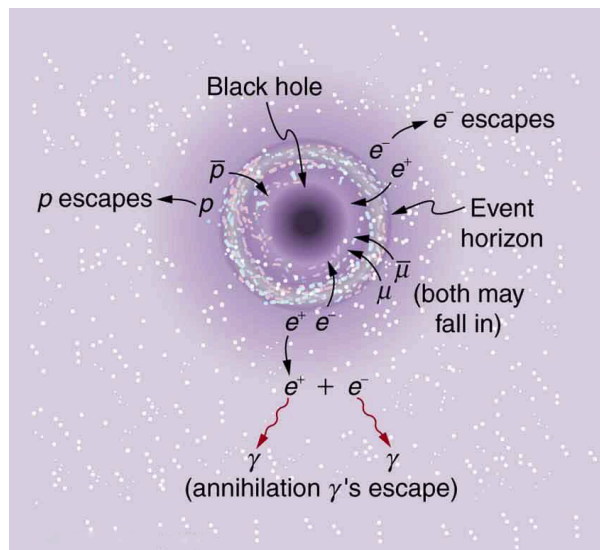


FIGURE 34.17 Gravity and quantum mechanics come into play when a black hole creates a particle-antiparticle pair from the energy in its gravitational field. One member of the pair falls into the hole while the other escapes, removing energy and shrinking the black hole. The search is on for the characteristic energy.

Wormholes and time travel The subject of time travel captures the imagination. Theoretical physicists, such as the American Kip Thorne, have treated the subject seriously, looking into the possibility that falling into a black hole could result in popping up in another time and place—a trip through a so-called wormhole. Time travel and

wormholes appear in innumerable science fiction dramatizations, but the consensus is that time travel is not possible in theory. While still debated, it appears that quantum gravity effects inside a black hole prevent time travel due to the creation of particle pairs. Direct evidence is elusive.

The shortest time Theoretical studies indicate that, at extremely high energies and correspondingly early in the universe, quantum fluctuations may make time intervals meaningful only down to some finite time limit. Early work indicated that this might be the case for times as long as 10^{-43} s, the time at which all forces were unified. If so, then it would be meaningless to consider the universe at times earlier than this. Subsequent studies indicate that the crucial time may be as short as 10^{-95} s. But the point remains—quantum gravity seems to imply that there is no such thing as a vanishingly short time. Time may, in fact, be grainy with no meaning to time intervals shorter than some tiny but finite size.

The future of quantum gravity Not only is quantum gravity in its infancy, no one knows how to get started on a theory of gravitons and unification of forces. The energies at which TOE should be valid may be so high (at least 10^{19} GeV) and the necessary particle separation so small (less than 10^{-35} m) that only indirect evidence can provide clues. For some time, the common lament of theoretical physicists was one so familiar to struggling students—how do you even get started? But Hawking and others have made a start, and the approach many theorists have taken is called Superstring theory, the topic of the [Superstrings](#).

34.3 Superstrings

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Define Superstring theory.
- Explain the relationship between Superstring theory and the Big Bang.

Introduced earlier in [GUTS: The Unification of Forces](#) **Superstring theory** is an attempt to unify gravity with the other three forces and, thus, must contain quantum gravity. The main tenet of Superstring theory is that fundamental particles, including the graviton that carries the gravitational force, act like one-dimensional vibrating strings. Since gravity affects the time and space in which all else exists, Superstring theory is an attempt at a Theory of Everything (TOE). Each independent quantum number is thought of as a separate dimension in some super space (analogous to the fact that the familiar dimensions of space are independent of one another) and is represented by a different type of Superstring. As the universe evolved after the Big Bang and forces became distinct (spontaneous symmetry breaking), some of the dimensions of superspace are imagined to have curled up and become unnoticed.

Forces are expected to be unified only at extremely high energies and at particle separations on the order of 10^{-35} m. This could mean that Superstrings must have dimensions or wavelengths of this size or smaller. Just as quantum gravity may imply that there are no time intervals shorter than some finite value, it also implies that there may be no sizes smaller than some tiny but finite value. That may be about 10^{-35} m. If so, and if Superstring theory can explain all it strives to, then the structures of Superstrings are at the lower limit of the smallest possible size and can have no further substructure. This would be the ultimate answer to the question the ancient Greeks considered. There is a finite lower limit to space.

Not only is Superstring theory in its infancy, it deals with dimensions about 17 orders of magnitude smaller than the 10^{-18} m details that we have been able to observe directly. It is thus relatively unconstrained by experiment, and there are a host of theoretical possibilities to choose from. This has led theorists to make choices subjectively (as always) on what is the most elegant theory, with less hope than usual that experiment will guide them. It has also led to speculation of alternate universes, with their Big Bangs creating each new universe with a random set of rules. These speculations may not be tested even in principle, since an alternate universe is by definition unattainable. It is something like exploring a self-consistent field of mathematics, with its axioms and rules of logic that are not consistent with nature. Such endeavors have often given insight to mathematicians and scientists alike and occasionally have been directly related to the description of new discoveries.

34.4 Dark Matter and Closure

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Discuss the existence of dark matter.
- Explain neutrino oscillations and their consequences.

One of the most exciting problems in physics today is the fact that there is far more matter in the universe than we can see. The motion of stars in galaxies and the motion of galaxies in clusters imply that there is about 10 times as much mass as in the luminous objects we can see. The indirectly observed non-luminous matter is called **dark matter**. Why is dark matter a problem? For one thing, we do not know what it is. It may well be 90% of all matter in the universe, yet there is a possibility that it is of a completely unknown form—a stunning discovery if verified. Dark matter has implications for particle physics. It may be possible that neutrinos actually have small masses or that there are completely unknown types of particles. Dark matter also has implications for cosmology, since there may be enough dark matter to stop the expansion of the universe. That is another problem related to dark matter—we do not know how much there is. We keep finding evidence for more matter in the universe, and we have an idea of how much it would take to eventually stop the expansion of the universe, but whether there is enough is still unknown.

Evidence

The first clues that there is more matter than meets the eye came from the Swiss-born American astronomer Fritz Zwicky in the 1930s; major work was also done by the American astronomer Vera Rubin. Zwicky measured the velocities of stars orbiting the galaxy, using the relativistic Doppler shift of their spectra (see [Figure 34.18\(a\)](#)). He found that velocity varied with distance from the center of the galaxy, as graphed in [Figure 34.18\(b\)](#). If the mass of the galaxy was concentrated in its center, as are its luminous stars, the velocities should decrease as the square root of the distance from the center. Instead, the velocity curve is almost flat, implying that there is a tremendous amount of matter in the galactic halo. Using instruments and methods that offered a greater degree of precision, Rubin investigated the movement of spiral galaxies and observed that their outermost reaches were rotating as quickly as their centers. She also calculated that the rotational velocity of galaxies should have been enough to cause them to fly apart, unless there was a significant discrepancy between their observable matter and their actual matter. This became known as the galaxy rotation problem, which can be "solved" by the presence of unobserved or dark matter. Although not immediately recognized for its significance, such measurements have now been made for many galaxies, with similar results. Further, studies of galactic clusters have also indicated that galaxies have a mass distribution greater than that obtained from their brightness (proportional to the number of stars), which also extends into large halos surrounding the luminous parts of galaxies. Observations of other EM wavelengths, such as radio waves and X rays, have similarly confirmed the existence of dark matter. Take, for example, X rays in the relatively dark space between galaxies, which indicates the presence of previously unobserved hot, ionized gas (see [Figure 34.18\(c\)](#)).

Theoretical Yearnings for Closure

Is the universe open or closed? That is, will the universe expand forever or will it stop, perhaps to contract? This, until recently, was a question of whether there is enough gravitation to stop the expansion of the universe. In the past few years, it has become a question of the combination of gravitation and what is called the **cosmological constant**. The cosmological constant was invented by Einstein to prohibit the expansion or contraction of the universe. At the time he developed general relativity, Einstein considered that an illogical possibility. The cosmological constant was discarded after Hubble discovered the expansion, but has been re-invoked in recent years.

Gravitational attraction between galaxies is slowing the expansion of the universe, but the amount of slowing down is not known directly. In fact, the cosmological constant can counteract gravity's effect. As recent measurements indicate, the universe is expanding *faster* now than in the past—perhaps a “modern inflationary era” in which the dark energy is thought to be causing the expansion of the present-day universe to accelerate. If the expansion rate were affected by gravity alone, we should be able to see that the expansion rate between distant galaxies was once greater than it is now. However, measurements show it was *less* than now. We can, however, calculate the amount of slowing based on the average density of matter we observe directly. Here we have a definite answer—there is far

less visible matter than needed to stop expansion. The **critical density** ρ_c is defined to be the density needed to just halt universal expansion in a universe with no cosmological constant. It is estimated to be about

$$\rho_c \approx 10^{-26} \text{ kg/m}^3. \quad 34.3$$

However, this estimate of ρ_c is only good to about a factor of two, due to uncertainties in the expansion rate of the universe. The critical density is equivalent to an average of only a few nucleons per cubic meter, remarkably small and indicative of how truly empty intergalactic space is. Luminous matter seems to account for roughly 0.5% to 2% of the critical density, far less than that needed for closure. Taking into account the amount of dark matter we detect indirectly and all other types of indirectly observed normal matter, there is only 10% to 40% of what is needed for closure. If we are able to refine the measurements of expansion rates now and in the past, we will have our answer regarding the curvature of space and we will determine a value for the cosmological constant to justify this observation. Finally, the most recent measurements of the CMBR have implications for the cosmological constant, so it is not simply a device concocted for a single purpose.

After the recent experimental discovery of the cosmological constant, most researchers feel that the universe should be just barely open. Since matter can be thought to curve the space around it, we call an open universe **negatively curved**. This means that you can in principle travel an unlimited distance in any direction. A universe that is closed is called **positively curved**. This means that if you travel far enough in any direction, you will return to your starting point, analogous to circumnavigating the Earth. In between these two is a **flat (zero curvature) universe**. The recent discovery of the cosmological constant has shown the universe is very close to flat, and will expand forever. Why do theorists feel the universe is flat? Flatness is a part of the inflationary scenario that helps explain the flatness of the microwave background. In fact, since general relativity implies that matter creates the space in which it exists, there is a special symmetry to a flat universe.

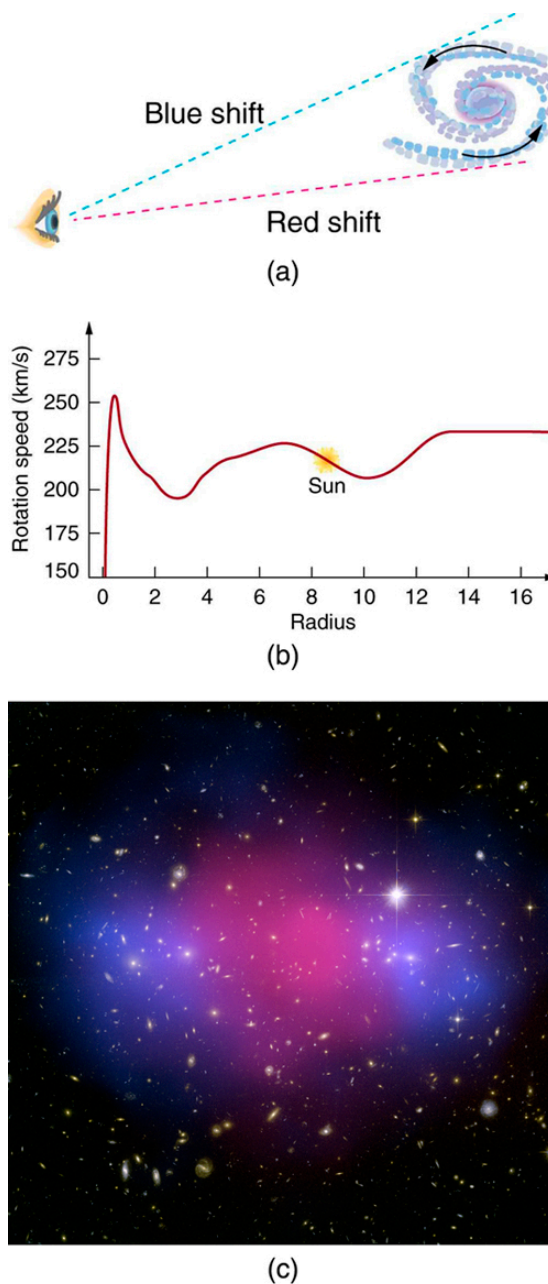


FIGURE 34.18 Evidence for dark matter: (a) We can measure the velocities of stars relative to their galaxies by observing the Doppler shift in emitted light, usually using the hydrogen spectrum. These measurements indicate the rotation of a spiral galaxy. (b) A graph of velocity versus distance from the galactic center shows that the velocity does not decrease as it would if the matter were concentrated in luminous stars. The flatness of the curve implies a massive galactic halo of dark matter extending beyond the visible stars. (c) This is a computer-generated image of X rays from a galactic cluster. The X rays indicate the presence of otherwise unseen hot clouds of ionized gas in the regions of space previously considered more empty. (credit: NASA, ESA, CXC, M. Bradac (University of California, Santa Barbara), and S. Allen (Stanford University))

What Is the Dark Matter We See Indirectly?

There is no doubt that dark matter exists, but its form and the amount in existence are two facts that are still being studied vigorously. As always, we seek to explain new observations in terms of known principles. However, as more discoveries are made, it is becoming more and more difficult to explain dark matter as a known type of matter.

One of the possibilities for normal matter is being explored using the Hubble Space Telescope and employing the lensing effect of gravity on light (see [Figure 34.19](#)). Stars glow because of nuclear fusion in them, but planets are visible primarily by reflected light. Jupiter, for example, is too small to ignite fusion in its core and become a star, but we can see sunlight reflected from it, since we are relatively close. If Jupiter orbited another star, we would not be

able to see it directly. The question is open as to how many planets or other bodies smaller than about 1/1000 the mass of the Sun are there. If such bodies pass between us and a star, they will not block the star's light, being too small, but they will form a gravitational lens, as discussed in [General Relativity and Quantum Gravity](#).

In a process called **microlensing**, light from the star is focused and the star appears to brighten in a characteristic manner. Searches for dark matter in this form are particularly interested in galactic halos because of the huge amount of mass that seems to be there. Such microlensing objects are thus called **massive compact halo objects**, or **MACHOs**. To date, a few MACHOs have been observed, but not predominantly in galactic halos, nor in the numbers needed to explain dark matter.

MACHOs are among the most conventional of unseen objects proposed to explain dark matter. Others being actively pursued are red dwarfs, which are small dim stars, but too few have been seen so far, even with the Hubble Telescope, to be of significance. Old remnants of stars called white dwarfs are also under consideration, since they contain about a solar mass, but are small as the Earth and may dim to the point that we ordinarily do not observe them. While white dwarfs are known, old dim ones are not. Yet another possibility is the existence of large numbers of smaller than stellar mass black holes left from the Big Bang—here evidence is entirely absent.

There is a very real possibility that dark matter is composed of the known neutrinos, which may have small, but finite, masses. As discussed earlier, neutrinos are thought to be massless, but we only have upper limits on their masses, rather than knowing they are exactly zero. So far, these upper limits come from difficult measurements of total energy emitted in the decays and reactions in which neutrinos are involved. There is an amusing possibility of proving that neutrinos have mass in a completely different way.

We have noted in [Particles, Patterns, and Conservation Laws](#) that there are three flavors of neutrinos (ν_e , ν_μ , and ν_τ) and that the weak interaction could change quark flavor. It should also change neutrino flavor—that is, any type of neutrino could change spontaneously into any other, a process called **neutrino oscillations**. However, this can occur only if neutrinos have a mass. Why? Crudely, because if neutrinos are massless, they must travel at the speed of light and time will not pass for them, so that they cannot change without an interaction. In 1999, results began to be published containing convincing evidence that neutrino oscillations do occur. Using the Super-Kamiokande detector in Japan, the oscillations have been observed and are being verified and further explored at present at the same facility and others.

Neutrino oscillations may also explain the low number of observed solar neutrinos. Detectors for observing solar neutrinos are specifically designed to detect electron neutrinos ν_e produced in huge numbers by fusion in the Sun. A large fraction of electron neutrinos ν_e may be changing flavor to muon neutrinos ν_μ on their way out of the Sun, possibly enhanced by specific interactions, reducing the flux of electron neutrinos to observed levels. There is also a discrepancy in observations of neutrinos produced in cosmic ray showers. While these showers of radiation produced by extremely energetic cosmic rays should contain twice as many ν_μ s as ν_e s, their numbers are nearly equal. This may be explained by neutrino oscillations from muon flavor to electron flavor. Massive neutrinos are a particularly appealing possibility for explaining dark matter, since their existence is consistent with a large body of known information and explains more than dark matter. The question is not settled at this writing.

The most radical proposal to explain dark matter is that it consists of previously unknown leptons (sometimes obtusely referred to as non-baryonic matter). These are called **weakly interacting massive particles**, or **WIMPs**, and would also be chargeless, thus interacting negligibly with normal matter, except through gravitation. One proposed group of WIMPs would have masses several orders of magnitude greater than nucleons and are sometimes called **neutralinos**. Others are called **axions** and would have masses about 10^{-10} that of an electron mass. Both neutralinos and axions would be gravitationally attached to galaxies, but because they are chargeless and only feel the weak force, they would be in a halo rather than interact and coalesce into spirals, and so on, like normal matter (see [Figure 34.20](#)).



FIGURE 34.19 The Hubble Space Telescope is producing exciting data with its corrected optics and with the absence of atmospheric distortion. It has observed some MACHOs, disks of material around stars thought to precede planet formation, black hole candidates, and collisions of comets with Jupiter. (credit: NASA (crew of STS-125))



FIGURE 34.20 Dark matter may shepherd normal matter gravitationally in space, as this stream moves the leaves. Dark matter may be invisible and even move through the normal matter, as neutrinos penetrate us without small-scale effect. (credit: Shinichi Sugiyama)

Some particle theorists have built WIMPs into their unified force theories and into the inflationary scenario of the evolution of the universe so popular today. These particles would have been produced in just the correct numbers to make the universe flat, shortly after the Big Bang. The proposal is radical in the sense that it invokes entirely new forms of matter, in fact *two* entirely new forms, in order to explain dark matter and other phenomena. WIMPs have the extra burden of automatically being very difficult to observe directly. This is somewhat analogous to quark confinement, which guarantees that quarks are there, but they can never be seen directly. One of the primary goals of the LHC at CERN, however, is to produce and detect WIMPs. At any rate, before WIMPs are accepted as the best explanation, all other possibilities utilizing known phenomena will have to be shown inferior. Should that occur, we will be in the unanticipated position of admitting that, to date, all we know is only 10% of what exists. A far cry from the days when people firmly believed themselves to be not only the center of the universe, but also the reason for its existence.

34.5 Complexity and Chaos

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain complex systems.
- Discuss chaotic behavior of different systems.

Much of what impresses us about physics is related to the underlying connections and basic simplicity of the laws we have discovered. The language of physics is precise and well defined because many basic systems we study are simple enough that we can perform controlled experiments and discover unambiguous relationships. Our most spectacular successes, such as the prediction of previously unobserved particles, come from the simple underlying patterns we have been able to recognize. But there are systems of interest to physicists that are inherently complex. The simple laws of physics apply, of course, but complex systems may reveal patterns that simple systems do not. The emerging field of **complexity** is devoted to the study of complex systems, including those outside the traditional bounds of physics. Of particular interest is the ability of complex systems to adapt and evolve.

What are some examples of complex adaptive systems? One is the primordial ocean. When the oceans first formed, they were a random mix of elements and compounds that obeyed the laws of physics and chemistry. In a relatively short geological time (about 500 million years), life had emerged. Laboratory simulations indicate that the emergence of life was far too fast to have come from random combinations of compounds, even if driven by lightning and heat. There must be an underlying ability of the complex system to organize itself, resulting in the self-replication we recognize as life. Living entities, even at the unicellular level, are highly organized and systematic. Systems of living organisms are themselves complex adaptive systems. The grandest of these evolved into the biological system we have today, leaving traces in the geological record of steps taken along the way.

Complexity as a discipline examines complex systems, how they adapt and evolve, looking for similarities with other complex adaptive systems. Can, for example, parallels be drawn between biological evolution and the evolution of *economic systems*? Economic systems do emerge quickly, they show tendencies for self-organization, they are complex (in the number and types of transactions), and they adapt and evolve. Biological systems do all the same types of things. There are other examples of complex adaptive systems being studied for fundamental similarities. *Cultures* show signs of adaptation and evolution. The comparison of different cultural evolutions may bear fruit as well as comparisons to biological evolution. *Science* also is a complex system of human interactions, like culture and economics, that adapts to new information and political pressure, and evolves, usually becoming more organized rather than less. Those who study *creative thinking* also see parallels with complex systems. Humans sometimes organize almost random pieces of information, often subconsciously while doing other things, and come up with brilliant creative insights. The development of *language* is another complex adaptive system that may show similar tendencies. *Artificial intelligence* is an overt attempt to devise an adaptive system that will self-organize and evolve in the same manner as an intelligent living being learns. These are a few of the broad range of topics being studied by those who investigate complexity. There are now institutes, journals, and meetings, as well as popularizations of the emerging topic of complexity.

In traditional physics, the discipline of complexity may yield insights in certain areas. Thermodynamics treats systems on the average, while statistical mechanics deals in some detail with complex systems of atoms and molecules in random thermal motion. Yet there is organization, adaptation, and evolution in those complex systems. Non-equilibrium phenomena, such as heat transfer and phase changes, are characteristically complex in detail, and new approaches to them may evolve from complexity as a discipline. Crystal growth is another example of self-organization spontaneously emerging in a complex system. Alloys are also inherently complex mixtures that show certain simple characteristics implying some self-organization. The organization of iron atoms into magnetic domains as they cool is another. Perhaps insights into these difficult areas will emerge from complexity. But at the minimum, the discipline of complexity is another example of human effort to understand and organize the universe around us, partly rooted in the discipline of physics.

A predecessor to complexity is the topic of chaos, which has been widely publicized and has become a discipline of its own. It is also based partly in physics and treats broad classes of phenomena from many disciplines. **Chaos** is a word used to describe systems whose outcomes are extremely sensitive to initial conditions. The orbit of the planet Pluto, for example, may be chaotic in that it can change tremendously due to small interactions with other planets. This makes its long-term behavior impossible to predict with precision, just as we cannot tell precisely where a decaying Earth satellite will land or how many pieces it will break into. But the discipline of chaos has found ways to deal with such systems and has been applied to apparently unrelated systems. For example, the heartbeat of people with certain types of potentially lethal arrhythmias seems to be chaotic, and this knowledge may allow more sophisticated monitoring and recognition of the need for intervention.

Chaos is related to complexity. Some chaotic systems are also inherently complex; for example, vortices in a fluid as opposed to a double pendulum. Both are chaotic and not predictable in the same sense as other systems. But there can be organization in chaos and it can also be quantified. Examples of chaotic systems are beautiful fractal patterns such as in [Figure 34.21](#). Some chaotic systems exhibit self-organization, a type of stable chaos. The orbits of the planets in our solar system, for example, may be chaotic (we are not certain yet). But they are definitely organized and systematic, with a simple formula describing the orbital radii of the first eight planets *and* the asteroid belt. Large-scale vortices in Jupiter's atmosphere are chaotic, but the Great Red Spot is a stable self-organization of rotational energy. (See [Figure 34.22](#).) The Great Red Spot has been in existence for at least 400 years and is a complex self-adaptive system.

The emerging field of complexity, like the now almost traditional field of chaos, is partly rooted in physics. Both attempt to see similar systematics in a very broad range of phenomena and, hence, generate a better understanding of them. Time will tell what impact these fields have on more traditional areas of physics as well as on the other disciplines they relate to.

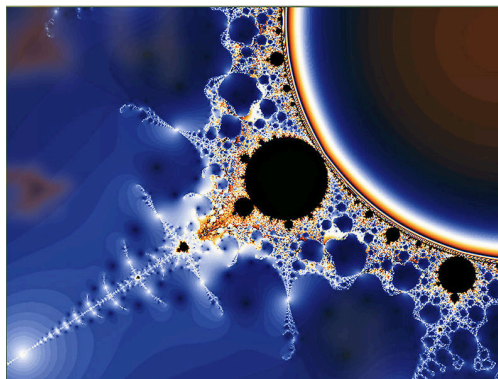


FIGURE 34.21 This image is related to the Mandelbrot set, a complex mathematical form that is chaotic. The patterns are infinitely fine as you look closer and closer, and they indicate order in the presence of chaos. (credit: Gilberto Santa Rosa)

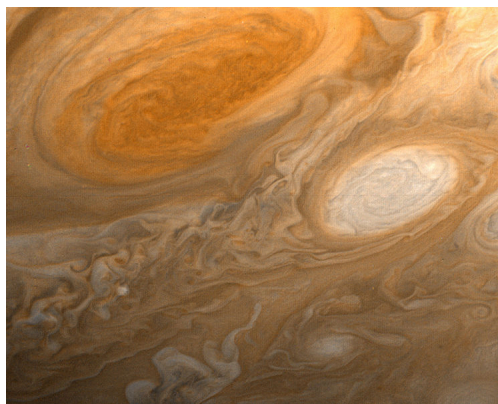


FIGURE 34.22 The Great Red Spot on Jupiter is an example of self-organization in a complex and chaotic system. Smaller vortices in Jupiter's atmosphere behave chaotically, but the triple-Earth-size spot is self-organized and stable for at least hundreds of years. (credit: NASA)

34.6 High-temperature Superconductors

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Identify superconductors and their uses.
- Discuss the need for a high- T_c superconductor.

Superconductors are materials with a resistivity of zero. They are familiar to the general public because of their practical applications and have been mentioned at a number of points in the text. Because the resistance of a piece of superconductor is zero, there are no heat losses for currents through them; they are used in magnets needing high currents, such as in MRI machines, and could cut energy losses in power transmission. But most superconductors must be cooled to temperatures only a few kelvin above absolute zero, a costly procedure limiting their practical applications. In the past decade, tremendous advances have been made in producing materials that become superconductors at relatively high temperatures. There is hope that room temperature superconductors may someday be manufactured.

Superconductivity was discovered accidentally in 1911 by the Dutch physicist H. Kamerlingh Onnes (1853–1926) when he used liquid helium to cool mercury. Onnes had been the first person to liquefy helium a few years earlier and was surprised to observe the resistivity of a mediocre conductor like mercury drop to zero at a temperature of 4.2 K. We define the temperature at which and below which a material becomes a superconductor to be its **critical temperature**, denoted by T_c . (See [Figure 34.23](#).) Progress in understanding how and why a material became a superconductor was relatively slow, with the first workable theory coming in 1957. Certain other elements were also

found to become superconductors, but all had T_c s less than 10 K, which are expensive to maintain. Although Onnes received a Nobel prize in 1913, it was primarily for his work with liquid helium.

In 1986, a breakthrough was announced—a ceramic compound was found to have an unprecedented T_c of 35 K. It looked as if much higher critical temperatures could be possible, and by early 1988 another ceramic (this of thallium, calcium, barium, copper, and oxygen) had been found to have $T_c = 125$ K (see [Figure 34.24](#).) The economic potential of perfect conductors saving electric energy is immense for T_c s above 77 K, since that is the temperature of liquid nitrogen. Although liquid helium has a boiling point of 4 K and can be used to make materials superconducting, it costs about \$5 per liter. Liquid nitrogen boils at 77 K, but only costs about \$0.30 per liter. There was general euphoria at the discovery of these complex ceramic superconductors, but this soon subsided with the sobering difficulty of forming them into usable wires. The first commercial use of a high temperature superconductor is in an electronic filter for cellular phones. High-temperature superconductors are used in experimental apparatus, and they are actively being researched, particularly in thin film applications.

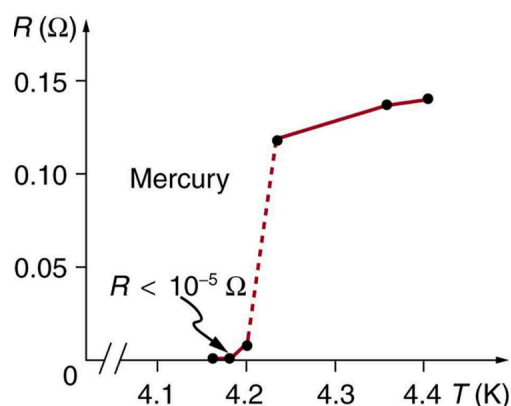


FIGURE 34.23 A graph of resistivity versus temperature for a superconductor shows a sharp transition to zero at the critical temperature T_c . High temperature superconductors have verifiable T_c s greater than 125 K, well above the easily achieved 77-K temperature of liquid nitrogen.



FIGURE 34.24 One characteristic of a superconductor is that it excludes magnetic flux and, thus, repels other magnets. The small magnet levitated above a high-temperature superconductor, which is cooled by liquid nitrogen, gives evidence that the material is superconducting. When the material warms and becomes conducting, magnetic flux can penetrate it, and the magnet will rest upon it. (credit: Saperaud)

The search is on for even higher T_c superconductors, many of complex and exotic copper oxide ceramics, sometimes including strontium, mercury, or yttrium as well as barium, calcium, and other elements. Room temperature (about 293 K) would be ideal, but any temperature close to room temperature is relatively cheap to produce and maintain. There are persistent reports of T_c s over 200 K and some in the vicinity of 270 K. Unfortunately, these observations are not routinely reproducible, with samples losing their superconducting nature once heated and re-cooled (cycled) a few times (see [Figure 34.25](#).) They are now called USOs or unidentified superconducting objects, out of frustration and the refusal of some samples to show high T_c even though produced in the same manner as others. Reproducibility is crucial to discovery, and researchers are justifiably reluctant to claim the breakthrough they all seek. Time will tell whether USOs are real or an experimental quirk.

The theory of ordinary superconductors is difficult, involving quantum effects for widely separated electrons traveling through a material. Electrons couple in a manner that allows them to get through the material without

losing energy to it, making it a superconductor. High- T_c superconductors are more difficult to understand theoretically, but theorists seem to be closing in on a workable theory. The difficulty of understanding how electrons can sneak through materials without losing energy in collisions is even greater at higher temperatures, where vibrating atoms should get in the way. Discoverers of high T_c may feel something analogous to what a politician once said upon an unexpected election victory—“I wonder what we did right?”

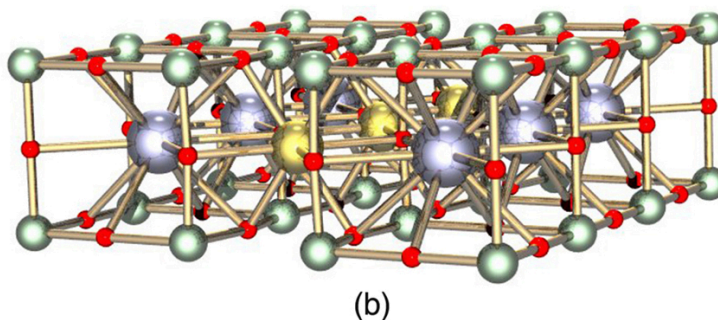
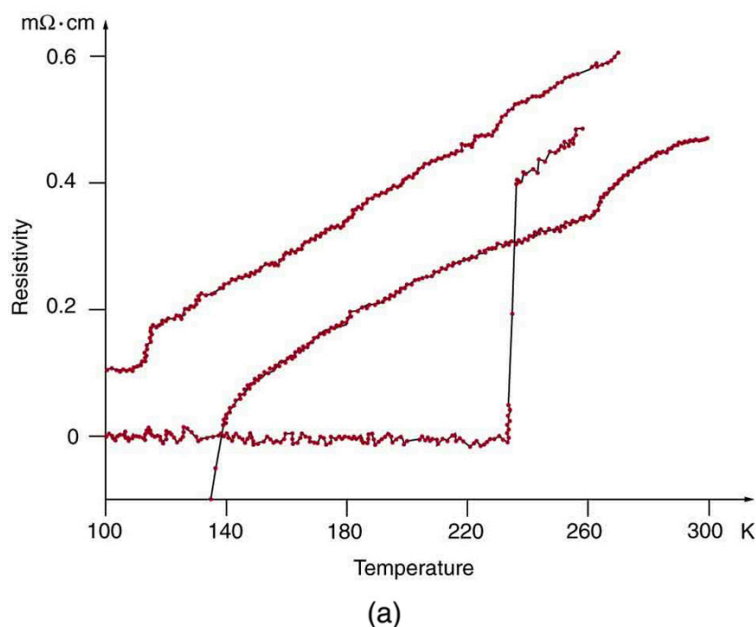


FIGURE 34.25 (a) This graph, adapted from an article in *Physics Today*, shows the behavior of a single sample of a high-temperature superconductor in three different trials. In one case the sample exhibited a T_c of about 230 K, whereas in the others it did not become superconducting at all. The lack of reproducibility is typical of forefront experiments and prohibits definitive conclusions. (b) This colorful diagram shows the complex but systematic nature of the lattice structure of a high-temperature superconducting ceramic. (credit: en:Cadmium, Wikimedia Commons)

34.7 Some Questions We Know to Ask

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Identify sample questions to be asked on the largest scales.
- Identify sample questions to be asked on the intermediate scale.
- Identify sample questions to be asked on the smallest scales.

Throughout the text we have noted how essential it is to be curious and to ask questions in order to first understand what is known, and then to go a little farther. Some questions may go unanswered for centuries; others may not have answers, but some bear delicious fruit. Part of discovery is knowing which questions to ask. You have to know something before you can even phrase a decent question. As you may have noticed, the mere act of asking a question can give you the answer. The following questions are a sample of those physicists now know to ask and are representative of the forefronts of physics. Although these questions are important, they will be replaced by others

if answers are found to them. The fun continues.

On the Largest Scale

1. *Is the universe open or closed?* Theorists would like it to be just barely closed and evidence is building toward that conclusion. Recent measurements in the expansion rate of the universe and in CMBR support a flat universe. There is a connection to small-scale physics in the type and number of particles that may contribute to closing the universe.
2. *What is dark matter?* It is definitely there, but we really do not know what it is. Conventional possibilities are being ruled out, but one of them still may explain it. The answer could reveal whole new realms of physics and the disturbing possibility that most of what is out there is unknown to us, a completely different form of matter.
3. *How do galaxies form?* They exist since very early in the evolution of the universe and it remains difficult to understand how they evolved so quickly. The recent finer measurements of fluctuations in the CMBR may yet allow us to explain galaxy formation.
4. *What is the nature of various-mass black holes?* Only recently have we become confident that many black hole candidates cannot be explained by other, less exotic possibilities. But we still do not know much about how they form, what their role in the history of galactic evolution has been, and the nature of space in their vicinity. However, so many black holes are now known that correlations between black hole mass and galactic nuclei characteristics are being studied.
5. *What is the mechanism for the energy output of quasars?* These distant and extraordinarily energetic objects now seem to be early stages of galactic evolution with a supermassive black-hole-devouring material. Connections are now being made with galaxies having energetic cores, and there is evidence consistent with less consuming, supermassive black holes at the center of older galaxies. New instruments are allowing us to see deeper into our own galaxy for evidence of our own massive black hole.
6. *Where do the γ bursts come from?* We see bursts of γ rays coming from all directions in space, indicating the sources are very distant objects rather than something associated with our own galaxy. Some γ bursts finally are being correlated with known sources so that the possibility they may originate in binary neutron star interactions or black holes eating a companion neutron star can be explored.

On the Intermediate Scale

1. *How do phase transitions take place on the microscopic scale?* We know a lot about phase transitions, such as water freezing, but the details of how they occur molecule by molecule are not well understood. Similar questions about specific heat a century ago led to early quantum mechanics. It is also an example of a complex adaptive system that may yield insights into other self-organizing systems.
2. *Is there a way to deal with nonlinear phenomena that reveals underlying connections?* Nonlinear phenomena lack a direct or linear proportionality that makes analysis and understanding a little easier. There are implications for nonlinear optics and broader topics such as chaos.
3. *How do high- T_c superconductors become resistanceless at such high temperatures?* Understanding how they work may help make them more practical or may result in surprises as unexpected as the discovery of superconductivity itself.
4. *There are magnetic effects in materials we do not understand—how do they work?* Although beyond the scope of this text, there is a great deal to learn in condensed matter physics (the physics of solids and liquids). We may find surprises analogous to lasing, the quantum Hall effect, and the quantization of magnetic flux. Complexity may play a role here, too.

On the Smallest Scale

1. *Are quarks and leptons fundamental, or do they have a substructure?* The higher energy accelerators that are just completed or being constructed may supply some answers, but there will also be input from cosmology and other systematics.
2. *Why do leptons have integral charge while quarks have fractional charge?* If both are fundamental and analogous as thought, this question deserves an answer. It is obviously related to the previous question.
3. *Why are there three families of quarks and leptons?* First, does this imply some relationship? Second, why three and only three families?

4. *Are all forces truly equal (unified) under certain circumstances?* They don't have to be equal just because we want them to be. The answer may have to be indirectly obtained because of the extreme energy at which we think they are unified.
5. *Are there other fundamental forces?* There was a flurry of activity with claims of a fifth and even a sixth force a few years ago. Interest has subsided, since those forces have not been detected consistently. Moreover, the proposed forces have strengths similar to gravity, making them extraordinarily difficult to detect in the presence of stronger forces. But the question remains; and if there are no other forces, we need to ask why only four and why these four.
6. *Is the proton stable?* We have discussed this in some detail, but the question is related to fundamental aspects of the unification of forces. We may never know from experiment that the proton is stable, only that it is very long lived.
7. *Are there magnetic monopoles?* Many particle theories call for very massive individual north- and south-pole particles—magnetic monopoles. If they exist, why are they so different in mass and elusiveness from electric charges, and if they do not exist, why not?
8. *Do neutrinos have mass?* Definitive evidence has emerged for neutrinos having mass. The implications are significant, as discussed in this chapter. There are effects on the closure of the universe and on the patterns in particle physics.
9. *What are the systematic characteristics of high- Z nuclei?* All elements with $Z = 118$ or less (with the exception of 115 and 117) have now been discovered. It has long been conjectured that there may be an island of relative stability near $Z = 114$, and the study of the most recently discovered nuclei will contribute to our understanding of nuclear forces.

These lists of questions are not meant to be complete or consistently important—you can no doubt add to it yourself. There are also important questions in topics not broached in this text, such as certain particle symmetries, that are of current interest to physicists. Hopefully, the point is clear that no matter how much we learn, there always seems to be more to know. Although we are fortunate to have the hard-won wisdom of those who preceded us, we can look forward to new enlightenment, undoubtedly sprinkled with surprise.

Glossary

axions a type of WIMPs having masses about 10^{-10} of an electron mass

Big Bang a gigantic explosion that threw out matter a few billion years ago

black holes objects having such large gravitational fields that things can fall in, but nothing, not even light, can escape

chaos word used to describe systems the outcomes of which are extremely sensitive to initial conditions

complexity an emerging field devoted to the study of complex systems

cosmic microwave background the spectrum of microwave radiation of cosmic origin

cosmological constant a theoretical construct intimately related to the expansion and closure of the universe

cosmological red shift the photon wavelength is stretched in transit from the source to the observer because of the expansion of space itself

cosmology the study of the character and evolution of the universe

critical density the density of matter needed to just halt universal expansion

critical temperature the temperature at which and below which a material becomes a superconductor

dark matter indirectly observed non-luminous matter

electroweak epoch the stage before 10^{-11} back to 10^{-34} after the Big Bang

escape velocity takeoff velocity when kinetic energy just cancels gravitational potential energy

event horizon the distance from the object at which the escape velocity is exactly the speed of light

flat (zero curvature) universe a universe that is infinite but not curved

general relativity Einstein's theory that describes all types of relative motion including accelerated motion and the effects of gravity

gravitational waves mass-created distortions in space that propagate at the speed of light and that are predicted by general relativity

GUT epoch the time period from 10^{-43} to 10^{-34} after the Big Bang, when Grand Unification Theory, in which all forces except gravity are identical, governed the universe

Hubble constant a central concept in cosmology

whose value is determined by taking the slope of a graph of velocity versus distance, obtained from red shift measurements

inflationary scenario the rapid expansion of the universe by an incredible factor of 10^{-50} for the brief time from 10^{-35} to about 10^{-32} s

MACHOs massive compact halo objects; microlensing objects of huge mass

microlensing a process in which light from a distant star is focused and the star appears to brighten in a characteristic manner, when a small body (smaller than about 1/1000 the mass of the Sun) passes between us and the star

negatively curved an open universe that expands forever

neutralinos a type of WIMPs having masses several orders of magnitude greater than nucleon masses

neutrino oscillations a process in which any type of neutrino could change spontaneously into any other

neutron stars literally a star composed of neutrons

positively curved a universe that is closed and eventually contracts

Quantum gravity the theory that deals with particle exchange of gravitons as the mechanism for the force

quasars the moderately distant galaxies that emit as much or more energy than a normal galaxy

Schwarzschild radius the radius of the event horizon

spontaneous symmetry breaking the transition from GUT to electroweak where the forces were no longer unified

Superconductors materials with resistivity of zero

superforce hypothetical unified force in TOE epoch

Superstring theory a theory to unify gravity with the other three forces in which the fundamental particles are considered to act like one-dimensional vibrating strings

thought experiment mental analysis of certain carefully and clearly defined situations to develop an idea

TOE epoch before 10^{-43} after the Big Bang

WIMPs weakly interacting massive particles; chargeless leptons (non-baryonic matter) interacting negligibly with normal matter

Section Summary

34.1 Cosmology and Particle Physics

- Cosmology is the study of the character and evolution of the universe.
- The two most important features of the universe

are the cosmological red shifts of its galaxies being proportional to distance and its cosmic microwave background (CMBR). Both support the notion that there was a gigantic explosion, known as the Big Bang that created the universe.

- Galaxies farther away than our local group have, on an average, a recessional velocity given by $v = H_0 d$, where d is the distance to the galaxy and H_0 is the Hubble constant, taken to have the average value $H_0 = 20 \text{ km/s} \cdot \text{Mly}$.
- Explanations of the large-scale characteristics of the universe are intimately tied to particle physics.
- The dominance of matter over antimatter and the smoothness of the CMBR are two characteristics that are tied to particle physics.
- The epochs of the universe are known back to very shortly after the Big Bang, based on known laws of physics.
- The earliest epochs are tied to the unification of forces, with the electroweak epoch being partially understood, the GUT epoch being speculative, and the TOE epoch being highly speculative since it involves an unknown single superforce.
- The transition from GUT to electroweak is called spontaneous symmetry breaking. It released energy that caused the inflationary scenario, which in turn explains the smoothness of the CMBR.

34.2 General Relativity and Quantum Gravity

- Einstein's theory of general relativity includes accelerated frames and, thus, encompasses special relativity and gravity. Created by use of careful thought experiments, it has been repeatedly verified by real experiments.
- One direct result of this behavior of nature is the gravitational lensing of light by massive objects, such as galaxies, also seen in the microlensing of light by smaller bodies in our galaxy.
- Another prediction is the existence of black holes, objects for which the escape velocity is greater than the speed of light and from which nothing can escape.
- The event horizon is the distance from the object at which the escape velocity equals the speed of light c . It is called the Schwarzschild radius R_S and is given by

$$R_S = \frac{2GM}{c^2},$$
 where G is the universal gravitational constant, and M is the mass of the body.
- Physics is unknown inside the event horizon, and the possibility of wormholes and time travel are being studied.
- Candidates for black holes may power the extremely energetic emissions of quasars, distant objects that seem to be early stages of galactic evolution.
- Neutron stars are stellar remnants, having the density of a nucleus, that hint that black holes could form from supernovas, too.
- Gravitational waves are wrinkles in space, predicted by general relativity but not yet observed, caused by changes in very massive objects.
- Quantum gravity is an incompletely developed theory that strives to include general relativity, quantum mechanics, and unification of forces (thus, a TOE).
- One unconfirmed connection between general relativity and quantum mechanics is the prediction of characteristic radiation from just outside black holes.

34.3 Superstrings

- Superstring theory holds that fundamental particles are one-dimensional vibrations analogous to those on strings and is an attempt at a theory of quantum gravity.

34.4 Dark Matter and Closure

- Dark matter is non-luminous matter detected in and around galaxies and galactic clusters.
- It may be 10 times the mass of the luminous matter in the universe, and its amount may determine whether the universe is open or closed (expands forever or eventually stops).
- The determining factor is the critical density of the universe and the cosmological constant, a theoretical construct intimately related to the expansion and closure of the universe.
- The critical density ρ_c is the density needed to just halt universal expansion. It is estimated to be approximately 10^{-26} kg/m^3 .
- An open universe is negatively curved, a closed universe is positively curved, whereas a universe with exactly the critical density is flat.
- Dark matter's composition is a major mystery, but it may be due to the suspected mass of neutrinos or a completely unknown type of leptonic matter.
- If neutrinos have mass, they will change families, a process known as neutrino oscillations, for which there is growing evidence.

34.5 Complexity and Chaos

- Complexity is an emerging field, rooted primarily in physics, that considers complex adaptive systems and their evolution, including self-organization.
- Complexity has applications in physics and many other disciplines, such as biological evolution.
- Chaos is a field that studies systems whose properties depend extremely sensitively on some variables and whose evolution is impossible to predict.
- Chaotic systems may be simple or complex.
- Studies of chaos have led to methods for understanding and predicting certain chaotic behaviors.

34.6 High-temperature Superconductors

- High-temperature superconductors are materials that become superconducting at temperatures

well above a few kelvin.

- The critical temperature T_c is the temperature below which a material is superconducting.
- Some high-temperature superconductors have verified T_c s above 125 K, and there are reports of T_c s as high as 250 K.

34.7 Some Questions We Know to Ask

- On the largest scale, the questions which can be asked may be about dark matter, dark energy, black holes, quasars, and other aspects of the universe.
- On the intermediate scale, we can query about gravity, phase transitions, nonlinear phenomena, high- T_c superconductors, and magnetic effects on materials.
- On the smallest scale, questions may be about quarks and leptons, fundamental forces, stability of protons, and existence of monopoles.

Conceptual Questions

34.1 Cosmology and Particle Physics

1. Explain why it only *appears* that we are at the center of expansion of the universe and why an observer in another galaxy would see the same relative motion of all but the closest galaxies away from her.
2. If there is no observable edge to the universe, can we determine where its center of expansion is? Explain.
3. If the universe is infinite, does it have a center? Discuss.
4. Another known cause of red shift in light is the source being in a high gravitational field. Discuss how this can be eliminated as the source of galactic red shifts, given that the shifts are proportional to distance and not to the size of the galaxy.
5. If some unknown cause of red shift—such as light becoming “tired” from traveling long distances through empty space—is discovered, what effect would there be on cosmology?
6. Olbers’s paradox poses an interesting question: If the universe is infinite, then any line of sight should eventually fall on a star’s surface. Why then is the sky dark at night? Discuss the commonly accepted evolution of the universe as a solution to this paradox.
7. If the cosmic microwave background radiation

(CMBR) is the remnant of the Big Bang’s fireball, we expect to see hot and cold regions in it. What are two causes of these wrinkles in the CMBR? Are the observed temperature variations greater or less than originally expected?

8. The decay of one type of K -meson is cited as evidence that nature favors matter over antimatter. Since mesons are composed of a quark and an antiquark, is it surprising that they would preferentially decay to one type over another? Is this an asymmetry in nature? Is the predominance of matter over antimatter an asymmetry?
9. Distances to local galaxies are determined by measuring the brightness of stars, called Cepheid variables, that can be observed individually and that have absolute brightnesses at a standard distance that are well known. Explain how the measured brightness would vary with distance as compared with the absolute brightness.
10. Distances to very remote galaxies are estimated based on their apparent type, which indicate the number of stars in the galaxy, and their measured brightness. Explain how the measured brightness would vary with distance. Would there be any correction necessary to compensate for the red shift of the galaxy (all distant galaxies have significant red shifts)? Discuss possible causes of uncertainties in these

measurements.

11. If the smallest meaningful time interval is greater than zero, will the lines in [Figure 34.9](#) ever meet?

34.2 General Relativity and Quantum Gravity

12. Quantum gravity, if developed, would be an improvement on both general relativity and quantum mechanics, but more mathematically difficult. Under what circumstances would it be necessary to use quantum gravity? Similarly, under what circumstances could general relativity be used? When could special relativity, quantum mechanics, or classical physics be used?
13. Does observed gravitational lensing correspond to a converging or diverging lens? Explain briefly.
14. Suppose you measure the red shifts of all the images produced by gravitational lensing, such as in [Figure 34.12](#). You find that the central image has a red shift less than the outer images, and those all have the same red shift. Discuss how this not only shows that the images are of the same object, but also implies that the red shift is not affected by taking different paths through space. Does it imply that cosmological red shifts are not caused by traveling through space (light getting tired, perhaps)?
15. What are gravitational waves, and have they yet been observed either directly or indirectly?
16. Is the event horizon of a black hole the actual physical surface of the object?
17. Suppose black holes radiate their mass away and the lifetime of a black hole created by a supernova is about 10^{67} years. How does this lifetime compare with the accepted age of the universe? Is it surprising that we do not observe the predicted characteristic radiation?

34.4 Dark Matter and Closure

18. Discuss the possibility that star velocities at the edges of galaxies being greater than expected is due to unknown properties of gravity rather than to the existence of dark matter. Would this mean, for example, that gravity is greater or smaller than expected at large distances? Are there other tests that could be made of gravity at large distances, such as observing the motions of neighboring galaxies?

19. How does relativistic time dilation prohibit neutrino oscillations if they are massless?
20. If neutrino oscillations do occur, will they violate conservation of the various lepton family numbers (L_e , L_μ , and L_τ)? Will neutrino oscillations violate conservation of the total number of leptons?
21. Lacking direct evidence of WIMPs as dark matter, why must we eliminate all other possible explanations based on the known forms of matter before we invoke their existence?

34.5 Complexity and Chaos

22. Must a complex system be adaptive to be of interest in the field of complexity? Give an example to support your answer.
23. State a necessary condition for a system to be chaotic.

34.6 High-temperature Superconductors

24. What is critical temperature T_c ? Do all materials have a critical temperature? Explain why or why not.
25. Explain how good thermal contact with liquid nitrogen can keep objects at a temperature of 77 K (liquid nitrogen's boiling point at atmospheric pressure).
26. Not only is liquid nitrogen a cheaper coolant than liquid helium, its boiling point is higher (77 K vs. 4.2 K). How does higher temperature help lower the cost of cooling a material? Explain in terms of the rate of heat transfer being related to the temperature difference between the sample and its surroundings.

34.7 Some Questions We Know to Ask

27. For experimental evidence, particularly of previously unobserved phenomena, to be taken seriously it must be reproducible or of sufficiently high quality that a single observation is meaningful. Supernova 1987A is not reproducible. How do we know observations of it were valid? The fifth force is not broadly accepted. Is this due to lack of reproducibility or poor-quality experiments (or both)? Discuss why forefront experiments are more subject to observational problems than those involving established phenomena.
28. Discuss whether you think there are limits to what humans can understand about the laws of

physics. Support your arguments.

Problems & Exercises

34.1 Cosmology and Particle Physics

- Find the approximate mass of the luminous matter in the Milky Way galaxy, given it has approximately 10^{11} stars of average mass 1.5 times that of our Sun.
- Find the approximate mass of the dark and luminous matter in the Milky Way galaxy. Assume the luminous matter is due to approximately 10^{11} stars of average mass 1.5 times that of our Sun, and take the dark matter to be 10 times as massive as the luminous matter.
- (a) Estimate the mass of the luminous matter in the known universe, given there are 10^{11} galaxies, each containing 10^{11} stars of average mass 1.5 times that of our Sun. (b) How many protons (the most abundant nuclide) are there in this mass? (c) Estimate the total number of particles in the observable universe by multiplying the answer to (b) by two, since there is an electron for each proton, and then by 10^9 , since there are far more particles (such as photons and neutrinos) in space than in luminous matter.
- If a galaxy is 500 Mly away from us, how fast do we expect it to be moving and in what direction?
- On average, how far away are galaxies that are moving away from us at 2.0% of the speed of light?
- Our solar system orbits the center of the Milky Way galaxy. Assuming a circular orbit 30,000 ly in radius and an orbital speed of 250 km/s, how many years does it take for one revolution? Note that this is approximate, assuming constant speed and circular orbit, but it is representative of the time for our system and local stars to make one revolution around the galaxy.
- (a) What is the approximate speed relative to us of a galaxy near the edge of the known universe, some 10 Gly away? (b) What fraction of the speed of light is this? Note that we have observed galaxies moving away from us at greater than $0.9c$.
- (a) Calculate the approximate age of the universe from the average value of the Hubble constant, $H_0 = 20 \text{ km/s} \cdot \text{Mly}$. To do this, calculate the time it would take to travel 1 Mly at a constant expansion rate of 20 km/s. (b) If deceleration is taken into account, would the actual age of the universe be greater or less than that found here? Explain.
- Assuming a circular orbit for the Sun about the center of the Milky Way galaxy, calculate its orbital speed using the following information: The mass of the galaxy is equivalent to a single mass 1.5×10^{11} times that of the Sun (or $3 \times 10^{41} \text{ kg}$), located 30,000 ly away.
- (a) What is the approximate force of gravity on a 70-kg person due to the Andromeda galaxy, assuming its total mass is 10^{13} that of our Sun and acts like a single mass 2 Mly away? (b) What is the ratio of this force to the person's weight? Note that Andromeda is the closest large galaxy.
- Andromeda galaxy is the closest large galaxy and is visible to the naked eye. Estimate its brightness relative to the Sun, assuming it has luminosity 10^{12} times that of the Sun and lies 2 Mly away.
- (a) A particle and its antiparticle are at rest relative to an observer and annihilate (completely destroying both masses), creating two γ rays of equal energy. What is the characteristic γ -ray energy you would look for if searching for evidence of proton-antiproton annihilation? (The fact that such radiation is rarely observed is evidence that there is very little antimatter in the universe.) (b) How does this compare with the 0.511-MeV energy associated with electron-positron annihilation?
- The average particle energy needed to observe unification of forces is estimated to be 10^{19} GeV . (a) What is the rest mass in kilograms of a particle that has a rest mass of $10^{19} \text{ GeV}/c^2$? (b) How many times the mass of a hydrogen atom is this?
- The peak intensity of the CMBR occurs at a wavelength of 1.1 mm. (a) What is the energy in eV of a 1.1-mm photon? (b) There are approximately 10^9 photons for each massive particle in deep space. Calculate the energy of 10^9 such photons. (c) If the average massive particle in space has a mass half that of a proton, what energy would be created by

converting its mass to energy? (d) Does this imply that space is “matter dominated”? Explain briefly.

15. (a) What Hubble constant corresponds to an approximate age of the universe of 10^{10} y? To get an approximate value, assume the expansion rate is constant and calculate the speed at which two galaxies must move apart to be separated by 1 Mly (present average galactic separation) in a time of 10^{10} y. (b) Similarly, what Hubble constant corresponds to a universe approximately 2×10^{10} -y old?
16. Show that the velocity of a star orbiting its galaxy in a circular orbit is inversely proportional to the square root of its orbital radius, assuming the mass of the stars inside its orbit acts like a single mass at the center of the galaxy. You may use an equation from a previous chapter to support your conclusion, but you must justify its use and define all terms used.
17. The core of a star collapses during a supernova, forming a neutron star. Angular momentum of the core is conserved, and so the neutron star spins rapidly. If the initial core radius is 5.0×10^5 km and it collapses to 10.0 km, find the neutron star's angular velocity in revolutions per second, given the core's angular velocity was originally 1 revolution per 30.0 days.
18. Using data from the previous problem, find the increase in rotational kinetic energy, given the core's mass is 1.3 times that of our Sun. Where does this increase in kinetic energy come from?
19. Distances to the nearest stars (up to 500 ly away) can be measured by a technique called parallax, as shown in Figure 34.26. What are the angles θ_1 and θ_2 relative to the plane of the Earth's orbit for a star 4.0 ly directly above the Sun?
20. (a) Use the Heisenberg uncertainty principle to calculate the uncertainty in energy for a corresponding time interval of 10^{-43} s. (b) Compare this energy with the 10^{19} GeV unification-of-forces energy and discuss why they are similar.
21. **Construct Your Own Problem**
Consider a star moving in a circular orbit at the edge of a galaxy. Construct a problem in which you calculate the mass of that galaxy in kg and in multiples of the solar mass based on the velocity

of the star and its distance from the center of the galaxy.

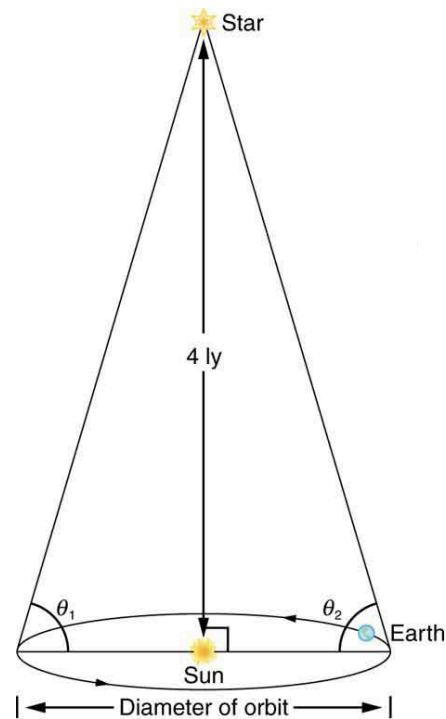


FIGURE 34.26 Distances to nearby stars are measured using triangulation, also called the parallax method. The angle of line of sight to the star is measured at intervals six months apart, and the distance is calculated by using the known diameter of the Earth's orbit. This can be done for stars up to about 500 ly away.

34.2 General Relativity and Quantum Gravity

22. What is the Schwarzschild radius of a black hole that has a mass eight times that of our Sun? Note that stars must be more massive than the Sun to form black holes as a result of a supernova.
23. Black holes with masses smaller than those formed in supernovas may have been created in the Big Bang. Calculate the radius of one that has a mass equal to the Earth's.
24. Supermassive black holes are thought to exist at the center of many galaxies.
(a) What is the radius of such an object if it has a mass of 10^9 Suns?
(b) What is this radius in light years?
25. **Construct Your Own Problem**
Consider a supermassive black hole near the center of a galaxy. Calculate the radius of such an object based on its mass. You must consider how much mass is reasonable for these large objects, and which is now nearly directly

observed. (Information on black holes posted on the Web by NASA and other agencies is reliable, for example.)

34.3 Superstrings

- 26.** The characteristic length of entities in Superstring theory is approximately 10^{-35} m.
- Find the energy in GeV of a photon of this wavelength.
 - Compare this with the average particle energy of 10^{19} GeV needed for unification of forces.

34.4 Dark Matter and Closure

- 27.** If the dark matter in the Milky Way were composed entirely of MACHOs (evidence shows it is not), approximately how many would there have to be? Assume the average mass of a MACHO is 1/1000 that of the Sun, and that dark matter has a mass 10 times that of the luminous Milky Way galaxy with its 10^{11} stars of average mass 1.5 times the Sun's mass.
- 28.** The critical mass density needed to just halt the expansion of the universe is approximately 10^{-26} kg/m³.
- Convert this to $\text{eV}/c^2 \cdot \text{m}^3$.
 - Find the number of neutrinos per cubic meter needed to close the universe if their average mass is $7 \text{ eV}/c^2$ and they have negligible kinetic energies.
- 29.** Assume the average density of the universe is 0.1 of the critical density needed for closure. What is the average number of protons per cubic meter, assuming the universe is composed mostly of hydrogen?
- 30.** To get an idea of how empty deep space is on the average, perform the following calculations:
- Find the volume our Sun would occupy if it

had an average density equal to the critical density of 10^{-26} kg/m³ thought necessary to halt the expansion of the universe.

- Find the radius of a sphere of this volume in light years.
- What would this radius be if the density were that of luminous matter, which is approximately 5% that of the critical density?
- Compare the radius found in part (c) with the 4-ly average separation of stars in the arms of the Milky Way.

34.6 High-temperature Superconductors

- 31.** A section of superconducting wire carries a current of 100 A and requires 1.00 L of liquid nitrogen per hour to keep it below its critical temperature. For it to be economically advantageous to use a superconducting wire, the cost of cooling the wire must be less than the cost of energy lost to heat in the wire. Assume that the cost of liquid nitrogen is \$0.30 per liter, and that electric energy costs \$0.10 per kW·h. What is the resistance of a normal wire that costs as much in wasted electric energy as the cost of liquid nitrogen for the superconductor?
- 32. Critical Thinking** Assume two galaxies are 2.00×10^2 Mly apart. (a) How fast are they separating due to space expansion? (b) Now suppose a third galaxy is located 2.00×10^2 Mly along a line that is a perpendicular bisector of the line segment between the first two galaxies. How fast is this galaxy separating from either of the other two galaxies? (c) Why would the third galaxy appear to be receding from both of the initially mentioned galaxies at the same rate? (d) How would the apparent motion be detected?

APPENDIX A

Atomic Masses

Atomic Number, Z	Name	Atomic Mass Number, A	Symbol	Atomic Mass (u)	Percent Abundance or Decay Mode	Half-life, t _{1/2}
0	neutron	1	<i>n</i>	1.008 665	β^-	10.37 min
1	Hydrogen	1	¹ H	1.007 825	99.985%	
	Deuterium	2	² H or D	2.014 102	0.015%	
	Tritium	3	³ H or T	3.016 050	β^-	12.33 y
2	Helium	3	³ He	3.016 030	1.38 × 10 ⁻⁴ %	
		4	⁴ He	4.002 603	≈100%	
3	Lithium	6	⁶ Li	6.015 121	7.5%	
		7	⁷ Li	7.016 003	92.5%	
4	Beryllium	7	⁷ Be	7.016 928	EC	53.29 d
		9	⁹ Be	9.012 182	100%	
5	Boron	10	¹⁰ B	10.012 937	19.9%	
		11	¹¹ B	11.009 305	80.1%	
6	Carbon	11	¹¹ C	11.011 432	EC, β^+	

TABLE A1 Atomic Masses

Atomic Number, Z	Name	Atomic Mass Number, A	Symbol	Atomic Mass (u)	Percent Abundance or Decay Mode	Half-life, $t_{1/2}$
		12	^{12}C	12.000 000	98.90%	
		13	^{13}C	13.003 355	1.10%	
		14	^{14}C	14.003 241	β^-	5730 y
7	Nitrogen	13	^{13}N	13.005 738	β^+	9.96 min
		14	^{14}N	14.003 074	99.63%	
		15	^{15}N	15.000 108	0.37%	
8	Oxygen	15	^{15}O	15.003 065	EC, β^+	122 s
		16	^{16}O	15.994 915	99.76%	
		18	^{18}O	17.999 160	0.200%	
9	Fluorine	18	^{18}F	18.000 937	EC, β^+	1.83 h
		19	^{19}F	18.998 403	100%	
10	Neon	20	^{20}Ne	19.992 435	90.51%	
		22	^{22}Ne	21.991 383	9.22%	
11	Sodium	22	^{22}Na	21.994 434	β^+	2.602 y
		23	^{23}Na	22.989 767	100%	
		24	^{24}Na	23.990 961	β^-	14.96 h

TABLE A1 Atomic Masses

Atomic Number, Z	Name	Atomic Mass Number, A	Symbol	Atomic Mass (u)	Percent Abundance or Decay Mode	Half-life, $t_{1/2}$
12	Magnesium	24	^{24}Mg	23.985 042	78.99%	
13	Aluminum	27	^{27}Al	26.981 539	100%	
14	Silicon	28	^{28}Si	27.976 927	92.23%	2.62h
		31	^{31}Si	30.975 362	β^-	
15	Phosphorus	31	^{31}P	30.973 762	100%	
		32	^{32}P	31.973 907	β^-	14.28 d
16	Sulfur	32	^{32}S	31.972 070	95.02%	
		35	^{35}S	34.969 031	β^-	87.4 d
17	Chlorine	35	^{35}Cl	34.968 852	75.77%	
		37	^{37}Cl	36.965 903	24.23%	
18	Argon	40	^{40}Ar	39.962 384	99.60%	
19	Potassium	39	^{39}K	38.963 707	93.26%	
		40	^{40}K	39.963 999	0.0117%, EC, β^-	$1.28 \times 10^9 \text{ y}$
20	Calcium	40	^{40}Ca	39.962 591	96.94%	
21	Scandium	45	^{45}Sc	44.955 910	100%	
22	Titanium	48	^{48}Ti	47.947 947	73.8%	

TABLE A1 Atomic Masses

Atomic Number, Z	Name	Atomic Mass Number, A	Symbol	Atomic Mass (u)	Percent Abundance or Decay Mode	Half-life, $t_{1/2}$
23	Vanadium	51	^{51}V	50.943 962	99.75%	
24	Chromium	52	^{52}Cr	51.940 509	83.79%	
25	Manganese	55	^{55}Mn	54.938 047	100%	
26	Iron	56	^{56}Fe	55.934 939	91.72%	
27	Cobalt	59	^{59}Co	58.933 198	100%	
		60	^{60}Co	59.933 819	β^-	5.271 y
28	Nickel	58	^{58}Ni	57.935 346	68.27%	
		60	^{60}Ni	59.930 788	26.10%	
29	Copper	63	^{63}Cu	62.939 598	69.17%	
		65	^{65}Cu	64.927 793	30.83%	
30	Zinc	64	^{64}Zn	63.929 145	48.6%	
		66	^{66}Zn	65.926 034	27.9%	
31	Gallium	69	^{69}Ga	68.925 580	60.1%	
32	Germanium	72	^{72}Ge	71.922 079	27.4%	
		74	^{74}Ge	73.921 177	36.5%	
33	Arsenic	75	^{75}As	74.921 594	100%	

TABLE A1 Atomic Masses

Atomic Number, Z	Name	Atomic Mass Number, A	Symbol	Atomic Mass (u)	Percent Abundance or Decay Mode	Half-life, $t_{1/2}$
34	Selenium	80	^{80}Se	79.916 520	49.7%	
35	Bromine	79	^{79}Br	78.918 336	50.69%	
36	Krypton	84	^{84}Kr	83.911 507	57.0%	
37	Rubidium	85	^{85}Rb	84.911 794	72.17%	
38	Strontium	86	^{86}Sr	85.909 267	9.86%	
		88	^{88}Sr	87.905 619	82.58%	
		90	^{90}Sr	89.907 738	β^-	28.8 y
39	Yttrium	89	^{89}Y	88.905 849	100%	
		90	^{90}Y	89.907 152	β^-	64.1 h
40	Zirconium	90	^{90}Zr	89.904 703	51.45%	
41	Niobium	93	^{93}Nb	92.906 377	100%	
42	Molybdenum	98	^{98}Mo	97.905 406	24.13%	
43	Technetium	98	^{98}Tc	97.907 215	β^-	4.2×10^6 y
44	Ruthenium	102	^{102}Ru	101.904 348	31.6%	
45	Rhodium	103	^{103}Rh	102.905 500	100%	
46	Palladium	106	^{106}Pd	105.903 478	27.33%	

TABLE A1 Atomic Masses

Atomic Number, Z	Name	Atomic Mass Number, A	Symbol	Atomic Mass (u)	Percent Abundance or Decay Mode	Half-life, $t_{1/2}$
47	Silver	107	^{107}Ag	106.905 092	51.84%	
		109	^{109}Ag	108.904 757	48.16%	
48	Cadmium	114	^{114}Cd	113.903 357	28.73%	
49	Indium	115	^{115}In	114.903 880	95.7%, β^-	$4.4 \times 10^{14}\text{y}$
50	Tin	120	^{120}Sn	119.902 200	32.59%	
51	Antimony	121	^{121}Sb	120.903 821	57.3%	
52	Tellurium	130	^{130}Te	129.906 229	33.8%, β^-	$2.5 \times 10^{21}\text{y}$
53	Iodine	127	^{127}I	126.904 473	100%	
		131	^{131}I	130.906 114	β^-	8.040 d
54	Xenon	132	^{132}Xe	131.904 144	26.9%	
		136	^{136}Xe	135.907 214	8.9%	
55	Cesium	133	^{133}Cs	132.905 429	100%	
		134	^{134}Cs	133.906 696	EC, β^-	2.06 y
56	Barium	137	^{137}Ba	136.905 812	11.23%	
		138	^{138}Ba	137.905 232	71.70%	
57	Lanthanum	139	^{139}La	138.906 346	99.91%	

TABLE A1 Atomic Masses

Atomic Number, Z	Name	Atomic Mass Number, A	Symbol	Atomic Mass (u)	Percent Abundance or Decay Mode	Half-life, $t_{1/2}$
58	Cerium	140	^{140}Ce	139.905 433	88.48%	
59	Praseodymium	141	^{141}Pr	140.907 647	100%	
60	Neodymium	142	^{142}Nd	141.907 719	27.13%	
61	Promethium	145	^{145}Pm	144.912 743	EC, α	17.7 y
62	Samarium	152	^{152}Sm	151.919 729	26.7%	
63	Europium	153	^{153}Eu	152.921 225	52.2%	
64	Gadolinium	158	^{158}Gd	157.924 099	24.84%	
65	Terbium	159	^{159}Tb	158.925 342	100%	
66	Dysprosium	164	^{164}Dy	163.929 171	28.2%	
67	Holmium	165	^{165}Ho	164.930 319	100%	
68	Erbium	166	^{166}Er	165.930 290	33.6%	
69	Thulium	169	^{169}Tm	168.934 212	100%	
70	Ytterbium	174	^{174}Yb	173.938 859	31.8%	
71	Lutecium	175	^{175}Lu	174.940 770	97.41%	
72	Hafnium	180	^{180}Hf	179.946 545	35.10%	
73	Tantalum	181	^{181}Ta	180.947 992	99.98%	

TABLE A1 Atomic Masses

Atomic Number, Z	Name	Atomic Mass Number, A	Symbol	Atomic Mass (u)	Percent Abundance or Decay Mode	Half-life, $t_{1/2}$
74	Tungsten	184	^{184}W	183.950 928	30.67%	
75	Rhenium	187	^{187}Re	186.955 744	62.6%, β^-	4.6×10^{10} y
76	Osmium	191	^{191}Os	190.960 920	β^-	15.4 d
		192	^{192}Os	191.961 467	41.0%	
77	Iridium	191	^{191}Ir	190.960 584	37.3%	
		193	^{193}Ir	192.962 917	62.7%	
78	Platinum	195	^{195}Pt	194.964 766	33.8%	
79	Gold	197	^{197}Au	196.966 543	100%	
		198	^{198}Au	197.968 217	β^-	2.696 d
80	Mercury	199	^{199}Hg	198.968 253	16.87%	
		202	^{202}Hg	201.970 617	29.86%	
81	Thallium	205	^{205}Tl	204.974 401	70.48%	
82	Lead	206	^{206}Pb	205.974 440	24.1%	
		207	^{207}Pb	206.975 872	22.1%	
		208	^{208}Pb	207.976 627	52.4%	
		210	^{210}Pb	209.984 163	α , β^-	22.3 y

TABLE A1 Atomic Masses

Atomic Number, Z	Name	Atomic Mass Number, A	Symbol	Atomic Mass (u)	Percent Abundance or Decay Mode	Half-life, $t_{1/2}$
		211	^{211}Pb	210.988 735	β^-	36.1 min
		212	^{212}Pb	211.991 871	β^-	10.64 h
83	Bismuth	209	^{209}Bi	208.980 374	100%	
		211	^{211}Bi	210.987 255	α, β^-	2.14 min
84	Polonium	210	^{210}Po	209.982 848	α	138.38 d
85	Astatine	218	^{218}At	218.008 684	α, β^-	1.6 s
86	Radon	222	^{222}Rn	222.017 570	α	3.82 d
87	Francium	223	^{223}Fr	223.019 733	α, β^-	21.8 min
88	Radium	226	^{226}Ra	226.025 402	α	$1.60 \times 10^3 \text{ y}$
89	Actinium	227	^{227}Ac	227.027 750	α, β^-	21.8 y
90	Thorium	228	^{228}Th	228.028 715	α	1.91 y
		232	^{232}Th	232.038 054	100%, α	$1.41 \times 10^{10} \text{ y}$
91	Protactinium	231	^{231}Pa	231.035 880	α	$3.28 \times 10^4 \text{ y}$
92	Uranium	233	^{233}U	233.039 628	α	$1.59 \times 10^3 \text{ y}$
		235	^{235}U	235.043 924	0.720%, α	$7.04 \times 10^8 \text{ y}$
		236	^{236}U	236.045 562	α	$2.34 \times 10^7 \text{ y}$

TABLE A1 Atomic Masses

Atomic Number, Z	Name	Atomic Mass Number, A	Symbol	Atomic Mass (u)	Percent Abundance or Decay Mode	Half-life, $t_{1/2}$
		238	^{238}U	238.050 784	99.2745%, α	$4.47 \times 10^9 \text{ y}$
		239	^{239}U	239.054 289	β^-	23.5 min
93	Neptunium	239	^{239}Np	239.052 933	β^-	2.355 d
94	Plutonium	239	^{239}Pu	239.052 157	α	$2.41 \times 10^4 \text{ y}$
95	Americium	243	^{243}Am	243.061 375	α , fission	$7.37 \times 10^3 \text{ y}$
96	Curium	245	^{245}Cm	245.065 483	α	$8.50 \times 10^3 \text{ y}$
97	Berkelium	247	^{247}Bk	247.070 300	α	$1.38 \times 10^3 \text{ y}$
98	Californium	249	^{249}Cf	249.074 844	α	351 y
99	Einsteinium	254	^{254}Es	254.088 019	α , β^-	276 d
100	Fermium	253	^{253}Fm	253.085 173	EC, α	3.00 d
101	Mendelevium	255	^{255}Md	255.091 081	EC, α	27 min
102	Nobelium	255	^{255}No	255.093 260	EC, α	3.1 min
103	Lawrencium	257	^{257}Lr	257.099 480	EC, α	0.646 s
104	Rutherfordium	261	^{261}Rf	261.108 690	α	1.08 min
105	Dubnium	262	^{262}Db	262.113 760	α , fission	34 s
106	Seaborgium	263	^{263}Sg	263.11 86	α , fission	0.8 s

TABLE A1 Atomic Masses

Atomic Number, Z	Name	Atomic Mass Number, A	Symbol	Atomic Mass (u)	Percent Abundance or Decay Mode	Half-life, t _{1/2}
107	Bohrium	262	²⁶² Bh	262.123 1	α	0.102 s
108	Hassium	264	²⁶⁴ Hs	264.128 5	α	0.08 ms
109	Meitnerium	266	²⁶⁶ Mt	266.137 8	α	3.4 ms

TABLE A1 Atomic Masses

APPENDIX B

Selected Radioactive Isotopes

Decay modes are α , β^- , β^+ , electron capture (EC) and isomeric transition (IT). EC results in the same daughter nucleus as would β^+ decay. IT is a transition from a metastable excited state. Energies for $\beta^\pm$ decays are the maxima; average energies are roughly one-half the maxima.

Isotope	$t_{1/2}$	DecayMode(s)	Energy(MeV)	Percent		γ -Ray Energy(MeV)	Percent
^3H	12.33 y	β^-	0.0186	100%			
^{14}C	5730 y	β^-	0.156	100%			
^{13}N	9.96 min	β^+	1.20	100%			
^{22}Na	2.602 y	β^+	0.55	90%	γ	1.27	100%
^{32}P	14.28 d	β^-	1.71	100%			
^{35}S	87.4 d	β^-	0.167	100%			
^{36}Cl	3.00×10^5 y	β^-	0.710	100%			
^{40}K	1.28×10^9 y	β^-	1.31	89%			
^{43}K	22.3 h	β^-	0.827	87%	γ s	0.373	87%
						0.618	87%
^{45}Ca	165 d	β^-	0.257	100%			
^{51}Cr	27.70 d	EC			γ	0.320	10%
^{52}Mn	5.59d	β^+	3.69	28%	γ s	1.33	28%
						1.43	28%
^{52}Fe	8.27 h	β^+	1.80	43%		0.169	43%
						0.378	43%
^{59}Fe	44.6 d	β^- s	0.273	45%	γ s	1.10	57%
			0.466	55%		1.29	43%
^{60}Co	5.271 y	β^-	0.318	100%	γ s	1.17	100%

TABLE B1 Selected Radioactive Isotopes

Isotope	$t_{1/2}$	DecayMode(s)	Energy(MeV)	Percent		γ -Ray Energy(MeV)	Percent
						1.33	100%
^{65}Zn	244.1 d	EC			γ	1.12	51%
^{67}Ga	78.3 h	EC			γ s	0.0933	70%
						0.185	35%
						0.300	19%
						others	
^{75}Se	118.5 d	EC			γ s	0.121	20%
						0.136	65%
						0.265	68%
						0.280	20%
						others	
^{86}Rb	18.8 d	β^- s	0.69	9%	γ	1.08	9%
			1.77	91%			
^{85}Sr	64.8 d	EC			γ	0.514	100%
^{90}Sr	28.8 y	β^-	0.546	100%			
^{90}Y	64.1 h	β^-	2.28	100%			
$^{99\text{m}}\text{Tc}$	6.02 h	IT			γ	0.142	100%
$^{113\text{m}}\text{In}$	99.5 min	IT			γ	0.392	100%
^{123}I	13.0 h	EC			γ	0.159	$\approx 100\%$
^{131}I	8.040 d	β^- s	0.248	7%	γ s	0.364	85%
			0.607	93%		others	
			others				
^{129}Cs	32.3 h	EC			γ s	0.0400	35%
						0.372	32%

TABLE B1 Selected Radioactive Isotopes

Isotope	$t_{1/2}$	DecayMode(s)	Energy(MeV)	Percent		γ -Ray Energy(MeV)	Percent
						0.411	25%
						others	
^{137}Cs	30.17 y	β^- s	0.511	95%	γ	0.662	95%
			1.17	5%			
^{140}Ba	12.79 d	β^-	1.035	$\approx 100\%$	γ s	0.030	25%
						0.044	65%
						0.537	24%
						others	
^{198}Au	2.696 d	β^-	1.161	$\approx 100\%$	γ	0.412	$\approx 100\%$
^{197}Hg	64.1 h	EC			γ	0.0733	100%
^{210}Po	138.38 d	α	5.41	100%			
^{226}Ra	1.60×10^3 y	α s	4.68	5%	γ	0.186	100%
			4.87	95%			
^{235}U	7.038×10^8 y	α	4.68	$\approx 100\%$	γ s	numerous	<0.400%
^{238}U	4.468×10^9 y	α s	4.22	23%	γ	0.050	23%
			4.27	77%			
^{237}Np	2.14×10^6 y	α s	numerous		γ s	numerous	<0.250%
			4.96 (max.)				
^{239}Pu	2.41×10^4 y	α s	5.19	11%	γ s	7.5×10^{-5}	73%
			5.23	15%		0.013	15%
			5.24	73%		0.052	10%
						others	
^{243}Am	7.37×10^3 y	α s	Max. 5.44		γ s	0.075	
			5.37	88%		others	

TABLE B1 Selected Radioactive Isotopes

Isotope	$t_{1/2}$	DecayMode(s)	Energy(MeV)	Percent		γ -Ray Energy(MeV)	Percent
			5.32	11%			
			others				

TABLE B1 Selected Radioactive Isotopes

APPENDIX C

Useful Information

This appendix is broken into several tables.

- [Table C1](#), Important Constants
- [Table C2](#), Submicroscopic Masses
- [Table C3](#), Solar System Data
- [Table C4](#), Metric Prefixes for Powers of Ten and Their Symbols
- [Table C5](#), The Greek Alphabet
- [Table C6](#), SI units
- [Table C7](#), Selected British Units
- [Table C8](#), Other Units
- [Table C9](#), Useful Formulae

Symbol	Meaning	Best Value	Approximate Value
c	Speed of light in vacuum	$2.99792458 \times 10^8 \text{ m/s}$	$3.00 \times 10^8 \text{ m/s}$
G	Gravitational constant	$6.67408(31) \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$	$6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$
N_A	Avogadro's number	$6.02214076 \times 10^{23}$	6.02×10^{23}
k	Boltzmann's constant	$1.380649 \times 10^{-23} \text{ J/K}$	$1.38 \times 10^{-23} \text{ J/K}$
R	Gas constant	$8.3144621(75) \text{ J/mol} \cdot \text{K}$	$8.31 \text{ J/mol} \cdot \text{K} = 1.99 \text{ cal/mol} \cdot \text{K}$ $= 0.0821 \text{ atm} \cdot \text{L/mol} \cdot \text{K}$
σ	Stefan-Boltzmann constant	$5.670373(21) \times 10^{-8} \text{ W/m}^2 \cdot \text{K}$	$5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}$
k	Coulomb force constant	$8.987551788... \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$	$8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$
q_e	Charge on electron	$-1.602176634 \times 10^{-19} \text{ C}$	$-1.60 \times 10^{-19} \text{ C}$
ϵ_0	Permittivity of free space	$8.854187817... \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2$	$8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2$
μ_0	Permeability of free space	$4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$	$1.26 \times 10^{-6} \text{ T} \cdot \text{m/A}$
h	Planck's constant	$6.62607015 \times 10^{-34} \text{ J} \cdot \text{s}$	$6.63 \times 10^{-34} \text{ J} \cdot \text{s}$

TABLE C1 Important Constants

1 Stated values are according to the National Institute of Standards and Technology Reference on Constants, Units, and Uncertainty, www.physics.nist.gov/cuu (accessed May 18, 2012). Values in parentheses are the uncertainties in the last digits. Numbers without uncertainties are exact as defined.

Symbol	Meaning	Best Value	Approximate Value
m_e	Electron mass	$9.10938291(40) \times 10^{-31} \text{ kg}$	$9.11 \times 10^{-31} \text{ kg}$
m_p	Proton mass	$1.672621777(74) \times 10^{-27} \text{ kg}$	$1.6726 \times 10^{-27} \text{ kg}$
m_n	Neutron mass	$1.674927351(74) \times 10^{-27} \text{ kg}$	$1.6749 \times 10^{-27} \text{ kg}$
u	Atomic mass unit	$1.660538921(73) \times 10^{-27} \text{ kg}$	$1.6605 \times 10^{-27} \text{ kg}$

TABLE C2 Submicroscopic Masses

Sun	mass	$1.99 \times 10^{30} \text{ kg}$
	average radius	$6.96 \times 10^8 \text{ m}$
	Earth-sun distance (average)	$1.496 \times 10^{11} \text{ m}$
Earth	mass	$5.9736 \times 10^{24} \text{ kg}$
	average radius	$6.376 \times 10^6 \text{ m}$
	orbital period	$3.16 \times 10^7 \text{ s}$
Moon	mass	$7.35 \times 10^{22} \text{ kg}$
	average radius	$1.74 \times 10^6 \text{ m}$
	orbital period (average)	$2.36 \times 10^6 \text{ s}$
	Earth-moon distance (average)	$3.84 \times 10^8 \text{ m}$

TABLE C3 Solar System Data

Prefix	Symbol	Value	Prefix	Symbol	Value
tera	T	10^{12}	deci	d	10^{-1}
giga	G	10^9	centi	c	10^{-2}
mega	M	10^6	milli	m	10^{-3}
kilo	k	10^3	micro	μ	10^{-6}
hecto	h	10^2	nano	n	10^{-9}

TABLE C4 Metric Prefixes for Powers of Ten and Their Symbols

2 Stated values are according to the National Institute of Standards and Technology Reference on Constants, Units, and Uncertainty, www.physics.nist.gov/cuu (accessed May 18, 2012). Values in parentheses are the uncertainties in the last digits. Numbers without uncertainties are exact as defined.

Prefix	Symbol	Value	Prefix	Symbol	Value
deka	da	10^1	pico	p	10^{-12}
—	—	$10^0 (= 1)$	femto	f	10^{-15}

TABLE C4 Metric Prefixes for Powers of Ten and Their Symbols

Alpha	A	α	Eta	H	η	Nu	N	ν	Tau	T	τ
Beta	B	β	Theta	Θ	θ	Xi	Ξ	ξ	Upsilon	Υ	υ
Gamma	Γ	γ	Iota	I	ι	Omicron	O	o	Phi	Φ	ϕ
Delta	Δ	δ	Kappa	K	κ	Pi	Π	π	Chi	X	χ
Epsilon	E	ε	Lambda	Λ	λ	Rho	P	ρ	Psi	Ψ	ψ
Zeta	Z	ζ	Mu	M	μ	Sigma	Σ	σ	Omega	Ω	ω

TABLE C5 The Greek Alphabet

	Entity	Abbreviation	Name
Fundamental units	Length	m	meter
	Mass	kg	kilogram
	Time	s	second
	Current	A	ampere
Supplementary unit	Angle	rad	radian
Derived units	Force	$N = kg \cdot m/s^2$	newton
	Energy	$J = kg \cdot m^2/s^2$	joule
	Power	$W = J/s$	watt
	Pressure	$Pa = N/m^2$	pascal
	Frequency	$Hz = 1/s$	hertz
	Electronic potential	$V = J/C$	volt
	Capacitance	$F = C/V$	farad
	Charge	$C = s \cdot A$	coulomb
	Resistance	$\Omega = V/A$	ohm

TABLE C6 SI Units

	Entity	Abbreviation	Name
	Magnetic field	$T = N/(A \cdot m)$	tesla
	Nuclear decay rate	$Bq = 1/s$	becquerel

TABLE C6 SI Units

Length	1 inch (in.)= 2.54 cm (exactly)
	1 foot (ft)= 0.3048 m
	1 mile (mi)= 1.609 km
Force	1 pound (lb)= 4.448 N
Energy	1 British thermal unit (Btu)= 1.055×10^3 J
Power	1 horsepower (hp)= 746 W
Pressure	$1 \text{ lb/in}^2 = 6.895 \times 10^3$ Pa

TABLE C7 Selected British Units

Length	1 light year (ly) = 9.46×10^{15} m
	1 astronomical unit (au) = 1.50×10^{11} m
	1 nautical mile = 1.852 km
	1 angstrom($\text{\AA}$) = 10^{-10} m
Area	1 acre (ac) = 4.05×10^3 m ²
	1 square foot (ft ²) = 9.29×10^{-2} m ²
	1 barn (b) = 10^{-28} m ²
Volume	1 liter (L) = 10^{-3} m ³
	1 U.S. gallon (gal) = 3.785×10^{-3} m ³
Mass	1 solar mass = 1.99×10^{30} kg
	1 metric ton = 10^3 kg
	1 atomic mass unit (u) = 1.6605×10^{-27} kg
Time	1 year (y) = 3.16×10^7 s
	1 day (d) = 86,400 s

TABLE C8 Other Units

Speed	1 mile per hour (mph) = 1.609 km/h
	1 nautical mile per hour (naut) = 1.852 km/h
Angle	1 degree ($^{\circ}$) = 1.745×10^{-2} rad
	1 minute of arc ($'$) = 1/60 degree
	1 second of arc ($''$) = 1/60 minute of arc
	1 grad = 1.571×10^{-2} rad
Energy	1 kiloton TNT (kT) = 4.2×10^{12} J
	1 kilowatt hour (kW · h) = 3.60×10^6 J
	1 food calorie (kcal) = 4186 J
	1 calorie (cal) = 4.186 J
	1 electron volt (eV) = 1.60×10^{-19} J
Pressure	1 atmosphere (atm) = 1.013×10^5 Pa
	1 millimeter of mercury (mm Hg) = 133.3 Pa
	1 torricelli (torr) = 1 mm Hg = 133.3 Pa
Nuclear decay rate	1 curie (Ci) = 3.70×10^{10} Bq

TABLE C8 Other Units

Circumference of a circle with radius r or diameter d	$C = 2\pi r = \pi d$
Area of a circle with radius r or diameter d	$A = \pi r^2 = \pi d^2/4$
Area of a sphere with radius r	$A = 4\pi r^2$
Volume of a sphere with radius r	$V = (4/3)(\pi r^3)$

TABLE C9 Useful Formulae

APPENDIX D

Glossary of Key Symbols and Notation

In this glossary, key symbols and notation are briefly defined.

Symbol	Definition
$\overline{\text{any symbol}}$	average (indicated by a bar over a symbol—e.g., $\overline{v}$ is average velocity)
$^{\circ}\text{C}$	Celsius degree
$^{\circ}\text{F}$	Fahrenheit degree
//	parallel
$\perp$	perpendicular
$\propto$	proportional to
$\pm$	plus or minus
0	zero as a subscript denotes an initial value
α	alpha rays
α	angular acceleration
α	temperature coefficient(s) of resistivity
β	beta rays
β	sound level
β	volume coefficient of expansion
β^{-}	electron emitted in nuclear beta decay
β^{+}	positron decay
γ	gamma rays
γ	surface tension
$\gamma = 1/\sqrt{1 - v^2/c^2}$	a constant used in relativity
Δ	change in whatever quantity follows

TABLE D1

Symbol	Definition
δ	uncertainty in whatever quantity follows
ΔE	change in energy between the initial and final orbits of an electron in an atom
ΔE	uncertainty in energy
Δm	difference in mass between initial and final products
ΔN	number of decays that occur
Δp	change in momentum
Δp	uncertainty in momentum
ΔPE_g	change in gravitational potential energy
$\Delta\theta$	rotation angle
Δs	distance traveled along a circular path
Δt	uncertainty in time
Δt_0	proper time as measured by an observer at rest relative to the process
ΔV	potential difference
Δx	uncertainty in position
ϵ_0	permittivity of free space
η	viscosity
θ	angle between the force vector and the displacement vector
θ	angle between two lines
θ	contact angle
θ	direction of the resultant
θ_b	Brewster's angle
θ_c	critical angle
κ	dielectric constant
λ	decay constant of a nuclide
λ	wavelength

TABLE D1

Symbol	Definition
λ_n	wavelength in a medium
μ_0	permeability of free space
μ_k	coefficient of kinetic friction
μ_s	coefficient of static friction
ν_e	electron neutrino
π^+	positive pion
π^-	negative pion
π^0	neutral pion
ρ	density
ρ_c	critical density, the density needed to just halt universal expansion
ρ_{fl}	fluid density
$\bar{\rho}_{obj}$	average density of an object
ρ/ρ_w	specific gravity
τ	characteristic time constant for a resistance and inductance (RL) or resistance and capacitance (RC) circuit
τ	characteristic time for a resistor and capacitor (RC) circuit
τ	torque
Υ	upsilon meson
Φ	magnetic flux
ϕ	phase angle
Ω	ohm (unit)
ω	angular velocity
A	ampere (current unit)
A	area
A	cross-sectional area

TABLE D1

Symbol	Definition
A	total number of nucleons
a	acceleration
a_B	Bohr radius
a_c	centripetal acceleration
a_t	tangential acceleration
AC	alternating current
AM	amplitude modulation
atm	atmosphere
B	baryon number
B	blue quark color
$\bar{B}$	antiblack (yellow) antiquark color
b	quark flavor bottom or beauty
B	bulk modulus
B	magnetic field strength
B_{int}	electron's intrinsic magnetic field
B_{orb}	orbital magnetic field
BE	binding energy of a nucleus—it is the energy required to completely disassemble it into separate protons and neutrons
BE/ A	binding energy per nucleon
Bq	becquerel—one decay per second
C	capacitance (amount of charge stored per volt)
C	coulomb (a fundamental SI unit of charge)
C_p	total capacitance in parallel
C_s	total capacitance in series
CG	center of gravity

TABLE D1

Symbol	Definition
CM	center of mass
c	quark flavor charm
c	specific heat
c	speed of light
Cal	kilocalorie
cal	calorie
COP_{hp}	heat pump's coefficient of performance
COP_{ref}	coefficient of performance for refrigerators and air conditioners
$\cos \theta$	cosine
$\cot \theta$	cotangent
$\csc \theta$	cosecant
D	diffusion constant
d	displacement
d	quark flavor down
dB	decibel
d_i	distance of an image from the center of a lens
d_o	distance of an object from the center of a lens
DC	direct current
E	electric field strength
ϵ	emf (voltage) or Hall electromotive force
emf	electromotive force
E	energy of a single photon
E	nuclear reaction energy
E	relativistic total energy
E	total energy

TABLE D1

Symbol	Definition
E_0	ground state energy for hydrogen
E_0	rest energy
EC	electron capture
E_{cap}	energy stored in a capacitor
Eff	efficiency—the useful work output divided by the energy input
Eff_C	Carnot efficiency
E_{in}	energy consumed (food digested in humans)
E_{ind}	energy stored in an inductor
E_{out}	energy output
e	emissivity of an object
e^+	antielectron or positron
eV	electron volt
F	farad (unit of capacitance, a coulomb per volt)
F	focal point of a lens
F	force
F	magnitude of a force
F	restoring force
F_B	buoyant force
F_c	centripetal force
F_i	force input
F _{net}	net force
F_o	force output
FM	frequency modulation
f	focal length
f	frequency

TABLE D1

Symbol	Definition
f_0	resonant frequency of a resistance, inductance, and capacitance (RLC) series circuit
f_0	threshold frequency for a particular material (photoelectric effect)
f_1	fundamental
f_2	first overtone
f_3	second overtone
f_B	beat frequency
f_k	magnitude of kinetic friction
f_s	magnitude of static friction
G	gravitational constant
G	green quark color
$\bar{G}$	antigreen (magenta) antiquark color
g	acceleration due to gravity
g	gluons (carrier particles for strong nuclear force)
h	change in vertical position
h	height above some reference point
h	maximum height of a projectile
h	Planck's constant
hf	photon energy
h_i	height of the image
h_o	height of the object
I	electric current
I	intensity
I	intensity of a transmitted wave
I	moment of inertia (also called rotational inertia)
I_0	intensity of a polarized wave before passing through a filter

TABLE D1

Symbol	Definition
I_{ave}	average intensity for a continuous sinusoidal electromagnetic wave
I_{rms}	average current
J	joule
J/Ψ	Joules/psi meson
K	kelvin
k	Boltzmann constant
k	force constant of a spring
K_{α}	x rays created when an electron falls into an $n = 1$ shell vacancy from the $n = 3$ shell
K_{β}	x rays created when an electron falls into an $n = 2$ shell vacancy from the $n = 3$ shell
kcal	kilocalorie
KE	translational kinetic energy
KE + PE	mechanical energy
KE_e	kinetic energy of an ejected electron
KE_{rel}	relativistic kinetic energy
KE_{rot}	rotational kinetic energy
$\overline{\text{KE}}$	thermal energy
kg	kilogram (a fundamental SI unit of mass)
L	angular momentum
L	liter
L	magnitude of angular momentum
L	self-inductance
ℓ	angular momentum quantum number
L_{α}	x rays created when an electron falls into an $n = 2$ shell
L_e	electron total family number
L_{μ}	muon family total number

TABLE D1

Symbol	Definition
L_τ	tau family total number
L_f	heat of fusion
L_f and L_v	latent heat coefficients
L_{orb}	orbital angular momentum
L_s	heat of sublimation
L_v	heat of vaporization
L_z	z - component of the angular momentum
M	angular magnification
M	mutual inductance
m	indicates metastable state
m	magnification
m	mass
m	mass of an object as measured by a person at rest relative to the object
m	meter (a fundamental SI unit of length)
m	order of interference
m	overall magnification (product of the individual magnifications)
$m(^A\text{X})$	atomic mass of a nuclide
MA	mechanical advantage
m_e	magnification of the eyepiece
m_e	mass of the electron
m_ℓ	angular momentum projection quantum number
m_n	mass of a neutron
m_o	magnification of the objective lens
mol	mole
m_p	mass of a proton

TABLE D1

Symbol	Definition
m_s	spin projection quantum number
N	magnitude of the normal force
N	newton
N	normal force
N	number of neutrons
n	index of refraction
n	number of free charges per unit volume
N_A	Avogadro's number
N_r	Reynolds number
$N \cdot m$	newton-meter (work-energy unit)
$N \cdot m$	newtons times meters (SI unit of torque)
OE	other energy
P	power
P	power of a lens
P	pressure
p	momentum
p	momentum magnitude
p	relativistic momentum
p _{tot}	total momentum
p' _{tot}	total momentum some time later
P_{abs}	absolute pressure
P_{atm}	atmospheric pressure
P_{atm}	standard atmospheric pressure
PE	potential energy
PE _{el}	elastic potential energy

TABLE D1

Symbol	Definition
PE_{elec}	electric potential energy
PE_s	potential energy of a spring
P_g	gauge pressure
P_{in}	power consumption or input
P_{out}	useful power output going into useful work or a desired, form of energy
Q	latent heat
Q	net heat transferred into a system
Q	flow rate—volume per unit time flowing past a point
$+Q$	positive charge
$-Q$	negative charge
q	electron charge
q_p	charge of a proton
q	test charge
QF	quality factor
R	activity, the rate of decay
R	radius of curvature of a spherical mirror
R	red quark color
$\bar{R}$	antired (cyan) quark color
R	resistance
R	resultant or total displacement
R	Rydberg constant
R	universal gas constant
r	distance from pivot point to the point where a force is applied
r	internal resistance
$r_{\perp}$	perpendicular lever arm

TABLE D1

Symbol	Definition
r	radius of a nucleus
r	radius of curvature
r	resistivity
r or rad	radiation dose unit
rem	roentgen equivalent man
rad	radian
RBE	relative biological effectiveness
RC	resistor and capacitor circuit
rms	root mean square
r_n	radius of the n th H-atom orbit
R_p	total resistance of a parallel connection
R_s	total resistance of a series connection
R_s	Schwarzschild radius
S	entropy
S	intrinsic spin (intrinsic angular momentum)
S	magnitude of the intrinsic (internal) spin angular momentum
S	shear modulus
S	strangeness quantum number
s	quark flavor strange
s	second (fundamental SI unit of time)
s	spin quantum number
s	total displacement
$\sec \theta$	secant
$\sin \theta$	sine
s_z	z-component of spin angular momentum

TABLE D1

Symbol	Definition
T	period—time to complete one oscillation
T	temperature
T_c	critical temperature—temperature below which a material becomes a superconductor
T	tension
T	tesla (magnetic field strength B)
t	quark flavor top or truth
t	time
$t_{1/2}$	half-life—the time in which half of the original nuclei decay
$\tan \theta$	tangent
U	internal energy
u	quark flavor up
u	unified atomic mass unit
$\mathbf{u}$	velocity of an object relative to an observer
$\mathbf{u}'$	velocity relative to another observer
V	electric potential
V	terminal voltage
V	volt (unit)
V	volume
$\mathbf{v}$	relative velocity between two observers
v	speed of light in a material
$\mathbf{v}$	velocity
$\bar{\mathbf{v}}$	average fluid velocity
$V_B - V_A$	change in potential
$\mathbf{v}_d$	drift velocity
V_p	transformer input voltage

TABLE D1

Symbol	Definition
V_{rms}	rms voltage
V_s	transformer output voltage
$\mathbf{v}_{\text{tot}}$	total velocity
v_w	propagation speed of sound or other wave
$\mathbf{v}_w$	wave velocity
W	work
$\mathcal{W}$	net work done by a system
W	watt
w	weight
w_{fl}	weight of the fluid displaced by an object
W_c	total work done by all conservative forces
W_{nc}	total work done by all nonconservative forces
W_{out}	useful work output
X	amplitude
X	symbol for an element
${}_A^Z X_N$	notation for a particular nuclide
x	deformation or displacement from equilibrium
x	displacement of a spring from its undeformed position
x	horizontal axis
X_C	capacitive reactance
X_L	inductive reactance
x_{rms}	root mean square diffusion distance
y	vertical axis
Y	elastic modulus or Young's modulus

TABLE D1

Symbol	Definition
Z	atomic number (number of protons in a nucleus)
Z	impedance

TABLE D1

ANSWER KEY

Chapter 1

Problems & Exercises

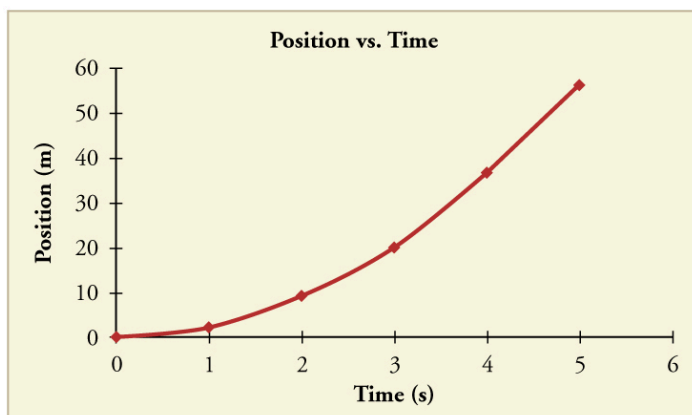
1. (a) 27.8 m/s
(b) 62.1 mph
3. $\frac{1.0 \text{ m}}{\text{s}} = \frac{1.0 \text{ m}}{\text{s}} \times \frac{3600 \text{ s}}{1 \text{ hr}} \times \frac{1 \text{ km}}{1000 \text{ m}}$
= 3.6 km/h.
5. length: 377 ft; 4.53×10^3 in. width: 280 ft; 3.3×10^3 in.
7. 8.847 km
9. (a) 1.3×10^{-9} m
(b) 40 km/My
11. 2 kg
13. (a) 85 to 95 km/h
(b) 53 to 59 mi/h
15. (a) 7.6×10^7 beats
(b) 7.57×10^7 beats
(c) 7.57×10^7 beats
17. (a) 3
(b) 3
(c) 3
19. (a) 2.2%
(b) 59 to 61 km/h
21. 80 ± 3 beats/min
23. 2.8 h
25. $11.4 \pm 0.6 \text{ cm}^3$
27. $12.06 \pm 0.04 \text{ m}^2$
29. Sample answer: 2×10^9 heartbeats
31. Sample answer: 2×10^{31} if an average human lifetime is taken to be about 70 years.
33. Sample answer: 50 atoms
35. Sample answers:
(a) 10^{12} cells/hummingbird
(b) 10^{16} cells/human

Chapter 2

Problems & Exercises

1. (a) 7 m
(b) 7 m
(c) +7 m
3. a. $8 \text{ m} + 2 \text{ m} + 3 \text{ m} = \underline{13 \text{ m}}$
b. $\underline{9 \text{ m}}$
c. $\Delta x = 11 \text{ m} - 2 \text{ m} = \underline{9 \text{ m}}$
5. (a) 3.0×10^4 m/s
(b) 0 m/s
7. 2×10^7 years
9. $34.689 \text{ m/s} = 124.88 \text{ km/h}$
11. (a) 40.0 km/h

- (b) 34.3 km/h, 25° S of E.
 (c) average speed = 3.20 km/h, $\bar{v} = 0$.
13. 384,000 km
15. (a) 6.61×10^{15} rev/s
 (b) 0 m/s
16. 4.29 m/s^2
18. (a) 1.43 s
 (b) -2.50 m/s^2
20. (a) 10.8 m/s
 (b)



21. 38.9 m/s (about 87 miles per hour)
23. (a) 16.5 s
 (b) 13.5 s
 (c) -2.68 m/s^2
25. (a) 20.0 m
 (b) -1.00 m/s
 (c) This result does not really make sense. If the runner starts at 9.00 m/s and decelerates at 2.00 m/s^2 , then she will have stopped after 4.50 s. If she continues to decelerate, she will be running backwards.
27. 0.799 m
29. (a) 28.0 m/s
 (b) 50.9 s
 (c) 7.68 km to accelerate and 713 m to decelerate
31. (a) 51.4 m
 (b) 17.1 s
33. (a) -80.4 m/s^2
 (b) $9.33 \times 10^{-2} \text{ s}$
35. (a) 7.7 m/s
 (b) $-15 \times 10^2 \text{ m/s}^2$. This is about 3 times the deceleration of the pilots, who were falling from thousands of meters high!
37. (a) 32.6 m/s^2
 (b) 162 m/s
 (c) $v > v_{\text{max}}$, because the assumption of constant acceleration is not valid for a dragster. A dragster changes gears, and would have a greater acceleration in first gear than second gear than third gear, etc. The acceleration would be greatest at the beginning, so it would not be accelerating at 32.6 m/s^2 during the last few meters, but substantially less, and the final velocity would be less than 162 m/s.
39. 104 s
40. (a) $v = 12.2 \text{ m/s}$; $a = 4.07 \text{ m/s}^2$
 (b) $v = 11.2 \text{ m/s}$
41. (a) $y_1 = 6.28 \text{ m}$; $v_1 = 10.1 \text{ m/s}$

- (b) $y_2 = 10.1 \text{ m}$; $v_2 = 5.20 \text{ m/s}$
 (c) $y_3 = 11.5 \text{ m}$; $v_3 = 0.300 \text{ m/s}$
 (d) $y_4 = 10.4 \text{ m}$; $v_4 = -4.60 \text{ m/s}$

43. $v_0 = 4.95 \text{ m/s}$

45. (a) $a = -9.80 \text{ m/s}^2$; $v_0 = 13.0 \text{ m/s}$; $y_0 = 0 \text{ m}$

(b) $v = 0 \text{ m/s}$. Unknown is distance y to top of trajectory, where velocity is zero. Use equation $v^2 = v_0^2 + 2a(y - y_0)$ because it contains all known values except for y , so we can solve for y . Solving for y gives

$$v^2 - v_0^2 = 2a(y - y_0)$$

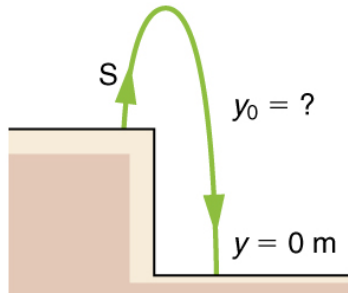
$$\frac{v^2 - v_0^2}{2a} = y - y_0$$

$$y = y_0 + \frac{v^2 - v_0^2}{2a} = 0 \text{ m} + \frac{(0 \text{ m/s})^2 - (13.0 \text{ m/s})^2}{2(-9.80 \text{ m/s}^2)} = 8.62 \text{ m}$$

Dolphins measure about 2 meters long and can jump several times their length out of the water, so this is a reasonable result.

(c) 2.65 s

47.



- (a) 8.26 m
 (b) 0.717 s

49. 1.91 s

51. (a) 94.0 m
 (b) 3.13 s

53. (a) -70.0 m/s (downward)
 (b) 6.10 s

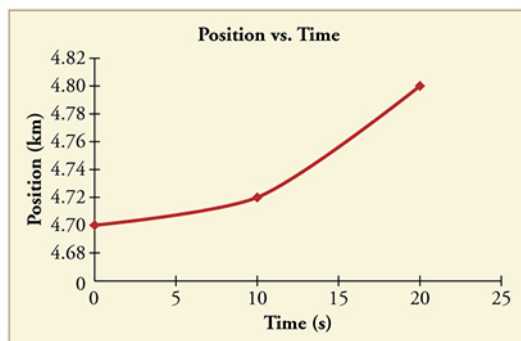
55. (a) 19.6 m
 (b) 18.5 m

57. (a) 305 m
 (b) 262 m, -29.2 m/s
 (c) 8.91 s

59. (a) 115 m/s
 (b) 5.0 m/s^2

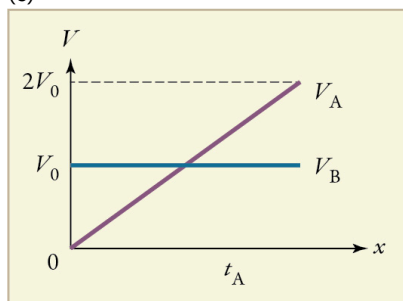
61. (a) At $t = 10$, $v = 0.208 \text{ m/s}$; at $t = 30$, $v = 0.238 \text{ m/s}$. (b) 0.0015 m/s^2

63.



65. (a) 6 m/s

- (b) 12 m/s
 (c) 3 m/s²
 (d) 10 s
67. (a) Car A is traveling faster at the checkpoint because it must go past the speed of car B to reach the same distance.
 (b) i. Yes, the equation is consistent with the answer because the speed of car A is only a constant away from the correct answer. ii. Yes, the equation makes sense because $V = 2V_0$.
 (c)



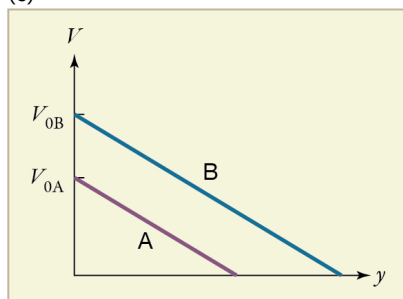
Chapter 3

Problems & Exercises

1. (a) 480 m
 (b) 379 m, 18.4° east of north
3. north component 3.21 km, east component 3.83 km
5. 19.5 m, 4.65° south of west
7. (a) 26.6 m, 65.1° north of east
 (b) 26.6 m, 65.1° south of west
9. 52.9 m, 90.1° with respect to the x-axis.
11. x-component 4.41 m/s
 y-component 5.07 m/s
13. (a) 1.56 km
 (b) 120 m east
15. North-component 87.0 km, east-component 87.0 km
17. 30.8 m, 35.8 west of north
19. (a) 30.8 m, 54.2° south of west
 (b) 30.8 m, 54.2° north of east
21. 18.4 km south, then 26.2 km west (b) 31.5 km at 45.0° south of west, then 5.56 km at 45.0° west of north
23. 7.34 km, 63.5° south of east
25. $x = 1.30 \text{ m} \times 10^2$
 $y = 30.9 \text{ m}$.
27. (a) 3.50 s
 (b) 28.6 m/s (c) 34.3 m/s
 (d) 44.7 m/s, 50.2° below horizontal
29. (a) 18.4°
 (b) The arrow will go over the branch.
31. $R = \frac{v_0^2}{\sin 2\theta_0 g}$
 For $\theta = 45^\circ$, $R = \frac{v_0^2}{g}$
 $R = 91.8 \text{ m}$ for $v_0 = 30 \text{ m/s}$; $R = 163 \text{ m}$ for $v_0 = 40 \text{ m/s}$; $R = 255 \text{ m}$ for $v_0 = 50 \text{ m/s}$.
33. (a) 560 m/s
 (b) $8.00 \times 10^3 \text{ m}$
 (c) 80.0 m. This error is not significant because it is only 1% of the answer in part (b).

35. 1.50 m, assuming launch angle of 45°
37. $\theta = 6.1^\circ$
yes, the ball lands at 5.3 m from the net
39. (a) -0.486 m
(b) The larger the muzzle velocity, the smaller the deviation in the vertical direction, because the time of flight would be smaller. Air resistance would have the effect of decreasing the time of flight, therefore increasing the vertical deviation.
41. 4.23 m. No, the owl is not lucky; he misses the nest.
43. No, the maximum range (neglecting air resistance) is about 92 m.
45. 15.0 m/s
47. (a) 24.2 m/s
(b) The ball travels a total of 57.4 m with the brief gust of wind.
49. $y - y_0 = 0 = v_{0y}t - \frac{1}{2}gt^2 = (v_0 \sin \theta)t - \frac{1}{2}gt^2$,
so that $t = \frac{2(v_0 \sin \theta)}{g}$
 $x - x_0 = v_{0x}t = (v_0 \cos \theta)t = R$, and substituting for t gives:
$$R = v_0 \cos \theta \left(\frac{2v_0 \sin \theta}{g} \right) = \frac{2v_0^2 \sin \theta \cos \theta}{g}$$

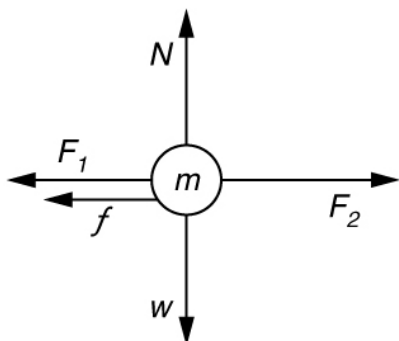
since $2 \sin \theta \cos \theta = \sin 2\theta$, the range is:
$$R = \frac{v_0^2 \sin 2\theta}{g}.$$
52. (a) 35.8 km, 45° south of east
(b) 5.53 m/s, 45° south of east
(c) 56.1 km, 45° south of east
54. (a) 0.70 m/s faster
(b) Second runner wins
(c) 4.17 m
56. 17.0 m/s, 22.1°
58. (a) 230 m/s, 8.0° south of west
(b) The wind should make the plane travel slower and more to the south, which is what was calculated.
60. (a) 63.5 m/s
(b) 29.6 m/s
62. 6.68 m/s, 53.3° south of west
64. (a) $H_{\text{average}} = 14.9 \frac{\text{km/s}}{\text{Mly}}$
(b) 20.2 billion years
66. 1.72 m/s, 42.3° north of east
71. (a) Since $\sin^2 50 > \sin^2 40$, then B reaches the greatest height.
(b) i. Yes, it is consistent because $\sin 50 > \sin 40$. ii. No, it does not make sense because y is proportional to $\sin^2 \theta$.
(c)



Chapter 4

Problems & Exercises

1. 265 N
3. 13.3 m/s^2
7. (a) 12 m/s^2 .
(b) The acceleration is not one-fourth of what it was with all rockets burning because the frictional force is still as large as it was with all rockets burning.
9. (a) The system is the child in the wagon plus the wagon.
(b)



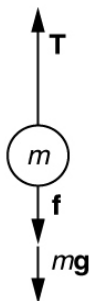
- (c) $a = 0.130 \text{ m/s}^2$ in the direction of the second child's push.
- (d) $a = 0.00 \text{ m/s}^2$
11. (a) $3.68 \times 10^3 \text{ N}$. This force is 5.00 times greater than his weight.
(b) 3750 N; 11.3° above horizontal
13. $1.5 \times 10^3 \text{ N}$, 150 kg, 150 kg
15. Force on shell: $2.64 \times 10^7 \text{ N}$
Force exerted on ship = $-2.64 \times 10^7 \text{ N}$, by Newton's third law
17. (a) 0.11 m/s^2
(b) $1.2 \times 10^4 \text{ N}$
19. (a) $7.84 \times 10^{-4} \text{ N}$
(b) $1.89 \times 10^{-3} \text{ N}$. This is 2.41 times the tension in the vertical strand.
21. Newton's second law applied in vertical direction gives

$$F_y = F - 2T \sin \theta = 0$$

$$F = 2T \sin \theta$$

$$T = \frac{F}{2 \sin \theta}.$$

23.



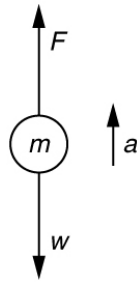
Using the free-body diagram:

$$F_{\text{net}} = T - f - mg = ma,$$

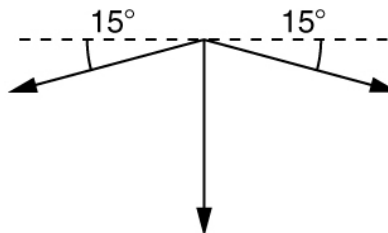
so that

$$a = \frac{T - f - mg}{m} = \frac{1.250 \times 10^7 \text{ N} - 4.50 \times 10^6 \text{ N} - (5.00 \times 10^5 \text{ kg})(9.80 \text{ m/s}^2)}{5.00 \times 10^5 \text{ kg}} = 6.20 \text{ m/s}^2.$$

25. 1. Use Newton's laws of motion.

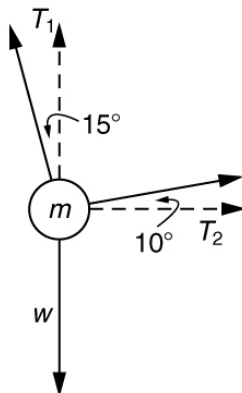


2. Given : $a = 4.00g = (4.00)(9.80 \text{ m/s}^2) = 39.2 \text{ m/s}^2$; $m = 70.0 \text{ kg}$,
Find: F .
3. $\sum F = +F - w = ma$, so that $F = ma + w = ma + mg = m(a + g)$.
 $F = (70.0 \text{ kg})[(39.2 \text{ m/s}^2) + (9.80 \text{ m/s}^2)] = 3.43 \times 10^3 \text{ N}$. The force exerted by the high-jumper is actually down on the ground, but F is up from the ground and makes him jump.
4. This result is reasonable, since it is quite possible for a person to exert a force of the magnitude of 10^3 N .
27. (a) $4.41 \times 10^5 \text{ N}$
(b) $1.50 \times 10^5 \text{ N}$
29. (a) 910 N
(b) $1.11 \times 10^3 \text{ N}$
31. $a = 0.139 \text{ m/s}$, $\theta = 12.4^\circ$ north of east
33. 1. Use Newton's laws since we are looking for forces.
2. Draw a free-body diagram:



3. The tension is given as $T = 25.0 \text{ N}$. Find F_{app} . Using Newton's laws gives: $\sum F_y = 0$, so that applied force is due to the y -components of the two tensions: $F_{\text{app}} = 2 T \sin \theta = 2(25.0 \text{ N})\sin(15^\circ) = 12.9 \text{ N}$
The x -components of the tension cancel. $\sum F_x = 0$.
4. This seems reasonable, since the applied tensions should be greater than the force applied to the tooth.
40. 10.2 m/s^2 , 4.67° from vertical

42.

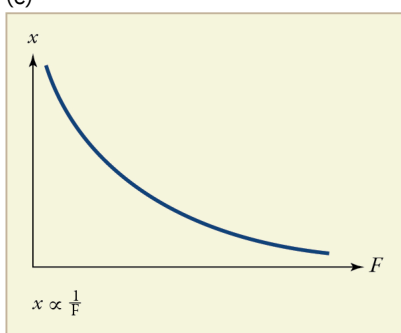


$$T_1 = 736 \text{ N}$$

$$T_2 = 194 \text{ N}$$

44. (a) 7.43 m/s

- (b) 2.97 m
46. (a) 4.20 m/s
(b) 29.4 m/s²
(c) 4.31×10^3 N
48. (a) 47.1 m/s
(b) 2.47×10^3 m/s²
(c) 6.18×10^3 N. The average force is 252 times the shell's weight.
52. (a) 1×10^{-13}
(b) 1×10^{-11}
54. 10^2
55. (a) Box A travels faster at the finishing distance since a greater force with equal mass results in a greater acceleration. Also, a greater acceleration over the same distance results in a greater final speed.
(b) i. Yes, it is consistent because a greater force results in a greater final speed. ii. It does not make sense because $V = \sqrt{2(f/m)x} = K\sqrt{(F)}$.
(c)

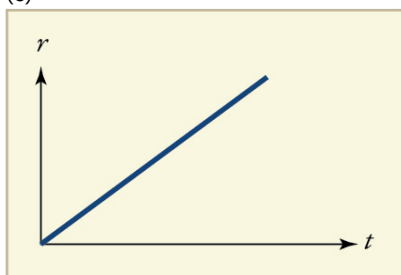


Chapter 5

Problems & Exercises

1. 5.00 N
4. (a) 588 N
(b) 1.96 m/s²
6. (a) 3.29 m/s²
(b) 3.52 m/s²
(c) 980 N; 945 N
10. 1.83 m/s²
14. (a) 4.20 m/s²
(b) 2.74 m/s²
(c) -0.195 m/s²
16. (a) 1.03×10^6 N
(b) 3.48×10^5 N
18. (a) 51.0 N
(b) 0.720 m/s²
20. 115 m/s; 414 km/hr
22. 25.1 m/s; 9.90 m/s
24. 2.9
26. $[\eta] = \frac{[F_s]}{[r][v]} = \frac{\text{kg} \cdot \text{m/s}^2}{\text{m} \cdot \text{m/s}} = \frac{\text{kg}}{\text{m} \cdot \text{s}}$
28. 0.76 kg/m · s
29. 1.90×10^{-3} cm
31. (a) 1 mm
(b) This does seem reasonable, since the lead does seem to shrink a little when you push on it.

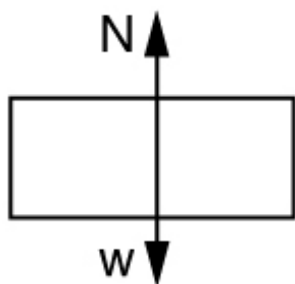
33. (a) 9 cm
(b) This seems reasonable for nylon climbing rope, since it is not supposed to stretch that much.
35. 8.59 mm
37. 1.49×10^{-7} m
39. (a) 3.99×10^{-7} m
(b) 9.67×10^{-8} m
41. 4×10^6 N/m². This is about 36 atm, greater than a typical jar can withstand.
43. 1.4 cm
45. (a) Bead A has the larger radius because both have the same weight and drag force. Since A takes longer to fall, its speed is less and, therefore, it must have larger area or radius.
(b) i. This equation is not consistent because it shows that the bead that took longer to fall has a smaller radius. ii. It does not make sense because r is proportional to t .
(c)



Chapter 6

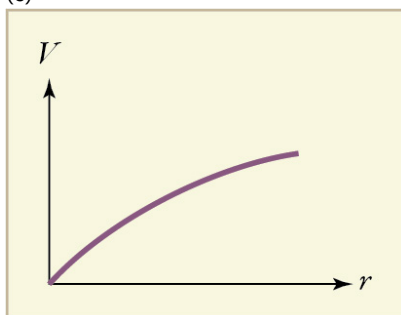
Problems & Exercises

1. 723 km
3. 5×10^7 rotations
5. 117 rad/s
7. 76.2 rad/s
728 rpm
8. (a) 33.3 rad/s
(b) 500 N
(c) 40.8 m
10. 12.9 rev/min
12. 4×10^{21} m
14. a) 3.47×10^4 m/s², 3.55×10^3 g
b) 51.1 m/s
16. a) 31.4 rad/s
b) 118 m/s
c) 384 m/s
d) The centripetal acceleration felt by Olympic skaters is 12 times larger than the acceleration due to gravity. That's quite a lot of acceleration in itself. The centripetal acceleration felt by Button's nose was 39.2 times larger than the acceleration due to gravity. It is no wonder that he ruptured small blood vessels in his spins.
18. a) 0.524 km/s
b) 29.7 km/s
20. (a) 1.35×10^3 rpm
(b) 8.47×10^3 m/s²
(c) 8.47×10^{-12} N
(d) 865
21. (a) 39.2 m/s
(b)



- (c) 2.93×10^3 N or $5.00 w$, that is, the normal force (upward) is five times her weight.
 (d) This answer seems reasonable, since she feels like she's being forced into the chair MUCH stronger than just by gravity.
22. a) 40.5 m/s^2
 b) 905 N
 c) The force in part (b) is very large. The acceleration in part (a) is too much, about 4 g.
 d) The speed of the swing is too large. At the given velocity at the bottom of the swing, there is enough kinetic energy to send the child all the way over the top, ignoring friction.
23. a) 483 N
 b) 17.4 N
 c) 2.24 times her weight, 0.0807 times her weight
25. 4.14°
27. a) 24.6 m
 b) 36.6 m/s^2
 c) $a_c = 3.73 g$. This does not seem too large, but it is clear that bobsledders feel a lot of force on them going through sharply banked turns.
29. a) 2.56 rad/s
 b) 5.71°
30. a) 16.2 m/s
 b) 0.234
32. a) 1.84
 b) A coefficient of friction this much greater than 1 is unreasonable.
 c) The assumed speed is too great for the tight curve.
33. a) $g = \frac{GM}{r^2} \Rightarrow M = \frac{r^2 g}{G} = 5.95296 \times 10^{24} \text{ kg}$
 b) This is slightly, but measurably, smaller than the accepted value, differing in the third significant digit. The equation we used to find g assumes a sphere, but we already noted that the radius at the pole is smaller than at the equator. The calculation gives the mass for a sphere of the radius we used—and ignores the additional mass outside that radius.
35. a) 1.62 m/s^2
 b) 3.75 m/s^2
37. a) $3.42 \times 10^{-5} \text{ m/s}^2$
 b) $3.34 \times 10^{-5} \text{ m/s}^2$
 The values are nearly identical. One would expect the gravitational force to be the same as the centripetal force at the core of the system.
39. a) $7.01 \times 10^{-7} \text{ N}$
 b) $1.35 \times 10^{-6} \text{ N}$, 0.521
41. a) $1.66 \times 10^{-10} \text{ m/s}^2$
 b) $2.17 \times 10^5 \text{ m/s}$
42. a) $2.937 \times 10^{17} \text{ kg}$
 b) 4.91×10^{-8}
 of the Earth's mass.
 c) The mass of the mountain and its fraction of the Earth's mass are too great.

- d) The gravitational force assumed to be exerted by the mountain is too great.
44. $1.98 \times 10^{30} \text{ kg}$
46. $\frac{M_J}{M_E} = 316$
48. a) 7910 m/s
b) $1.12 \times 10^4 \text{ m/s}$
c) $5.36 \times 10^{-7} \text{ s}$
d) $1.04478 \times 10^7 \text{ N}$
e) $3.14 \times 10^4 \text{ J}$
49. a) $5.08 \times 10^3 \text{ km}$
b) This radius is unreasonable because it is less than the radius of earth.
c) The premise of a one-hour orbit is inconsistent with the known radius of the earth.
51. (a) The normal force is greatest at point B. At point A, the normal force and the weight are the net force; at point B, only the normal force points in the same direction as the net force.
(b) i. This is not consistent because a larger radius should decrease the net force and tension, not increase it.
ii. This does not make sense because $F_c = mv^2/r$.
(c)



Chapter 7

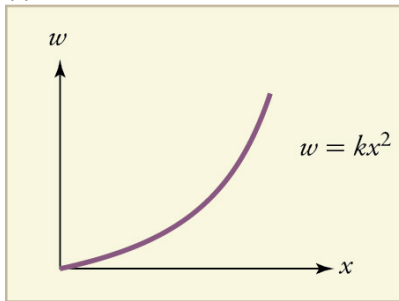
Problems & Exercises

1. $3.00 \text{ J} = 7.17 \times 10^{-4} \text{ kcal}$
3. (a) $5.92 \times 10^5 \text{ J}$
(b) $-5.88 \times 10^5 \text{ J}$
(c) The net force is zero.
5. $3.14 \times 10^3 \text{ J}$
7. (a) -700 J
(b) 0
(c) 700 J
(d) 38.6 N
(e) 0
9. 1/250
11. $1.1 \times 10^{10} \text{ J}$
13. $2.8 \times 10^3 \text{ N}$
15. 102 N
16. (a) $1.96 \times 10^{16} \text{ J}$
(b) The ratio of gravitational potential energy in the lake to the energy stored in the bomb is 0.52. That is, the energy stored in the lake is approximately half that in a 9-megaton fusion bomb.
18. (a) 1.8 J
(b) 8.6 J
20. $v_f = \sqrt{2gh + v_0^2} = \sqrt{2(9.80 \text{ m/s}^2)(-0.180 \text{ m}) + (2.00 \text{ m/s})^2} = 0.687 \text{ m/s}$
22. $7.81 \times 10^5 \text{ N/m}$

24. 9.46 m/s
26. 4×10^4 molecules
27. Equating ΔPE_g and ΔKE , we obtain $v = \sqrt{2gh + v_0^2} = \sqrt{2(9.80 \text{ m/s}^2)(20.0 \text{ m}) + (15.0 \text{ m/s})^2} = 24.8 \text{ m/s}$
29. (a) 25×10^6 years
(b) This is much, much longer than human time scales.
30. 2×10^{-10}
32. (a) 40
(b) 8 million
34. \$149
36. (a) 208 W
(b) 141 s
38. (a) 3.20 s
(b) 4.04 s
40. (a) $9.46 \times 10^7 \text{ J}$
(b) 2.54 y
42. Identify knowns: $m = 950 \text{ kg}$, slope angle $\theta = 2.00^\circ$, $v = 3.00 \text{ m/s}$, $f = 600 \text{ N}$
Identify unknowns: power P of the car, force F that car applies to road
Solve for unknown:
 $P = \frac{W}{t} = \frac{Fd}{t} = F\left(\frac{d}{t}\right) = Fv$,
where F is parallel to the incline and must oppose the resistive forces and the force of gravity:
 $F = f + w = 600 \text{ N} + mg \sin \theta$
Insert this into the expression for power and solve:
 $P = (f + mg \sin \theta)v$
 $= [600 \text{ N} + (950 \text{ kg})(9.80 \text{ m/s}^2) \sin 2^\circ](30.0 \text{ m/s})$
 $= 2.77 \times 10^4 \text{ W}$
About 28 kW (or about 37 hp) is reasonable for a car to climb a gentle incline.
44. (a) 9.5 min
(b) 69 flights of stairs
46. 641 W, 0.860 hp
48. 31 g
50. 14.3%
52. (a) $3.21 \times 10^4 \text{ N}$
(b) $2.35 \times 10^3 \text{ N}$
(c) Ratio of net force to weight of person is 41.0 in part (a); 3.00 in part (b)
54. (a) 108 kJ
(b) 599 W
56. (a) 144 J
(b) 288 W
58. (a) $2.50 \times 10^{12} \text{ J}$
(b) 2.52%
(c) $1.4 \times 10^4 \text{ kg}$ (14 metric tons)
60. (a) 294 N
(b) 118 J
(c) 49.0 W
62. (a) 0.500 m/s^2
(b) 62.5 N
(c) Assuming the acceleration of the swimmer decreases linearly with time over the 5.00 s interval, the frictional force must therefore be increasing linearly with time, since $f = F - ma$. If the acceleration

decreases linearly with time, the velocity will contain a term dependent on time squared (t^2). Therefore, the water resistance will not depend linearly on the velocity.

64. (a) 16.1×10^3 N
 (b) 3.22×10^5 J
 (c) 5.66 m/s
 (d) 4.00 kJ
66. (a) 4.65×10^3 kcal
 (b) 38.8 kcal/min
 (c) This power output is higher than the highest value on [Table 7.5](#), which is about 35 kcal/min (corresponding to 2415 watts) for sprinting.
 (d) It would be impossible to maintain this power output for 2 hours (imagine sprinting for 2 hours!).
69. (a) 4.32 m/s
 (b) 3.47×10^3 N
 (c) 8.93 kW
70. (a) Box A has the greatest speed since it has the greatest work done; work equals change in kinetic energy and thus fastest speed.
 (b) i. Yes, this equation is consistent. An increased force or distance increases the work done, and thus increases the change in kinesthetic energy. ii. No, this equation does not make sense.
 (c)



Chapter 8

Problems & Exercises

1. (a) 1.50×10^4 kg · m/s
 (b) 625 to 1
 (c) 6.66×10^2 kg · m/s
3. (a) 8.00×10^4 m/s
 (b) 1.20×10^6 kg · m/s
 (c) Because the momentum of the airplane is 3 orders of magnitude smaller than of the ship, the ship will not recoil very much. The recoil would be -0.0100 m/s, which is probably not noticeable.
5. 54 s
7. 9.00×10^3 N
9. a) 2.40×10^3 N toward the leg
 b) The force on each hand would have the same magnitude as that found in part (a) (but in opposite directions by Newton's third law) because the change in momentum and the time interval are the same.
11. a) 800 kg · m/s away from the wall
 b) 1.20 m/s away from the wall
13. (a) 1.50×10^6 N away from the dashboard
 (b) 1.00×10^5 N away from the dashboard
15. 4.69×10^5 N in the boat's original direction of motion
17. 2.10×10^3 N away from the wall

$$\mathbf{p} = m\mathbf{v} \Rightarrow p^2 = m^2 v^2 \Rightarrow \frac{p^2}{m} = mv^2$$

$$19. \Rightarrow \frac{p^2}{2m} = \frac{1}{2}mv^2 = KE$$

$$KE = \frac{p^2}{2m}$$

$$21. 60.0 \text{ g}$$

$$23. 0.122 \text{ m/s}$$

25. In a collision with an identical car, momentum is conserved. Afterwards $v_f = 0$ for both cars. The change in momentum will be the same as in the crash with the tree. However, the force on the body is not determined since the time is not known. A padded stop will reduce injurious force on body.

27. 22.4 m/s in the same direction as the original motion

$$29. 0.250 \text{ m/s}$$

31. (a) 86.4 N perpendicularly away from the bumper

$$(b) 0.389 \text{ J}$$

$$(c) 64.0\%$$

33. (a) 8.06 m/s

$$(b) -56.0 \text{ J}$$

$$(c)(i) 7.88 \text{ m/s}; (ii) -223 \text{ J}$$

35. (a) 0.163 m/s in the direction of motion of the more massive satellite

$$(b) 81.6 \text{ J}$$

(c) $8.70 \times 10^{-2} \text{ m/s}$ in the direction of motion of the less massive satellite, 81.5 J. Because there are no external forces, the velocity of the center of mass of the two-satellite system is unchanged by the collision. The two velocities calculated above are the velocity of the center of mass in each of the two different individual reference frames. The loss in KE is the same in both reference frames because the KE lost to internal forces (heat, friction, etc.) is the same regardless of the coordinate system chosen.

$$37. 0.704 \text{ m/s}$$

$$-2.25 \text{ m/s}$$

38. (a) 4.58 m/s away from the bullet

$$(b) 31.5 \text{ J}$$

$$(c) -0.491 \text{ m/s}$$

$$(d) 3.38 \text{ J}$$

40. (a) $1.02 \times 10^{-6} \text{ m/s}$

$$(b) 5.63 \times 10^{20} \text{ J (almost all KE lost)}$$

(c) Recoil speed is $6.79 \times 10^{-17} \text{ m/s}$, energy lost is $6.25 \times 10^9 \text{ J}$. The plume will not affect the momentum result because the plume is still part of the Moon system. The plume may affect the kinetic energy result because a significant part of the initial kinetic energy may be transferred to the kinetic energy of the plume particles.

$$42. 24.8 \text{ m/s}$$

44. (a) 4.00 kg

$$(b) 210 \text{ J}$$

(c) The clown does work to throw the barbell, so the kinetic energy comes from the muscles of the clown. The muscles convert the chemical potential energy of ATP into kinetic energy.

45. (a) 3.00 m/s, 60° below x -axis

$$(b) \text{ Find speed of first puck after collision: } 0 = mv'_1 \sin 30^\circ - mv'_2 \sin 60^\circ \Rightarrow v'_1 = v'_2 \frac{\sin 60^\circ}{\sin 30^\circ} = 5.196 \text{ m/s}$$

$$\left. \begin{array}{l} \text{Verify that ratio of initial to final KE equals one: } KE = \frac{1}{2}mv_1^2 = 18m \text{ J} \\ KE = \frac{1}{2}mv'_1{}^2 + \frac{1}{2}mv'_2{}^2 = 18m \text{ J} \end{array} \right\} \frac{KE}{KE'} = 1.00$$

47. (a) -2.26 m/s

$$(b) 7.63 \times 10^3 \text{ J}$$

(c) The ground will exert a normal force to oppose recoil of the cannon in the vertical direction. The momentum in the vertical direction is transferred to the earth. The energy is transferred into the ground, making a dent where the cannon is. After long barrages, cannon have erratic aim because the ground is full of

divots.

49. (a) $5.36 \times 10^5 \text{ m/s}$ at -29.5°

(b) $7.52 \times 10^{-13} \text{ J}$

51. We are given that $m_1 = m_2 \equiv m$. The given equations then become:

$$v_1 = v'_1 \cos \theta_1 + v'_2 \cos \theta_2$$

and

$$0 = v'_1 \sin \theta_1 + v'_2 \sin \theta_2.$$

Square each equation to get

$$v_1^2 = v'^2_1 \cos^2 \theta_1 + v'^2_2 \cos^2 \theta_2 + 2v'_1 v'_2 \cos \theta_1 \cos \theta_2$$

$$0 = v'^2_1 \sin^2 \theta_1 + v'^2_2 \sin^2 \theta_2 + 2v'_1 v'_2 \sin \theta_1 \sin \theta_2.$$

Add these two equations and simplify:

$$\begin{aligned} v_1^2 &= v'^2_1 + v'^2_2 + 2v'_1 v'_2 (\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2) \\ &= v'^2_1 + v'^2_2 + 2v'_1 v'_2 \left[\frac{1}{2} \cos (\theta_1 - \theta_2) + \frac{1}{2} \cos (\theta_1 + \theta_2) + \frac{1}{2} \cos (\theta_1 - \theta_2) - \frac{1}{2} \cos (\theta_1 + \theta_2) \right] \\ &= v'^2_1 + v'^2_2 + 2v'_1 v'_2 \cos (\theta_1 - \theta_2). \end{aligned}$$

Multiply the entire equation by $\frac{1}{2}m$ to recover the kinetic energy:

$$\frac{1}{2}mv_1^2 = \frac{1}{2}mv'^2_1 + \frac{1}{2}mv'^2_2 + mv'_1 v'_2 \cos (\theta_1 - \theta_2)$$

53. 39.2 m/s^2

55. $4.16 \times 10^3 \text{ m/s}$

57. The force needed to give a small mass Δm an acceleration $a_{\Delta m}$ is $F = \Delta m a_{\Delta m}$. To accelerate this mass in the small time interval Δt requires $v_e = a_{\Delta m} \Delta t$, so $F = v_e \frac{\Delta m}{\Delta t}$. By Newton's third law, this force is equal in magnitude to the thrust force acting on the rocket, so $F_{\text{thrust}} = v_e \frac{\Delta m}{\Delta t}$, where all quantities are positive. Applying Newton's second law to the rocket gives $F_{\text{thrust}} - mg = ma \Rightarrow a = \frac{v_e}{m} \frac{\Delta m}{\Delta t} - g$, where m is the mass of the rocket and unburnt fuel.

60. $2.63 \times 10^3 \text{ kg}$

61. (a) 0.421 m/s away from the ejected fluid.
(b) 0.237 J .

Chapter 9

Problems & Exercises

1. a) $46.8 \text{ N}\cdot\text{m}$

b) It does not matter at what height you push. The torque depends on only the magnitude of the force applied and the perpendicular distance of the force's application from the hinges. (Children don't have a tougher time opening a door because they push lower than adults, they have a tougher time because they don't push far enough from the hinges.)

3. 23.3 N

5. Given:

$$m_1 = 26.0 \text{ kg}, m_2 = 32.0 \text{ kg}, m_s = 12.0 \text{ kg},$$

$$r_1 = 1.60 \text{ m}, r_s = 0.160 \text{ m}, \text{ find (a) } r_2, \text{ (b) } F_p$$

a) Since children are balancing:

$$\text{net } \tau_{\text{cw}} = -\text{net } \tau_{\text{ccw}}$$

$$\Rightarrow w_1 r_1 + m_s g r_s = w_2 r_2$$

So, solving for r_2 gives:

$$\begin{aligned} r_2 &= \frac{w_1 r_1 + m_s g r_s}{w_2} = \frac{m_1 g r_1 + m_s g r_s}{m_2 g} = \frac{m_1 r_1 + m_s r_s}{m_2} \\ &= \frac{(26.0 \text{ kg})(1.60 \text{ m}) + (12.0 \text{ kg})(0.160 \text{ m})}{32.0 \text{ kg}} \\ &= 1.36 \text{ m} \end{aligned}$$

b) Since the children are not moving:

$$\text{net } F = 0 = F_p - w_1 - w_2 - w_s$$

$$\Rightarrow F_p = w_1 + w_2 + w_s$$

So that

$$\begin{aligned} F_p &= (26.0 \text{ kg} + 32.0 \text{ kg} + 12.0 \text{ kg})(9.80 \text{ m/s}^2) \\ &= 686 \text{ N} \end{aligned}$$

6. $F_{\text{wall}} = 1.43 \times 10^3 \text{ N}$
8. a) $2.55 \times 10^3 \text{ N}$, 16.3° to the left of vertical (i.e., toward the wall)
b) 0.292
10. $F_B = 2.12 \times 10^4 \text{ N}$
12. a) 0.167, or about one-sixth of the weight is supported by the opposite shore.
b) $F = 2.0 \times 10^4 \text{ N}$, straight up.
14. a) 21.6 N
b) 21.6 N
16. 350 N directly upwards
19. 25
50 N
21. a) $MA = 18.5$
b) $F_i = 29.1 \text{ N}$
c) 510 N downward
23. $1.3 \times 10^3 \text{ N}$
25. a) $T = 299 \text{ N}$
b) 897 N upward
 $F_B = 470 \text{ N}$; $r_1 = 4.00 \text{ cm}$; $w_a = 2.50 \text{ kg}$; $r_2 = 16.0 \text{ cm}$; $w_b = 4.00 \text{ kg}$; $r_3 = 38.0 \text{ cm}$
$$F_E = w_a \left(\frac{r_2}{r_1} - 1 \right) + w_b \left(\frac{r_3}{r_1} - 1 \right)$$
26.
$$\begin{aligned} &= (2.50 \text{ kg})(9.80 \text{ m/s}^2) \left(\frac{16.0 \text{ cm}}{4.0 \text{ cm}} - 1 \right) \\ &\quad + (4.00 \text{ kg})(9.80 \text{ m/s}^2) \left(\frac{38.0 \text{ cm}}{4.00 \text{ cm}} - 1 \right) \\ &= 407 \text{ N} \\ &\quad 1.1 \times 10^3 \text{ N} \end{aligned}$$
28. $\theta = 190^\circ$ ccw from positive x axis
30. $F_V = 97 \text{ N}$, $\theta = 59^\circ$
32. (a) 25 N downward
(b) 75 N upward
33. (a) $F_A = 2.21 \times 10^3 \text{ N}$ upward
(b) $F_B = 2.94 \times 10^3 \text{ N}$ downward
35. (a) $F_{\text{teeth on bullet}} = 1.2 \times 10^2 \text{ N}$ upward
(b) $F_j = 84 \text{ N}$ downward
37. (a) 147 N downward
(b) 1680 N, 3.4 times her weight
(c) 118 J
(d) 49.0 W
39. a) $\bar{x}_2 = 2.33 \text{ m}$
b) The seesaw is 3.0 m long, and hence, there is only 1.50 m of board on the other side of the pivot. The second child is off the board.
c) The position of the first child must be shortened, i.e. brought closer to the pivot.

Chapter 10

Problems & Exercises

1. $\omega = 0.737 \text{ rev/s}$
3. (a) -0.26 rad/s^2

- (b) 27 rev
5. (a) 80 rad/s²
(b) 1.0 rev
7. (a) 45.7 s
(b) 116 rev
9. a) 600 rad/s²
b) 450 rad/s^s
c) 21.0 m/s^s
10. (a) 0.338 s
(b) 0.0403 rev
(c) 0.313 s
12. 0.50 kg · m²
14. (a) 50.4 N · m
(b) 17.1 rad/s²
(c) 17.0 rad/s²
16. 3.96×10^{18} s
or 1.26×10^{11} y
18.
$$I_{\text{end}} = I_{\text{center}} + m\left(\frac{l}{2}\right)^2$$

Thus, $I_{\text{center}} = I_{\text{end}} - \frac{1}{4}ml^2 = \frac{1}{3}ml^2 - \frac{1}{4}ml^2 = \frac{1}{12}ml^2$
19. (a) 2.0 ms
(b) The time interval is too short.
(c) The moment of inertia is much too small, by one to two orders of magnitude. A torque of 500 N · m is reasonable.
20. (a) 17,500 rpm
(b) This angular velocity is very high for a disk of this size and mass. The radial acceleration at the edge of the disk is > 50,000 gs.
(c) Flywheel mass and radius should both be much greater, allowing for a lower spin rate (angular velocity).
21. (a) 185 J
(b) 0.0785 rev
(c) $W = 9.81$ N
23. (a) 2.57×10^{29} J
(b) $\text{KE}_{\text{rot}} = 2.65 \times 10^{33}$ J
25. $\text{KE}_{\text{rot}} = 434$ J
27. (a) 128 rad/s
(b) 19.9 m
29. (a) 10.4 rad/s²
(b) net $W = 6.11$ J
34. (a) 1.49 kJ
(b) 2.52×10^4 N
36. (a) 2.66×10^{40} kg · m²/s
(b) 7.07×10^{33} kg · m²/s
The angular momentum of the Earth in its orbit around the Sun is 3.77×10^6 times larger than the angular momentum of the Earth around its axis.
38. 22.5 kg · m²/s
40. 25.3 rpm
43. (a) 0.156 rad/s
(b) 1.17×10^{-2} J
(c) 0.188 kg · m/s
45. (a) 3.13 rad/s
(b) Initial KE = 438 J, final KE = 438 J
47. (a) 1.70 rad/s
(b) Initial KE = 22.5 J, final KE = 2.04 J

- (c) $1.50 \text{ kg} \cdot \text{m/s}$
 48. (a) $5.64 \times 10^{33} \text{ kg} \cdot \text{m}^2/\text{s}$
 (b) $1.39 \times 10^{22} \text{ N} \cdot \text{m}$
 (c) $2.17 \times 10^{15} \text{ N}$

Chapter 11

Problems & Exercises

1. 1.610 cm^3
3. (a) 2.58 g
 (b) The volume of your body increases by the volume of air you inhale. The average density of your body decreases when you take a deep breath, because the density of air is substantially smaller than the average density of the body before you took the deep breath.
4. 2.70 g/cm^3
6. (a) 0.163 m
 (b) Equivalent to 19.4 gallons, which is reasonable
8. $7.9 \times 10^2 \text{ kg/m}^3$
9. 15.8 g/cm^3
10. (a) 10^{18} kg/m^3
 (b) $2 \times 10^4 \text{ m}$
11. $3.59 \times 10^6 \text{ Pa}$; or 521 lb/in^2
13. $2.36 \times 10^3 \text{ N}$
14. 0.760 m
 $(h\rho g)_{\text{units}} = (\text{m})(\text{kg/m}^3)(\text{m/s}^2) = (\text{kg} \cdot \text{m}^2)/(\text{m}^3 \cdot \text{s}^2)$
16. $= (\text{kg} \cdot \text{m/s}^2)(1/\text{m}^2)$
 $= \text{N/m}^2$
18. (a) 20.5 mm Hg
 (b) The range of pressures in the eye is 12–24 mm Hg, so the result in part (a) is within that range
20. $1.09 \times 10^3 \text{ N/m}^2$
22. 24.0 N
24. $2.55 \times 10^7 \text{ Pa}$; or 251 atm
26. $5.76 \times 10^3 \text{ N}$ extra force
28. (a) $V = d_i A_i = d_o A_o \Rightarrow d_o = d_i \left(\frac{A_i}{A_o} \right)$.
 Now, using equation:
 $\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow F_o = F_i \left(\frac{A_o}{A_i} \right)$.
 Finally,
 $W_o = F_o d_o = \left(\frac{F_i A_o}{A_i} \right) \left(\frac{d_i A_i}{A_o} \right) = F_i d_i = W_i$.
 In other words, the work output equals the work input.
 (b) If the system is not moving, friction would not play a role. With friction, we know there are losses, so that $W_{\text{out}} = W_{\text{in}} - W_f$; therefore, the work output is less than the work input. In other words, with friction, you need to push harder on the input piston than was calculated for the nonfriction case.
29. Balloon:
 $P_g = 5.00 \text{ cm H}_2\text{O}$,
 $P_{\text{abs}} = 1.035 \times 10^3 \text{ cm H}_2\text{O}$.
 Jar:
 $P_g = -50.0 \text{ mm Hg}$,
 $P_{\text{abs}} = 710 \text{ mm Hg}$.
31. 4.08 m

- $\Delta P = 38.7 \text{ mm Hg}$,
33. Leg blood pressure = $\frac{159}{119}$.
35. 22.4 cm^2
36. 91.7%
38. 815 kg/m^3
40. (a) 41.4 g
(b) 41.4 cm^3
(c) 1.09 g/cm^3
42. (a) 39.5 g
(b) 50 cm^3
(c) 0.79 g/cm^3
It is ethyl alcohol.
44. 8.21 N
46. (a) 960 kg/m^3
(b) 6.34%
She indeed floats more in seawater.
48. (a) 0.24
(b) 0.68
(c) Yes, the cork will float because $\rho_{\text{obj}} < \rho_{\text{ethyl alcohol}} (0.678 \text{ g/cm}^3 < 0.79 \text{ g/cm}^3)$
50. The difference is 0.006%.
52. $F_{\text{net}} = F_2 - F_1 = P_2 A - P_1 A = (P_2 - P_1) A$
 $= (h_2 \rho_{\text{fl}} g - h_1 \rho_{\text{fl}} g) A$
 $= (h_2 - h_1) \rho_{\text{fl}} g A$
 where ρ_{fl} = density of fluid. Therefore,
 $F_{\text{net}} = (h_2 - h_1) A \rho_{\text{fl}} g = V_{\text{fl}} \rho_{\text{fl}} g = m_{\text{fl}} g = w_{\text{fl}}$
 where is w_{fl} the weight of the fluid displaced.
54. 592 N/m^2
56. $2.23 \times 10^{-2} \text{ mm Hg}$
58. (a) $1.65 \times 10^{-3} \text{ m}$
(b) $3.71 \times 10^{-4} \text{ m}$
60. $6.32 \times 10^{-2} \text{ N/m}$
Based on the values in table, the fluid is probably glycerin.
 $P_{\text{w}} = 14.6 \text{ N/m}^2$,
62. $P_{\text{a}} = 4.46 \text{ N/m}^2$,
 $P_{\text{sw}} = 7.40 \text{ N/m}^2$.
 Alcohol forms the most stable bubble, since the absolute pressure inside is closest to atmospheric pressure.
64. 5.1°
This is near the value of $\theta = 0^\circ$ for most organic liquids.
66. -2.78
The ratio is negative because water is raised whereas mercury is lowered.
68. 479 N
70. 1.96 N
71. $-63.0 \text{ cm H}_2\text{O}$
73. (a) $3.81 \times 10^3 \text{ N/m}^2$
(b) 28.7 mm Hg, which is sufficient to trigger micturition reflex
75. (a) 13.6 m water
(b) 76.5 cm water
77. (a) $3.98 \times 10^6 \text{ Pa}$
(b) $2.1 \times 10^{-3} \text{ cm}$
79. (a) 2.97 cm
(b) $3.39 \times 10^{-6} \text{ J}$

(c) Work is done by the surface tension force through an effective distance $h/2$ to raise the column of water.

81. (a) 2.01×10^4 N
 (b) 1.17×10^{-3} m
 (c) 2.56×10^{10} N/m²
83. (a) 1.38×10^4 N
 (b) 2.81×10^7 N/m²
 (c) 283 N
85. (a) 867 N
 (b) This is too much force to exert with a hand pump.
 (c) The assumed radius of the pump is too large; it would be nearly two inches in diameter—too large for a pump or even a master cylinder. The pressure is reasonable for bicycle tires.

Chapter 12

Problems & Exercises

1. $2.78 \text{ cm}^3/\text{s}$
 3. 27 cm/s
 5. (a) 0.75 m/s
 (b) 0.13 m/s
 7. (a) 40.0 cm^2
 (b) 5.09×10^7
 9. (a) 22 h
 (b) 0.016 s
 11. (a) 12.6 m/s
 (b) $0.0800 \text{ m}^3/\text{s}$
 (c) No, independent of density.
 13. (a) 0.402 L/s
 (b) 0.584 cm
 15. (a) $127 \text{ cm}^3/\text{s}$
 (b) 0.890 cm

$$P = \frac{\text{Force}}{\text{Area}},$$

 17. $(P)_{\text{units}} = \text{N/m}^2 = \text{N} \cdot \text{m/m}^3 = \text{J/m}^3$
 $= \text{energy/volume}$
 19. 184 mm Hg
 21. 2.54×10^5 N
 23. (a) $1.58 \times 10^6 \text{ N/m}^2$
 (b) 163 m
 25. (a) $9.56 \times 10^8 \text{ W}$
 (b) 1.41
 27. 1.26 W
 29. (a) $3.02 \times 10^{-3} \text{ N}$
 (b) 1.03×10^{-3}
 31. $1.60 \text{ cm}^3/\text{min}$
 33. $8.7 \times 10^{-11} \text{ m}^3/\text{s}$
 35. 0.316
 37. (a) 1.52
 (b) Turbulence will decrease the flow rate of the blood, which would require an even larger increase in the pressure difference, leading to higher blood pressure.
 39. $225 \text{ mPa} \cdot \text{s}$
 41. $0.138 \text{ Pa} \cdot \text{s}$,
 or
 Olive oil.

43. (a) $1.62 \times 10^4 \text{ N/m}^2$
 (b) $0.111 \text{ cm}^3/\text{s}$
 (c) 10.6 cm
45. 1.59
47. $2.95 \times 10^6 \text{ N/m}^2$ (gauge pressure)
51. $N_R = 1.99 \times 10^3 < 2000$
53. (a) nozzle: 1.27×10^5 , not laminar
 (b) hose: 3.51×10^4 , not laminar.
55. $2.54 \ll 2000$, laminar.
57. 1.02 m/s
 $1.28 \times 10^{-2} \text{ L/s}$
59. (a) $\geq 13.0 \text{ m}$
 (b) $2.68 \times 10^{-6} \text{ N/m}^2$
61. (a) 23.7 atm or 344 lb/in^2
 (b) The pressure is much too high.
 (c) The assumed flow rate is very high for a garden hose.
 (d) $5.27 \times 10^6 > 3000$, turbulent, contrary to the assumption of laminar flow when using this equation.
62. $1.41 \times 10^{-3} \text{ m}$
64. $1.3 \times 10^2 \text{ s}$
66. 0.391 s

Chapter 13

Problems & Exercises

1. 102°F
3. 20.0°C and 25.6°C
5. 9890°F
7. (a) 22.2°C
 $\Delta T(^{\circ}\text{F}) = T_2(^{\circ}\text{F}) - T_1(^{\circ}\text{F})$
 (b) $= \frac{9}{5}T_2(^{\circ}\text{C}) + 32.0^\circ - \left(\frac{9}{5}T_1(^{\circ}\text{C}) + 32.0^\circ\right)$
 $= \frac{9}{5}(T_2(^{\circ}\text{C}) - T_1(^{\circ}\text{C})) = \frac{9}{5}\Delta T(^{\circ}\text{C})$
9. 169.98 m
11. $5.4 \times 10^{-6} \text{ m}$
13. Because the area gets smaller, the price of the land DECREASES by $\sim \$17,000$.
 $V = V_0 + \Delta V = V_0(1 + \beta\Delta T)$
15. $= (60.00 \text{ L})[1 + (950 \times 10^{-6}/^{\circ}\text{C})(35.0^\circ\text{C} - 15.0^\circ\text{C})]$
 $= 61.1 \text{ L}$
17. (a) 9.35 mL
 (b) 7.56 mL
19. 0.832 mm
21. We know how the length changes with temperature: $\Delta L = \alpha L_0 \Delta T$. Also we know that the volume of a cube is related to its length by $V = L^3$, so the final volume is then $V = V_0 + \Delta V = (L_0 + \Delta L)^3$. Substituting for ΔL gives
 $V = (L_0 + \alpha L_0 \Delta T)^3 = L_0^3(1 + \alpha \Delta T)^3$.
 Now, because $\alpha \Delta T$ is small, we can use the binomial expansion:
 $V \approx L_0^3(1 + 3\alpha \Delta T) = L_0^3 + 3\alpha L_0^3 \Delta T$.
 So writing the length terms in terms of volumes gives $V = V_0 + \Delta V \approx V_0 + 3\alpha V_0 \Delta T$, and so
 $\Delta V = \beta V_0 \Delta T \approx 3\alpha V_0 \Delta T$, or $\beta \approx 3\alpha$.
22. 1.62 atm
24. (a) 0.136 atm
 (b) 0.135 atm . The difference between this value and the value from part (a) is negligible.

26. (a) $nRT = (\text{mol})(\text{J/mol} \cdot \text{K})(\text{K}) = \text{J}$
 (b) $nRT = (\text{mol})(\text{cal/mol} \cdot \text{K})(\text{K}) = \text{cal}$
 $nRT = (\text{mol})(\text{L} \cdot \text{atm/mol} \cdot \text{K})(\text{K})$
 (c) $= \text{L} \cdot \text{atm} = (\text{m}^3)(\text{N/m}^2)$
 $= \text{N} \cdot \text{m} = \text{J}$
28. $7.86 \times 10^{-2} \text{ mol}$
30. (a) $6.02 \times 10^5 \text{ km}^3$
 (b) $6.02 \times 10^8 \text{ km}$
32. -73.9°C
34. (a) $9.14 \times 10^6 \text{ N/m}^2$
 (b) $8.23 \times 10^6 \text{ N/m}^2$
 (c) 2.16 K
 (d) No. The final temperature needed is much too low to be easily achieved for a large object.
36. 41 km
38. (a) $3.7 \times 10^{-17} \text{ Pa}$
 (b) $6.0 \times 10^{17} \text{ m}^3$
 (c) $8.4 \times 10^2 \text{ km}$
39. $1.25 \times 10^3 \text{ m/s}$
41. (a) $1.20 \times 10^{-19} \text{ J}$
 (b) $1.24 \times 10^{-17} \text{ J}$
43. 458 K
45. $1.95 \times 10^7 \text{ K}$
47. $6.09 \times 10^5 \text{ m/s}$
49. $7.89 \times 10^4 \text{ Pa}$
51. (a) $1.99 \times 10^5 \text{ Pa}$
 (b) 0.97 atm
53. $3.12 \times 10^4 \text{ Pa}$
55. 78.3%
57. (a) $2.12 \times 10^4 \text{ Pa}$
 (b) 1.06%
59. (a) $8.80 \times 10^{-2} \text{ g}$
 (b) $6.30 \times 10^3 \text{ Pa}$; the two values are nearly identical.
61. 82.3%
63. 4.77°C
65. 38.3 m
67. $\frac{(F_B/w_{\text{Cu}})}{(F_B/w_{\text{Cu}})} = 1.02$. The buoyant force supports nearly the exact same amount of force on the copper block in both circumstances.
69. (a) $4.41 \times 10^{10} \text{ mol/m}^3$
 (b) It's unreasonably large.
 (c) At high pressures such as these, the ideal gas law can no longer be applied. As a result, unreasonable answers come up when it is used.
71. (a) $7.03 \times 10^8 \text{ m/s}$
 (b) The velocity is too high—it's greater than the speed of light.
 (c) The assumption that hydrogen inside a supernova behaves as an idea gas is responsible, because of the great temperature and density in the core of a star. Furthermore, when a velocity greater than the speed of light is obtained, classical physics must be replaced by relativity, a subject not yet covered.

Chapter 14

Problems & Exercises

1. $5.02 \times 10^8 \text{ J}$
3. $3.07 \times 10^3 \text{ J}$
5. 0.171°C
7. 10.8
9. 617 W
11. 35.9 kcal
13. (a) 591 kcal
(b) $4.94 \times 10^3 \text{ s}$
15. 13.5 W
17. (a) 148 kcal
(b) 0.418 s, 3.34 s, 4.19 s, 22.6 s, 0.456 s
19. 33.0 g
20. (a) 9.67 L
(b) Crude oil is less dense than water, so it floats on top of the water, thereby exposing it to the oxygen in the air, which it uses to burn. Also, if the water is under the oil, it is less efficient in absorbing the heat generated by the oil.
22. a) 319 kcal
b) 2.00°C
24. 20.6°C
26. 4.38 kg
28. (a) $1.57 \times 10^4 \text{ kcal}$
(b) $18.3 \text{ kW} \cdot \text{h}$
(c) $1.29 \times 10^4 \text{ kcal}$
30. (a) $1.01 \times 10^3 \text{ W}$
(b) One
32. 84.0 W
34. 2.59 kg
36. (a) 39.7 W
(b) 820 kcal
38. 35 to 1, window to wall
40. $1.05 \times 10^3 \text{ K}$
42. (a) 83 W
(b) 24 times that of a double pane window.
44. 20.0 W, 17.2% of 2400 kcal per day
45. 10 m/s
47. 85.7°C
49. 1.48 kg
51. $2 \times 10^4 \text{ MW}$
53. (a) 97.2 J
(b) 29.2 W
(c) 9.49 W
(d) The total rate of heat loss would be $29.2 \text{ W} + 9.49 \text{ W} = 38.7 \text{ W}$. While sleeping, our body consumes 83 W of power, while sitting it consumes 120 to 210 W. Therefore, the total rate of heat loss from breathing will not be a major form of heat loss for this person.
55. -21.7 kW
Note that the negative answer implies heat loss to the surroundings.
57. -266 kW
59. -36.0 W
61. (a) 1.31%
(b) 20.5%

63. (a) -15.0 kW
(b) 4.2 cm
65. (a) 48.5°C
(b) A pure white object reflects more of the radiant energy that hits it, so a white tent would prevent more of the sunlight from heating up the inside of the tent, and the white tunic would prevent that heat which entered the tent from heating the rider. Therefore, with a white tent, the temperature would be lower than 48.5°C , and the rate of radiant heat transferred to the rider would be less than 20.0 W .
67. (a) $3 \times 10^{17} \text{ J}$
(b) $1 \times 10^{13} \text{ kg}$
(c) When a large meteor hits the ocean, it causes great tidal waves, dissipating large amount of its energy in the form of kinetic energy of the water.
69. (a) $3.44 \times 10^5 \text{ m}^3/\text{s}$
(b) This is equivalent to 12 million cubic feet of air per second. That is tremendous. This is too large to be dissipated by heating the air by only 5°C . Many of these cooling towers use the circulation of cooler air over warmer water to increase the rate of evaporation. This would allow much smaller amounts of air necessary to remove such a large amount of heat because evaporation removes larger quantities of heat than was considered in part (a).
71. 20.9 min
73. (a) $3.96 \times 10^{-2} \text{ g}$
(b) 96.2 J
(c) 16.0 W
75. (a) 1.102
(b) $2.79 \times 10^4 \text{ J}$
(c) 12.6 J . This will not cause a significant cooling of the air because it is much less than the energy found in part (b), which is the energy required to warm the air from 20.0°C to 50.0°C .
76. (a) 36°C
(b) Any temperature increase greater than about 3°C would be unreasonably large. In this case the final temperature of the person would rise to 73°C (163°F).
(c) The assumption of 95% heat retention is unreasonable.
78. (a) 1.46 kW
(b) Very high power loss through a window. An electric heater of this power can keep an entire room warm.
(c) The surface temperatures of the window do not differ by as great an amount as assumed. The inner surface will be warmer, and the outer surface will be cooler.

Chapter 15

Problems & Exercises

1. $1.6 \times 10^9 \text{ J}$
3. $-9.30 \times 10^8 \text{ J}$
5. (a) $-1.0 \times 10^4 \text{ J}$, or -2.39 kcal
(b) 5.00%
7. (a) 122 W
(b) $2.10 \times 10^6 \text{ J}$
(c) Work done by the motor is $1.61 \times 10^7 \text{ J}$; thus the motor produces 7.67 times the work done by the man
9. (a) 492 kJ
(b) This amount of heat is consistent with the fact that you warm quickly when exercising. Since the body is inefficient, the excess heat produced must be dissipated through sweating, breathing, etc.
10. $2.09 \times 10^4 \text{ J}$
12. (a) $W = P\Delta V = 1.76 \times 10^5 \text{ J}$
(b) $W = Fd = 1.76 \times 10^5 \text{ J}$. Yes, the answer is the same.
14. $W = 4.5 \times 10^3 \text{ J}$
16. W is not equal to the difference between the heat input and the heat output.

20. (a) 18.5 kJ
(b) 54.1%
22. (a) 1.32×10^9 J
(b) 4.68×10^9 J
24. (a) 3.80×10^9 J
(b) 0.667 barrels
26. (a) 8.30×10^{12} J, which is 3.32% of 2.50×10^{14} J.
(b) -8.30×10^{12} J, where the negative sign indicates a reduction in heat transfer to the environment.
28. 403°C
30. (a) 244°C
(b) 477°C
(c) Yes, since automobiles engines cannot get too hot without overheating, their efficiency is limited.
32. (a) $Eff_1 = 1 - \frac{T_{c,1}}{T_{h,1}} = 1 - \frac{543 \text{ K}}{723 \text{ K}} = 0.249$ or 24.9%
(b) $Eff_2 = 1 - \frac{423 \text{ K}}{543 \text{ K}} = 0.221$ or 22.1%
(c) $Eff_1 = 1 - \frac{T_{c,1}}{T_{h,1}} \Rightarrow T_{c,1} = T_{h,1}(1 - Eff_1)$ similarly, $T_{c,2} = T_{h,2}(1 - Eff_2)$

$$T_{c,2} = T_{h,1}(1 - Eff_1)(1 - Eff_2) \equiv T_{h,1}(1 - Eff_{\text{overall}})$$
using $T_{h,2} = T_{c,1}$ in above equation gives $\therefore (1 - Eff_{\text{overall}}) = (1 - Eff_1)(1 - Eff_2)$

$$Eff_{\text{overall}} = 1 - (1 - 0.249)(1 - 0.221) = 41.5\%$$
(d) $Eff_{\text{overall}} = 1 - \frac{423 \text{ K}}{723 \text{ K}} = 0.415$ or 41.5%
34. The heat transfer to the cold reservoir is $Q_c = Q_h - W = 25 \text{ kJ} - 12 \text{ kJ} = 13 \text{ kJ}$, so the efficiency is
 $Eff = 1 - \frac{Q_c}{Q_h} = 1 - \frac{13 \text{ kJ}}{25 \text{ kJ}} = 0.48$. The Carnot efficiency is $Eff_C = 1 - \frac{T_c}{T_h} = 1 - \frac{300 \text{ K}}{600 \text{ K}} = 0.50$. The actual efficiency is 96% of the Carnot efficiency, which is much higher than the best-ever achieved of about 70%, so her scheme is likely to be fraudulent.
36. (a) -56.3°C
(b) The temperature is too cold for the output of a steam engine (the local environment). It is below the freezing point of water.
(c) The assumed efficiency is too high.
37. 4.82
39. 0.311
41. (a) 4.61
(b) 1.66×10^8 J or 3.97×10^4 kcal
(c) To transfer 1.66×10^8 J, heat pump costs \$1.00, natural gas costs \$1.34.
43. 27.6°C
45. (a) 1.44×10^7 J
(b) 40 cents
(c) This cost seems quite realistic; it says that running an air conditioner all day would cost \$9.59 (if it ran continuously).
47. (a) 9.78×10^4 J/K
(b) In order to gain more energy, we must generate it from things within the house, like a heat pump, human bodies, and other appliances. As you know, we use a lot of energy to keep our houses warm in the winter because of the loss of heat to the outside.
49. 8.01×10^5 J
51. (a) 1.04×10^{31} J/K
(b) 3.28×10^{31} J
53. 199 J/K
55. (a) 2.47×10^{14} J
(b) 1.60×10^{14} J
(c) 2.85×10^{10} J/K
(d) 8.29×10^{12} J

57. It should happen twice in every 1.27×10^{30} s or once in every 6.35×10^{29} s
 $(6.35 \times 10^{29} \text{ s}) \left(\frac{1 \text{ h}}{3600 \text{ s}} \right) \left(\frac{1 \text{ d}}{24 \text{ h}} \right) \left(\frac{1 \text{ y}}{365.25 \text{ d}} \right)$
 $= 2.0 \times 10^{22} \text{ y}$
59. (a) 3.0×10^{29}
 (b) 24%
61. (a) $-2.38 \times 10^{-23} \text{ J/K}$
 (b) 5.6 times more likely
 (c) If you were betting on two heads and 8 tails, the odds of breaking even are 252 to 45, so on average you would break even. So, no, you wouldn't bet on odds of 252 to 45.
63. (b) 7
 (c) 64
 (d) 9.38%
 (e) 3.33 times more likely (20 to 6)

Chapter 16

Problems & Exercises

1. (a) $1.23 \times 10^3 \text{ N/m}$
 (b) 6.88 kg
 (c) 4.00 mm
3. (a) 889 N/m
 (b) 133 N
5. (a) $6.53 \times 10^3 \text{ N/m}$
 (b) Yes
7. 16.7 ms
8. 0.400 s/beats
9. 400 Hz
10. 12,500 Hz
11. 1.50 kHz
12. (a) 93.8 m/s
 (b) $11.3 \times 10^3 \text{ rev/min}$
13. 2.37 N/m
15. 0.389 kg
18. 94.7 kg
21. 1.94 s
22. 6.21 cm
24. 2.01 s
26. 2.23 Hz
28. (a) 2.99541 s
 (b) Since the period is related to the square root of the acceleration of gravity, when the acceleration changes by 1% the period changes by $(0.01)^2 = 0.01\%$ so it is necessary to have at least 4 digits after the decimal to see the changes.
30. (a) Period increases by a factor of 1.41 ($\sqrt{2}$)
 (b) Period decreases to 97.5% of old period
32. Slow by a factor of 2.45
34. length must increase by 0.0116%.
35. (a) 1.99 Hz
 (b) 50.2 cm
 (c) 1.40 Hz, 71.0 cm
36. (a) $3.95 \times 10^6 \text{ N/m}$
 (b) $7.90 \times 10^6 \text{ J}$

37. a). 0.266 m/s
b). 3.00 J
39. $\pm \frac{\sqrt{3}}{2}$
42. 384 J
44. (a). 0.123 m
(b). -0.600 J
(c). 0.300 J. The rest of the energy may go into heat caused by friction and other damping forces.
46. (a) 5.00×10^5 J
(b) 1.20×10^3 s
47. $t = 9.26$ d
49. $f = 40.0$ Hz
51. $v_w = 16.0$ m/s
53. $\lambda = 700$ m
55. $d = 34.0$ cm
57. $f = 4$ Hz
59. 462 Hz,
4 Hz
61. (a) 3.33 m/s
(b) 1.25 Hz
63. 0.225 W
65. 7.07
67. 16.0 d
68. 2.50 kW
70. 3.38×10^{-5} W/m²

Chapter 17

Problems & Exercises

1. 0.288 m
3. 332 m/s
5.
$$v_w = (331 \text{ m/s})\sqrt{\frac{T}{273 \text{ K}}} = (331 \text{ m/s})\sqrt{\frac{293 \text{ K}}{273 \text{ K}}}$$

$$= 343 \text{ m/s}$$
7. 0.223
9. (a) 7.70 m
(b) This means that sonar is good for spotting and locating large objects, but it isn't able to resolve smaller objects, or detect the detailed shapes of objects. Objects like ships or large pieces of airplanes can be found by sonar, while smaller pieces must be found by other means.
11. (a) 18.0 ms, 17.1 ms
(b) 5.00%
(c) This uncertainty could definitely cause difficulties for the bat, if it didn't continue to use sound as it closed in on its prey. A 5% uncertainty could be the difference between catching the prey around the neck or around the chest, which means that it could miss grabbing its prey.
12. 3.16×10^{-4} W/m²
14. 3.04×10^{-4} W/m²
16. 106 dB
18. (a) 93 dB
(b) 83 dB
20. (a) 50.1
(b) 5.01×10^{-3} or $\frac{1}{200}$
22. 70.0 dB
24. 100

26. $1.45 \times 10^{-3} \text{ J}$
28. 28.2 dB
30. (a) 878 Hz
(b) 735 Hz
32. $3.79 \times 10^3 \text{ Hz}$
34. (a) 12.9 m/s
(b) 193 Hz
36. First eagle hears $4.23 \times 10^3 \text{ Hz}$
Second eagle hears $3.56 \times 10^3 \text{ Hz}$
38. 0.7 Hz
40. 0.3 Hz, 0.2 Hz, 0.5 Hz
42. (a) 256 Hz
(b) 512 Hz
44. 180 Hz, 270 Hz, 360 Hz
46. 1.56 m
48. (a) 0.334 m
(b) 259 Hz
50. 3.39 to 4.90 kHz
52. (a) 367 Hz
(b) 1.07 kHz
54. (a) $f_n = n(47.6 \text{ Hz})$, $n = 1, 3, 5, \dots, 419$
(b) $f_n = n(95.3 \text{ Hz})$, $n = 1, 2, 3, \dots, 210$
55. $1 \times 10^6 \text{ km}$
57. 498.5 or 501.5 Hz
59. 82 dB
61. approximately 48, 9, 0, -7, and 20 dB, respectively
63. (a) 23 dB
(b) 70 dB
65. Five factors of 10
67. (a) $2 \times 10^{-10} \text{ W/m}^2$
(b) $2 \times 10^{-13} \text{ W/m}^2$
69. 2.5
71. 1.26
72. 170 dB
74. 103 dB
76. (a) 1.00
(b) 0.823
(c) Gel is used to facilitate the transmission of the ultrasound between the transducer and the patient's body.
78. (a) $77.0 \text{ } \mu\text{m}$
(b) Effective penetration depth = 3.85 cm, which is enough to examine the eye.
(c) $16.6 \text{ } \mu\text{m}$
80. (a) $5.78 \times 10^{-4} \text{ m}$
(b) $2.67 \times 10^6 \text{ Hz}$
82. (a) $v_w = 1540 \text{ m/s} = f\lambda \Rightarrow \lambda = \frac{1540 \text{ m/s}}{100 \times 10^3 \text{ Hz}} = 0.0154 \text{ m} < 3.50 \text{ m}$. Because the wavelength is much shorter than the distance in question, the wavelength is not the limiting factor.
(b) 4.55 ms
84. 974 Hz
(Note: extra digits were retained in order to show the difference.)

Chapter 18

Problems & Exercises

1. (a) 1.25×10^{10}
(b) 3.13×10^{12}
3. -600 C
5. 1.03×10^{12}
7. 9.09×10^{-13}
9. 1.48×10^8 C
11. (a) 0.263 N
(b) If the charges are distributed over some area, there will be a concentration of charge along the side closest to the oppositely charged object. This effect will increase the net force.
13. The separation decreased by a factor of 5.

$$F = k \frac{|q_1 q_2|}{r^2} = ma \Rightarrow a = \frac{kq^2}{mr^2}$$
17.
$$= \frac{(9.00 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(1.60 \times 10^{-19} \text{ m})^2}{(1.67 \times 10^{-27} \text{ kg})(2.00 \times 10^{-9} \text{ m})^2}$$

$$= 3.45 \times 10^{16} \text{ m/s}^2$$
18. (a) 3.2
(b) If the distance increases by 3.2, then the force will decrease by a factor of 10 ; if the distance decreases by 3.2, then the force will increase by a factor of 10. Either way, the force changes by a factor of 10.
20. (a) 1.04×10^{-9} C
(b) This charge is approximately 1 nC, which is consistent with the magnitude of charge typical for static electricity
23. 1.02×10^{-11}
25. a. 0.859 m beyond negative charge on line connecting two charges
b. 0.109 m from lesser charge on line connecting two charges
28. 8.75×10^{-4} N
30. (a) 6.94×10^{-8} C
(b) 6.25 N/C
32. (a) 300 N/C (east)
(b) 4.80×10^{-17} N (east)
42. (a) 2.12×10^5 N/C
(b) one charge of $+q$
44. (a) 0.252 N to the left
(b) $x = 6.07$ cm
46. (a) The electric field at the center of the square will be straight up, since q_a and q_b are positive and q_c and q_d are negative and all have the same magnitude.
(b) 2.04×10^7 N/C (upward)
48. 0.102 N, in the $-y$ direction
50. (a) $\vec{E} = 4.36 \times 10^3$ N/C, 35.0° , below the horizontal.
(b) No
52. (a) 5.58×10^{-11} N/C
(b) the coulomb force is extraordinarily stronger than gravity
54. (a) -6.76×10^5 C
(b) 2.63×10^{13} m/s² (upward)
(c) 2.45×10^{-18} kg
56. The charge q_2 is 9 times greater than q_1 .
69. (a) The forces are balanced in the x-direction, so the net force is vertical. It is composed of the sum of the vertical components of the Coulomb force and the gravitational force.

$$F_y = -2 \left(\frac{kq_1 q_2}{r^2} + G \frac{m_1 m_2}{r^2} \right) \cos 45^\circ =$$

$$-2 \left(\frac{8.99 \times 10^9 (1.00 \times 10^{-9})(2.00 \times 10^{-9})}{8} + 6.67 \times 10^{-11} \frac{(10.0)(10.0)}{8} \right)$$

$$\cos 45^\circ = -4.36 \times 10^{-9} \text{ N}$$

(b) No, it is in a metastable position. Since it cannot move horizontally, it cannot traverse any part of the track.

$$F_x = - \left(\frac{kq_1 q_2}{4} + G \frac{m_1 m_2}{4} \right) \frac{1}{2} + \left(\frac{kq_1 q_2}{12} + G \frac{m_1 m_2}{12} \right) \frac{\sqrt{3}}{2} =$$

$$(c) - \left(\frac{8.99 \times 10^9 (-1.00 \times 10^{-9})(2.00 \times 10^{-9})}{4} \frac{1}{2} - 6.67 \times 10^{-11} \frac{(10.0)(10.0)}{4} \frac{1}{2} \right)$$

$$+ \left(\frac{8.99 \times 10^9 (-1.00 \times 10^{-9})(2.00 \times 10^{-9})}{12} \frac{\sqrt{3}}{2} - 6.67 \times 10^{-11} \frac{(10.0)(10.0)}{12} \frac{\sqrt{3}}{2} \right) \text{N} = 2.60 \times 10^{-9} \text{ N}$$

$$F_y = - \left(\frac{kq_1 q_2}{4} + G \frac{m_1 m_2}{4} \right) \frac{\sqrt{3}}{2} - \left(\frac{kq_1 q_2}{12} + G \frac{m_1 m_2}{12} \right) \frac{1}{2} = -2.08 \times 10^{-9} \text{ N}$$

$$- \left(\frac{8.99 \times 10^9 (1.00 \times 10^{-9})(2.00 \times 10^{-9})}{4} \frac{\sqrt{3}}{2} - 6.67 \times 10^{-11} \frac{(10.0)(10.0)}{4} \frac{\sqrt{3}}{2} \right)$$

$$- \left(\frac{8.99 \times 10^9 (1.00 \times 10^{-9})(2.00 \times 10^{-9})}{12} \frac{1}{2} + 6.67 \times 10^{-11} \frac{(10.0)(10.0)}{12} \frac{1}{2} \right) \text{N} = -6.36 \times 10^{-9} \text{ N}$$

(d) Yes.

(e) It will reach $(1, -\sqrt{3})$ before it will change direction.

(f) The metastable positions where there is no component in one direction are $(0, 0)$, $(4, 0)$, $(2, 2)$, and $(2, -2)$. They number 4.

Chapter 19

Problems & Exercises

1. 42.8
4. $1.00 \times 10^5 \text{ K}$
6. (a) $4 \times 10^4 \text{ W}$
(b) A defibrillator does not cause serious burns because the skin conducts electricity well at high voltages, like those used in defibrillators. The gel used aids in the transfer of energy to the body, and the skin doesn't absorb the energy, but rather lets it pass through to the heart.
8. (a) $7.40 \times 10^3 \text{ C}$
(b) 1.54×10^{20} electrons per second
9. $3.89 \times 10^6 \text{ C}$
11. (a) $1.44 \times 10^{12} \text{ V}$
(b) This voltage is very high. A 10.0 cm diameter sphere could never maintain this voltage; it would discharge.
(c) An 8.00 C charge is more charge than can reasonably be accumulated on a sphere of that size.
15. (a) 3.00 kV
(b) 750 V
17. (a) No. The electric field strength between the plates is $2.5 \times 10^6 \text{ V/m}$, which is lower than the breakdown strength for air ($3.0 \times 10^6 \text{ V/m}$).
(b) 1.7 mm
19. 44.0 mV
21. 15 kV
23. (a) 800 KeV
(b) 25.0 km
24. 144 V
26. (a) 1.80 km

- (b) A charge of 1 C is a very large amount of charge; a sphere of radius 1.80 km is not practical.
28. $-2.22 \times 10^{-13} \text{ C}$
30. (a) $3.31 \times 10^6 \text{ V}$
(b) 152 MeV
32. (a) $2.78 \times 10^{-7} \text{ C}$
(b) $2.00 \times 10^{-10} \text{ C}$
35. (a) $2.96 \times 10^9 \text{ m/s}$
(b) This velocity is far too great. It is faster than the speed of light.
(c) The assumption that the speed of the electron is far less than that of light and that the problem does not require a relativistic treatment produces an answer greater than the speed of light.
46. 21.6 mC
48. 80.0 mC
50. 20.0 kV
52. 667 pF
54. (a) 4.4 μF
(b) $4.0 \times 10^{-5} \text{ C}$
56. (a) 14.2 kV
(b) The voltage is unreasonably large, more than 100 times the breakdown voltage of nylon.
(c) The assumed charge is unreasonably large and cannot be stored in a capacitor of these dimensions.
57. 0.293 μF
59. 3.08 μF in series combination, 13.0 μF in parallel combination
60. 2.79 μF
62. (a) $-3.00 \mu\text{F}$
(b) You cannot have a negative value of capacitance.
(c) The assumption that the capacitors were hooked up in parallel, rather than in series, was incorrect. A parallel connection always produces a greater capacitance, while here a smaller capacitance was assumed. This could happen only if the capacitors are connected in series.
63. (a) 405 J
(b) 90.0 mC
64. (a) 3.16 kV
(b) 25.3 mC
66. (a) $1.42 \times 10^{-5} \text{ C}$, $6.38 \times 10^{-5} \text{ J}$
(b) $8.46 \times 10^{-5} \text{ C}$, $3.81 \times 10^{-4} \text{ J}$
67. (a) $4.43 \times 10^{-9} \text{ F}$
(b) 0.452 V
(c) $4.52 \times 10^{-10} \text{ J}$
70. (a) 133 F
(b) Such a capacitor would be too large to carry with a truck. The size of the capacitor would be enormous.
(c) It is unreasonable to assume that a capacitor can store the amount of energy needed.

Chapter 20

Problems & Exercises

1. 0.278 mA
3. 0.250 A
5. 1.50ms
7. (a) 1.67k Ω
(b) If a 50 times larger resistance existed, keeping the current about the same, the power would be increased by a factor of about 50 (based on the equation $P = I^2 R$), causing much more energy to be transferred to the skin, which could cause serious burns. The gel used reduces the resistance, and therefore reduces the power transferred to the skin.
9. (a) 0.120 C

- (b) 7.50×10^{17} electrons
11. 96.3 s
13. (a) 7.81×10^{14} He⁺⁺ nuclei/s
(b) 4.00×10^3 s
(c) 7.71×10^8 s
15. -1.13×10^{-4} m/s
17. 9.42×10^{13} electrons
18. 0.833 A
20. $7.33 \times 10^{-2} \Omega$
22. (a) 0.300 V
(b) 1.50 V
(c) The voltage supplied to whatever appliance is being used is reduced because the total voltage drop from the wall to the final output of the appliance is fixed. Thus, if the voltage drop across the extension cord is large, the voltage drop across the appliance is significantly decreased, so the power output by the appliance can be significantly decreased, reducing the ability of the appliance to work properly.
24. 0.104 Ω
26. 2.81×10^{-4} m
28. 1.10×10^{-3} A
30. -5°C to 45°C
32. 1.03
34. 0.06%
36. -17°C
38. (a) 4.7 Ω (total)
(b) 3.0% decrease
40. 2.00×10^{12} W
44. (a) 1.50 W
(b) 7.50 W
46. $\frac{V^2}{\Omega} = \frac{V^2}{V/A} = AV = \left(\frac{C}{s}\right)\left(\frac{J}{C}\right) = \frac{J}{s} = 1 \text{ W}$
48. $1 \text{ kW} \cdot \text{h} = \left(\frac{1 \times 10^3 \text{ J}}{1 \text{ s}}\right)(1 \text{ h})\left(\frac{3600 \text{ s}}{1 \text{ h}}\right) = 3.60 \times 10^6 \text{ J}$
50. \$438/y
52. \$6.25
54. 1.58 h
56. \$3.94 billion/year
58. 25.5 W
60. (a) 2.00×10^9 J
(b) 769 kg
62. 45.0 s
64. (a) 343 A
(b) 2.17×10^3 A
(c) 1.10×10^3 A
66. (a) 1.23×10^3 kg
(b) 2.64×10^3 kg
69. (a) 2.08×10^5 A
(b) 4.33×10^4 MW
(c) The transmission lines dissipate more power than they are supposed to transmit.
(d) A voltage of 480 V is unreasonably low for a transmission voltage. Long-distance transmission lines are

kept at much higher voltages (often hundreds of kilovolts) to reduce power losses.

73. 480 V
 75. 2.50 ms
 77. (a) 4.00 kA
 (b) 16.0 MW
 (c) 16.0%
 79. 2.40 kW
 81. (a) 4.0
 (b) 0.50
 (c) 4.0
 83. (a) 1.39 ms
 (b) 4.17 ms
 (c) 8.33 ms
 85. (a) 194 kW
 (b) 880 A
 87. (a) 0.400 mA, no effect
 (b) 26.7 mA, muscular contraction for duration of the shock (can't let go)
 89. $1.20 \times 10^5 \Omega$
 91. (a) 1.00Ω
 (b) 14.4 kW
 93. Temperature increases 860°C . It is very likely to be damaging.
 95. 80 beats/minute

$$\frac{3.25 \times 10^{-3}}{1} \text{ J/s} = \text{Power} = I^2 R = \frac{1}{r^2} \frac{(2.50)^2 (1.72 \times 10^{-8}) (1.25)}{\pi} \text{ Jm}^2/\text{s}$$

97. (a)
$$r = \sqrt{\frac{(2.50)^2 (1.72 \times 10^{-8}) (1.25)}{0.00325 \pi}} = 3.63 \text{ mm}$$

 (b) $\text{Power} = \frac{(2.50)^2 (1.74 \times 10^{-8}) (1.25)}{(3.63 \times 10^{-3})^2 \pi} \text{ J/s}$, so 3.28 J of energy is expended in 1 s. Since α is specified in the chart to only two significant figures, round to 3.3 J.
 (c) Since the energy dissipation is directly proportional to the resistivity, the increase of 0.01% in energy dissipation is when the resistivity increases 1%. Or, $1.01 = 1 + 0.0039 \Delta T$. $T = 23^\circ \text{C}$.

Chapter 21

Problems & Exercises

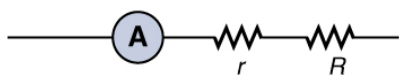
1. (a) $2.75 \text{ k}\Omega$
 (b) 27.5Ω
 3. (a) 786Ω
 (b) 20.3Ω
 5. 29.6 W
 7. (a) 0.74 A
 (b) 0.742 A
 9. (a) 60.8 W
 (b) 3.18 kW

$$R_s = R_1 + R_2$$

 11. (a) $\Rightarrow R_s \approx R_1 (R_1 \gg R_2)$
 (b) $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{R_1 + R_2}{R_1 R_2}$,
 so that
$$R_p = \frac{R_1 R_2}{R_1 + R_2} \approx \frac{R_1 R_2}{R_1} = R_2 (R_1 \gg R_2).$$

 13. (a) $-400 \text{ k}\Omega$

- (b) Resistance cannot be negative.
 (c) Series resistance is said to be less than one of the resistors, but it must be greater than any of the resistors.
14. 2.00 V
 16. 2.9994 V
 18. $0.375\ \Omega$
 21. (a) 0.658 A
 (b) 0.997 W
 (c) 0.997 W; yes
 23. (a) 200 A
 (b) 10.0 V
 (c) 2.00 kW
 (d) $0.1000\ \Omega$; 80.0 A, 4.0 V, 320 W
 25. (a) $0.400\ \Omega$
 (b) No, there is only one independent equation, so only r can be found.
 29. (a) $-0.120\ \text{V}$
 (b) $-1.41 \times 10^{-2}\ \Omega$
 (c) Negative terminal voltage; negative load resistance.
 (d) The assumption that such a cell could provide 8.50 A is inconsistent with its internal resistance.
31. $-I_2 R_2 + \text{emf}_1 - I_2 r_1 + I_3 R_3 + I_3 r_2 - \text{emf}_2 = 0$
 35. $I_3 = I_1 + I_2$
 37. $\text{emf}_2 - I_2 r_2 - I_2 R_2 + I_1 R_5 + I_1 r_1 - \text{emf}_1 + I_1 R_1 = 0$
 39. (a) $I_1 = 4.75\ \text{A}$
 (b) $I_2 = -3.5\ \text{A}$
 (c) $I_3 = 8.25\ \text{A}$
 41. (a) No, you would get inconsistent equations to solve.
 (b) $I_1 \neq I_2 + I_3$. The assumed currents violate the junction rule.
 42. $30\ \mu\text{A}$
 44. $1.98\ \text{k}\Omega$
 46. $1.25 \times 10^{-4}\ \Omega$
 48. (a) $3.00\ \text{M}\Omega$
 (b) $2.99\ \text{k}\Omega$
 50. (a) 1.58 mA
 (b) 1.5848 V (need four digits to see the difference)
 (c) 0.99990 (need five digits to see the difference from unity)
 52. $15.0\ \mu\text{A}$
 54. (a)



- (b) $10.02\ \Omega$
 (c) 0.9980, or a 2.0×10^{-1} percent decrease
 (d) 1.002, or a 2.0×10^{-1} percent increase
 (e) Not significant.
56. (a) $-66.7\ \Omega$
 (b) You can't have negative resistance.
 (c) It is unreasonable that I_G is greater than I_{tot} (see [Figure 21.30](#)). You cannot achieve a full-scale deflection using a current less than the sensitivity of the galvanometer.
57. 24.0 V
 59. $1.56\ \text{k}\Omega$
 61. (a) 2.00 V
 (b) $9.68\ \Omega$

62. Range = $5.00\ \Omega$ to $5.00\ \text{k}\Omega$
 63. range 4.00 to $30.0\ \text{M}\Omega$
 65. (a) $2.50\ \mu\text{F}$
 (b) $2.00\ \text{s}$
 67. 86.5%
 69. (a) $1.25\ \text{k}\Omega$
 (b) $30.0\ \text{ms}$
 71. (a) $20.0\ \text{s}$
 (b) $120\ \text{s}$
 (c) $16.0\ \text{ms}$
 73. $1.73 \times 10^{-2}\ \text{s}$
 74. $3.33 \times 10^{-3}\ \Omega$
 76. (a) $4.99\ \text{s}$
 (b) 3.87°C
 (c) $31.1\ \text{k}\Omega$
 (d) No
 80. (a) $P = \frac{V^2}{R} = \frac{(1.00 \times 10^2)^2}{2.50 \times 10^3}\ \text{W} = 4.00\ \text{W}$
 $\frac{1}{R_{eq}} = \frac{2}{R_1}$
 (b) $R_{eq} = \frac{R_1}{2} = 1.25 \times 10^3\ \Omega$
 $P = \frac{V^2}{R} = \frac{(1.00 \times 10^2)^2}{1.25 \times 10^3}\ \text{W} = 8\ \text{W}$
 $R_{eq} = R_1 + R_2 = 2R_1$
 (c) $P = \frac{V^2}{R} = \frac{(1.00 \times 10^2)^2}{5.00 \times 10^3}\ \text{W} = 2\ \text{W}$

(d) In the parallel case current passes through each branch, so more current is passed through the resistor system. In the series, the higher resistance of the circuit allows less current to flow. $P = IV$ is proportional to the current when the voltage does not change.

(e) The parallel combination delivers more energy in a given time period.

Chapter 22

Problems & Exercises

- (a) Left (West)
 (b) Into the page
 (c) Up (North)
 (d) No force
 (e) Right (East)
 (f) Down (South)
- (a) East (right)
 (b) Into page
 (c) South (down)
- (a) Into page
 (b) West (left)
 (c) Out of page
- $7.50 \times 10^{-7}\ \text{N}$ perpendicular to both the magnetic field lines and the velocity
- (a) $3.01 \times 10^{-5}\ \text{T}$
 (b) This is slightly less than the magnetic field strength of $5 \times 10^{-5}\ \text{T}$ at the surface of the Earth, so it is consistent.
- (a) $6.67 \times 10^{-10}\ \text{C}$ (taking the Earth's field to be $5.00 \times 10^{-5}\ \text{T}$)
 (b) Less than typical static, therefore difficult

12. 4.27 m
14. (a) 0.261 T
(b) This strength is definitely obtainable with today's technology. Magnetic field strengths of 0.500 T are obtainable with permanent magnets.
16. 4.36×10^{-4} m
18. (a) 3.00 kV/m
(b) 30.0 V
20. 0.173 m
22. 7.50×10^{-4} V
24. (a) 1.18×10^3 m/s
(b) Once established, the Hall emf pushes charges one direction and the magnetic force acts in the opposite direction resulting in no net force on the charges. Therefore, no current flows in the direction of the Hall emf. This is the same as in a current-carrying conductor—current does not flow in the direction of the Hall emf.
26. 11.3 mV
28. 1.16 μ V
30. 2.00 T
31. (a) west (left)
(b) into page
(c) north (up)
(d) no force
(e) east (right)
(f) south (down)
33. (a) into page
(b) west (left)
(c) out of page
35. (a) 2.50 N
(b) This is about half a pound of force per 100 m of wire, which is much less than the weight of the wire itself. Therefore, it does not cause any special concerns.
37. 1.80 T
39. (a) 30°
(b) 4.80 N
41. (a) τ decreases by 5.00% if B decreases by 5.00%
(b) 5.26% increase
43. 10.0 A
45. $A \cdot m^2 \cdot T = A \cdot m^2 \left(\frac{N}{A \cdot m} \right) = N \cdot m$.
47. 3.48×10^{-26} N · m
49. (a) 0.666 N · m west
(b) This is not a very significant torque, so practical use would be limited. Also, the current would need to be alternated to make the loop rotate (otherwise it would oscillate).
50. (a) 8.53 N, repulsive
(b) This force is repulsive and therefore there is never a risk that the two wires will touch and short circuit.
52. 400 A in the opposite direction
54. (a) 1.67×10^{-3} N/m
(b) 3.33×10^{-3} N/m
(c) Repulsive
(d) No, these are very small forces
56. (a) Top wire: 2.65×10^{-4} N/m s, 10.9° to left of up
(b) Lower left wire: 3.61×10^{-4} N/m, 13.9° down from right
(c) Lower right wire: 3.46×10^{-4} N/m, 30.0° down from left
58. (a) right-into page, left-out of page
(b) right-out of page, left-into page
(c) right-out of page, left-into page
60. (a) clockwise

- (b) clockwise as seen from the left
(c) clockwise as seen from the right
61. $1.01 \times 10^{13} \text{ T}$
63. (a) $4.80 \times 10^{-4} \text{ T}$
(b) Zero
(c) If the wires are not paired, the field is about 10 times stronger than Earth's magnetic field and so could severely disrupt the use of a compass.
65. 39.8 A
67. (a) $3.14 \times 10^{-5} \text{ T}$
(b) 0.314 T
69. $7.55 \times 10^{-5} \text{ T}$, 23.4°
71. 10.0 A
73. (a) $9.09 \times 10^{-7} \text{ N}$ upward
(b) $3.03 \times 10^{-5} \text{ m/s}^2$
75. 60.2 cm
77. (a) $1.02 \times 10^3 \text{ N/m}^2$
(b) Not a significant fraction of an atmosphere
79. $17.0 \times 10^{-4} \text{ }^\circ\text{C}$
81. 18.3 MHz
83. (a) Straight up
(b) $6.00 \times 10^{-4} \text{ N/m}$
(c) $94.1 \text{ } \mu\text{m}$
(d) $2.47 \text{ } \Omega\text{/m}$, 49.4 V/m
85. (a) 571 C
(b) Impossible to have such a large separated charge on such a small object.
(c) The 1.00-N force is much too great to be realistic in the Earth's field.
87. (a) $2.40 \times 10^6 \text{ m/s}$
(b) The speed is too high to be practical $\leq 1\%$ speed of light
(c) The assumption that you could reasonably generate such a voltage with a single wire in the Earth's field is unreasonable
89. (a) 25.0 kA
(b) This current is unreasonably high. It implies a total power delivery in the line of $50.0 \times 10^9 \text{ W}$, which is much too high for standard transmission lines.
(c) 100 meters is a long distance to obtain the required field strength. Also coaxial cables are used for transmission lines so that there is virtually no field for DC power lines, because of cancellation from opposing currents. The surveyor's concerns are not a problem for his magnetic field measurements.
92. (a) $F = qvB \sin \theta$
Since $\sin \theta = 0$, $F = 0$ for the first electron.
For the second and third electrons, $\sin \theta = 1$.
So, $F = (1.60 \times 10^{-19} \text{ C})(5.00 \times 10^7 \text{ m/s})(2.00 \times 10^{-7} \text{ T}) = 1.60 \times 10^{-18} \text{ N}$.
(b) The first electron moves in a straight line. The other two electrons take a circular path.
(c) The first electron will never return to the origin. The other two will return.
- $$r = \frac{(9.11 \times 10^{-31} \text{ kg})(5.00 \times 10^7 \text{ m/s})}{(1.60 \times 10^{-19} \text{ C})(2.00 \times 10^{-7} \text{ T})} \text{ m}$$
- $$v = \frac{r}{t}$$
- $$t = \frac{(9.11 \times 10^{-31} \text{ kg})}{(1.60 \times 10^{-19} \text{ C})(2.00 \times 10^{-7} \text{ T})} \text{ s} = 2.85 \times 10^{15} \text{ s}$$

Chapter 23

Problems & Exercises

1. Zero

3. (a) CCW
(b) CW
(c) No current induced
5. (a) 1 CCW, 2 CCW, 3 CW
(b) 1, 2, and 3 no current induced
(c) 1 CW, 2 CW, 3 CCW
9. (a) 3.04 mV
(b) As a lower limit on the ring, estimate $R = 1.00 \text{ m}\Omega$. The heat transferred will be 2.31 mJ. This is not a significant amount of heat.
11. 0.157 V
13. proportional to $\frac{1}{r}$
17. (a) 0.630 V
(b) No, this is a very small emf.
19. 2.22 m/s
25. (a) 10.0 N
(b) $2.81 \times 10^8 \text{ J}$
(c) 0.36 m/s
(d) For a week-long mission (168 hours), the change in velocity will be 60 m/s, or approximately 1%. In general, a decrease in velocity would cause the orbit to start spiraling inward because the velocity would no longer be sufficient to keep the circular orbit. The long-term consequences are that the shuttle would require a little more fuel to maintain the desired speed, otherwise the orbit would spiral slightly inward.
28. 474 V
30. 0.247 V
32. (a) 50
(b) yes
34. (a) 0.477 T
(b) This field strength is small enough that it can be obtained using either a permanent magnet or an electromagnet.
36. (a) 5.89 V
(b) At $t=0$
(c) 0.393 s
(d) 0.785 s
38. (a) $1.92 \times 10^6 \text{ rad/s}$
(b) This angular velocity is unreasonably high, higher than can be obtained for any mechanical system.
(c) The assumption that a voltage as great as 12.0 kV could be obtained is unreasonable.
39. (a) $12.00 \text{ }\Omega$
(b) 1.67 A
41. 72.0 V
43. $0.100 \text{ }\Omega$
44. (a) 30.0
(b) $9.75 \times 10^{-2} \text{ A}$
46. (a) 20.0 mA
(b) 2.40 W
(c) Yes, this amount of power is quite reasonable for a small appliance.
48. (a) 0.063 A
(b) Greater input current needed.
50. (a) 2.2
(b) 0.45
(c) 0.20, or 20.0%
52. (a) 335 MV
(b) way too high, well beyond the breakdown voltage of air over reasonable distances
(c) input voltage is too high
54. (a) 15.0 V

- (b) 75.0 A
- (c) yes
- 55. 1.80 mH
- 57. 3.60 V
- 61. (a) 31.3 kV
- (b) 125 kJ
- (c) 1.56 MW
- (d) No, it is not surprising since this power is very high.
- 63. (a) 1.39 mH
- (b) 3.33 V
- (c) Zero
- 65. 60.0 mH
- 67. (a) 200 H
- (b) 5.00°C
- 69. 500 H
- 71. 50.0 Ω
- 73. 1.00×10^{-18} s to 0.100 s
- 75. 95.0%
- 77. (a) 24.6 ms
- (b) 26.7 ms
- (c) 9% difference, which is greater than the inherent uncertainty in the given parameters.
- 79. 531 Hz
- 81. 1.33 nF
- 83. (a) 2.55 A
- (b) 1.53 mA
- 85. 63.7 μ H
- 87. (a) 21.2 mH
- (b) 8.00 Ω
- 89. (a) 3.18 mF
- (b) 16.7 Ω
- 92. (a) 40.02 Ω at 60.0 Hz, 193 Ω at 10.0 kHz
- (b) At 60 Hz, with a capacitor, $Z=531 \Omega$, over 13 times as high as without the capacitor. The capacitor makes a large difference at low frequencies. At 10 kHz, with a capacitor $Z=190 \Omega$, about the same as without the capacitor. The capacitor has a smaller effect at high frequencies.
- 94. (a) 529 Ω at 60.0 Hz, 185 Ω at 10.0 kHz
- (b) These values are close to those obtained in [Example 23.12](#) because at low frequency the capacitor dominates and at high frequency the inductor dominates. So in both cases the resistor makes little contribution to the total impedance.
- 96. 9.30 nF to 101 nF
- 98. 3.17 pF
- 100. (a) 1.31 μ H
- (b) 1.66 pF
- 102. (a) 12.8 k Ω
- (b) 1.31 k Ω
- (c) 31.9 mA at 500 Hz, 312 mA at 7.50 kHz
- (d) 82.2 kHz
- (e) 0.408 A
- 104. (a) 0.159
- (b) 80.9°
- (c) 26.4 W
- (d) 166 W
- 106. 16.0 W
- 107. (a) $|\text{emf}| = 1 \frac{\Delta\phi}{\Delta t} = \frac{1.000(0.02000)}{100} \text{ V} = 0.0002000 \text{ V}$

$$(b) |\text{emf}| = 2 \frac{\Delta\phi}{\Delta t} = \frac{2(0.2500)(0.02000)}{100} \text{ V} = 0.0001000 \text{ V}$$

(c) The average emf is 0 since the loops are made by a wire that crosses, so the voltage from each loop cancels the emf from the other.

(d) No.

Chapter 24

Problems & Exercises

3. 150 kV/m

6. (a) 33.3 cm (900 MHz) 11.7 cm (2560 MHz)

(b) The microwave oven with the smaller wavelength would produce smaller hot spots in foods, corresponding to the one with the frequency 2560 MHz.

8. 26.96 MHz

10. $5.0 \times 10^{14} \text{ Hz}$

$$12. \lambda = \frac{c}{f} = \frac{3.00 \times 10^8 \text{ m/s}}{1.20 \times 10^{15} \text{ Hz}} = 2.50 \times 10^{-7} \text{ m}$$

14. 0.600 m

$$16. (a) f = \frac{c}{\lambda} = \frac{3.00 \times 10^8 \text{ m/s}}{1 \times 10^{-10} \text{ m}} = 3 \times 10^{18} \text{ Hz}$$

(b) X-rays

19. (a) $6.00 \times 10^6 \text{ m}$

(b) $4.33 \times 10^{-5} \text{ T}$

21. (a) $1.50 \times 10^6 \text{ Hz}$, AM band

(b) The resonance of currents on an antenna that is 1/4 their wavelength is analogous to the fundamental resonant mode of an air column closed at one end, since the tube also has a length equal to 1/4 the wavelength of the fundamental oscillation.

23. (a) $1.55 \times 10^{15} \text{ Hz}$

(b) The shortest wavelength of visible light is 380 nm, so that

$$\begin{aligned} \frac{\lambda_{\text{visible}}}{\lambda_{\text{UV}}} &= \frac{380 \text{ nm}}{193 \text{ nm}} \\ &= 1.97. \end{aligned}$$

In other words, the UV radiation is 97% more accurate than the shortest wavelength of visible light, or almost twice as accurate!

25. $3.90 \times 10^8 \text{ m}$

27. (a) $1.50 \times 10^{11} \text{ m}$

(b) $0.500 \mu\text{s}$

(c) 66.7 ns

29. (a) $-3.5 \times 10^2 \text{ W/m}^2$

(b) 88%

(c) $1.7 \mu\text{T}$

$$I = \frac{c\epsilon_0 E_0^2}{2}$$

$$\begin{aligned} 30. &= \frac{(3.00 \times 10^8 \text{ m/s})(8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2)(125 \text{ V/m})^2}{2} \\ &= 20.7 \text{ W/m}^2 \end{aligned}$$

$$32. (a) I = \frac{P}{A} = \frac{P}{\pi r^2} = \frac{0.250 \times 10^{-3} \text{ W}}{\pi(0.500 \times 10^{-3} \text{ m})^2} = 318 \text{ W/m}^2$$

$$\begin{aligned}
 I_{\text{ave}} &= \frac{cB_0^2}{2\mu_0} \Rightarrow B_0 = \left(\frac{2\mu_0 I}{c} \right)^{1/2} \\
 \text{(b)} \quad &= \left(\frac{2(4\pi \times 10^{-7} \text{ T}\cdot\text{m/A})(318.3 \text{ W/m}^2)}{3.00 \times 10^8 \text{ m/s}} \right)^{1/2} \\
 &= 1.63 \times 10^{-6} \text{ T} \\
 E_0 &= cB_0 = (3.00 \times 10^8 \text{ m/s})(1.63 \times 10^{-6} \text{ T}) \\
 \text{(c)} \quad &= 4.90 \times 10^2 \text{ V/m}
 \end{aligned}$$

34. (a) 89.2 cm
 (b) 27.4 V/m
 36. (a) 333 T
 (b) $1.33 \times 10^{19} \text{ W/m}^2$
 (c) 13.3 kJ
 38. (a) $I = \frac{P}{A} = \frac{P}{4\pi r^2} \propto \frac{1}{r^2}$
 (b) $I \propto E_0^2, B_0^2 \Rightarrow E_0^2, B_0^2 \propto \frac{1}{r^2} \Rightarrow E_0, B_0 \propto \frac{1}{r}$
 40. 13.5 pF
 42. (a) 4.07 kW/m²
 (b) 1.75 kV/m
 (c) 5.84 μT
 (d) 2 min 19 s
 44. (a) $5.00 \times 10^3 \text{ W/m}^2$
 (b) $3.88 \times 10^{-6} \text{ N}$
 (c) $5.18 \times 10^{-12} \text{ N}$
 46. (a) $t = 0$
 (b) $7.50 \times 10^{-10} \text{ s}$
 (c) $1.00 \times 10^{-9} \text{ s}$
 48. (a) $1.01 \times 10^6 \text{ W/m}^2$
 (b) Much too great for an oven.
 (c) The assumed magnetic field is unreasonably large.
 50. (a) $2.53 \times 10^{-20} \text{ H}$
 (b) L is much too small.
 (c) The wavelength is unreasonably small.
 $B = \frac{E}{c}$
 53. (a) $B = \frac{1250}{3.00 \times 10^8} \text{ T} = 4.17 \times 10^{-6} \text{ T}$
 (b) E would increase.
 (c) It would be by 1/c.

Chapter 25

Problems & Exercises

- Top 1.715 m from floor, bottom 0.825 m from floor. Height of mirror is 0.890 m, or precisely one-half the height of the person.
- $2.25 \times 10^8 \text{ m/s}$ in water
 $2.04 \times 10^8 \text{ m/s}$ in glycerine
- 1.490, polystyrene
- 1.28 s
- 1.03 ns

13. $n = 1.46$, fused quartz
17. (a) 0.898
(b) Can't have $n < 1.00$ since this would imply a speed greater than c .
(c) Refracted angle is too big relative to the angle of incidence.
19. (a) $\frac{c}{5.00}$
(b) Speed of light too slow, since index is much greater than that of diamond.
(c) Angle of refraction is unreasonable relative to the angle of incidence.
22. 66.3°
24. > 1.414
26. 1.50, benzene
29. 46.5° , red; 46.0° , violet
31. (a) 0.043°
(b) 1.33 m
33. 71.3°
35. 53.5° , red; 55.2° , violet
37. 5.00 to 12.5 D
39. -0.222 m
41. (a) 3.43 m
(b) 0.800 by 1.20 m
42. (a) -1.35 m (on the object side of the lens).
(b) $+10.0$
(c) 5.00 cm
43. 44.4 cm
45. (a) 6.60 cm
(b) -0.500
47. (a) $+7.50$ cm
(b) 13.3 D
(c) Much greater
49. (a) $+6.67$
(b) $+20.0$
(c) The magnification increases without limit (to infinity) as the object distance increases to the limit of the focal distance.
51. -0.933 mm
53. $+0.667$ m
55. (a) -1.5×10^{-2} m
(b) -66.7 D
57. $+0.360$ m (concave)
59. (a) $+0.111$
(b) -0.334 cm (behind "mirror")
(c) 0.752 cm
61. $m = \frac{h_i}{h_o} = -\frac{d_i}{d_o} = -\frac{-d_o}{d_o} = \frac{d_o}{d_o} = 1 \Rightarrow h_i = h_o$
63. 6.82 kW/m^2
65. (a) $v_{610} = \frac{3.00 \times 10^8}{1.530} \text{ m/s}$, $v_{410} = \frac{3.00 \times 10^8}{1.514} \text{ m/s}$
 $3.00 \times 10^8 \left(\frac{1}{1.514} - \frac{1}{1.530} \right) \text{ m/s} = 2.07 \times 10^6 \text{ m/s}$
 (b) No.
 (c) No.
 (d) Yes.

Chapter 26

Problems & Exercises

1. 52.0 D
3. (a) -0.233 mm
(b) The size of the rods and the cones is smaller than the image height, so we can distinguish letters on a page.
5. (a) $+62.5$ D
(b) -0.250 mm
(c) -0.0800 mm
6. 2.00 m
8. (a) ± 0.45 D
(b) The person was nearsighted because the patient was myopic and the power was reduced.
10. 0.143 m
12. 1.00 m
14. 20.0 cm
16. -5.00 D
18. 25.0 cm
20. -0.198 D
22. 30.8 cm
24. -0.444 D
26. (a) 4.00
(b) 1600
28. (a) 0.501 cm
(b) Eyepiece should be 204 cm behind the objective lens.
(c) No, a distance of 205 cm (which is over 2 m) is unrealistic for the tube of a microscope.
30. (a) $+18.3$ cm (on the eyepiece side of the objective lens)
(b) -60.0
(c) -11.3 cm (on the objective side of the eyepiece)
(d) $+6.67$
(e) -400
33. -40.0
35. -167
37. $+10.0$ cm
39. (a) 0.251 μm
(b) Yes, this thickness implies that the shape of the cornea can be very finely controlled, producing normal distant vision in more than 90% of patients.

$$\frac{1}{d_i} + \frac{1}{d_0} = \frac{1}{f} \frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_0} = \frac{1}{4.00 \text{ mm}} - \frac{1}{6.00 \text{ mm}}$$

$$d_i = 12.0 \text{ mm}$$
40. (a) $m_0 = -\frac{d_i}{d_0} = -\frac{12.0}{6.00} = -2.00$

$$\frac{1}{d'_i} + \frac{1}{d'_0} = \frac{1}{f} \frac{1}{d'_i} = \frac{1}{f} - \frac{1}{d'_0} = \frac{1}{50 \text{ mm}} - \frac{1}{13.0 \text{ mm}}$$

$$d'_i = -17.6 \text{ mm}$$

$$m_e = -\frac{-17.6}{13} = 1.35$$

$$m = -2(1.35) = -2.70$$

$$\frac{1}{d_i} + \frac{1}{d_0} = \frac{1}{f} \frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_0} = \frac{1}{4.00 \text{ mm}} - \frac{1}{5.00 \text{ mm}}$$

$$d_i = 20.0 \text{ mm}$$

$$m_0 = -\frac{d_i}{d_o} = -\frac{20.0}{5.00} = -4.00$$

$$\frac{1}{d'_i} + \frac{1}{d'_o} = \frac{1}{f} \quad \frac{1}{d'_i} = \frac{1}{f} - \frac{1}{d'_o} = \frac{1}{50 \text{ mm}} - \frac{1}{5.00 \text{ mm}}$$

$$d'_i = -5.56 \text{ mm}$$

$$m_e = -\frac{-5.56}{5} = 1.11$$

$$m = -4(1.11) = -4.44$$

(b) It increased.

(c) The magnification should decrease.

(d) In both cases the image is inverted.

Chapter 27

Problems & Exercises

1. $1/1.333 = 0.750$
3. 1.49, Polystyrene
5. 0.877 glass to water
6. 0.516°
8. $1.22 \times 10^{-6} \text{ m}$
10. 600 nm
12. 2.06°
14. 1200 nm (not visible)
16. (a) 760 nm
(b) 1520 nm
18. For small angles $\sin \theta - \tan \theta \approx \theta$ (in radians).
For two adjacent fringes we have,
 $d \sin \theta_m = m\lambda$
and
 $d \sin \theta_{m+1} = (m+1)\lambda$
Subtracting these equations gives
 $d(\sin \theta_{m+1} - \sin \theta_m) = [(m+1) - m]\lambda$
 $d(\theta_{m+1} - \theta_m) = \lambda$
 $\tan \theta_m = \frac{y_m}{x} \approx \theta_m \Rightarrow d\left(\frac{y_{m+1}}{x} - \frac{y_m}{x}\right) = \lambda$
 $d\frac{\Delta y}{x} = \lambda \Rightarrow \Delta y = \frac{x\lambda}{d}$
20. 450 nm
21. 5.97°
23. 8.99×10^3
25. 707 nm
27. (a) $11.8^\circ, 12.5^\circ, 14.1^\circ, 19.2^\circ$
(b) $24.2^\circ, 25.7^\circ, 29.1^\circ, 41.0^\circ$
(c) Decreasing the number of lines per centimeter by a factor of x means that the angle for the x -order maximum is the same as the original angle for the first-order maximum.
29. 589.1 nm and 589.6 nm
31. 28.7°
33. 43.2°
35. 90.0°
37. (a) The longest wavelength is 333.3 nm, which is not visible.
(b) 333 nm (UV)
(c) $6.58 \times 10^3 \text{ cm}$
39. $1.13 \times 10^{-2} \text{ m}$
41. (a) 42.3 nm

(b) Not a visible wavelength

The number of slits in this diffraction grating is too large. Etching in integrated circuits can be done to a resolution of 50 nm, so slit separations of 400 nm are at the limit of what we can do today. This line spacing is too small to produce diffraction of light.

43. (a) 33.4°
(b) No
45. (a) 1.35×10^{-6} m
(b) 69.9°
47. 750 nm
49. (a) 9.04°
(b) 12
51. (a) 0.0150°
(b) 0.262 mm
(c) This distance is not easily measured by human eye, but under a microscope or magnifying glass it is quite easily measurable.
53. (a) 30.1°
(b) 48.7°
(c) No
(d) $2\theta_1 = (2)(14.5^\circ) = 29^\circ$, $\theta_2 - \theta_1 = 30.05^\circ - 14.5^\circ = 15.56^\circ$. Thus, $29^\circ \approx (2)(15.56^\circ) = 31.1^\circ$.
55. 23.6° and 53.1°
57. (a) 1.63×10^{-4} rad
(b) 326 ly
59. 1.46×10^{-5} rad
61. (a) 3.04×10^{-7} rad
(b) Diameter of 235 m
63. 5.15 cm
65. (a) Yes. Should easily be able to discern.
(b) The fact that it is just barely possible to discern that these are separate bodies indicates the severity of atmospheric aberrations.
70. 532 nm (green)
72. 83.9 nm
74. 620 nm (orange)
76. 380 nm
78. 33.9 nm
80. 4.42×10^{-5} m
82. The oil film will appear black, since the reflected light is not in the visible part of the spectrum.
84. 45.0°
86. 45.7 mW/m^2
88. 90.0%
90. I_0
92. 48.8°
94. 41.2°
96. (a) 1.92, not diamond (Zircon)
(b) 55.2°
98. $B_2 = 0.707 B_1$
100. (a) $2.07 \times 10^{-2} \text{ }^\circ\text{C/s}$
(b) Yes, the polarizing filters get hot because they absorb some of the lost energy from the sunlight.
101. (a) $\sin \theta = \frac{4.00 \times 10^{-7}}{10^{-6}} = 0.400$ $\theta = 0.400 = 23.6^\circ$
(b) $\sin \theta = \frac{(2)(4.00 \times 10^{-7})}{10^{-6}} = 0.800$ $\theta = 0.800 = 53.1^\circ$
(c) $\sin 23.9^\circ = \frac{x}{10^{-6} \text{ cm}}$ $x = \sin 23.9^\circ (10^{-6}) = 4.05 \times 10^{-7} = 405 \text{ nm}$

$$(d) \sin \theta = \frac{4.00 \times 10^{-7}}{2.00 \times 10^{-6}} = 0.200 \quad \theta = 0.200 = 11.5^\circ$$

(e) No.

Chapter 28

Problems & Exercises

1. (a) 1.0328
(b) 1.15
3. 5.96×10^{-8} s
5. $0.800c$
7. $0.140c$
9. (a) $0.745c$
(b) $0.99995c$ (to five digits to show effect)
11. (a) 0.996
(b) γ cannot be less than 1.
(c) Assumption that time is longer in moving ship is unreasonable.
12. 48.6 m
14. (a) 1.387 km = 1.39 km
(b) 0.433 km
(c) $L = \frac{L_0}{\gamma} = \frac{1.387 \times 10^3 \text{ m}}{3.20} = 433.4 \text{ m} = 0.433 \text{ km}$
Thus, the distances in parts (a) and (b) are related when $\gamma = 3.20$.
16. (a) 4.303 y (to four digits to show any effect)
(b) 0.1434 y
(c) $\Delta t = \gamma \Delta t_0 \Rightarrow \gamma = \frac{\Delta t}{\Delta t_0} = \frac{4.303 \text{ y}}{0.1434 \text{ y}} = 30.0$
Thus, the two times are related when $\gamma = 30.00$.
18. (a) 0.250
(b) γ must be ≥ 1
(c) The Earth-bound observer must measure a shorter length, so it is unreasonable to assume a longer length.
20. (a) $0.909c$
(b) $0.400c$
22. $0.198c$
24. a) 658 nm
b) red
c) $v/c = 9.92 \times 10^{-5}$ (negligible)
26. $0.991c$
28. $-0.696c$
30. $0.01324c$
32. $u' = c$, so
$$u = \frac{v+u'}{1+(vu'/c^2)} = \frac{v+c}{1+(vc/c^2)} = \frac{v+c}{1+(v/c)}$$
$$= \frac{c(v+c)}{c+v} = c$$
34. a) $0.99947c$
b) 1.2064×10^{11} y
c) 1.2058×10^{11} y (all to sufficient digits to show effects)
35. 4.09×10^{-19} kg · m/s
37. (a) $3.000000015 \times 10^{13}$ kg · m/s.
(b) Ratio of relativistic to classical momenta equals 1.000000005 (extra digits to show small effects)
39. 2.9957×10^8 m/s
41. (a) 1.121×10^{-8} m/s
(b) The small speed tells us that the mass of a proton is substantially smaller than that of even a tiny amount

of macroscopic matter!

43. 8.20×10^{-14} J
0.512 MeV
45. 2.3×10^{-30} kg
47. (a) 1.11×10^{27} kg
(b) 5.56×10^{-5}
49. 7.1×10^{-3} kg
 7.1×10^{-3}
The ratio is greater for hydrogen.
51. 208
0.999988c
53. 6.92×10^5 J
1.54
55. (a) 0.914c
(b) The rest mass energy of an electron is 0.511 MeV, so the kinetic energy is approximately 150% of the rest mass energy. The electron should be traveling close to the speed of light.
57. 90.0 MeV
 $E^2 = p^2 c^2 + m^2 c^4 = \gamma^2 m^2 c^4$, so that
59. (a) $p^2 c^2 = (\gamma^2 - 1) m^2 c^4$, and therefore
$$\frac{(pc)^2}{(mc^2)^2} = \gamma^2 - 1$$

(b) yes
61. 1.07×10^3
63. 6.56×10^{-8} kg
 4.37×10^{-10}
65. 0.314c
0.99995c
67. (a) 1.00 kg
(b) This much mass would be measurable, but probably not observable just by looking because it is 0.01% of the total mass.
69. (a) 6.3×10^{11} kg/s
(b) 4.5×10^{10} y
(c) 4.44×10^9 kg
(d) 0.32%
73. (a) $L = 1000.0 \sqrt{1 - \frac{(0.6c)^2}{c^2}} = 800.00$ m
(b) $L = 1000.0 \sqrt{1 - \frac{(0.6c)^2}{c^2}} = 800.00$ m
(c) No.
(d) Yes.

Chapter 29

Problems & Exercises

1. (a) 0.070 eV
(b) 14
3. (a) 2.21×10^{34} J
(b) 2.26×10^{34}
(c) No
4. 263 nm
6. 3.69 eV

8. 0.483 eV
10. 2.25 eV
12. (a) 264 nm
(b) Ultraviolet
14. 1.95×10^6 m/s
16. (a) 4.02×10^{15} /s
(b) 0.256 mW
18. (a) -1.90 eV
(b) Negative kinetic energy
(c) That the electrons would be knocked free.
20. 6.34×10^{-9} eV, 1.01×10^{-27} J
22. 2.42×10^{20} Hz
24.
$$hc = (6.62607 \times 10^{-34} \text{ J} \cdot \text{s})(2.99792 \times 10^8 \text{ m/s}) \left(\frac{10^9 \text{ nm}}{1 \text{ m}} \right) \left(\frac{1.00000 \text{ eV}}{1.60218 \times 10^{-19} \text{ J}} \right)$$

$$= 1239.84 \text{ eV} \cdot \text{nm}$$

$$\approx 1240 \text{ eV} \cdot \text{nm}$$
26. (a) 0.0829 eV
(b) 121
(c) 1.24 MeV
(d) 1.24×10^5
28. (a) 25.0×10^3 eV
(b) 6.04×10^{18} Hz
30. (a) 2.69
(b) 0.371
32. (a) 1.25×10^{13} photons/s
(b) 997 km
34. 8.33×10^{13} photons/s
36. 181 km
38. (a) 1.66×10^{-32} kg · m/s
(b) The wavelength of microwave photons is large, so the momentum they carry is very small.
40. (a) 13.3 μm
(b) 9.38×10^{-2} eV
42. (a) 2.65×10^{-28} kg · m/s
(b) 291 m/s
(c) electron 3.86×10^{-26} J, photon 7.96×10^{-20} J, ratio 2.06×10^6
44. (a) 1.32×10^{-13} m
(b) 9.39 MeV
(c) 4.70×10^{-2} MeV
46. $E = \gamma mc^2$ and $P = \gamma mu$, so

$$\frac{E}{P} = \frac{\gamma mc^2}{\gamma mu} = \frac{c^2}{u}.$$

 As the mass of particle approaches zero, its velocity u will approach c , so that the ratio of energy to momentum in this limit is

$$\lim_{m \rightarrow 0} \frac{E}{P} = \frac{c^2}{c} = c$$

 which is consistent with the equation for photon energy.
48. (a) 3.00×10^6 W
(b) Headlights are way too bright.
(c) Force is too large.
49. 7.28×10^{-4} m
51. 6.62×10^7 m/s
53. 1.32×10^{-13} m

55. (a) 6.62×10^7 m/s
(b) 22.9 MeV
57. 15.1 keV
59. (a) 5.29 fm
(b) 4.70×10^{-12} J
(c) 29.4 MV
61. (a) 7.28×10^{12} m/s
(b) This is thousands of times the speed of light (an impossibility).
(c) The assumption that the electron is non-relativistic is unreasonable at this wavelength.
62. (a) 57.9 m/s
(b) 9.55×10^{-9} eV
(c) From [Table 29.1](#), we see that typical molecular binding energies range from about 1eV to 10 eV, therefore the result in part (b) is approximately 9 orders of magnitude smaller than typical molecular binding energies.
64. 29 nm,
290 times greater
66. 1.10×10^{-13} eV
68. 3.3×10^{-22} s
70. 2.66×10^{-46} kg
72. 0.395 nm
74. (a) 1.3×10^{-19} J
(b) 2.1×10^{23}
(c) 1.4×10^2 s
76. (a) 3.35×10^5 J
(b) 1.12×10^{-3} kg · m/s
(c) 1.12×10^{-3} m/s
(d) 6.23×10^{-7} J
78. (a) 1.06×10^3
(b) 5.33×10^{-16} kg · m/s
(c) 1.24×10^{-18} m
80. (a) 1.62×10^3 m/s
(b) 4.42×10^{-19} J for photon, 1.19×10^{-24} J for electron, photon energy is 3.71×10^5 times greater
(c) The light is easier to make because 450-nm light is blue light and therefore easy to make. Creating electrons with 7.43 μ eV of energy would not be difficult, but would require a vacuum.
81. (a) 2.30×10^{-6} m
(b) 3.20×10^{-12} m
83. 3.69×10^{-4} °C
85. (a) 2.00 kJ
(b) 1.33×10^{-5} kg · m/s
(c) 1.33×10^{-5} N
(d) yes
87. (a) $p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{550 \times 10^{-9}} \text{ kg m/s} = 1.21 \text{ kg} \times 10^{-27} \text{ m/s}$
(b) $p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{650 \times 10^{-9}} \text{ kg m/s} = 1.02 \times 10^{-27} \text{ kg m/s}$
(c) Yes, conservation of momentum applies.
(d) The photon with the longer wavelength has less momentum than the one with the shorter wavelength.

Chapter 30

Problems & Exercises

1. 1.84×10^3
3. 50 km
4. 6×10^{20} kg/m³

6. (a) $10.0\ \mu\text{m}$
 (b) It isn't hard to make one of approximately this size. It would be harder to make it exactly $10.0\ \mu\text{m}$.
7. $\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) \Rightarrow \lambda = \frac{1}{R} \left[\frac{(n_i \cdot n_f)^2}{n_i^2 - n_f^2} \right]; n_i = 2, n_f = 1$, so that
 $\lambda = \left(\frac{\text{m}}{1.097 \times 10^7} \right) \left[\frac{(2 \times 1)^2}{2^2 - 1^2} \right] = 1.22 \times 10^{-7}\ \text{m} = 122\ \text{nm}$, which is UV radiation.
9. $a_B = \frac{h^2}{4\pi^2 m_e k Z q_e^2} = \frac{(6.626 \times 10^{-34}\ \text{J}\cdot\text{s})^2}{4\pi^2 (9.109 \times 10^{-31}\ \text{kg})(8.988 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2)(1)(1.602 \times 10^{-19}\ \text{C})^2} = 0.529 \times 10^{-10}\ \text{m}$
11. $0.850\ \text{eV}$
13. $2.12 \times 10^{-10}\ \text{m}$
15. $365\ \text{nm}$
 It is in the ultraviolet.
17. No overlap
 $365\ \text{nm}$
 $122\ \text{nm}$
19. 7
21. (a) 2
 (b) $54.4\ \text{eV}$
23. $\frac{kZq_e^2}{r_n^2} = \frac{m_e V^2}{r_n}$, so that $r_n = \frac{kZq_e^2}{m_e V^2} = \frac{kZq_e^2}{m_e} \frac{1}{V^2}$. From the equation $m_e v r_n = n \frac{h}{2\pi}$, we can substitute for the velocity, giving: $r_n = \frac{kZq_e^2}{m_e} \cdot \frac{4\pi^2 m_e^2 r_n^2}{n^2 h^2}$ so that $r_n = \frac{n^2}{Z} \frac{h^2}{4\pi^2 m_e k q_e^2} = \frac{n^2}{Z} a_B$, where $a_B = \frac{h^2}{4\pi^2 m_e k q_e^2}$.
25. (a) $0.248 \times 10^{-10}\ \text{m}$
 (b) $50.0\ \text{keV}$
 (c) The photon energy is simply the applied voltage times the electron charge, so the value of the voltage in volts is the same as the value of the energy in electron volts.
27. (a) $100 \times 10^3\ \text{eV}$, $1.60 \times 10^{-14}\ \text{J}$
 (b) $0.124 \times 10^{-10}\ \text{m}$
29. (a) $8.00\ \text{keV}$
 (b) $9.48\ \text{keV}$
30. (a) $1.96\ \text{eV}$
 (b) $(1240\ \text{eV}\cdot\text{nm})/(1.96\ \text{eV}) = 633\ \text{nm}$
 (c) $60.0\ \text{nm}$
32. $693\ \text{nm}$
34. (a) $590\ \text{nm}$
 (b) $(1240\ \text{eV}\cdot\text{nm})/(1.17\ \text{eV}) = 1.06\ \mu\text{m}$
35. $l = 4, 3$ are possible since $l < n$ and $|m_l| \leq l$.
37. $n = 4 \Rightarrow l = 3, 2, 1, 0 \Rightarrow m_l = \pm 3, \pm 2, \pm 1, 0$ are possible.
39. (a) $1.49 \times 10^{-34}\ \text{J}\cdot\text{s}$
 (b) $1.06 \times 10^{-34}\ \text{J}\cdot\text{s}$
41. (a) $3.66 \times 10^{-34}\ \text{J}\cdot\text{s}$
 (b) $s = 9.13 \times 10^{-35}\ \text{J}\cdot\text{s}$
 (c) $\frac{L}{S} = \frac{\sqrt{12}}{\sqrt{3/4}} = 4$
43. $\theta = 54.7^\circ, 125.3^\circ$
44. (a) 32. (b) 2 in s , 6 in p , 10 in d , and 14 in f , for a total of 32.
46. (a) 2
 (b) $3d^9$
48. (b) $n \geq l$ is violated,
 (c) cannot have 3 electrons in s subshell since $3 > (2l + 1) = 2$
 (d) cannot have 7 electrons in p subshell since $7 > (2l + 1) = 2(2 + 1) = 6$
50. (a) The number of different values of m_l is $\pm l, \pm(l - 1), \dots, 0$ for each $l > 0$ and one for $l = 0 \Rightarrow (2l + 1)$. Also

an overall factor of 2 since each m_l can have m_s equal to either $+1/2$ or $-1/2 \Rightarrow 2(2l + 1)$.

(b) for each value of l , you get $2(2l + 1)$

$$= 0, 1, 2, \dots, (n-1) \Rightarrow 2\{[(2)(0)+1]+[(2)(1)+1]+\dots+[(2)(n-1)+1]\} = 2[1+3+\dots+(2n-3)+(2n-1)]$$

$n \text{ terms}$

to see that the expression in the box is $= n^2$, imagine taking $(n-1)$ from the last term and adding it to first term $= 2[1+(n-1)+3+\dots+(2n-3)+(2n-1)-(n-1)] = 2[n+3+\dots+(2n-3)+n]$. Now take $(n-3)$ from penultimate term and add to the second term $2[\underbrace{n+n+\dots+n+n}_{n \text{ terms}}] = 2n^2$.

52. The electric force on the electron is up (toward the positively charged plate). The magnetic force is down (by the RHR).
54. 401 nm
56. (a) 6.54×10^{-16} kg
(b) 5.54×10^{-7} m
58. 1.76×10^{11} C/kg, which agrees with the known value of 1.759×10^{11} C/kg to within the precision of the measurement
60. (a) 2.78 fm
(b) 0.37 of the nuclear radius.
62. (a) 1.34×10^{23}
(b) 2.52 MW
64. (a) 6.42 eV
(b) 7.27×10^{-20} J/molecule
(c) 0.454 eV, 14.1 times less than a single UV photon. Therefore, each photon will evaporate approximately 14 molecules of tissue. This gives the surgeon a rather precise method of removing corneal tissue from the surface of the eye.
66. 91.18 nm to 91.22 nm
68. (a) 1.24×10^{11} V
(b) The voltage is extremely large compared with any practical value.
(c) The assumption of such a short wavelength by this method is unreasonable.

$$E = -\frac{ze^2}{n^2}$$

72. (a) $E_3 - E_2 = -\left(\frac{6^2}{3^2} - \frac{6^2}{2^2}\right)13.6\text{eV} = 68.0\text{eV}$
 $E_3 - E_2 = -\left(\frac{6^2}{2^2} - \frac{6^2}{1^2}\right)13.6\text{eV} = 367\text{eV}$
 $E = 435\text{eV}$
 (b) $E_3 - E_2 = -\left(\frac{6^2}{3^2} - \frac{6^2}{1^2}\right)13.6\text{eV} = 435\text{eV}$
 (c) Yes.

Chapter 31

Problems & Exercises

1. 1.67×10^4
5. $m = \rho V = \rho d^3 \Rightarrow a = \left(\frac{m}{\rho}\right)^{1/3} = \left(\frac{2.3 \times 10^{17} \text{ kg}}{1000 \text{ kg/m}^3}\right)^{1/3}$
 $= 61 \times 10^3 \text{ m} = 61 \text{ km}$
7. 1.9 fm
9. (a) 4.6 fm
(b) 0.61 to 1
11. 85.4 to 1
13. 12.4 GeV
15. 19.3 to 1

17. ${}^3_1\text{H}_2 \rightarrow {}^3_2\text{He}_1 + \beta^- + \bar{\nu}_e$
19. ${}^{50}_{25}\text{Mn}_{25} \rightarrow {}^{50}_{24}\text{Cr}_{26} + \beta^+ + \nu_e$
21. ${}^7_4\text{Be}_3 + e^- \rightarrow {}^7_3\text{Li}_4 + \nu_e$
23. ${}^{210}_{84}\text{Po}_{126} \rightarrow {}^{206}_{82}\text{Pb}_{124} + {}^4_2\text{He}_2$
25. ${}^{137}_{55}\text{Cs}_{82} \rightarrow {}^{137}_{56}\text{Ba}_{81} + \beta^- + \bar{\nu}_e$
27. ${}^{232}_{90}\text{Th}_{142} \rightarrow {}^{228}_{88}\text{Ra}_{140} + {}^4_2\text{He}_2$
29. (a) charge: $(+1)+(-1)=0$; electron family number: $(+1)+(-1)=0$; $A: 0+0=0$
 (b) 0.511 MeV
 (c) The two γ rays must travel in exactly opposite directions in order to conserve momentum, since initially there is zero momentum if the center of mass is initially at rest.
31. $Z=(Z+1)-1$; $A=A$; efn: $0=(+1)+(-1)$
33. $Z-1=Z-1$; $A=A$; efn: $(+1)=(+1)$
35. (a) ${}^{226}_{88}\text{Ra}_{138} \rightarrow {}^{222}_{86}\text{Rn}_{136} + {}^4_2\text{He}_2$
 (b) 4.87 MeV
37. (a) $n \rightarrow p + \beta^- + \bar{\nu}_e$
 (b) 0.783 MeV
39. 1.82 MeV
41. (a) 4.274 MeV
 (b) 1.927×10^{-5}
 (c) Since U-238 is a slowly decaying substance, only a very small number of nuclei decay on human timescales; therefore, although those nuclei that decay lose a noticeable fraction of their mass, the change in the total mass of the sample is not detectable for a macroscopic sample.
43. (a) ${}^{15}_8\text{O}_7 + e^- \rightarrow {}^{15}_7\text{N}_8 + \nu_e$
 (b) 2.754 MeV
44. 57,300 y
46. (a) 0.988 Ci
 (b) The half-life of ${}^{226}\text{Ra}$ is now better known.
48. 1.22×10^3 Bq
50. (a) 16.0 mg
 (b) 0.0114%
52. 1.48×10^{17} y
54. 5.6×10^4 y
56. 2.71 y
58. (a) 1.56 mg
 (b) 11.3 Ci
60. (a) 1.23×10^{-3}
 (b) Only part of the emitted radiation goes in the direction of the detector. Only a fraction of that causes a response in the detector. Some of the emitted radiation (mostly α particles) is observed within the source. Some is absorbed within the source, some is absorbed by the detector, and some does not penetrate the detector.
62. (a) 1.68×10^{-5} Ci
 (b) 8.65×10^{10} J
 (c) $\$2.9 \times 10^3$
64. (a) 6.97×10^{15} Bq
 (b) 6.24 kW
 (c) 5.67 kW
68. (a) 84.5 Ci
 (b) An extremely large activity, many orders of magnitude greater than permitted for home use.
 (c) The assumption of 1.00 μA is unreasonably large. Other methods can detect much smaller decay rates.
69. 1.112 MeV, consistent with graph
71. 7.848 MeV, consistent with graph

73. (a) 7.680 MeV, consistent with graph
 (b) 7.520 MeV, consistent with graph. Not significantly different from value for ^{12}C , but sufficiently lower to allow decay into another nuclide that is more tightly bound.
75. (a) 1.46×10^{-8} u vs. 1.007825 u for ^1H
 (b) 0.000549 u
 (c) 2.66×10^{-5}
76. (a) -9.315 MeV
 (b) The negative binding energy implies an unbound system.
 (c) This assumption that it is two bound neutrons is incorrect.
78. 22.8 cm
79. (a) $^{235}_{92}\text{U}_{143} \rightarrow ^{231}_{90}\text{Th}_{141} + ^4_2\text{He}_2$
 (b) 4.679 MeV
 (c) 4.599 MeV
81. a) 2.4×10^8 u
 (b) The greatest known atomic masses are about 260. This result found in (a) is extremely large.
 (c) The assumed radius is much too large to be reasonable.
82. (a) -1.805 MeV
 (b) Negative energy implies energy input is necessary and the reaction cannot be spontaneous.
 (c) Although all conservation laws are obeyed, energy must be supplied, so the assumption of spontaneous decay is incorrect.

$$r = r_0 A^{1/3} = 1.2(235)^{1/3} = 7.4 \text{ fm}$$
84. (a) $V = \frac{4\pi r^3}{3} = 1700 \text{ fm}^3$

$$\rho = \frac{m}{v} = 0.14 \text{ u/fm}^3$$

 (b) For barium: $r = r_0 A^{1/3} = 1.2(142)^{1/3} = 6.3 \text{ fm}$

$$V = \frac{4\pi r^3}{3} = 1047 \text{ fm}^3$$

$$\rho = \frac{m}{v} = 0.14 \text{ u/fm}^3$$

 For krypton: $r = r_0 A^{1/3} = 1.2(92)^{1/3} = 5.4 \text{ fm}$

$$V = \frac{4\pi r^3}{3} = 660 \text{ fm}^3$$

$$\rho = \frac{m}{v} = 0.14 \text{ u/fm}^3$$

 To two significant figures, they are all alike.

Chapter 32

Problems & Exercises

1. 5.701 MeV
3. $^{99}_{42}\text{Mo}_{57} \rightarrow ^{99}_{43}\text{Tc}_{56} + \beta^- + \bar{\nu}_e$
5. 1.43×10^{-9} g
7. (a) 6.958 MeV
 (b) 5.7×10^{-10} g
8. (a) 100 mSv
 (b) 80 mSv
 (c) ~30 mSv
10. ~2 Gy
12. 1.69 mm
14. 1.24 MeV
16. 7.44×10^8
18. 4.92×10^{-4} Sv
20. 4.43 g
22. 0.010 g

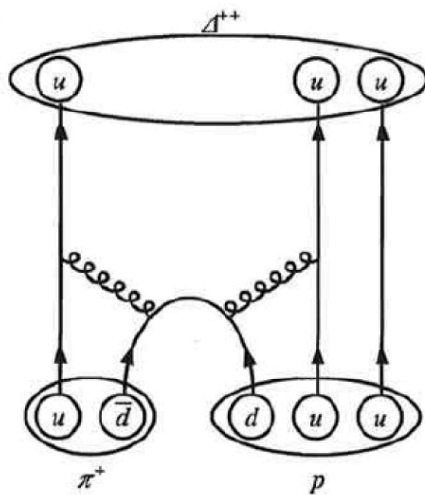
24. 95%
26. (a) $A=1+1=2$, $Z=1+1=1+1$, $efn = 0 = -1 + 1$
 (b) $A=1+2=3$, $Z=1+1=2$, $efn=0=0$
 (c) $A=3+3=4+1+1$, $Z=2+2=2+1+1$, $efn=0=0$
 $E = (m_i - m_f)c^2$
 $= [4m(^1\text{H}) - m(^4\text{He})]c^2$
28. $= [4(1.007825) - 4.002603](931.5 \text{ MeV})$
 $= 26.73 \text{ MeV}$
30. $3.12 \times 10^5 \text{ kg}$ (about 200 tons)
 $E = (m_i - m_f)c^2$
 $E_1 = (1.008665 + 3.016030 - 4.002603)(931.5 \text{ MeV})$
32. $= 20.58 \text{ MeV}$
 $E_2 = (1.008665 + 1.007825 - 2.014102)(931.5 \text{ MeV})$
 $= 2.224 \text{ MeV}$
 ^4He is more tightly bound, since this reaction gives off more energy per nucleon.
34. $1.19 \times 10^4 \text{ kg}$
36. $2e^- + 4^1\text{H} \rightarrow ^4\text{He} + 7\gamma + 2\nu_e$
38. (a) $A=12+1=13$, $Z=6+1=7$, $efn = 0 = 0$
 (b) $A=13=13$, $Z=7=6+1$, $efn = 0 = -1 + 1$
 (c) $A=13 + 1=14$, $Z=6+1=7$, $efn = 0 = 0$
 (d) $A=14 + 1=15$, $Z=7+1=8$, $efn = 0 = 0$
 (e) $A=15=15$, $Z=8=7+1$, $efn = 0 = -1 + 1$
 (f) $A=15 + 1=12 + 4$, $Z=7+1=6 + 2$, $efn = 0 = 0$
40. $E_\gamma = 20.6 \text{ MeV}$
 $E_{^4\text{He}} = 5.68 \times 10^{-2} \text{ MeV}$
42. (a) $3 \times 10^9 \text{ y}$
 (b) This is approximately half the lifetime of the Earth.
43. (a) 177.1 MeV
 (b) Because the gain of an external neutron yields about 6 MeV, which is the average BE/A for heavy nuclei.
 (c) $A = 1 + 238 = 96 + 140 + 1 + 1 + 1$, $Z = 92 = 38 + 53$, $efn = 0 = 0$
45. (a) 180.6 MeV
 (b) $A = 1 + 239 = 96 + 140 + 1 + 1 + 1 + 1$, $Z = 94 = 38 + 56$, $efn = 0 = 0$
47. $^{238}\text{U} + n \rightarrow ^{239}\text{U} + \gamma$ 4.81 MeV
 $^{239}\text{U} \rightarrow ^{239}\text{Np} + \beta^- + \nu_e$ 0.753 MeV
 $^{239}\text{Np} \rightarrow ^{239}\text{Pu} + \beta^- + \nu_e$ 0.211 MeV
49. (a) $2.57 \times 10^3 \text{ MW}$
 (b) $8.03 \times 10^{19} \text{ fission/s}$
 (c) 991 kg
51. 0.56 g
53. 4.781 MeV
55. (a) Blast yields $2.1 \times 10^{12} \text{ J}$ to $8.4 \times 10^{11} \text{ J}$, or 2.5 to 1, conventional to radiation enhanced.
 (b) Prompt radiation yields $6.3 \times 10^{11} \text{ J}$ to $2.1 \times 10^{11} \text{ J}$, or 3 to 1, radiation enhanced to conventional.
57. (a) 1.1×10^{25} fissions, 4.4 kg
 (b) 3.2×10^{26} fusions, 2.7 kg
 (c) The nuclear fuel totals only 6 kg, so it is quite reasonable that some missiles carry 10 overheads. The mass of the fuel would only be 60 kg and therefore the mass of the 10 warheads, weighing about 10 times the nuclear fuel, would be only 1500 lbs. If the fuel for the missiles weighs 5 times the total weight of the warheads, the missile would weigh about 9000 lbs or 4.5 tons. This is not an unreasonable weight for a missile.
59. $7 \times 10^4 \text{ g}$
61. (a) $4.86 \times 10^9 \text{ W}$

- (b) 11.0 y
62. (a) $(-0.500, 2.00, 3.00)$ and $(0.500, 2.00, 3.00)$
 (b) $(0.000, 1.50, 3.00)$ and $(0.000, 2.50, 3.00)$
 (c) They travel in opposite directions along the same line.
 (d) Yes.

Chapter 33

Problems & Exercises

1. 3×10^{-39} s
3. 1.99×10^{-16} m (0.2 fm)
4. (a) 10^{-11} to 1, weak to EM
 (b) 1 to 1
6. (a) 2.09×10^{-5} s
 (b) 4.77×10^4 Hz
8. 78.0 cm
10. 1.40×10^6
12. 100 GeV
13. 67.5 MeV
15. (a) 1×10^{14}
 (b) 2×10^{17}
17. (a) 1671 MeV
 (b) $Q = 1, Q' = 1 + 0 + 0 = 1. L_\tau = -1; L'\tau = -1; L_\mu = 0; L'\mu = -1 + 1 = 0$
 (c) $\tau^- \rightarrow \mu^- + \nu_\mu + \bar{\nu}_\tau$
 $\Rightarrow \mu^-$ antiparticle of μ^+ ; ν_μ of $\bar{\nu}_\mu$; $\bar{\nu}_\tau$ of ν_τ
19. (a) 3.9 eV
 (b) 2.9×10^{-8}
21. (a) The uud composition is the same as for a proton.
 (b) 3.3×10^{-24} s
 (c) Strong (short lifetime)
23. a) $\Delta^{++}(uuu); B = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$
 b)



25. (a) +1
 (b) $B = 1 = 1 + 0, Z = 0 + (-1)$, all lepton numbers are 0 before and after
 (c) $(sss) \rightarrow (uds) + (\bar{u}s)$
27. (a) $(u\bar{u} + d\bar{d}) \rightarrow (u\bar{u} + d\bar{d}) + (u\bar{u} + d\bar{d})$

- (b) 277.9 MeV
(c) 547.9 MeV
29. No. Charge = -1 is conserved. $L_{e_i} = 0 \neq L_{e_f} = 2$ is not conserved. $L_\mu = 1$ is conserved.
31. (a) Yes. $Z = -1 = 0 + (-1)$, $B = 1 = 1 + 0$, all lepton family numbers are 0 before and after, spontaneous since mass greater before reaction.
(b) $dds \rightarrow udd + \bar{u}d$
33. (a) 216
(b) There are more baryons observed because we have the 6 antiquarks and various mixtures of quarks (as for the π -meson) as well.
 $\Omega^+(\bar{s}\bar{s}\bar{s})$
 $B = -\frac{1}{3} - \frac{1}{3} - \frac{1}{3} = -1$,
35. $L_e, \mu, \tau = 0 + 0 + 0 = 0$,
 $Q = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$,
 $S = 1 + 1 + 1 = 3$.
37. (a) 803 MeV
(b) 938.8 MeV
(c) The annihilation energy of an extra electron is included in the total energy.
39. $c\bar{d}$
41. a) The antiproton
b) $\bar{p} \rightarrow \pi^0 + e^-$
43. (a) 5×10^{10}
(b) 5×10^4 particles/m²
45. 2.5×10^{-17} m
47. (a) 33.9 MeV
(b) Muon antineutrino 29.8 MeV, muon 4.1 MeV (kinetic energy)
49. (a) 7.2×10^5 kg
(b) 7.2×10^2 m³
(c) 100 months
53. (a) $2(1.67 \times 10^{-27} \text{ kg}) = 3.34 \times 10^{-27} \text{ kg}$
(b) $3.34 \times 10^{-27} (3.00 \times 10^8)^2 \text{ joules} = 3.01 \times 10^{-10} \text{ joules}$
(c) $3.01 \times 10^{-10} \text{ joules}$
(d) The proton has a positive charge, and the antiproton has a negative charge.

Chapter 34

Problems & Exercises

1. 3×10^{41} kg
3. (a) 3×10^{52} kg
(b) 2×10^{79}
(c) 4×10^{88}
5. 0.30 Gly
7. (a) 2.0×10^5 km/s
(b) $0.67c$
9. 2.7×10^5 m/s
11. 6×10^{-11} (an overestimate, since some of the light from Andromeda is blocked by gas and dust within that galaxy)
13. (a) 2×10^{-8} kg
(b) 1×10^{19}
15. (a) 30 km/s · Mly
(b) 15 km/s · Mly
17. 960 rev/s

19. 89.999773° (many digits are used to show the difference between 90°)
22. 23.6 km
24. (a) 2.95×10^{12} m
(b) 3.12×10^{-4} ly
26. (a) 1×10^{20}
(b) 10 times greater
27. 1.5×10^{15}
29. 0.6 m^{-3}
31. 0.30Ω
32. (a) $v = H_0 d = (20 \text{ km/s)/Mly } (2.00 \times 10^2 \text{ Mly}) = 4.00 \times 10^3 \text{ km/s}$

$$d = \sqrt{(1.00 \times 10^2)^2 + (2.00 \times 10^2)^2} \text{ Mly}$$

 (b) $= 2.24 \times 10^2 \text{ Mly}$

$$v = (20 \text{ km/s)/Mly } (2.24 \times 10^2 \text{ Mly}) = 4.48 \times 10^3 \text{ km/s}$$

 (c) The galaxies are separating mostly due to the expansion of space.
 (d) Light from the third galaxy would have a red shift when detected in either of the other galaxies.

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